

Own-Cluster Reuse Bias in Orthogonal Panel Estimation*

Zixi Zhou[†]

Abstract

Estimating a unit coordinate from the same observations used to evaluate an orthogonal panel score can produce a nonzero asymptotic bias. We study this effect in a partially linear model. For a smooth scalar moment coordinate, the order- T^{-1} score drift has three components: derivative-influence covariance, first-step bias propagation, and curvature. The profiled score makes the first-step-bias component vanish as the nuisance sieve grows. The other two components can shift the $\sqrt{NT}$ limit when N/T converges to a positive constant. We construct feasible bias corrections and standard errors under independent units, deterministic quasi-uniform coordinates, independent uniform covariates, bounded fixed-range dependence, and tensor splines. A regular unique-root result covers smooth scalar moment coordinates; a separate sample-mean result permits a wider sieve window under additional structure. For a sample-mean coordinate and a fixed sieve, buffered splitting removes the covariance component but doubles the leading curvature coefficient under an equal split. Simulations examine both bias mechanisms and feasible inference, including a ratio coordinate with a biased first step.

1 Introduction

Many panel estimators combine two statistically different first steps. Global regression functions are learned from observations pooled across units, while a persistent unit coordinate is learned from repeated observations on the same unit. Such a coordinate may represent baseline risk, productivity, ability, or another persistent latent characteristic measured through repeated proxies. A natural implementation uses all T observations of unit i to estimate its coordinate and then uses those same observations to evaluate the target score. This implementation uses the full unit history. It may also be combined with cross-fitting of the global regression functions. Nevertheless, it creates a dependence that ordinary cross-fitting across units does not remove.

We call this dependence *own-cluster reuse*. The issue is not that the score ceases to be orthogonal with respect to its global nuisance functions. Rather, the point at which the score is evaluated is itself estimated from the same cluster. Without additional cross-unit structure, the coordinate for unit i must be learned from the history of unit i ; leaving out other units does not break the reuse. Within-unit splitting can break it, but only by estimating the coordinate from a shorter history. This paper characterizes the bias that remains when an orthogonal score is evaluated at a unit coordinate estimated from the same cluster, and develops a feasible correction. We study this problem in the partially linear panel model

$$\begin{aligned} Z_{it} &= \alpha_i + \zeta_{it}, \\ D_{it} &= m_0(X_{it}, \alpha_i) + v_{it}, \\ Y_{it} &= \theta_0 D_{it} + h_0(X_{it}, \alpha_i) + \varepsilon_{it}. \end{aligned}$$

The scalar θ_0 is the target. The functions m_0 and h_0 are global nuisances, whereas α_i is a persistent unit coordinate. The measurement equation normalizes α_i as the within-unit conditional mean of Z_{it} , suggesting the sample-mean estimator

$$\check{\alpha}_i = \frac{1}{T} \sum_{s=1}^T Z_{is}, \quad \Delta_i = \check{\alpha}_i - \alpha_i = \frac{1}{T} \sum_{s=1}^T \zeta_{is}.$$

After projecting D_{it} and Y_{it} on a sieve in (X_{it}, α_i) , we form the usual residual-on-residual target score. The profile normal equations remove the first-order effect of estimating the global sieve coefficients. They do not

*This version: September 20, 2026. This work is preliminary and any comments are welcome.

[†]Department of Economics, Duke University. zixi.zhou@duke.edu.

eliminate the second-order drift from evaluating the score at an estimated unit coordinate. In particular, a zero population coordinate derivative need not eliminate the derivative–influence covariance or curvature terms.

The remaining effect is visible from a local expansion. Let K be the sieve dimension, let ρ collect the target and global sieve coefficients, and let $\bar{f}_{i,K}^\theta(\rho, a) := \frac{1}{T} \sum_{t=1}^T f_{it,K}^\theta(\rho, a)$ denote the unit-average target score at coordinate a . Around the true coordinate,

$$\bar{f}_{i,K}^\theta(\rho, \check{\alpha}_i) - \bar{f}_{i,K}^\theta(\rho, \alpha_i) = \partial_a \bar{f}_{i,K}^\theta(\rho, \alpha_i) \Delta_i + \frac{1}{2} \partial_{aa} \bar{f}_{i,K}^\theta(\rho, \alpha_i) \Delta_i^2 + R_{f,iT}^\theta.$$

Here $R_{f,iT}^\theta$ is the third-order remainder. The expectation of the first term is not generally zero. The empirical score derivative and Δ_i are constructed from the same time series, producing a long-run derivative–influence covariance. The second term combines score curvature with the long-run variance of the coordinate noise. Under weak within-unit dependence, both contributions to the score’s mean have order T^{-1} .

This order is asymptotically consequential. Under rectangular sequences with $N/T \rightarrow \kappa_0 \in (0, \infty)$,

$$\sqrt{NT} T^{-1} = \sqrt{N/T} \rightarrow \sqrt{\kappa_0}.$$

Consequently, either channel can generate a nonzero centering term in the $\sqrt{NT}$ limit. The channels may reinforce, have opposite signs and cancel, or vanish separately under additional structure. Thus a small total bias does not establish that own-cluster reuse is harmless: it may conceal two large offsetting components that respond differently to changes in dependence, curvature, or sample splitting.

Our results are not a generic failure theorem for orthogonal estimation. Neyman orthogonality continues to control perturbations of the nuisance directions for which it is imposed (Neyman, 1959; Bickel et al., 1993; Newey, 1994; Chernozhukov et al., 2018, 2022). The problem is more specific: the score is evaluated at a cluster-level random coordinate whose estimation error is shared by every observation in that cluster. Our correction retains the original profiled score and estimates the resulting drift. It should therefore be distinguished from constructing an estimating equation that is orthogonal to a higher order in all nuisance directions.

The paper makes three contributions.

Exact decomposition of own-cluster reuse. We first derive an exact fixed-sieve expansion. It separates the derivative–influence and curvature channels from three other objects that must not be conflated with them: finite-sieve target drift, the quadratic profile remainder, and higher-order score-expansion terms. We distinguish the exact finite- T coefficient for the raw sample-mean coordinate, the corresponding coefficient after clipping, its long-run two-channel limit, and the feasible sample analogue. The fixed-sieve analysis isolates the reuse mechanism without requiring the growing-sieve benchmark used later for feasible inference. For a smooth scalar moment-generated coordinate, the same expansion contains a third term: the bias of the first step multiplied by the population coordinate derivative of the target score. In our partially linear model, that multiplier tends to zero as the sieve approximates the nuisance functions. Thus the two surviving channels are a property of the profiled score, rather than a consequence of an exactly unbiased sample mean. The total population coefficient also respects smooth re-expression of the same generated coordinate and downstream score.

Feasible correction and inference. We estimate the drift using fitted coordinate derivatives of the target score and first-step lag products, and construct a cluster long-run variance estimator that retains the sampling effect of the generated coordinate. The theory has two levels. For the sample mean, we retain the wider growing-sieve window obtained from its exact linear representation and a mixed third-moment restriction. For a smooth scalar moment coordinate with a regular, observably selected unique root, we prove feasible correction and studentization under the sufficient restrictions $K\mathfrak{h}_K^{-4}/T \rightarrow 0$ and $T\mathfrak{h}_K^{\tau_m+\tau_h} \rightarrow 0$, without that moment cancellation. Both results use independent units, deterministic quasi-uniform coordinates, independent uniform covariates, bounded fixed-range dependence, and tensor splines. Subtracting the estimated drift restores centered inference.

What sample splitting does and does not remove. For a sample-mean coordinate and a fixed sieve, we study buffered within-unit splitting because it breaks the dependence between the coordinate-training block and the score-evaluation block. Buffering removes the derivative–influence covariance under fixed-range dependence. It does not remove curvature. Moreover, an equal split uses only half of the time observations to estimate each fold-specific coordinate, doubling its leading variance and hence doubling the leading curvature coefficient. Orthogonalization, within-unit splitting, and second-order bias correction therefore address different components of the problem.

Relation to existing work. Locally robust and double-machine-learning procedures construct moments that are first-order insensitive to global nuisance estimation and commonly use cross-fitting to control empirical dependence (Chernozhukov et al., 2018, 2022). Higher-order influence functions and higher-order orthogonal methods go further by adding higher-order U -statistics or by forcing additional expected nuisance derivatives to vanish (Robins et al., 2008; van der Vaart, 2014; Mackey et al., 2018; Bonhomme et al., 2026a,b). Our procedure is different. It does not replace the target score by a generally higher-order orthogonal score. It estimates the second-order drift induced by a particular cluster-level coordinate-estimation operation, including the first-step-bias term when the moment equation is nonlinear or has a random Jacobian.

The paper is also related to two-step and generated-regressor asymptotics. Classical results characterize how a preliminary estimator changes an influence function or asymptotic variance (Newey, 1984; Pagan, 1984; Murphy and Topel, 1985; Hahn and Ridder, 2013). Later work studies semiparametrically generated covariates and first steps containing many regressors (Mammen et al., 2016; Cattaneo et al., 2019). In our setting, one coordinate error is shared by all T score contributions from a unit. The resulting drift can therefore remain first order even when the sieve dimension is fixed. Closely related recent work studies locally robust GMM with machine-learning-generated regressors under independent sampling and cross-sectional moments with fixed effects estimated from auxiliary panel data (Escanciano and Pérez-Izquierdo, 2026; Huang, 2026). Here the time series used to estimate the coordinate is also the time series entering the target score.

An especially close conceptual precursor is Cattaneo et al. (2013), who decompose the bias of a kernel weighted-average-derivative estimator into a leave-in component, removed by a leave-one-out construction, and a nonlinearity component that remains. That decomposition supplies an analogy rather than an equivalence: our derivative–influence and curvature channels arise because a coordinate error estimated from a dependent unit history is reused in all T score contributions, making the two coefficients serial long-run objects whose drift enters at order T^{-1} . Related panel work develops large- T corrections for two-step fixed-effects estimators and for cross-sectional functionals of unit-specific GMM estimates (Fernández-Val and Vella, 2011; Fernández-Val and Lee, 2013), while Hahn et al. (2024) compare analytical, bootstrap, leave-one-out, and split-sample corrections and their higher-order variances. Freeman and Kristensen (2026) instead combine moments orthogonal to a nonparametric regression with an adjustment that prevents estimated two-way fixed effects from generating incidental-parameter bias. Thus our novelty is not the general observation that first-order orthogonality can leave higher-order bias. It is the exact characterization of the bias induced by estimating and reusing a smooth unit coordinate within the same dependent cluster and the feasible estimation of both serial long-run components while retaining the full unit history. For a sample-mean coordinate and a fixed sieve, buffered splitting removes the derivative–influence component but an equal split doubles the leading curvature coefficient. The analysis also complements panel DML methods that accommodate fixed effects or dependent sampling (Chiang et al., 2022; Clarke and Polselli, 2026), as well as work on inference from noisy unit-level estimates (Jochmans and Weidner, 2024).

Simulation evidence and scope. A fixed-sieve experiment separates the two channels using nuisance functions that lie in the spline span. A second experiment uses nonlinear nuisance functions and a growing sieve to evaluate bias correction and studentized inference. A ratio-coordinate experiment implements the full smooth moment procedure. In that design, the first step is biased but its population multiplier in the profiled score is zero. The simulations also report the accuracy of standard errors and Gaussian calibration across cluster sizes.

The feasible results use a supplied fixed dependence bound and deterministic quasi-uniform coordinates. The smooth moment theorem is scalar and just identified, with a known moment and a regular unique root on a known interval. Data-selected dependence ranges, random coordinate designs, long-range dependence, and nonsmooth or overidentified first steps are outside these results.

Section 2 defines the model, sieve targets, and profiled score. Section 3 develops the three-channel mechanism, its partially linear and sample-mean specializations, and buffered splitting. Sections 4 and 5 give the feasible sample-mean procedure and its wider sieve window. Section 6 establishes feasible inference for regular smooth scalar moment coordinates. Section 7 reports the simulation evidence. The appendices provide the fixed-sieve, spline, localization, correction, and limit arguments for the two procedures.

2 Setup and Orthogonal Moments

2.1 Panel Model and Generated Unit Coordinate

For units $i = 1, \dots, N$ and periods $t = 1, \dots, T$, observe $W_{it} = (Y_{it}, D_{it}, X_{it}, Z_{it})$. Let $\mathcal{C}_i$ be the conditioning σ -field generated by persistent unit-level variables, including α_i , and write $\mathbb{E}_i$ for integration with respect to the conditional law of the i th unit array given $\mathcal{C}_i$. Units are independent in the sense that the sigma-fields $\mathcal{C}_i \vee \sigma(W_{it} : t \in \mathbb{Z})$ are independent across i . Within-unit observations may be serially dependent. We begin with a sample-mean coordinate; Section 6 replaces its measurement equation by a known scalar moment. The baseline model is

$$Z_{it} = \alpha_i + \zeta_{it}. \quad (2.1)$$

$$D_{it} = m_0(X_{it}, \alpha_i) + v_{it}. \quad (2.2)$$

$$Y_{it} = \theta_0 D_{it} + h_0(X_{it}, \alpha_i) + \varepsilon_{it}. \quad (2.3)$$

Here $\theta_0 \in \mathbb{R}$ is the target and $\alpha_i \in \mathbb{R}$ is a persistent unit coordinate. The measurement equation normalizes the scale of α_i . We define $\alpha_i := \mathbb{E}_i[Z_{it}]$. This location normalization gives $\mathbb{E}_i[\zeta_{it}] = 0$ for every t . We impose the conditional-mean restrictions

$$\mathbb{E}[v_{it} \mid X_{it}, \mathcal{C}_i] = 0, \quad \mathbb{E}[\varepsilon_{it} \mid D_{it}, X_{it}, \mathcal{C}_i] = 0. \quad (2.4)$$

Thus persistent information in $\mathcal{C}_i$ beyond α_i may affect higher-order and dynamic features of the unit law, but not the conditional means in (2.2)–(2.3).

For the measurement equation (2.1), the raw sample-mean coordinate is

$$\check{\alpha}_i = \frac{1}{T} \sum_{s=1}^T Z_{is}, \quad \Delta_i := \check{\alpha}_i - \alpha_i = \frac{1}{T} \sum_{s=1}^T \zeta_{is}. \quad (2.5)$$

Fixed- K results use $\hat{\alpha}_i = \check{\alpha}_i$. The growing-sieve result clips $\check{\alpha}_i$ to an interior interval. We keep those two coordinates distinct because clipping changes exact finite- T expectations.

For any array q_{it} , define

$$\mathbb{P}_{NT} q_{it} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T q_{it}, \quad \mathbb{E}_{N,T}[q_{it}] = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_i \left[\frac{1}{T} \sum_{t=1}^T q_{it} \right]. \quad (2.6)$$

Write $\mathcal{C}_N = \sigma(\mathcal{C}_1, \dots, \mathcal{C}_N)$. Conditional on $\mathcal{C}_N$, the unit innovation arrays remain independent. Whenever an integrand contains $\mathcal{C}_N$ -measurable coordinates, coefficients, or targets, $\mathbb{E}_i$ integrates only over the i th conditional innovation law, holding all such $\mathcal{C}_N$ -measurable quantities fixed. Write $\mathbb{P}_i$ for probability under that unit law. For a scalar or Euclidean vector Q , let $\|Q\|_{L_i^r} = (\mathbb{E}_i \|Q\|^r)^{1/r}$, using absolute value for scalars. For an observation array, let $\|Q\|_{NT,2} = (\mathbb{P}_{NT} \|Q_{it}\|^2)^{1/2}$.

2.2 Sieve Targets and Approximation Bias

Let $p_K(x, a) \in \mathbb{R}^K$ be a differentiable basis. The panel-grid projection coefficients satisfy

$$\begin{aligned} \gamma_{0,K,N} &= \arg \min_{\gamma} \mathbb{E}_{N,T} [\{D_{it} - p_K(X_{it}, \alpha_i)^\top \gamma\}^2], \\ \delta_{0,K,N} &= \arg \min_{\delta} \mathbb{E}_{N,T} [\{Y_{it} - p_K(X_{it}, \alpha_i)^\top \delta\}^2]. \end{aligned}$$

Write $m_{K,N}(x, a) = p_K(x, a)^\top \gamma_{0,K,N}$, $g_{K,N}(x, a) = p_K(x, a)^\top \delta_{0,K,N}$. The projection of the structural nuisance is $h_{K,N}^0 = g_{K,N} - \theta_0 m_{K,N}$. In the tensor-spline benchmark, $m_{K,N} = \Pi_{K,N}^g m_0$ and $h_{K,N}^0 = \Pi_{K,N}^g h_0$.

The continuous projection Π_K^c uses Lebesgue measure on $[0, 1]^{d_x+1}$, whereas $\Pi_{K,N}^g$ uses Lebesgue measure in x and the empirical quasi-uniform coordinate design in a . We keep these operators distinct. Appendix C.1 bounds their difference using quadrature.

Let $e_{m,it,K} = m_0(X_{it}, \alpha_i) - m_{K,N}(X_{it}, \alpha_i)$ and $e_{h,it,K} = h_0(X_{it}, \alpha_i) - h_{K,N}^0(X_{it}, \alpha_i)$ denote the projection errors. Define the residuals $u_{D,it,K} = D_{it} - m_{K,N}(X_{it}, \alpha_i)$ and $u_{Y,it,K} = Y_{it} - g_{K,N}(X_{it}, \alpha_i)$. Assume $V_K := \mathbb{E}_{N,T}[u_{D,it,K}^2] > 0$. Whenever the finite-sieve target is used, define $\theta_{0,K,N} = V_K^{-1} \mathbb{E}_{N,T}[u_{D,it,K} u_{Y,it,K}]$ and $\Delta_{\theta,K,N} = \theta_{0,K,N} - \theta_0$.

Proposition 2.1 (Finite-sieve target drift). *Suppose $V_K > 0$ and*

$$\mathbb{E}[v_{it} \mid X_{it}, \mathcal{C}_i] = \mathbb{E}[\varepsilon_{it} \mid X_{it}, \mathcal{C}_i] = 0, \quad \mathbb{E}_i[v_{it} \varepsilon_{it}] = 0.$$

These conditions follow from (2.4). Then

$$\Delta_{\theta,K,N} = \frac{\mathbb{E}_{N,T}[e_{m,it,K} e_{h,it,K}]}{V_K}. \quad (2.7)$$

If $V_K \geq c_V > 0$ and the two grid-projection errors have L^2 rates $O(\mathfrak{h}_K^{\tau_m})$ and $O(\mathfrak{h}_K^{\tau_h})$, then

$$|\Delta_{\theta,K,N}| \leq C \mathfrak{h}_K^{\tau_m + \tau_h}. \quad (2.8)$$

Proof. Substitute $u_D = v + e_m$ and $u_Y = \theta_0 u_D + e_h + \varepsilon$ into the target numerator:

$$\begin{aligned} V_K \Delta_{\theta,K,N} &= \mathbb{E}_{N,T}[u_D(u_Y - \theta_0 u_D)] \\ &= \mathbb{E}_{N,T}[(v + e_m)(e_h + \varepsilon)] \\ &= \mathbb{E}_{N,T}[e_m e_h] + \mathbb{E}_{N,T}[v e_h] + \mathbb{E}_{N,T}[e_m \varepsilon] + \mathbb{E}_{N,T}[v \varepsilon]. \end{aligned}$$

Under the conditional-law convention following (2.6), the projection targets are held fixed and e_m, e_h are functions of $(X_{it}, \mathcal{C}_i)$. Hence

$$\begin{aligned} \mathbb{E}_i[v_{it} e_{h,it,K}] &= \mathbb{E}_i[e_{h,it,K} \mathbb{E}[v_{it} \mid X_{it}, \mathcal{C}_i]] = 0, \\ \mathbb{E}_i[e_{m,it,K} \varepsilon_{it}] &= \mathbb{E}_i[e_{m,it,K} \mathbb{E}[\varepsilon_{it} \mid X_{it}, \mathcal{C}_i]] = 0. \end{aligned}$$

The remaining cross term vanishes by $\mathbb{E}_i[v_{it} \varepsilon_{it}] = 0$. Dividing by V_K proves (2.7). The same orthogonality gives

$$\begin{aligned} V_K &= \mathbb{E}_{N,T}[v^2] + 2 \mathbb{E}_{N,T}[v e_m] + \mathbb{E}_{N,T}[e_m^2] = \mathbb{E}_{N,T}[v^2] + \mathbb{E}_{N,T}[e_m^2], \\ |\Delta_{\theta,K,N}| &\leq V_K^{-1} \{\mathbb{E}_{N,T}[e_m^2]\}^{1/2} \{\mathbb{E}_{N,T}[e_h^2]\}^{1/2} \leq C \mathfrak{h}_K^{\tau_m + \tau_h}, \end{aligned}$$

which proves (2.8). $\square$

2.3 Stacked Moments and the Plug-In Estimator

For $\rho = (\theta, \eta_K^\top)^\top$ with $\eta_K = (\gamma^\top, \delta^\top)^\top$, set $u_{D,it}(\gamma, a) = D_{it} - p_K(X_{it}, a)^\top \gamma$ and $u_{Y,it}(\delta, a) = Y_{it} - p_K(X_{it}, a)^\top \delta$. Write $\rho_{0,K,N} = (\theta_{0,K,N}, \gamma_{0,K,N}^\top, \delta_{0,K,N}^\top)^\top$. The target score and the full $(1 + 2K)$ -dimensional moment vector are

$$\begin{aligned} f_{it,K}^\theta(\rho, a) &= u_{D,it}(\gamma, a) \{u_{Y,it}(\delta, a) - \theta u_{D,it}(\gamma, a)\}, \\ f_{it,K}(\rho, a) &= \begin{pmatrix} f_{it,K}^\theta(\rho, a) \\ p_K(X_{it}, a) u_{D,it}(\gamma, a) \\ p_K(X_{it}, a) u_{Y,it}(\delta, a) \end{pmatrix}. \end{aligned} \quad (2.9)$$

The last two blocks identify the grid projections. At the population coefficients, their normal equations give the nuisance orthogonality

$$\begin{aligned} \partial_\gamma \mathbb{E}_{N,T}[f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i)] &= -\mathbb{E}_{N,T}[p_K u_{Y,K}] + 2\theta_{0,K,N} \mathbb{E}_{N,T}[p_K u_{D,K}] = 0, \\ \partial_\delta \mathbb{E}_{N,T}[f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i)] &= -\mathbb{E}_{N,T}[p_K u_{D,K}] = 0. \end{aligned}$$

Here every basis and residual is evaluated at the true coordinate. The sample counterparts are also the normal equations that produce the exact profile cancellation used in Section 5. For the fixed- K analysis below, let $\hat{\rho}_K$ denote the globally measurable extension constructed in Appendix C.5. On the regular-root

event $\mathcal{G}_K$ it is the profiled solution to $\mathbb{P}_{NT}f_{it,K}(\hat{\rho}_K, \check{\alpha}_i) = 0$; off that event it equals a fixed deterministic vector. Let $\hat{\theta}_K$ be its first coordinate. Section 4 uses the same construction at the clipped coordinate for the growing-sieve analysis.

Define the coordinate derivatives and score residual by

$$\begin{aligned} p_{a,it,K}(a) &= \partial_a p_K(X_{it}, a), & p_{aa,it,K}(a) &= \partial_{aa} p_K(X_{it}, a), \\ q_{it}(\rho, a) &= u_{Y,it}(\delta, a) - \theta u_{D,it}(\gamma, a). \end{aligned}$$

Then

$$\dot{f}_{it,K}^\theta(\rho, a) = -\{p_{a,it,K}(a)^\top \gamma\} q_{it}(\rho, a) - u_{D,it}(\gamma, a) p_{a,it,K}(a)^\top (\delta - \theta \gamma). \quad (2.10)$$

$$\begin{aligned} \ddot{f}_{it,K}^\theta(\rho, a) &= -\{p_{aa,it,K}(a)^\top \gamma\} q_{it}(\rho, a) - u_{D,it}(\gamma, a) p_{aa,it,K}(a)^\top (\delta - \theta \gamma) \\ &\quad + 2\{p_{a,it,K}(a)^\top \gamma\} \{p_{a,it,K}(a)^\top (\delta - \theta \gamma)\}. \end{aligned} \quad (2.11)$$

The first and second derivatives generate the derivative–influence and curvature channels, respectively.

3 Generated-Coordinate Bias

Profile orthogonality removes first-order perturbations of the global sieve coefficients. A smooth generated coordinate also contributes its own first-order influence and second-order bias. We first separate the resulting three score terms, then show how the partially linear structure and the sample-mean specialization each reduce them.

3.1 Smooth Scalar Moment Coordinates and Three Bias Channels

A sample mean is one way to generate the unit coordinate. More generally, let a known scalar moment ψ identify α_i through $\mathbb{E}_i[\psi(W_{i0}, \alpha_i)] = 0$, and consider a regular root of

$$\Psi_{iT}(a) = \frac{1}{T} \sum_{t=1}^T \psi(W_{it}, a) = 0. \quad (3.1)$$

We call this a *smooth scalar moment-generated coordinate*. The moment is just identified and contains no additional estimated nuisance. Section 6 specifies a fully observable unique-root rule; the local branch in Appendix B.1 suffices for the fixed-sieve population calculation.

Define the population Jacobian, Hessian, and first-order coordinate influence by

$$G_i = \mathbb{E}_i[\partial_a \psi(W_{i0}, \alpha_i)], \quad H_i^\psi = \mathbb{E}_i[\partial_{aa} \psi(W_{i0}, \alpha_i)], \quad \xi_{it} = -G_i^{-1} \psi(W_{it}, \alpha_i).$$

Thus $\mathbb{E}_i[\xi_{it}] = 0$. For a supplied dependence bound L , put

$$\begin{aligned} \Omega_{\xi,i} &= \sum_{|\ell| \leq L} \mathbb{E}_i[\xi_{i0} \xi_{i\ell}], \\ C_{G\xi,i} &= \sum_{|\ell| \leq L} \mathbb{E}_i[(\partial_a \psi(W_{i0}, \alpha_i) - G_i) \xi_{i\ell}]. \end{aligned}$$

The first-step bias coefficient is

$$\beta_i = -G_i^{-1} \left(C_{G\xi,i} + \frac{1}{2} H_i^\psi \Omega_{\xi,i} \right). \quad (3.2)$$

The two terms arise from the random sample Jacobian and the curvature of the first-step moment, respectively. For example, if $\psi((U, R), a) = U - aR$ and R is bounded away from zero, then

$$\begin{aligned} \check{\alpha}_i &= \frac{\bar{U}_i}{\bar{R}_i}, & \xi_{it} &= \frac{U_{it} - \alpha_i R_{it}}{\mathbb{E}_i[R_{i0}]}, \\ \beta_i &= -\frac{\sum_{|\ell| \leq L} \text{Cov}_i(R_{i0}, U_{i\ell} - \alpha_i R_{i\ell})}{\mathbb{E}_i[R_{i0}]^2}. \end{aligned}$$

The first-step moment is affine in a , but its random Jacobian can still produce a nonzero β_i .

The expansion proved in Lemma F.1 is

$$\hat{\alpha}_i - \alpha_i = \bar{\xi}_i + \frac{\beta_i}{T} + q_{iT}^c + r_{iT}^M, \quad \mathbb{E}_i[q_{iT}^c] = 0, \quad (3.3)$$

where $\bar{\xi}_i = T^{-1} \sum_t \xi_{it}$, $\|q_{iT}^c\|_{L_i^r} = O(T^{-1})$, and $\|r_{iT}^M\|_{L_i^r} = O(T^{-3/2})$ for $2 \leq r \leq 8$. The centered second-order term cannot be discarded within a unit. Its average across independent units is negligible at the $\sqrt{NT}$ scale.

Let $\mu_{\dot{f},i,K}$ and $\mu_{\ddot{f},i,K}$ be the expectations of the first and second derivatives in (2.10)–(2.11) at $(\rho_{0,K,N}, \alpha_i)$. The resulting population coefficient is

$$\begin{aligned} B_{\theta,K,N}^M &= B_{\text{reuse},K,N}^M + B_{\text{first},K,N}^M + B_{\text{curv},K,N}^M, \\ B_{\text{reuse},K,N}^M &= \frac{1}{N} \sum_{i=1}^N \sum_{|\ell| \leq L} \mathbb{E}_i[\dot{f}_{i0,K}^\theta(\rho_{0,K,N}, \alpha_i) \xi_{i\ell}], \\ B_{\text{first},K,N}^M &= \frac{1}{N} \sum_{i=1}^N \mu_{\dot{f},i,K} \beta_i, \\ B_{\text{curv},K,N}^M &= \frac{1}{2N} \sum_{i=1}^N \mu_{\ddot{f},i,K} \Omega_{\xi,i}. \end{aligned} \quad (3.4)$$

Lemma F.2 proves that this coefficient approximates the exact clipped-coordinate drift with error $O(T^{-1/2})$, uniformly along the growing-sieve sequences specified below. The same Taylor calculation applies to any scalar downstream score with uniformly bounded first three coordinate derivatives and the stated finite-range dependence. Appendix B.1 gives the fixed-sieve version under local identification.

The first-step channel in the partially linear score. The coefficient multiplying β_i is small in this model. Write $e_m(x, a) = m_0(x, a) - m_{K,N}(x, a)$ and $e_h(x, a) = h_0(x, a) - h_{K,N}^0(x, a)$. Under either growing-sieve benchmark in the paper, the product rule and centering of v and ε give

$$\mu_{\dot{f},i,K} = \mathbb{E}_X[-\partial_a m_{K,N} e_h - e_m \partial_a h_{K,N}^0 + 2\Delta_{\theta,K,N} e_m \partial_a m_{K,N}], \quad (3.5)$$

where the functions inside the expectation are evaluated at (X, α_i) . Uniform projection approximation and boundedness of β_i give

$$\begin{aligned} \sup_i |\mu_{\dot{f},i,K}| &\leq C \mathfrak{h}_K^{\min\{\tau_m, \tau_h\}}, \\ |B_{\text{first},K,N}^M| &\leq \sup_i |\mu_{\dot{f},i,K}| \frac{1}{N} \sum_i |\beta_i| \leq C \mathfrak{h}_K^{\min\{\tau_m, \tau_h\}} \rightarrow 0. \end{aligned}$$

Thus reuse and curvature remain the two possible asymptotic drift channels even for a biased smooth moment coordinate. At finite K , and for other downstream scores, the first-step channel need not vanish. The feasible procedure retains all three terms.

Coordinate changes. The total coefficient respects a smooth change of coordinate. Let $\alpha_i^* = h(\alpha_i)$ and $\hat{\alpha}_i^* = h(\hat{\alpha}_i)$, where h is a C^3 diffeomorphism on a compact neighborhood containing the coordinate paths, with bounded derivatives and $|h'|$ bounded away from zero. Write $h'_i = h'(\alpha_i)$ and $h''_i = h''(\alpha_i)$. Taylor's formula applied to (3.3) gives

$$\begin{aligned} \hat{\alpha}_i^* - \alpha_i^* &= h'_i(\hat{\alpha}_i - \alpha_i) + \frac{1}{2} h''_i(\hat{\alpha}_i - \alpha_i)^2 + O_{L_i^2}(T^{-3/2}), \\ \xi_{it}^* &= h'_i \xi_{it}, \quad \Omega_{\xi,i}^* = (h'_i)^2 \Omega_{\xi,i}, \quad \beta_i^* = h'_i \beta_i + \frac{1}{2} h''_i \Omega_{\xi,i}. \end{aligned}$$

The last identity uses $T \mathbb{E}_i[\hat{\alpha}_i - \alpha_i] = \beta_i + O(T^{-1/2})$ and $T \mathbb{E}_i[(\hat{\alpha}_i - \alpha_i)^2] = \Omega_{\xi,i} + O(T^{-1/2})$, obtained from the root expansion and the finite-window covariance identity. For the same scalar score expressed as

$f_{it}^*(a^*) = f_{it}(h^{-1}(a^*))$, the chain rule gives, at the true coordinate,

$$\partial_{a^*} f_{it}^* = \frac{\partial_a f_{it}}{h'_i}, \quad \partial_{a^* a^*} f_{it}^* = \frac{\partial_{aa} f_{it}}{(h'_i)^2} - \frac{(\partial_a f_{it}) h''_i}{(h'_i)^3}.$$

The reuse term is unchanged because $\mathbb{E}_i[(\partial_{a^*} f_{i0}^*) \xi_{i\ell}^*] = \mathbb{E}_i[(\partial_a f_{i0}) \xi_{i\ell}]$. Writing $\lambda_i = \mathbb{E}_i[\partial_a f_{i0}]$ and $\mu_{H,i} = \mathbb{E}_i[\partial_{aa} f_{i0}]$, the other two terms add to

$$\begin{aligned} \frac{\lambda_i}{h'_i} \beta_i^* + \frac{1}{2} \left\{ \frac{\mu_{H,i}}{(h'_i)^2} - \frac{\lambda_i h''_i}{(h'_i)^3} \right\} \Omega_{\xi,i}^* &= \lambda_i \beta_i + \frac{\lambda_i h''_i}{2 h'_i} \Omega_{\xi,i} + \frac{1}{2} \mu_{H,i} \Omega_{\xi,i} - \frac{\lambda_i h''_i}{2 h'_i} \Omega_{\xi,i} \\ &= \lambda_i \beta_i + \frac{1}{2} \mu_{H,i} \Omega_{\xi,i}. \end{aligned}$$

Thus the total bias is unchanged. Starting from a sample mean sets $\beta_i = 0$ and shows why a nonlinear re-expression generally requires the first-step channel. This identity concerns the same score and generated coordinate. Choosing a new spline space after transformation changes the finite-sieve projection and is a different operation.

3.2 Two Bias Channels for a Sample-Mean Coordinate

For fixed K , recall the derivative means

$$\mu_{\tilde{f},i,K} = \mathbb{E}_i[\dot{f}_{i0,K}^\theta(\rho_{0,K,N}, \alpha_i)], \quad \mu_{\tilde{f},i,K} = \mathbb{E}_i[\ddot{f}_{i0,K}^\theta(\rho_{0,K,N}, \alpha_i)].$$

Also set $\Omega_{\zeta,i} = \sum_{\ell \in \mathbb{Z}} \mathbb{E}_i[\zeta_{i0} \zeta_{i\ell}]$. The long-run channels for the raw sample-mean coordinate are

$$\begin{aligned} B_{\zeta,\theta,K,N} &= \frac{1}{N} \sum_{i=1}^N \sum_{\ell \in \mathbb{Z}} \mathbb{E}_i[\dot{f}_{i0,K}^\theta(\rho_{0,K,N}, \alpha_i) \zeta_{i\ell}]. \\ B_{\zeta\zeta,\theta,K,N} &= \frac{1}{2N} \sum_{i=1}^N \mu_{\tilde{f},i,K} \Omega_{\zeta,i}, \\ B_{\theta,K,N}^\infty &= \underbrace{B_{\zeta,\theta,K,N}}_{\text{derivative-influence}} + \underbrace{B_{\zeta\zeta,\theta,K,N}}_{\text{curvature}}. \end{aligned} \tag{3.6}$$

The exact raw finite- T counterpart is

$$\mathcal{B}_{\theta,K,N,T}^{\text{raw}} = T \mathbb{E}_{N,T} [f_{it,K}^\theta(\rho_{0,K,N}, \tilde{\alpha}_i) - f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i)].$$

The fixed- K decomposition is based on a local Taylor expansion and finite-range moment bounds under the following regularity conditions.

Assumption 3.1 (Fixed-sieve sample-mean regularity). *The following conditions hold for a common fixed $\delta > 0$ and a common compact interval $\mathcal{A}$. Write $I_i = [\alpha_i - \delta, \alpha_i + \delta]$.*

(i) Sampling and dependence. *For fixed K , the unit arrays are independent across i . Conditional on $\mathcal{C}_i$, the joint time array containing ζ_{it} , the full stacked score map $a \mapsto f_{it,K}(\rho_{0,K,N}, a)$ on $\mathcal{A}$, its first three coordinate derivatives on I_i , and the map $a \mapsto p_K(X_{it}, a)$ with its first coordinate derivative on $\mathcal{A}$ is strictly stationary and ℓ_0 -dependent for a fixed ℓ_0 .*

(ii) Local smoothness. *The variables in part (i) are uniformly bounded on their displayed domains, the highest displayed coordinate derivative is uniformly Lipschitz on I_i , and $Z_{it} \in \mathcal{A}$ almost surely. Moreover, $I_i \subset \mathcal{A}$ uniformly in i .*

(iii) Coordinate localization. *The generated coordinate is the raw sample mean in (2.5). The true coordinates remain in a compact interior set, and there are constants $C, c > 0$, independent of i, N , and T , such that*

$$\sup_{1 \leq i \leq N} \mathbb{P}(\tilde{\alpha}_i \notin I_i | \mathcal{C}_i) \leq C \exp(-cT). \tag{3.7}$$

The local conditions in Assumption 3.1 specialize Assumption B.1 by setting $\psi(W_{it}, a) = Z_{it} - a$, $G_i = -1$, and $\xi_{it} = \zeta_{it}$; local identification then holds automatically. The bounded score envelope on $\mathcal{A}$ controls the exponentially rare event on which a raw block mean leaves I_i .

Proposition 3.2 (Two bias channels for a sample-mean coordinate). *Suppose K is fixed and Assumption 3.1 holds with constants uniform in i and N . As $T \rightarrow \infty$, allowing $N = N_T$,*

$$\mathcal{B}_{\theta,K,N,T}^{\text{raw}} = B_{\theta,K,N}^{\infty} + o(1). \quad (3.8)$$

Thus the exact finite- T expectation and the long-run two-channel coefficient agree after multiplication by T . The proposition concerns population centering. The next subsection adds the centered sampling fluctuation needed for the estimator expansion.

Appendix B.2 proves the proposition. For $\psi(W, a) = Z - a$, both the random-Jacobian fluctuation and the moment Hessian vanish. Thus the two second-order first-step components in Section 3.1 are identically zero before clipping. Section 6 treats smooth scalar moment coordinates whose first-step bias need not vanish.

3.3 Consequences for the Uncorrected Estimator

Define the centered theta influence

$$x_{\theta,it,K} = f_{it,K}^{\theta}(\rho_{0,K,N}, \alpha_i) - \mathbb{E}_i[f_{i0,K}^{\theta}(\rho_{0,K,N}, \alpha_i)] + \mu_{f,i,K} \zeta_{it}, \quad \psi_{\theta,it,K} = V_K^{-1} x_{\theta,it,K}. \quad (3.9)$$

The final term in $x_{\theta,it,K}$ is the first-order sampling effect of estimating α_i . It is distinct from the mean reuse bias in (3.6): the influence term contributes to variance, while the two channel coefficients determine centering.

Proposition 3.3 (Centered fixed-sieve sampling expansion). *Suppose K is fixed, Assumption 3.1 holds, and $N, T \rightarrow \infty$ with $N/T \rightarrow \kappa \in (0, \infty)$. Then*

$$\mathbb{P}_{NT} f_{it,K}^{\theta}(\rho_{0,K,N}, \check{\alpha}_i) - \mathbb{E}_{N,T}[f_{it,K}^{\theta}(\rho_{0,K,N}, \check{\alpha}_i)] = \mathbb{P}_{NT} x_{\theta,it,K} + o_p((NT)^{-1/2}). \quad (3.10)$$

The next proposition inserts this centered score expansion into the profiled estimator and isolates the resulting noncentral drift.

Proposition 3.4 (First-order bias of the uncorrected estimator). *Suppose K is fixed, Assumptions 3.1 and B.10 hold, $N, T \rightarrow \infty$ with $N/T \rightarrow \kappa \in (0, \infty)$, and let $\hat{\rho}_K$ be the globally reported root defined in Appendix C.5. Then*

$$\sqrt{NT}(\hat{\theta}_K - \theta_{0,K,N}) = \frac{1}{\sqrt{NT}} \sum_{i=1}^N \sum_{t=1}^T \psi_{\theta,it,K} + \sqrt{\frac{N}{T}} V_K^{-1} B_{\theta,K,N}^{\infty} + o_p(1). \quad (3.11)$$

Thus an $O(T^{-1})$ coordinate drift is first order when $(NT)^{-1/2} \asymp T^{-1}$.

Appendix B.4 proves Propositions 3.3 and 3.4.

Example 3.5 (A nonzero derivative-influence channel). *Let $D_{it} = v_{it}$ and $Y_{it} = \theta_0 v_{it} + \alpha_i + \varepsilon_{it}$. Suppose the sieve reproduces constants and $a \mapsto a$. If (v_{it}, ζ_{it}) is jointly stationary MA(1), ε_{it} is independent of it, and $\Omega_{v\zeta} = \sum_{\ell} \mathbb{E}[v_{i0} \zeta_{i\ell}] \neq 0$, then at the oracle parameter, $f_{it,K}^{\theta}(\rho_{0,K,N}, a) = v_{it}(\varepsilon_{it} + \alpha_i - a)$, $\dot{f}_{it,K}^{\theta} = -v_{it}$, and $\ddot{f}_{it,K}^{\theta} = 0$. Hence $B_{\theta,K,N}^{\infty} = -\Omega_{v\zeta}$. In particular, $\mathbb{E}[\dot{f}_{it,K}^{\theta}] = 0$ does not force the derivative-influence channel to vanish: the relevant object is its covariance with the coordinate influence.*

3.4 What Buffered Splitting and Cross-Fitting Remove

To isolate what splitting changes, we analyze a fixed-sieve population comparison using two contiguous folds separated by a fixed buffer wider than the dependence range.

Let ℓ_0 be the fixed dependence range in Assumption 3.1. Define

$$\mathcal{I}_{1,T} = \{1, \dots, \lfloor T/2 \rfloor - \ell_0\}, \quad \mathcal{I}_{2,T} = \{\lfloor T/2 \rfloor + \ell_0 + 1, \dots, T\}.$$

With sizes T_1, T_2 , let $T_o = T_1 + T_2$. For evaluation on fold j , estimate the coordinate on the other fold:

$$\check{\alpha}_i^{(-1)} = \frac{1}{T_2} \sum_{s \in \mathcal{I}_{2,T}} Z_{is}, \quad \check{\alpha}_i^{(-2)} = \frac{1}{T_1} \sum_{s \in \mathcal{I}_{1,T}} Z_{is}.$$

At the fixed-sieve target, define the two-fold cross-fitted score average

$$\bar{f}_{i,K}^{\theta,\text{cf}} = \frac{1}{T_o} \left\{ \sum_{t \in \mathcal{I}_{1,T}} f_{it,K}^{\theta}(\rho_{0,K,N}, \check{\alpha}_i^{(-1)}) + \sum_{t \in \mathcal{I}_{2,T}} f_{it,K}^{\theta}(\rho_{0,K,N}, \check{\alpha}_i^{(-2)}) \right\}.$$

and let $\bar{f}_{i,K}^{\theta,\text{cf},0}$ replace both generated coordinates by α_i .

Proposition 3.6 (Buffered splitting removes reuse covariance, not curvature). *Suppose K is fixed and Assumption 3.1 holds. Then the first-order derivative terms in the two folds have mean zero exactly, whereas*

$$T \frac{1}{N} \sum_{i=1}^N \mathbb{E}_i \left[\bar{f}_{i,K}^{\theta,\text{cf}} - \bar{f}_{i,K}^{\theta,\text{cf},0} \right] = 2B_{\zeta\zeta,\theta,K,N} + o(1). \quad (3.12)$$

More generally, a one-way buffered split whose coordinate-training block has size $T_{\text{tr}}/T \rightarrow \lambda \in (0,1)$ has coefficient $\lambda^{-1}B_{\zeta\zeta,\theta,K,N} + o(1)$. Thus buffered splitting or cross-fitting removes the derivative-influence channel but does not remove curvature. An equal split doubles the curvature coefficient because the coordinate is estimated from only half of the periods.

Appendix B.3 proves the proposition. Proposition 3.6 shows precisely what splitting accomplishes: it eliminates reuse covariance but amplifies curvature under equal folds. The full-sample two-channel correction in (4.2) removes both channels without reducing the coordinate-training sample.

4 Bias-Corrected Inference

For the sample-mean coordinate, we first construct a globally measurable root and then define feasible bias and variance estimators relative to it. The profile identity and corrected-estimator decomposition used in the analysis appear in Section 5.

4.1 Profile Estimation and Acceptance

Both coordinate constructions use the same candidate profile regression. At generated coordinates $\hat{\alpha}_i$, set $\hat{p}_{it,K} = p_K(X_{it}, \hat{\alpha}_i)$ and $\hat{\Sigma}_{p,K} = \mathbb{P}_{NT} \hat{p}_{it,K} \hat{p}_{it,K}^{\top}$. For positive thresholds $c_{\Sigma,T}, c_{V,T}$, compute

$$\begin{aligned} \gamma_K^{\text{pre}} &= \hat{\Sigma}_{p,K}^{-1} \mathbb{P}_{NT} \hat{p}_{it,K} D_{it}, & \delta_K^{\text{pre}} &= \hat{\Sigma}_{p,K}^{-1} \mathbb{P}_{NT} \hat{p}_{it,K} Y_{it}, \\ u_{D,it}^{\text{pre}} &= D_{it} - \hat{p}_{it,K}^{\top} \gamma_K^{\text{pre}}, & u_{Y,it}^{\text{pre}} &= Y_{it} - \hat{p}_{it,K}^{\top} \delta_K^{\text{pre}}, \\ V_K^{\text{pre}} &= \mathbb{P}_{NT} (u_{D,it}^{\text{pre}})^2 \end{aligned} \quad (4.1)$$

when $\lambda_{\min}(\hat{\Sigma}_{p,K}) \geq c_{\Sigma,T}$; otherwise set $V_K^{\text{pre}} = 0$. Accept the candidate when this Gram test and $V_K^{\text{pre}} \geq c_{V,T}$ both hold. On acceptance, report the candidate coefficients and

$$\hat{\theta}_K = (V_K^{\text{pre}})^{-1} \mathbb{P}_{NT} u_{D,it}^{\text{pre}} u_{Y,it}^{\text{pre}}.$$

The sample-mean procedure uses fixed thresholds $c_{\Sigma,T} = c_{\Sigma}$ and $c_{V,T} = c_V/2$, chosen in (5.9); its acceptance event is $\mathcal{G}_K$. On rejection, report a fixed deterministic vector $\rho_K^{\dagger} = (\theta_K^{\dagger}, \gamma_K^{\dagger\top}, \delta_K^{\dagger\top})^{\top}$. Using the reported coefficients on the whole sample space, define $\hat{u}_{D,it,K} = D_{it} - \hat{p}_{it,K}^{\top} \hat{\gamma}_K$, $\hat{u}_{Y,it,K} = Y_{it} - \hat{p}_{it,K}^{\top} \hat{\delta}_K$, and $\hat{V}_K = \mathbb{P}_{NT} \hat{u}_{D,it,K}^2$. The correction and variance formulas below apply on $\mathcal{G}_K$ and receive their stated defaults otherwise. Appendix C.5 proves that acceptance has probability approaching one. The smooth moment procedure in Section 6 instead uses declining thresholds and its own rejection defaults; these finite-sample conventions remain distinct.

4.2 Estimating the Two Sample-Mean Bias Channels

For an integer ℓ , let $I_T(\ell) = \{t : 1 \leq t \leq T, 1 \leq t + \ell \leq T\}$. Let $L_{B,T}$ be the correction cutoff and let k_B be even. Set $\hat{\zeta}_{it} = Z_{it} - \hat{\alpha}_i$. Let $\hat{f}_{it,K}$ and $\hat{\dot{f}}_{it,K}$ denote the corresponding derivatives evaluated at $(\hat{\rho}_K, \hat{\alpha}_i)$, and

let $\widehat{\widehat{f}}_{i,K} = \frac{1}{T} \sum_t \widehat{f}_{it,K}$. The feasible channel estimators are

$$\begin{aligned}\widehat{B}_{\zeta,\theta,K} &= \frac{1}{NT} \sum_i \sum_{|\ell| \leq L_{B,T}} k_B(\ell/L_{B,T}) \sum_{t \in I_T(\ell)} \widehat{f}_{it,K} \widehat{\zeta}_{i,t+\ell}, \\ \widehat{\Omega}_{\zeta,i} &= \frac{1}{T} \sum_{|\ell| \leq L_{B,T}} k_B(\ell/L_{B,T}) \sum_{t \in I_T(\ell)} \widehat{\zeta}_{it} \widehat{\zeta}_{i,t+\ell}, \\ \widehat{B}_{\zeta\zeta,\theta,K} &= \frac{1}{2N} \sum_i \widehat{\widehat{f}}_{i,K} \widehat{\Omega}_{\zeta,i}, \quad \widehat{B}_{\theta,K} = \widehat{B}_{\zeta,\theta,K} + \widehat{B}_{\zeta\zeta,\theta,K}.\end{aligned}$$

The corrected estimator is defined on the good event by

$$\widehat{\theta}_K^c = \widehat{\theta}_K - \frac{1}{T} \widehat{V}_K^{-1} \widehat{B}_{\theta,K}, \quad (4.2)$$

and equals the deterministic default $\theta_K^\dagger$ otherwise. Since $\partial_\theta f^\theta = -u_D^2$, a positive score drift raises the uncorrected root by $\frac{1}{T} V_K^{-1} B_{\theta,K}$ and must be subtracted.

4.3 Cluster Variance Estimation for the Corrected Score

Let $L_{V,T}$ be the variance cutoff. It is distinct from $L_{B,T}$. On $\mathcal{G}_K$, define $\widehat{\widehat{f}}_{i,K} = \frac{1}{T} \sum_t \widehat{f}_{it,K}$, $\widehat{\varphi}_{\theta,it,K} = f_{it,K}^\theta(\widehat{\rho}_K, \widehat{\alpha}_i) + \widehat{\widehat{f}}_{i,K} \widehat{\zeta}_{it}$, $\widehat{\chi}_{\theta,it,K} = \widehat{V}_K^{-1} \widehat{\varphi}_{\theta,it,K}$, and $\widetilde{\chi}_{\theta,it,K} = \widehat{\chi}_{\theta,it,K} - \frac{1}{T} \sum_s \widehat{\chi}_{\theta,is,K}$. Then

$$\widehat{\Gamma}_{\theta,i,K}(\ell) = \frac{1}{T} \sum_{t \in I_T(\ell)} \widetilde{\chi}_{\theta,it,K} \widetilde{\chi}_{\theta,i,t+\ell,K}, \quad \widehat{\sigma}_{\theta,K}^2 = \frac{1}{N} \sum_i \sum_{|\ell| \leq L_{V,T}} k_V(\ell/L_{V,T}) \widehat{\Gamma}_{\theta,i,K}(\ell).$$

The added term $\widehat{\widehat{f}}_{i,K} \widehat{\zeta}_{it}$ estimates the sampling effect of the generated coordinate in (3.9). The within-unit demeaning in $\widetilde{\chi}_{\theta,it,K}$ removes the estimated cluster mean before the fixed-lag covariance calculation. On $\mathcal{G}_K$, set $\widehat{\sigma}_{\theta,K} = \sqrt{\widehat{\sigma}_{\theta,K}^2}$ when the squared estimate is positive and set it equal to 1 otherwise. On $\mathcal{G}_K^c$, set $\widehat{\widehat{f}}_{i,K} = \widehat{\varphi}_{\theta,it,K} = \widehat{\chi}_{\theta,it,K} = \widetilde{\chi}_{\theta,it,K} = 0$, $\widehat{\Gamma}_{\theta,i,K}(\ell) = 0$, and $\widehat{\sigma}_{\theta,K}^2 = \widehat{\sigma}_{\theta,K} = 1$.

For its finite- T population target, write $\chi_{\theta,it,K} = V_K^{-1} x_{\theta,it,K}$ and define

$$\sigma_{\theta,K,N}^2 = \frac{1}{N} \sum_{i=1}^N \sum_{|\ell| \leq L_{V,T}} \left(1 - \frac{|\ell|}{T}\right) \text{Cov}_i(\chi_{\theta,i0,K}, \chi_{\theta,i\ell,K}).$$

Section 5.1 shows how the bias correction and variance estimator enter the limiting distribution.

5 Theoretical Guarantees

For the clipped sample-mean coordinate under the tensor-spline fixed-range benchmark, the results below establish exact correction identities, accommodate a nonzero derivative-influence channel, prove consistency of the bias and variance estimators, and yield centered studentized inference under a common rate window.

5.1 Exact Profile and Correction Identities

Let $e_\gamma = \widehat{\gamma}_K - \gamma_{0,K,N}$, $e_\delta = \widehat{\delta}_K - \delta_{0,K,N}$, and $e_\pi = e_\delta - \theta_{0,K,N} e_\gamma$.

Lemma 5.1 (Exact profile identity). *For any generated-coordinate array, on an event where the profile normal equations hold and $\widehat{V}_K > 0$,*

$$\mathbb{P}_{NT} f_{it,K}^\theta(\rho_{0,K,N}, \widehat{\alpha}_i) = \widehat{V}_K (\widehat{\theta}_K - \theta_{0,K,N}) + e_\gamma^\top \widehat{\Sigma}_{p,K} e_\pi. \quad (5.1)$$

For the sample-mean procedure this event is $\mathcal{G}_K$; for the smooth moment procedure it is $\mathcal{O}_T$, defined in (6.3). Multiplication by its indicator gives the corresponding unconditional identity.

Proof. We first express the target-coefficient residuals in terms of the estimated residuals and coefficient errors. We then derive the four normal-equation cancellations available on the event in the lemma. Expanding the target score produces six products; those cancellations remove four of them and leave the denominator term and the quadratic profile term in the lemma.

Define the target-coefficient residuals at the generated coordinate, $\tilde{u}_{D,it,K} = D_{it} - \hat{p}_{it,K}^\top \gamma_{0,K,N}$ and $\tilde{u}_{Y,it,K} = Y_{it} - \hat{p}_{it,K}^\top \delta_{0,K,N}$. They satisfy

$$\begin{aligned}\tilde{u}_{D,it,K} &= \hat{u}_{D,it,K} + \hat{p}_{it,K}^\top e_\gamma, \\ \tilde{u}_{Y,it,K} - \theta_{0,K,N} \tilde{u}_{D,it,K} &= \hat{q}_{it,K} + (\hat{\theta}_K - \theta_{0,K,N}) \hat{u}_{D,it,K} + \hat{p}_{it,K}^\top e_\pi,\end{aligned}\tag{5.2}$$

where $\hat{q}_{it,K} = \hat{u}_{Y,it,K} - \hat{\theta}_K \hat{u}_{D,it,K}$.

The profile normal equations imply

$$\mathbb{P}_{NT} \hat{p}_{it,K} \hat{u}_{D,it,K} = 0, \quad \mathbb{P}_{NT} \hat{p}_{it,K} \hat{u}_{Y,it,K} = 0, \quad \mathbb{P}_{NT} \hat{u}_{D,it,K} \hat{q}_{it,K} = 0.$$

The first two equations also imply

$$\mathbb{P}_{NT} \hat{p}_{it,K} \hat{q}_{it,K} = \mathbb{P}_{NT} \hat{p}_{it,K} \hat{u}_{Y,it,K} - \hat{\theta}_K \mathbb{P}_{NT} \hat{p}_{it,K} \hat{u}_{D,it,K} = 0.$$

The target score is $\mathbb{P}_{NT} \tilde{u}_{D,it,K} (\tilde{u}_{Y,it,K} - \theta_{0,K,N} \tilde{u}_{D,it,K})$. Substituting (5.2) and expanding every product gives

$$\begin{aligned}\mathbb{P}_{NT} f_{it,K}^\theta(\rho_{0,K,N}, \hat{\alpha}_i) &= \mathbb{P}_{NT} \hat{u}_{D,it,K} \hat{q}_{it,K} + (\hat{\theta}_K - \theta_{0,K,N}) \mathbb{P}_{NT} \hat{u}_{D,it,K}^2 + e_\pi^\top \mathbb{P}_{NT} \hat{p}_{it,K} \hat{u}_{D,it,K} \\ &\quad + e_\gamma^\top \mathbb{P}_{NT} \hat{p}_{it,K} \hat{q}_{it,K} + (\hat{\theta}_K - \theta_{0,K,N}) e_\gamma^\top \mathbb{P}_{NT} \hat{p}_{it,K} \hat{u}_{D,it,K} \\ &\quad + e_\gamma^\top (\mathbb{P}_{NT} \hat{p}_{it,K} \hat{p}_{it,K}^\top) e_\pi.\end{aligned}$$

The first, third, fourth, and fifth terms are zero by the displayed normal equations. The second term equals $\hat{V}_K(\hat{\theta}_K - \theta_{0,K,N})$, and the sixth equals $e_\gamma^\top \hat{\Sigma}_{p,K} e_\pi$. These are exactly the two terms in (5.1). $\square$

For the clipped coordinate, define

$$\mathcal{B}_{\theta,K}^{\text{cl}} = T \mathbb{E}_{N,T} [f_{it,K}^\theta(\rho_{0,K,N}, \hat{\alpha}_i) - f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i)].\tag{5.3}$$

Appendix D.1 proves the exact score expansion

$$\mathbb{P}_{NT} f_{it,K}^\theta(\rho_{0,K,N}, \hat{\alpha}_i) = \bar{Z}_{\theta,K} + \frac{1}{T} \mathcal{B}_{\theta,K}^{\text{cl}} + r_{\theta,F,K},\tag{5.4}$$

where $\bar{Z}_{\theta,K} = \mathbb{P}_{NT} x_{\theta,it,K}$ and $\sqrt{NT} |r_{\theta,F,K}| = o_p(1)$ under the rate window below.

To expose each term, substitute the score expansion (5.4) into the left side of the profile identity (5.1), solve for $\hat{\theta}_K - \theta_{0,K,N}$, and then subtract the correction in (4.2). On $\mathcal{G}_K$, these three operations give

$$\hat{\theta}_K^c - \theta_{0,K,N} = \underbrace{\hat{V}_K^{-1} \bar{Z}_{\theta,K}}_{\text{sampling}} + \underbrace{\hat{V}_K^{-1} r_{\theta,F,K}}_{\text{score-expansion remainder}} - \underbrace{\hat{V}_K^{-1} e_\gamma^\top \hat{\Sigma}_{p,K} e_\pi}_{\text{profile remainder}} + \underbrace{\frac{1}{T} \hat{V}_K^{-1} (\mathcal{B}_{\theta,K}^{\text{cl}} - \hat{B}_{\theta,K})}_{\text{correction error}}.\tag{5.5}$$

Under the rate window below, the last three terms are $o_p((NT)^{-1/2})$, whereas the sampling term satisfies a cluster central limit theorem.

Four bias objects remain distinct: the exact raw-coordinate coefficient $\mathcal{B}_{\theta,K,N,T}^{\text{raw}}$, the exact clipped-coordinate coefficient $\mathcal{B}_{\theta,K}^{\text{cl}}$, the long-run coefficient $B_{\theta,K,N}^\infty$, and its feasible counterpart $\hat{B}_{\theta,K}$.

5.2 Benchmark Conditions and Sieve Rates

We now impose a sufficient benchmark under which the fixed-sieve mechanism can be turned into feasible growing-sieve inference. The assumptions play three distinct roles. The coordinate design and spline conditions control projection and grid quadrature; bounded fixed-range dependence supplies the uniform concentration

bounds; and the mixed third-moment restriction simplifies the second-order nuisance expansion used for the wider sample-mean sieve window. It is not needed for the existence of the two leading scalar-score channels. Section 6 establishes feasible inference without this cancellation under a smaller sieve window. The fixed-sieve reuse-bias decomposition itself does not depend on this benchmark or on the particular spline construction.

Let $W^+ = [0, 1]^{d_X} \times [0, 1]$, let $d_W = d_X + 1$ be fixed, and set $b_T = C_b \sqrt{\log N/T}$.

Assumption 5.2 (Growing-sieve conditions). *The following conditions hold.*

(i) **Coordinate design and covariates.** *The coordinates are deterministic, and the benchmark has no additional persistent unit-level randomness: $C_i = \sigma(\alpha_i)$. Thus $\mathbb{E}_i$ integrates over the common covariate and disturbance law specified below. Let $\alpha_{(1)} \leq \dots \leq \alpha_{(N)}$ be their ordered values and let F_N be their empirical distribution function. All coordinates lie in $[2b_T, 1 - 2b_T]$. If F_T^I denotes the distribution function of the uniform law on $[2b_T, 1 - 2b_T]$, then, for fixed constants $0 < c_\alpha < C_\alpha < \infty$,*

$$\begin{aligned} \frac{c_\alpha}{N} &\leq \alpha_{(j+1)} - \alpha_{(j)} \leq \frac{C_\alpha}{N}, \quad j = 1, \dots, N-1, \\ \sup_{a \in \mathbb{R}} |F_N(a) - F_T^I(a)| &\leq \frac{C_\alpha}{N}. \end{aligned} \tag{5.6}$$

The variables in $\{X_{it}\}$ are independent and uniformly distributed on $[0, 1]^{d_X}$ and are independent of the coordinate design and all disturbance processes.

(ii) **Independent units and within-unit dependence.** *Fix an integer $q_0 \geq 1$ that does not vary with (N, T) and write $\Lambda_{q_0} = \{-q_0, \dots, q_0\}$. Viewed as random elements indexed by $\mathbb{Z}$, the joint disturbance histories $\{(\zeta_{it}, v_{it}, \varepsilon_{it}) : t \in \mathbb{Z}\}$ are independent across i and have the same law. The bivariate process $\{(\zeta_{it}, v_{it}) : t \in \mathbb{Z}\}$ is bounded, centered, strictly stationary, and q_0 -dependent: for every integer s , the σ -fields generated by dates $t \leq s$ and $t \geq s + q_0 + 1$ are independent. The process $\{\varepsilon_{it} : t \in \mathbb{Z}\}$ is bounded, centered, strictly stationary, and q_0 -dependent, and is independent of the entire (ζ, v) process. Assume $\sigma_v^2 = \mathbb{E}[v_{it}^2] > 0$ and $0 < \text{Var}(\varepsilon_{it}) < \infty$. Assume the mixed third-moment restriction*

$$\mathbb{E}[v_{i0}\zeta_{i\ell_1}\zeta_{i\ell_2}] = 0, \quad (\ell_1, \ell_2) \in \mathbb{Z}^2. \tag{5.7}$$

(iii) **Generated coordinate.** *The data satisfy (2.1)–(2.3). The raw mean $\check{\alpha}_i$ is clipped as $\hat{\alpha}_i = \Pi_{[b_T, 1-b_T]}(\check{\alpha}_i)$. Write $d_i = \hat{\alpha}_i - \alpha_i$ and $r_i^{\text{cl}} = d_i - \Delta_i$. The finite-range Hoeffding constant c_{fm} satisfies $c_{\text{fm}} C_b^2 > 1$.*

(iv) **Smoothness and sieve.** *The functions satisfy $m_0 \in C^{\tau_m}(W^+)$ and $h_0 \in C^{\tau_h}(W^+)$ with $3 < \tau_m, \tau_h \leq p + 1$. The basis is an open-clamped, maximum-smoothness tensor-product B-spline basis of degree $p \geq 4$, with quasi-uniform positive knot spans and Lebesgue- L^2 normalization. Write $p_K(x, a) = \psi_{K_x}(x) \otimes \phi_{K_a}(a)$, where $K = K_x K_a$, $K_x \asymp \mathfrak{h}_K^{-d_X}$, and $K_a \asymp \mathfrak{h}_K^{-1}$. When $d_X = 0$, take $K_x = 1$. Thus $K \asymp \mathfrak{h}_K^{-d_W}$.*

(v) **Asymptotic growth.**

$$\begin{aligned} \frac{N}{T} &\rightarrow \kappa_0 \in (0, \infty), & N\mathfrak{h}_K &\rightarrow \infty, \\ \frac{b_T}{\mathfrak{h}_K} &\rightarrow 0, & \frac{K \log K}{NT} &\rightarrow 0. \end{aligned} \tag{5.8}$$

Tuning convention. For the sample-mean growing-sieve results and their supporting appendices, use $L_{B,T} = L_{V,T} = q_0$ for the correction and variance estimators and set $k_B(\ell/q_0) = k_V(\ell/q_0) = 1$ for $\ell \in \Lambda_{q_0}$. Both kernels vanish at every other integer lag. Thus the theorem treats a fixed upper bound q_0 as supplied to the procedure. Selecting the dependence range from the data is outside the result; the simulations use the known MA(1) bound.

Let $c_{\text{gr}} > 0$ denote the uniform lower eigenvalue constant for the population grid Gram established in Lemma C.10. Choose the deterministic estimator thresholds so that

$$0 < c_\Sigma < c_{\text{gr}}/2, \quad 0 < c_V < \sigma_v^2. \tag{5.9}$$

Coordinate design. Condition (5.6) is a low-discrepancy quasi-uniform design condition. The exact endpoint grid $2b_T + (i-1)(1-4b_T)/(N-1)$ is one sufficient example, but grid-knot alignment is not imposed.

Mixed third moment. Condition (5.7) is the only odd-moment cancellation used in the proof. Under q_0 -dependence, it reduces to finitely many distinct lag configurations. The cancellation is used only in the

growing-sieve Hessian calculation: it prevents a treatment–coordinate–coordinate third-order interaction from entering the feasible curvature target. It neither eliminates the derivative–influence channel nor imposes independence between v and ζ . Joint central symmetry of the entire (ζ, v) process under simultaneous sign reversal is sufficient but strictly stronger. Marginal skewness and unrelated third moments are allowed.

Dependence and smoothness. The integer q_0 need only be a known fixed upper bound on the true dependence range. Retaining finitely many additional zero-covariance lags leaves every rate unchanged. The upper bounds $\tau_m, \tau_h \leq p+1$ prevent a fixed-degree spline from being assigned an unattainable approximation order.

Example 5.3 (Finite moving averages, including MA(1)). *Let $\{U_{it} = (u_{it}^\zeta, u_{it}^v)^\top\}$ be a bounded, centered array that is independent and identically distributed over (i, t) , and let $\{u_{it}^\varepsilon\}$ be a bounded, centered array that is independent and identically distributed over (i, t) and independent of the entire U process. For fixed $q_c \leq q_0$, coefficients $\rho_{c,0} = 1$, and $c \in \{\zeta, v, \varepsilon\}$, set*

$$c_{it} = \sum_{j=0}^{q_c} \rho_{c,j} u_{i,t-j}^c.$$

If $\mathbb{E}[u_{it}^v (u_{it}^\zeta)^2] = 0$, then (5.7) holds. Moreover,

$$\gamma_{v\zeta}(\ell) = \mathbb{E}[u_{it}^v u_{it}^\zeta] \sum_{j \in \mathbb{Z}} \rho_{v,j} \rho_{\zeta,j+\ell},$$

where coefficients outside their filter ranges are zero. Taking $q_\zeta, q_v, q_\varepsilon \leq 1$ recovers the bounded joint MA(1) benchmark.

Thus the benchmark permits an active derivative–influence channel while retaining the sole mixed third-moment cancellation used below.

5.3 When the Derivative–Influence Channel Is Nonzero

Define $\gamma_{v\zeta}(\ell) = \mathbb{E}[v_{i0} \zeta_{i\ell}]$, $\Omega_{v\zeta} = \sum_{\ell \in \Lambda_{q_0}} \gamma_{v\zeta}(\ell)$, $\gamma_\zeta(\ell) = \mathbb{E}[\zeta_{i0} \zeta_{i\ell}]$, and $\Omega_\zeta = \sum_{\ell \in \Lambda_{q_0}} \gamma_\zeta(\ell)$. Also set

$$\tilde{A}_{it,K}^\theta = \dot{f}_{it,K}^\theta(\rho_{0,K,N}, \alpha_i) - \mathbb{E}_i[\dot{f}_{i0,K}^\theta(\rho_{0,K,N}, \alpha_i)], \quad B_{\zeta,\theta,K}^J = \frac{1}{N} \sum_i \sum_{\ell \in \Lambda_{q_0}} \mathbb{E}_i[\tilde{A}_{i0,K}^\theta \zeta_{i\ell}].$$

Define the joint-law curvature channel by

$$B_{\zeta\zeta,\theta,K}^J = \frac{\Omega_\zeta}{2N} \sum_i \mathbb{E}_i[\ddot{f}_{i0,K}^\theta(\rho_{0,K,N}, \alpha_i)].$$

Under Assumption 5.2, the centered measurement error and finite-range covariance support imply $B_{\theta,K,N}^\infty = B_{\zeta,\theta,K}^J + B_{\zeta\zeta,\theta,K}^J$. Here $\mathbb{E}_X$ denotes expectation under the uniform covariate law, and a subscript a denotes the coordinate derivative ∂_a . Also set

$$\bar{a}_{v,K} = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_X[2\Delta_{\theta,K,N} m_{K,N,a}(X, \alpha_i) - h_{K,N,a}^0(X, \alpha_i)].$$

Finally, let $\mathbf{b}_h = \int_{[0,1]^d} \{h_0(x, 1) - h_0(x, 0)\} dx$.

The next result isolates the derivative–influence channel. Its limit combines the long-run v – ζ covariance with an average boundary contrast of the outcome nuisance.

Proposition 5.4 (Derivative–influence bias limit). *Under Assumption 5.2 and $\mathbf{h}_K \rightarrow 0$,*

$$B_{\zeta,\theta,K}^J = \Omega_{v\zeta} \bar{a}_{v,K}, \quad \bar{a}_{v,K} \longrightarrow -\mathbf{b}_h. \quad (5.10)$$

Therefore $B_{\zeta,\theta,K}^J \rightarrow -\Omega_{v\zeta} \mathbf{b}_h$, which is nonzero exactly when $\Omega_{v\zeta} \mathbf{b}_h \neq 0$.

Appendix D.5 proves (5.10) by an exact lag calculation and proves the boundary limit by grid quadrature. The curvature channel supplies the second component of the limiting reuse drift. Define

$$\begin{aligned}\mathbf{c}_{mh} &= \int_{[0,1]^{d_X+1}} \partial_a m_0(x, a) \partial_a h_0(x, a) dx da, \\ B_{\theta,J} &= -\Omega_{v\zeta} \mathbf{b}_h + \Omega_\zeta \mathbf{c}_{mh}.\end{aligned}$$

Proposition 5.5 (Primitive limits of the reuse-bias channels). *Under Assumption 5.2 and $\mathfrak{h}_K \rightarrow 0$, the exact decomposition $B_{\theta,K,N}^\infty = B_{\zeta,\theta,K}^J + B_{\zeta\zeta,\theta,K}^J$ holds. Moreover,*

$$B_{\zeta,\theta,K}^J \longrightarrow -\Omega_{v\zeta} \mathbf{b}_h.$$

$$B_{\zeta\zeta,\theta,K}^J \longrightarrow \Omega_\zeta \mathbf{c}_{mh}. \quad (5.11)$$

$$B_{\theta,K,N}^\infty \longrightarrow B_{\theta,J}. \quad (5.12)$$

$$\mathcal{B}_{\theta,K}^{\text{cl}} \longrightarrow B_{\theta,J}. \quad (5.13)$$

Thus the limiting reuse drift is the sum of the derivative-influence and curvature limits. It vanishes exactly when these two limiting terms sum to zero.

Appendix D.6 proves these limits.

5.4 Supporting Rate Conditions

The channel limits identify the centering error. We next collect the rate conditions used to control nuisance estimation, the feasible correction, and studentization at the $\sqrt{NT}$ scale. Let $m = \max\{d_W, 3\}$ and $\bar{r}_{\eta,K} = \mathfrak{h}_K^{-m/2}/T$. The sufficient remainder conditions are

$$\mathfrak{h}_K^{-m}/T \rightarrow 0, \quad \mathfrak{h}_K^{-2} \bar{r}_{\eta,K} \rightarrow 0, \quad K_x \log(NK_x)/T \rightarrow 0. \quad (5.14)$$

Let $\gamma_c(\ell) = \mathbb{E}[c_{i0}c_{i\ell}]$ for $c \in \{v, \varepsilon\}$, let $\Psi_{\theta,iT,K} = T^{-1/2} \sum_{t=1}^T \chi_{\theta,it,K}$, and define

$$\sigma_{\theta,J}^2 = \frac{\sum_{\ell \in \Lambda_{q_0}} \gamma_v(\ell) \gamma_\varepsilon(\ell)}{\gamma_v(0)^2}. \quad (5.15)$$

Under (5.14), the appendices establish the good-root event, the nuisance and profile-remainder rates, consistency of both bias channels, and feasible variance consistency. Along growing-sieve sequences with $\mathfrak{h}_K \rightarrow 0$, they also show that the residual denominator and variance target converge to strictly positive limits and prove the cluster central limit theorem. Appendix E.1 collects these supporting propositions with their proofs.

5.5 Asymptotic Normality and Valid Inference

The two theorems below use the common rate window

$$\frac{d_W}{\tau_m + \tau_h} < \nu < \min \left\{ 1, \frac{2d_W}{\max\{d_W, 3\} + 4} \right\}. \quad (5.16)$$

The first theorem translates the limiting reuse drift into the distribution of the uncorrected estimator.

Theorem 5.6 (Noncentral limit of the uncorrected growing-sieve estimator). *Suppose Assumption 5.2 holds and $K \asymp T^\nu$ for some ν satisfying (5.16). Then the uncorrected estimator admits the representation*

$$\sqrt{NT}(\hat{\theta}_K - \theta_0) = N^{-1/2} \sum_{i=1}^N \Psi_{\theta,iT,K} + \sqrt{\kappa_0} \gamma_v(0)^{-1} B_{\theta,J} + o_p(1). \quad (5.17)$$

Consequently,

$$\sqrt{NT}(\hat{\theta}_K - \theta_0) \rightsquigarrow \mathcal{N}\left(\sqrt{\kappa_0} \frac{B_{\theta,J}}{\gamma_v(0)}, \sigma_{\theta,J}^2\right).$$

In particular, standard centered inference based on $\hat{\theta}_K$ is invalid whenever $B_{\theta,J} \neq 0$.

The next theorem shows that correcting both channels restores centered feasible inference under the same rate window.

Theorem 5.7 (Centered inference after the feasible two-channel correction). *Suppose Assumption 5.2 holds and $K \asymp T^\nu$ for some ν satisfying (5.16). Then the corrected estimator admits the centered representation*

$$\sqrt{NT}(\hat{\theta}_K^c - \theta_0) = N^{-1/2} \sum_{i=1}^N \Psi_{\theta,iT,K} + o_p(1).$$

It follows that

$$\sqrt{NT}(\hat{\theta}_K^c - \theta_0) \rightsquigarrow \mathcal{N}(0, \sigma_{\theta,J}^2).$$

The feasible variance estimator satisfies

$$\hat{\sigma}_{\theta,K}^2 \xrightarrow{p} \sigma_{\theta,J}^2.$$

Therefore,

$$\frac{\sqrt{NT}(\hat{\theta}_K^c - \theta_0)}{\hat{\sigma}_{\theta,K}} \rightsquigarrow \mathcal{N}(0, 1),$$

where $\sigma_{\theta,J}^2$ is defined in (5.15).

Sections E.2 and E.3 prove the two theorems.

The lower bound in (5.16) is equivalent to $T\mathfrak{h}_K^{\tau_m + \tau_h} \rightarrow 0$ and therefore makes the finite-sieve target drift negligible. The upper restrictions control, respectively, the Gram and nuisance estimation error, the profile quadratic remainder, and the fitted coordinate derivatives used by the correction. Substituting $\mathfrak{h}_K \asymp T^{-\nu/d_W}$ yields the upper bound in (5.16). The rate window is nonempty exactly when

$$\tau_m + \tau_h > \max\left\{d_W, \frac{\max\{d_W, 3\} + 4}{2}\right\}.$$

The comparison between Theorems 5.6 and 5.7 is therefore a comparison of centering, not of first-order dispersion. The correction removes the deterministic local shift $\sqrt{\kappa_0}B_{\theta,J}/\gamma_v(0)$ while leaving the leading cluster influence sum and its asymptotic variance unchanged. This wider-window statement uses the sample-mean coordinate and Assumption 5.2. The next section proves a separate feasible theorem for a smooth scalar moment coordinate under a smaller sieve window and without the mixed third-moment restriction.

Corollary 5.8 (Serially independent specialization). *Under the conditions of Theorem 5.7, suppose additionally that the processes $\{\zeta_{it}\}_t$, $\{v_{it}\}_t$, and $\{\varepsilon_{it}\}_t$ are mutually independent and that each is independent and identically distributed over t . Then $B_{\zeta,\theta,K}^J = 0$ and all conclusions of Theorem 5.7 hold with $\sigma_{\theta,J}^2 = \text{Var}(\varepsilon_{it})/\text{Var}(v_{it})$.*

Appendix E.4 proves the corollary.

6 Feasible Inference with Smooth Scalar Moment Coordinates

We now replace the sample mean by a smooth scalar moment estimator. The partially linear equations, grid projections, and target score are unchanged. The conditions below replace the measurement equation and the sample-mean benchmark for this section. They permit dependence between the first-step moment and both regression disturbances, and do not impose (5.7). The sufficient sieve window is more restrictive than (5.16).

6.1 Regular Unique Roots and Sieve Conditions

The observations W_{it} may now contain any variables needed to evaluate the known moment $\psi(W_{it}, a)$. In particular, a ratio coordinate requires observations (U_{it}, R_{it}) , rather than the measurement Z_{it} alone. Write $V = \mathbb{E}[v_{i0}^2]$ and define

$$\sigma_M^2 = V^{-2} \sum_{|\ell| \leq L} \mathbb{E}[v_{i0} \varepsilon_{i0} v_{i\ell} \varepsilon_{i\ell}]. \quad (6.1)$$

Assumption 6.1 (Smooth scalar moment benchmark). *All constants below are uniform in i, N, T .*

(i) Sampling and model. *The unit histories are independent. The coordinates are deterministic and satisfy (5.6), with $b_T = C_b \sqrt{\log N/T}$ and $C_i = \sigma(\alpha_i)$. The covariates X_{it} are independent and uniformly distributed on $[0, 1]^{d_X}$, and independent of the entire (v, ε) histories. Equations (2.2)–(2.3) hold. The bounded, strictly stationary (v, ε) histories have a common law that does not vary with N, T , with $\mathbb{E}[v_{i0}] = \mathbb{E}[\varepsilon_{i0}] = \mathbb{E}[v_{i0} \varepsilon_{i0}] = 0$, $V > 0$, and $\sigma_M^2 > 0$. The joint process containing $X_{it}, v_{it}, \varepsilon_{it}$ and the maps $a \mapsto \partial_a^r \psi(W_{it}, a)$, $r = 0, 1, 2, 3$, is strictly stationary and L -dependent for a known fixed integer $L \geq 0$. For date sets whose minimum separation exceeds L , their generated σ -fields are independent. The moment may depend on X, v, ε .*

(ii) Moment and observable coordinate. *There is a known compact interval $\mathcal{A} = [-\delta, 1 + \delta]$, $\delta > 0$, such that*

$$\mathbb{E}_i[\psi(W_{i0}, \alpha_i)] = 0, \quad \inf_{i, a \in \mathcal{A}} |\mathbb{E}_i[\partial_a \psi(W_{i0}, a)]| \geq c_G > 0.$$

For each unit the population derivative does not change sign on $\mathcal{A}$. The known moment and its first three derivatives are uniformly bounded there, and its third derivative is uniformly Lipschitz. Let $\check{\alpha}_i$ be the unique zero of (3.1) in $\mathcal{A}$ if there is exactly one, and set $\check{\alpha}_i = 1/2$ otherwise. Use $\hat{\alpha}_i = \Pi_{[b_T, 1-b_T]}(\check{\alpha}_i)$. Choose C_b so that $cC_b^2 > 16$, where c is a uniform constant in the root tail bound in Lemma F.1.

(iii) Spline and smoothness. *Use the normalized tensor splines of Assumption 5.2(iv), with fixed $d_W = d_X + 1$ and degree $p \geq 4$. The functions m_0, h_0 have respective smoothness $3 < \tau_m, \tau_h \leq p + 1$ on W^+ , in the extension sense of Lemma C.4: their extensions to a fixed open rectangle containing W^+ have bounded Hölder norms.*

(iv) Growth. $N/T \rightarrow \kappa_0 \in (0, \infty)$, $\mathfrak{h}_K \rightarrow 0$, $K \asymp \mathfrak{h}_K^{-d_W}$, and

$$\frac{K \mathfrak{h}_K^{-4}}{T} \rightarrow 0, \quad T \mathfrak{h}_K^{\tau_m + \tau_h} \rightarrow 0. \quad (6.2)$$

The root rule uses neither the true coordinate nor an oracle neighborhood. Its population monotonicity requirement is stronger than local identification. The first-step equation is scalar, just identified, smooth, and known. The moment conditions in part (i) suffice for the grid projections and target-drift identity; conditional mean independence of ε from D is not needed in this benchmark.

6.2 Observable Correction and Standard Error

Write a superscript M for reported second-stage objects in this section. Use the candidate profile fits in (4.1), with $c_{\Sigma, T} = c_{V, T} = a_T$, where $a_T = c_{\text{tol}} T^{-1/4}$ and $c_{\text{tol}} > 0$ is known and fixed. This threshold is also used for the first-step inverse and variance floor below; $c_{\text{tol}} = 1$ is sufficient. The observable acceptance event is

$$\mathcal{O}_T = \{\lambda_{\min}(\hat{\Sigma}_{p, K}) \geq a_T, V_K^{\text{pre}} \geq a_T\}. \quad (6.3)$$

On this event, report the candidate coefficients as $\hat{\gamma}_K^M, \hat{\delta}_K^M$, form residuals $\hat{u}_{D, it, K}^M$ and $\hat{u}_{Y, it, K}^M$ as before, and set $\hat{V}_K^M = V_K^{\text{pre}}$ and report the profile slope defined in Section 4.1 as $\hat{\theta}_K^M$. Let $\hat{f}_{it, K}^M, \hat{f}_{it, K}^M$, and $\hat{f}_{it, K}^M$ be the score and the two explicit derivatives in (2.9)–(2.11) at these coefficients and coordinates. A bar denotes their within-unit average.

For any two within-unit arrays, define the fixed-lag sum

$$\mathcal{L}_{i, T}(U, V) = \sum_{|\ell| \leq L} \frac{1}{T} \sum_{t \in I_T(\ell)} U_{it} V_{i, t+\ell}.$$

The denominator remains T , so the finite-window factor is $1 - |\ell|/T$. Estimate the first-step objects by

$$\begin{aligned}\widehat{G}_{it}^\psi &= \partial_a \psi(W_{it}, \widehat{\alpha}_i), & \widehat{G}_i^\psi &= \frac{1}{T} \sum_t \widehat{G}_{it}^\psi, & \widehat{H}_i^\psi &= \frac{1}{T} \sum_t \partial_{aa} \psi(W_{it}, \widehat{\alpha}_i), \\ (\widehat{G}_i^\psi)^\ominus &= \begin{cases} 1/\widehat{G}_i^\psi, & |\widehat{G}_i^\psi| \geq a_T, \\ 0, & |\widehat{G}_i^\psi| < a_T, \end{cases}\end{aligned}$$

and explicitly center the estimated influence:

$$\widehat{\xi}_{it} = -(\widehat{G}_i^\psi)^\ominus \left\{ \psi(W_{it}, \widehat{\alpha}_i) - \frac{1}{T} \sum_s \psi(W_{is}, \widehat{\alpha}_i) \right\}. \quad (6.4)$$

This definition also applies when root selection or clipping binds. Set

$$\begin{aligned}\widehat{\Omega}_{\xi,i} &= \mathcal{L}_{i,T}(\widehat{\xi}, \widehat{\xi}), \\ \widehat{C}_{G\xi,i} &= \mathcal{L}_{i,T}(\widehat{G}^\psi - \widehat{G}_i^\psi, \widehat{\xi}), \\ \widehat{\beta}_i &= -(\widehat{G}_i^\psi)^\ominus \left(\widehat{C}_{G\xi,i} + \frac{1}{2} \widehat{H}_i^\psi \widehat{\Omega}_{\xi,i} \right).\end{aligned} \quad (6.5)$$

The three-channel correction is

$$\begin{aligned}\widehat{B}_{\theta,K,N}^M &= \frac{1}{N} \sum_{i=1}^N \left\{ \mathcal{L}_{i,T}(\widehat{f}_K^M, \widehat{\xi}) + \widehat{f}_{i,K}^M \widehat{\beta}_i + \frac{1}{2} \widehat{f}_{i,K}^M \widehat{\Omega}_{\xi,i} \right\}, \\ \widehat{\theta}_K^{M,c} &= \widehat{\theta}_K^M - \frac{\widehat{B}_{\theta,K,N}^M}{T \widehat{V}_K^M}.\end{aligned} \quad (6.6)$$

For the variance, retain the first-order sampling contribution of the coordinate. Define

$$\begin{aligned}\widehat{\chi}_{it,K}^M &= (\widehat{V}_K^M)^{-1} \left(\widehat{f}_{it,K}^M + \widehat{f}_{i,K}^M \widehat{\xi}_{it} \right), & \widetilde{\chi}_{it,K}^M &= \widehat{\chi}_{it,K}^M - \frac{1}{T} \sum_s \widehat{\chi}_{is,K}^M, \\ (\widehat{\sigma}_K^M)^2 &= \max \left\{ a_T, \frac{1}{N} \sum_{i=1}^N \mathcal{L}_{i,T}(\widetilde{\chi}_K^M, \widetilde{\chi}_K^M) \right\}.\end{aligned}$$

The floor makes the standard error well defined even when a finite-sample rectangular lag sum is nonpositive. On $\mathcal{O}_T^c$, report $\widehat{\theta}_K^M = \widehat{\theta}_K^{M,c} = 0$, $\widehat{V}_K^M = (\widehat{\sigma}_K^M)^2 = 1$, and $\widehat{B}_{\theta,K,N}^M = 0$; set second-stage coefficients and uncalculated second-stage arrays to zero. The first-step statistics retain their own definitions. All normal equations are asserted only on $\mathcal{O}_T$.

6.3 Growing-Sieve Theorem

Define the centered oracle observation and its influence by

$$\begin{aligned}x_{it,K}^M &= f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i) - \mathbb{E}_i[f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i)] + \mu_{f,i,K} \xi_{it}, \\ \chi_{it,K}^M &= V_K^{-1} x_{it,K}^M.\end{aligned} \quad (6.7)$$

Theorem 6.2 (Feasible inference for a smooth scalar moment coordinate). *Under Assumption 6.1, the procedure with acceptance event (6.3) satisfies $\mathbb{P}(\mathcal{O}_T) \rightarrow 1$, $\widehat{B}_{\theta,K,N}^M - B_{\theta,K,N}^M = o_p(1)$, and*

$$\sqrt{NT}(\widehat{\theta}_K^{M,c} - \theta_0) = \frac{1}{\sqrt{NT}} \sum_{i,t} \chi_{it,K}^M + o_p(1) \rightsquigarrow \mathcal{N}(0, \sigma_M^2). \quad (6.8)$$

Moreover,

$$(\widehat{\sigma}_K^M)^2 \xrightarrow{p} \sigma_M^2, \quad \frac{\sqrt{NT}(\widehat{\theta}_K^{M,c} - \theta_0)}{\widehat{\sigma}_K^M} \rightsquigarrow \mathcal{N}(0, 1).$$

The uncorrected estimator has the representation

$$\sqrt{NT}(\hat{\theta}_K^M - \theta_0) = \frac{1}{\sqrt{NT}} \sum_{i,t} \chi_{it,K}^M + \sqrt{N/T} V_K^{-1} B_{\theta,K,N}^M + o_p(1). \quad (6.9)$$

If additionally $B_{\theta,K,N}^M \rightarrow B_\infty^M$, its limit is $\mathcal{N}(\sqrt{\kappa_0} B_\infty^M / V, \sigma_M^2)$. The variance is given by (6.1). The centered conclusion requires no convergence of the bias coefficient itself.

Appendix F proves the theorem from the moment equation, uniform Gram perturbation, and the nuisance normal equations. In particular, it derives

$$\begin{aligned} \|\hat{\gamma}_K^M - \gamma_{0,K,N}\| + \|\hat{\delta}_K^M - \delta_{0,K,N}\| &= O_p\left(\frac{\sqrt{K} \mathfrak{h}_K^{-2}}{T}\right), \\ \sqrt{NT} |e_\gamma^\top \hat{\Sigma}_{p,K} e_\pi| &= O_p\left(\frac{K \mathfrak{h}_K^{-4}}{T}\right) = o_p(1). \end{aligned}$$

Here $e_\gamma = \hat{\gamma}_K^M - \gamma_{0,K,N}$ and $e_\pi = (\hat{\delta}_K^M - \delta_{0,K,N}) - \theta_{0,K,N} e_\gamma$. The standard error is $\hat{\sigma}_K^M / \sqrt{NT}$.

Table 1 puts the two sufficient results on the same mesh scale. Write $d = d_W$, $s = \tau_m + \tau_h$, $m = \max\{d, 3\}$, and $\mathfrak{h}_K \asymp T^{-c}$. Since $K \asymp \mathfrak{h}_K^{-d}$, the sample-mean exponent in (5.16) is $\nu = cd$. The larger window compares these two sufficient theorems; neither result claims an optimal boundary.

Table 1: Two sufficient sieve windows

Coordinate and additional structure	Sufficient exponent for $\mathfrak{h}_K \asymp T^{-c}$
Sample mean; exact linear expansion and mixed-third cancellation	$\frac{1}{s} < c < \min\left\{\frac{1}{d}, \frac{2}{m+4}\right\}$
Smooth scalar moment; regular unique root	$\frac{1}{s} < c < \frac{1}{d+4}$

More generally, a bandwidth sequence satisfying the two restrictions exists if and only if $\tau_m + \tau_h > d_W + 4$. Indeed, the first restriction is $T \mathfrak{h}_K^{d+4} \rightarrow \infty$. If $s \leq d + 4$, then $T \mathfrak{h}_K^s \geq T \mathfrak{h}_K^{d+4}$ eventually, contradicting the second restriction. If $s > d + 4$, the interval in Table 1 supplies a sequence. For example, $d_W = 2$, $\tau_m = \tau_h = 4$, and $\mathfrak{h}_K \asymp T^{-1/7}$ give both remainders of order $T^{-1/7}$. The fixed-degree restriction also requires $d_W < 2p - 2$ for such a smoothness pair to exist.

For $\psi(W, a) = Z - a$, the population quantities reduce to $G_i = -1$, $H_i^\psi = 0$, $\xi_{it} = \zeta_{it}$, and $\beta_i = 0$. The feasible first-step bias is identically zero, and the correction reduces to two channels, up to the already controlled clipping and demeaning differences. Theorems 5.6 and 5.7 retain the wider sample-mean window (5.16) under their additional structure. Theorem 6.2 establishes a second sufficient route: a regular smooth scalar coordinate, a smaller sieve window, and no mixed-third-moment cancellation.

7 Simulation Evidence

Common design. We simulate the panel model in Section 2 with $\theta_0 = 1$, independent units, and X_{it} independent and uniformly distributed on $[0, 1]$. For $c \in \{\zeta, v, \varepsilon\}$, let

$$q_{it}^c = \frac{u_{it}^c + 0.5 u_{i,t-1}^c}{\sqrt{1.25}},$$

where (u_{it}^ζ, u_{it}^v) are joint Rademacher innovations with correlation $\rho_{v\zeta}$, independent and identically distributed over (i, t) , and u_{it}^ε is an independent Rademacher sequence, independent of the entire (u^ζ, u^v) process. We set $\zeta_{it} = 0.1 q_{it}^\zeta$, $v_{it} = q_{it}^v$, and $\varepsilon_{it} = q_{it}^\varepsilon$. The deterministic α_i values are equally spaced on $[2b_{NT}, 1 - 2b_{NT}]$, where $b_{NT} = \min\{0.25\sqrt{\log(N)/T}, 0.20\}$, and the generated coordinate is $\hat{\alpha}_i = \Pi_{[b_{NT}, 1-b_{NT}]}(\check{\alpha}_i)$ for the full-sample estimators.

Layer A: mechanism isolation. Layer A isolates the two channels using

$$m_0(x, a) = 0.5(x - 0.5) + \beta_m(a - 0.5), \quad h_0(x, a) = 0.5(x - 0.5)^2 + \beta_h(a - 0.5).$$

The five cases pair each $(\beta_m, \beta_h, \rho_{v\zeta})$ tuple with its analytic derivative and curvature coefficients: null, $(5, 0, 0.5)$ with $(0, 0)$; derivative only, $(0, 5, 0.5)$ with $(-0.45, 0)$; curvature only, $(5, 5, 0)$ with $(0, 0.45)$; reinforcement, $(-5, 5, 0.5)$ with $(-0.45, -0.45)$; and cancellation, $(5, 5, 0.5)$ with $(-0.45, 0.45)$. We fix five open-clamped, Lebesgue- L^2 -normalized degree-four B-splines in each direction ($K = 25$). At $T = (50, 100, 200)$, respectively, we use $N = (12, 50, 200)$, $(25, 100, 400)$, and $(50, 200, 800)$, corresponding to nominal aspect ratios 1/4, 1, and 4. Each cell uses 500 replications; the $N = T = 200$ headline cells use 2,000. Layer A compares the oracle-coordinate and uncorrected estimators with the oracle bias-corrected estimator, the two partial corrections, the full correction, buffered splitting, and the split-panel jackknife. The last two enter as point-estimate diagnostics. For the split-panel jackknife (SPJ), let $\hat{\theta}_K^{(1)}$ and $\hat{\theta}_K^{(2)}$ denote the profile estimates from the first and second halves of the time series. We compute

$$\hat{\theta}_K^{\text{SPJ}} = 2\hat{\theta}_K - \frac{\hat{\theta}_K^{(1)} + \hat{\theta}_K^{(2)}}{2}.$$

Each half reestimates the coordinate and both nuisance regressions using the full-sample spline basis and clipping interval. The halves are contiguous and have no intervening buffer. We report no standard errors or coverage for either split estimator.

Layer B: growing-sieve inference. Layer B uses nonlinear nuisance functions,

$$\begin{aligned} m_0(x, a) &= 0.6 \sin(2\pi x) + 2.5 \sin(\pi a) + 1.5xa, \\ h_0(x, a) &= 0.5 \cos(2\pi x) + 4a + 2 \sin(\pi a) + 1.25 \sin(2\pi x) \sin(\pi a), \end{aligned}$$

with $\rho_{v\zeta} = -0.5$, so the two channels reinforce. This layer uses the same (N, T) grid, 1,000 replications per cell, and $(6, 7, 8)$ normalized basis functions per direction at $T = (50, 100, 200)$, giving $K = (36, 49, 64)$. It compares the oracle-coordinate, uncorrected, two partial-correction, full two-channel, and split-panel-jackknife estimators. We compute $\theta_{0,K,N}$ and the long-run finite- (K, N) channel targets by 32-node Gauss-Legendre integration over x and a discrete average over the deterministic α_i grid, and report errors relative to both θ_0 and $\theta_{0,K,N}$. Both feasible bias channels and the cluster variance use the uniform lag-one kernel dictated by the moving-average design.

Bias mechanisms. Figure 1 reports the broad 500-replication grid. The high-precision $N = T = 200$ cells sharpen the comparison. Across the five scenarios, the largest discrepancy between the uncorrected $\sqrt{NT}$ -scaled bias and its two-channel prediction is 1.84 Monte Carlo standard errors. The largest corrected bias is 1.38 Monte Carlo standard errors from zero. In the reinforcement cell, the uncorrected bias is -0.896 versus a prediction of -0.900 , and the corrected bias is -0.030 . In the cancellation cell, the estimated derivative and curvature channels are -0.440 and 0.440 ; their sum leaves the uncorrected bias at 0.003 , even though buffered splitting exposes the remaining doubled curvature with bias 0.893 versus a prediction of 0.900 . The analogous buffered-split comparisons across all five cells are within 1.55 Monte Carlo standard errors. No replication in either the broad or high-precision Layer A runs invokes the root or variance default, and the maximum headline clipping rate is 0.002%.

Growing-sieve inference. Table 2 reports the nonlinear $N = T = 200$ cell. The uncorrected $\sqrt{NT}$ -scaled bias is 0.744, compared with the finite-sieve prediction 0.778; the gap is 0.94 Monte Carlo standard errors. Correcting both channels reduces the $\sqrt{NT}$ -scaled bias to 0.008, or 0.21 Monte Carlo standard errors from zero. The sieve-target drift contributes only 0.0002 on this scale.

The corrected SE-to-SD ratio is 0.993, and its studentized statistic has mean 0.007, standard deviation 1.007, and empirical 2.5% and 97.5% quantiles -1.805 and 1.916 . Coverage is 95.9%, with a Wilson interval $[94.5\%, 97.0\%]$. No replication in this cell invokes the root or variance default, and coordinate clipping is never active.

Calibration is strongest when the number of independent clusters is moderate or large. At $T = 200$ and $N/T = 1/4$ ($N = 50$), corrected coverage is 93.4%, with a Wilson interval $[91.7\%, 94.8\%]$. The corrected

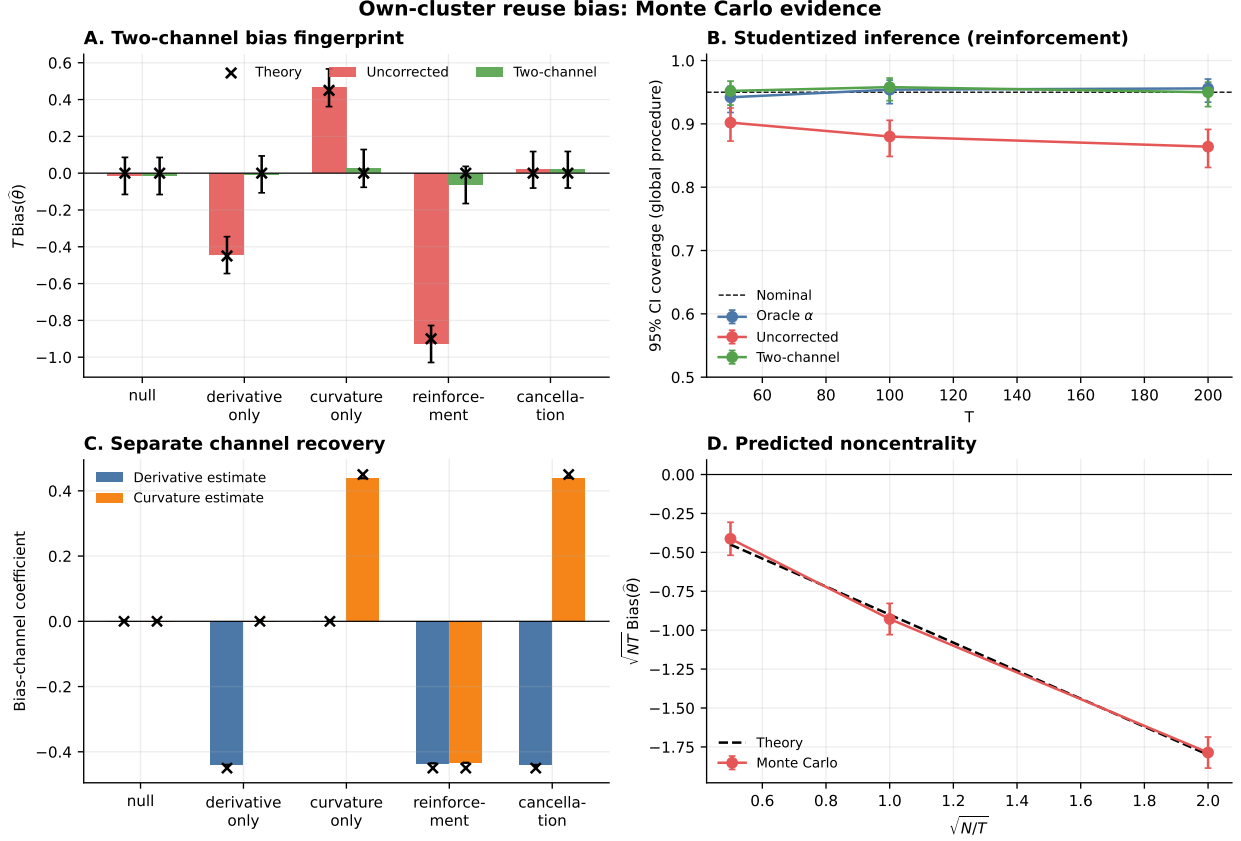


Figure 1: Layer A mechanism evidence. The broad grid uses 500 replications per cell. Panels A and C use $N = T = 200$; bars show Monte Carlo means with 95% Monte Carlo intervals and crosses show finite-sieve predictions. Panel B tracks global-procedure coverage in the reinforcement design at the aspect ratio nearest one; vertical bars are Wilson intervals. Panel D varies the nominal aspect ratio at $T = 200$ and compares the uncorrected $\sqrt{NT}$ -scaled bias with its predicted noncentral shift.

$\sqrt{NT}$ -scaled bias is only 0.002, so the remaining undercoverage has little connection to residual bias in this cell. The SE-to-SD ratio is 0.952 and the studentized standard deviation is 1.051, indicating that the estimated standard errors are too small on average. These summaries do not separate variance estimation error from error in the Gaussian approximation with 50 independent clusters. Appendix A reports all nine Layer B cells in Table A.1.

Layers A and B use the sample-mean benchmark, including its mixed third-moment restriction. The ratio experiment below uses the smooth scalar moment-coordinate procedure and permits a nonzero corresponding mixed third moment. All experiments retain a supplied dependence bound and deterministic coordinates.

A ratio coordinate. We add a small experiment for the complete procedure in Section 6. There are no X covariates ($d_W = 1$). Let S_{it}, Z_{it}^*, E_{it} be independent Rademacher variables across all indices and components, and set

$$\begin{aligned} R_{it} &= 1 + \frac{1}{4}S_{it}, & U_{it} &= \alpha_i R_{it} + 0.1\{\frac{1}{4}S_{it} + R_{it}Z_{it}^*\}, \\ D_{it} &= 10(\alpha_i - \frac{1}{2}) + S_{it}, & Y_{it} &= D_{it} + 10(\alpha_i - \frac{1}{2}) + E_{it}. \end{aligned}$$

Here $\theta_0 = 1$ and the known first-step moment is $\psi((U, R), a) = U - aR$ on $[-2, 3]$. Since $R_{it} \geq 3/4$, its unique root is $\bar{U}_i/\bar{R}_i$. We use $b_T = \sqrt{\log N/T}$, equally spaced coordinates on $[2b_T, 1 - 2b_T]$, and clipping to

Table 2: Layer B growing-sieve inference at $N = T = 200$

Method	$\sqrt{NT}$ -scaled bias	SE/SD	Stud. SD	Coverage [Wilson]
Oracle α	0.023 (0.036)	0.987	1.013	0.954 [0.939, 0.965]
Uncorrected	0.744 (0.037)	0.996	1.004	0.895 [0.874, 0.913]
Derivative correction	0.396 (0.037)	0.995	1.005	0.933 [0.916, 0.947]
Curvature correction	0.356 (0.037)	0.994	1.006	0.935 [0.918, 0.949]
Two-channel correction	0.008 (0.037)	0.993	1.007	0.959 [0.945, 0.970]
SPJ	0.047 (0.037)	–	–	–

Notes: 1,000 replications; $K = 64$. Parentheses are Monte Carlo standard errors; brackets are Wilson intervals for global coverage. The split-panel jackknife is a point-estimate benchmark.

Table 3: Feasible inference with a ratio coordinate

$N = T$	Method	$\sqrt{NT}$ -scaled bias	SE/SD	Coverage [Wilson]
500	Uncorrected	0.830 (0.032)	0.992	0.863 [0.840, 0.883]
	Full correction	−0.029 (0.032)	0.990	0.950 [0.935, 0.962]
1000	Uncorrected	0.859 (0.032)	1.000	0.862 [0.839, 0.882]
	Full correction	−0.010 (0.032)	1.000	0.963 [0.949, 0.973]

Notes: 1,000 replications per cell, seed 20260913, $K = 7$. Parentheses report Monte Carlo standard errors of scaled bias; brackets report 95% Wilson intervals for coverage of nominal 95% intervals. SE/SD is the mean estimated standard error divided by the empirical standard deviation of the corresponding estimator. All statistics include the globally defined fallback branches. The first-step Jacobian, bias channels, and variance are estimated from the simulated observations.

$[b_T, 1 - b_T]$. The numerator error has range $1/4$, so Hoeffding’s inequality and $\bar{R}_i \geq 3/4$ give

$$\mathbb{P}_i(|\tilde{\alpha}_i - \alpha_i| > x) \leq 2 \exp(-18Tx^2).$$

Thus this clipping constant satisfies the requirement $cC_b^2 > 16$ in Assumption 6.1.

We use normalized degree-four B-splines with $\lceil T^{1/7} \rceil$ equal positive knot spans, the known lag bound $L = 0$, and $c_{\text{tol}} = 10^{-8}$ in the observable thresholds. At $N = T = 500$ and 1000 this gives $K = 7$; each cell uses 1,000 replications. The nuisance functions lie exactly in these spline spaces, so $\theta_{0,K,N} = \theta_0$. The two reported sample sizes therefore hold the sieve dimension fixed and introduce no approximation error. Direct enumeration of the finite state law gives

$$\begin{aligned} \beta_i &= -0.00625, & \Omega_{\xi,i} &= 0.01125, & \mathbb{E}[S_{it}\xi_{it}^2] &= 0.005, \\ B_{\text{reuse},K,N}^M &= -0.25, & B_{\text{first},K,N}^M &= 0, & B_{\text{curv},K,N}^M &= 1.125. \end{aligned}$$

Indeed $\dot{f}_{it,K}^\theta = -10(E_{it} + S_{it})$, $\ddot{f}_{it,K}^\theta = 200$, and $\mu_{f,i,K} = 0$. Thus the first-step bias is nonzero, its population score contribution is zero, the corresponding mixed third moment is nonzero, and the predicted scaled root shift is 0.875, with $V = \sigma_M^2 = 1$.

Table 3 reports the uncorrected and fully corrected procedures, using the same feasible coordinate-adjusted standard error. At $N = T = 1000$, correction reduces scaled bias from 0.859 to -0.010 ; the latter is 0.32 Monte Carlo standard errors from zero. The estimated reuse, first-step, and curvature coefficients average -0.249 , -0.00014 , and 1.118 , respectively. The average estimate of β_i is -0.006248 . Coverage rises from 86.2% to 96.3%; its Wilson interval contains 95%. No replication at either reported sample size activates root fallback, clipping, second-stage fallback, or the variance floor. This experiment isolates the implementation of the feasible correction and variance estimator for a nonlinear first step. The growing-sieve comparison with approximation error is supplied by Layer B; here the first-step bias has zero population contribution to the profiled score.

8 Conclusion

A coordinate estimated from the same unit history contributes three second-order terms to a smooth downstream score: derivative–influence covariance, first-step bias, and curvature. In the partially linear panel model, the population derivative multiplying first-step bias tends to zero with sieve approximation. Reuse

and curvature remain as the two possible sources of asymptotic miscentering. For a sample-mean coordinate and a fixed sieve, buffered splitting removes covariance, while an equal split doubles the leading curvature coefficient.

Feasible bias correction and studentization follow for a smooth scalar moment coordinate with a regular unique root. The sufficient rate window is more restrictive than the sample-mean window, which we retain under its additional structure. Both results use deterministic quasi-uniform designs, tensor splines, and a supplied bound on fixed-range dependence. The ratio experiment implements the complete smooth moment procedure and separates a biased first step from its vanishing population contribution to the profiled score. These results establish a feasible correction within the stated scalar setting.

References

- Bickel, P. J., Klaassen, C. A. J., Ritov, Y., and Wellner, J. A. (1993). *Efficient and Adaptive Estimation for Semiparametric Models*. Johns Hopkins Series in the Mathematical Sciences. Johns Hopkins University Press, Baltimore.
- Bonhomme, S., Jochmans, K., Newey, W. K., and Weidner, M. (2026a). Higher-order Neyman orthogonality in moment-condition models. arXiv:2605.10842.
- Bonhomme, S., Jochmans, K., and Weidner, M. (2026b). A Neyman-orthogonalization approach to the incidental parameter problem in likelihood models. arXiv:2412.10304; originally circulated in 2024; revised August 2026.
- Cattaneo, M. D., Crump, R. K., and Jansson, M. (2013). Generalized jackknife estimators of weighted average derivatives. *Journal of the American Statistical Association*, 108(504):1243–1256.
- Cattaneo, M. D., Jansson, M., and Ma, X. (2019). Two-step estimation and inference with possibly many included covariates. *The Review of Economic Studies*, 86(3):1095–1122.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.
- Chernozhukov, V., Escanciano, J. C., Ichimura, H., Newey, W. K., and Robins, J. M. (2022). Locally robust semiparametric estimation. *Econometrica*, 90(4):1501–1535.
- Chiang, H. D., Kato, K., Ma, Y., and Sasaki, Y. (2022). Multiway cluster robust double/debiased machine learning. *Journal of Business & Economic Statistics*, 40(3):1046–1056.
- Clarke, P. S. and Polsell, A. (2026). Double machine learning for static panel models with fixed effects. *The Econometrics Journal*, 29(1):69–86.
- de Boor, C. (2001). *A Practical Guide to Splines*. Springer, New York, revised edition.
- Escanciano, J. C. and Pérez-Izquierdo, T. (2026). Automatic locally robust GMM with machine-learning-generated regressors. arXiv:2301.10643; version 4, revised March 2026.
- Fernández-Val, I. and Lee, J. (2013). Panel data models with nonadditive unobserved heterogeneity: Estimation and inference. *Quantitative Economics*, 4(3):453–481.
- Fernández-Val, I. and Vella, F. (2011). Bias corrections for two-step fixed effects panel data estimators. *Journal of Econometrics*, 163(2):144–162.
- Freeman, H. and Kristensen, D. (2026). Inference on linear regressions with two-way unobserved heterogeneity. arXiv:2605.06491.
- Hahn, J., Hughes, D. W., Kuersteiner, G., and Newey, W. K. (2024). Efficient bias correction for cross-section and panel data. *Quantitative Economics*, 15(3):783–816.

- Hahn, J. and Ridder, G. (2013). Asymptotic variance of semiparametric estimators with generated regressors. *Econometrica*, 81(1):315–340.
- Huang, J. (2026). Fixed effects as generated regressors. arXiv:2602.08899.
- Jochmans, K. and Weidner, M. (2024). Inference on a distribution from noisy draws. *Econometric Theory*, 40(1):60–97.
- Mackey, L., Syrgkanis, V., and Zadik, I. (2018). Orthogonal machine learning: Power and limitations. In Dy, J. and Krause, A., editors, *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 3375–3383. PMLR.
- Mammen, E., Rothe, C., and Schienle, M. (2016). Semiparametric estimation with generated covariates. *Econometric Theory*, 32(5):1140–1177.
- Murphy, K. M. and Topel, R. H. (1985). Estimation and inference in two-step econometric models. *Journal of Business & Economic Statistics*, 3(4):370–379.
- Newey, W. K. (1984). A method of moments interpretation of sequential estimators. *Economics Letters*, 14(2–3):201–206.
- Newey, W. K. (1994). The asymptotic variance of semiparametric estimators. *Econometrica*, 62(6):1349–1382.
- Neyman, J. (1959). Optimal asymptotic tests of composite statistical hypotheses. In Grenander, U., editor, *Probability and Statistics: The Harald Cramér Volume*, pages 213–234. Almqvist & Wiksell, Uppsala.
- Pagan, A. (1984). Econometric issues in the analysis of regressions with generated regressors. *International Economic Review*, 25(1):221–247.
- Robins, J. M., Li, L., Tchetgen Tchetgen, E. J., and van der Vaart, A. W. (2008). Higher order influence functions and minimax estimation of nonlinear functionals. In Nolan, D. and Speed, T., editors, *Probability and Statistics: Essays in Honor of David A. Freedman*, volume 2 of *IMS Collections*, pages 335–421. Institute of Mathematical Statistics.
- Schumaker, L. L. (2007). *Spline Functions: Basic Theory*. Cambridge University Press, Cambridge, third edition.
- Tropp, J. A. (2012). User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics*, 12(4):389–434.
- van der Vaart, A. W. (2014). Higher order tangent spaces and influence functions. *Statistical Science*, 29(4):679–686.

Appendix roadmap. Appendix A reports the complete simulation grid. Appendix B develops the fixed-sieve bias mechanism and sampling expansion, including its smooth-coordinate and split-sample extensions. Appendix C proceeds from deterministic spline properties and generated-coordinate clipping to finite-range moment tools, nuisance and Gram rates, estimator localization, and exact profile identities. Appendix D derives the theta-score expansion, proves consistency of the bias correction and variance estimator, and identifies the primitive bias limits. Appendix E assembles the supporting rates and proves the sample-mean growing-sieve results. Appendix F proves the observable-root expansion, uniform Gram bound, nuisance rate, feasible correction, and studentization for smooth scalar moment coordinates.

A Additional Simulation Results

Table A.1 reports inference for the corrected estimator in every Layer B design cell. The $\sqrt{NT}$ -scaled bias is small by $T = 200$ at all three aspect ratios. Coverage is weakest in designs with few independent clusters and is otherwise close to nominal. In the $T = 200, N = 50$ cell, the corrected $\sqrt{NT}$ -scaled bias is 0.002 but coverage is 93.4%. The SE-to-SD ratio of 0.952 and studentized standard deviation of 1.051 show that small estimated standard errors accompany the undercoverage. These summaries do not identify the separate contributions of variance estimation and the Gaussian approximation with 50 independent clusters.

Table A.1: Layer B corrected inference across the full design grid

T	N/T	N	K	$\sqrt{NT}$ -scaled bias	SE/SD	Stud. SD	Coverage [Wilson]	Defaults (root/var.)
50	0.24	12	36	-0.242 (0.039)	0.915	1.102	0.915 [0.896, 0.931]	0.000/0.000
50	1.00	50	36	-0.050 (0.038)	0.948	1.056	0.939 [0.922, 0.952]	0.000/0.000
50	4.00	200	36	0.050 (0.037)	0.978	1.022	0.940 [0.924, 0.953]	0.000/0.000
100	0.25	25	49	-0.204 (0.038)	0.960	1.043	0.939 [0.922, 0.952]	0.000/0.000
100	1.00	100	49	-0.003 (0.036)	1.009	0.991	0.954 [0.939, 0.965]	0.000/0.000
100	4.00	400	49	0.070 (0.036)	1.004	0.996	0.956 [0.941, 0.967]	0.000/0.000
200	0.25	50	64	0.002 (0.038)	0.952	1.051	0.934 [0.917, 0.948]	0.000/0.000
200	1.00	200	64	0.008 (0.037)	0.993	1.007	0.959 [0.945, 0.970]	0.000/0.000
200	4.00	800	64	0.019 (0.037)	0.993	1.008	0.950 [0.935, 0.962]	0.000/0.000

Notes: Each cell uses 1,000 replications. Parentheses report Monte Carlo standard errors for the $\sqrt{NT}$ -scaled bias. Coverage is unconditional; brackets give Wilson intervals. The final column reports the root- and variance-default rates. The basis dimension is $K = 36, 49, 64$ at $T = 50, 100, 200$, respectively.

B Fixed-Sieve Bias Mechanism and Extensions

B.1 Smooth Local M -Estimated Coordinates

For fixed K , this subsection derives the three-channel population expansion under local identification, using an oracle branch. The regular unique-root assumptions in Section 6 provide an observable procedure; Appendix F proves its growing-sieve guarantees. We retain the oracle calculation here because its local assumptions also support the fixed-sieve sample-mean bridge.

B.1.1 Oracle Local Root

For clarity, the local coordinate is scalar. Let

$$\Psi_{iT}(a) = \frac{1}{T} \sum_{t=1}^T \psi(W_{it}, a), \quad \mathbb{E}_i[\psi(W_{i0}, \alpha_i)] = 0, \quad (\text{B.1})$$

and define, at the true coordinate, $s_{it} = \psi(W_{it}, \alpha_i)$, $G_{it} = \partial_a \psi(W_{it}, \alpha_i)$, and $H_{it}^\psi = \partial_{aa} \psi(W_{it}, \alpha_i)$. Also set $G_i = \mathbb{E}_i[G_{i0}]$, $\bar{s}_i = \frac{1}{T} \sum_t s_{it}$, and $M_{iT} = \frac{1}{T} \sum_t G_{it} - G_i$. The first-order influence and its time average are $\xi_{it} = -G_i^{-1} s_{it}$ and $\bar{\xi}_i = \frac{1}{T} \sum_t \xi_{it} = -G_i^{-1} \bar{s}_i$. For a scalar or vector random variable Q , write $\|Q\|_{L_i^r} :=$

$\{\mathbb{E}_i[\|Q\|^r]\}^{1/r}$, where the inner norm is absolute value in the scalar case and Euclidean norm otherwise. All conditional norm and probability bounds in the oracle-root and second-order arguments below hold for almost every realization of $\mathcal{C}_i$, with deterministic constants uniform in i and independent of N ; their unconditional counterparts follow by integration.

The next assumption provides sufficient conditions for a unique oracle local root outside an exponentially rare event and for the uniform second-order expansion used below.

Assumption B.1 (Local M -score regularity). *The following conditions hold.*

(i) **Sampling and dependence.** *The unit arrays are independent across i . Conditional on the unit information $\mathcal{C}_i$, for each i the joint array containing $\psi(W_{it}, a)$, its first three a -derivatives, the fixed- K stacked moment, and its first three a -derivatives is strictly stationary and ℓ_0 -dependent for a fixed integer ℓ_0 . Here ℓ_0 -dependence is meant in the sigma-field sense: for finite index sets $\mathcal{T}_1, \mathcal{T}_2 \subset \mathbb{Z}$ satisfying $\min_{s \in \mathcal{T}_1, t \in \mathcal{T}_2} |s - t| > \ell_0$, the sigma-fields generated by all listed variables at dates in $\mathcal{T}_1$ and $\mathcal{T}_2$ are independent conditional on $\mathcal{C}_i$.*

(ii) **Local smoothness.** *For a common fixed $\delta > 0$, these variables are uniformly bounded on $I_i = [\alpha_i - \delta, \alpha_i + \delta]$, and their highest displayed derivatives are uniformly Lipschitz there.*

(iii) **Local identification.**

$$0 < c_G \leq |G_i| \leq C_G, \quad \inf_{a \in I_i} |\mathbb{E}_i[\partial_a \psi(W_{i0}, a)]| \geq c_G,$$

and, for each i , the map $a \mapsto \mathbb{E}_i[\partial_a \psi(W_{i0}, a)]$ does not change sign on I_i .

Oracle local branch. On the event that Ψ_{iT} has exactly one zero in I_i , let $\tilde{\alpha}_i$ be that zero; otherwise set $\tilde{\alpha}_i = \alpha_i$. Let $\mathcal{D}_{iT}$ denote this fallback event, namely, the event that Ψ_{iT} does not have exactly one zero in I_i . Because both I_i and the default use the true coordinate, this is an oracle local branch, not a feasible root-selection algorithm. An observable selector taking values in I_i inherits the expansion below if its conditional probability of differing from this branch is at most $C \exp(-cT)$, uniformly in i . Indeed, the two selectors then differ by at most 2δ , so their conditional L_i^r distance is at most $2\delta C^{1/r} \exp(-cT/r) = o(T^{-m})$ for every fixed $m > 0$. The oracle default is not asserted to solve (B.1).

The next lemma localizes the oracle root and the sample-Jacobian fluctuation at the parametric rate.

Lemma B.2 (Oracle-branch localization). *Under Assumption B.1, for every fixed $r \geq 2$, the following bounds hold uniformly in i :*

$$\|\tilde{\alpha}_i - \alpha_i\|_{L_i^r} = O(T^{-1/2}), \quad \|M_{iT}\|_{L_i^r} = O(T^{-1/2}). \quad (\text{B.2})$$

The conditional probability of $\mathcal{D}_{iT}$ is $O(\exp(-cT))$, uniformly in i .

Proof. Fix i and condition on $\mathcal{C}_i$. Residue-class concentration and tail integration give moment bounds for the score and the Jacobian. Extending the Jacobian bound to I_i then locates the unique root and transfers the score rate to the coordinate error. All constants below are deterministic and uniform in i .

Set $q = \ell_0 + 1$ and partition $\{1, \dots, T\}$ into $\mathcal{J}_j = \{t : t \equiv j \pmod{q}\}$, $j = 0, \dots, q-1$, with $n_j = \#\mathcal{J}_j \geq T/(2q)$ for sufficiently large T . The sigma-field condition in Assumption B.1(i) implies joint conditional independence within each class: successively separating each new date from all preceding dates leaves a gap of at least $q > \ell_0$. For any conditionally centered scalar sequence X_{it} from the joint array with $|X_{it}| \leq B$, the full average satisfies

$$\left| \frac{1}{T} \sum_{t=1}^T X_{it} \right| \leq \sum_{j=0}^{q-1} \frac{n_j}{T} \left| \frac{1}{n_j} \sum_{t \in \mathcal{J}_j} X_{it} \right| \leq \max_{0 \leq j \leq q-1} \left| \frac{1}{n_j} \sum_{t \in \mathcal{J}_j} X_{it} \right|.$$

Conditional Hoeffding's inequality in each class and a union bound therefore give, for $x > 0$,

$$\mathbb{P} \left(\left| \frac{1}{T} \sum_{t=1}^T X_{it} \right| \geq x \mid \mathcal{C}_i \right) \leq 2 \sum_{j=0}^{q-1} \exp \left(-\frac{n_j x^2}{2B^2} \right) \leq 2q \exp \left(-\frac{T x^2}{4qB^2} \right).$$

Stationarity and the population moment equation center s_{it} and $G_{it} - G_i$, so this inequality yields

$$\mathbb{P}(|\tilde{s}_i| \geq x \mid \mathcal{C}_i) + \mathbb{P}(|M_{iT}| \geq x \mid \mathcal{C}_i) \leq C \exp(-cT x^2).$$

For Y_{iT} equal to either $\bar{s}_i$ or M_{iT} , tail integration with $u = \sqrt{cT}x$ gives, for every fixed $r \geq 2$,

$$\begin{aligned}\mathbb{E}_i[|Y_{iT}|^r] &= r \int_0^\infty x^{r-1} \mathbb{P}(|Y_{iT}| > x \mid \mathcal{C}_i) dx \\ &\leq Cr \int_0^\infty x^{r-1} \exp(-cTx^2) dx = \frac{Cr}{(cT)^{r/2}} \int_0^\infty u^{r-1} \exp(-u^2) du \leq C_r T^{-r/2}.\end{aligned}$$

Consequently, $\|\bar{s}_i\|_{L_i^r} + \|M_{iT}\|_{L_i^r} \leq C_r T^{-1/2}$.

Conditional on $\mathcal{C}_i$, the interval I_i is fixed. Choose $J_T \leq CT^2$ grid points so that every $a \in I_i$ lies within T^{-2} of a point a_j . At each grid point the Jacobian discrepancy is the average of the bounded, conditionally centered sequence $\partial_a \psi(W_{iT}, a_j) - \mathbb{E}_i[\partial_a \psi(W_{i0}, a_j)]$. The preceding concentration bound at $x = c_G/4$ gives

$$\begin{aligned}\mathbb{P}\left(\max_{1 \leq j \leq J_T} |\partial_a \Psi_{iT}(a_j) - \mathbb{E}_i[\partial_a \psi(W_{i0}, a_j)]| \geq \frac{c_G}{4} \mid \mathcal{C}_i\right) \\ \leq CJ_T \exp(-cT) \leq CT^2 \exp(-cT) \leq C \exp(-c'T)\end{aligned}$$

for some $c' > 0$. The uniform bound on $\partial_{aa} \psi$ supplies a deterministic Lipschitz constant L for both the sample and population Jacobians. For a nearest grid point a_j , the triangle inequality gives

$$\begin{aligned}|\partial_a \Psi_{iT}(a) - \mathbb{E}_i[\partial_a \psi(W_{i0}, a)]| \\ \leq |\partial_a \Psi_{iT}(a_j) - \mathbb{E}_i[\partial_a \psi(W_{i0}, a_j)]| + |\partial_a \Psi_{iT}(a) - \partial_a \Psi_{iT}(a_j)| \\ + |\mathbb{E}_i[\partial_a \psi(W_{i0}, a_j)] - \mathbb{E}_i[\partial_a \psi(W_{i0}, a)]| \\ \leq |\partial_a \Psi_{iT}(a_j) - \mathbb{E}_i[\partial_a \psi(W_{i0}, a_j)]| + 2LT^{-2}.\end{aligned}$$

For sufficiently large T , $2LT^{-2} < c_G/4$. Outside the grid exceptional event, we therefore have

$$\sup_{a \in I_i} |\partial_a \Psi_{iT}(a) - \mathbb{E}_i[\partial_a \psi(W_{i0}, a)]| < \frac{c_G}{2}. \quad (\text{B.3})$$

Let $\mathcal{E}_{iT}^{\text{loc}}$ be the event on which $|\bar{s}_i| < c_G \delta/2$ and (B.3) hold. The score and grid bounds imply

$$\begin{aligned}\mathbb{P}((\mathcal{E}_{iT}^{\text{loc}})^c \mid \mathcal{C}_i) &\leq \mathbb{P}\left(|\bar{s}_i| \geq \frac{c_G \delta}{2} \mid \mathcal{C}_i\right) \\ &+ \mathbb{P}\left(\sup_{a \in I_i} |\partial_a \Psi_{iT}(a) - \mathbb{E}_i[\partial_a \psi(W_{i0}, a)]| \geq \frac{c_G}{2} \mid \mathcal{C}_i\right) \leq C \exp(-cT).\end{aligned}$$

To treat both possible signs of the population derivative, set $\varsigma_i = \text{sgn}(G_i)$. Assumption B.1(iii) and (B.3) imply, on $\mathcal{E}_{iT}^{\text{loc}}$,

$$\varsigma_i \partial_a \Psi_{iT}(a) \geq |\mathbb{E}_i[\partial_a \psi(W_{i0}, a)]| - |\partial_a \Psi_{iT}(a) - \mathbb{E}_i[\partial_a \psi(W_{i0}, a)]| > \frac{c_G}{2}, \quad a \in I_i.$$

Multiplication by ς_i leaves the zeros unchanged and makes the sample moment strictly increasing. Since $\Psi_{iT}(\alpha_i) = \bar{s}_i$, its signed endpoint values satisfy

$$\begin{aligned}\varsigma_i \Psi_{iT}(\alpha_i - \delta) &= \varsigma_i \bar{s}_i - \int_{\alpha_i - \delta}^{\alpha_i} \varsigma_i \partial_a \Psi_{iT}(a) da \leq |\bar{s}_i| - \frac{c_G \delta}{2} < 0, \\ \varsigma_i \Psi_{iT}(\alpha_i + \delta) &= \varsigma_i \bar{s}_i + \int_{\alpha_i}^{\alpha_i + \delta} \varsigma_i \partial_a \Psi_{iT}(a) da \geq -|\bar{s}_i| + \frac{c_G \delta}{2} > 0.\end{aligned}$$

Continuity and strict monotonicity give exactly one zero in I_i . Thus $\mathcal{E}_{iT}^{\text{loc}} \subseteq \mathcal{D}_{iT}^c$ and $\mathbb{P}(\mathcal{D}_{iT} \mid \mathcal{C}_i) \leq C \exp(-cT)$.

On $\mathcal{E}_{iT}^{\text{loc}}$, the oracle branch selects this zero. If it equals α_i , the bound below holds immediately; otherwise, the mean-value theorem gives a point α_i° between them with

$$\begin{aligned}-\bar{s}_i &= \Psi_{iT}(\tilde{\alpha}_i) - \Psi_{iT}(\alpha_i) = \partial_a \Psi_{iT}(\alpha_i^\circ)(\tilde{\alpha}_i - \alpha_i), \\ |\tilde{\alpha}_i - \alpha_i| &= \frac{|\bar{s}_i|}{|\partial_a \Psi_{iT}(\alpha_i^\circ)|} \leq \frac{2}{c_G} |\bar{s}_i|.\end{aligned}$$

On the complementary event, either a unique zero is selected in I_i or the oracle default is α_i , so the error is at most δ . Combining the events and applying Minkowski's inequality gives

$$\begin{aligned} |\tilde{\alpha}_i - \alpha_i| &\leq \frac{2}{c_G} |\bar{s}_i| \mathbf{1}\{\mathcal{E}_{iT}^{\text{loc}}\} + \delta \mathbf{1}\{(\mathcal{E}_{iT}^{\text{loc}})^c\}, \\ \|\tilde{\alpha}_i - \alpha_i\|_{L_i^r} &\leq \frac{2}{c_G} \|\bar{s}_i\|_{L_i^r} + \delta \mathbb{P}((\mathcal{E}_{iT}^{\text{loc}})^c \mid \mathcal{C}_i)^{1/r} \\ &\leq C_r T^{-1/2} + \delta C^{1/r} \exp(-cT/r) = O(T^{-1/2}). \end{aligned}$$

The exponential term is negligible for each fixed r . Together with the bound for M_{iT} , this proves (B.2). $\square$

B.1.2 Second-Order Expansion

Write $\bar{H}_i^\psi = \frac{1}{T} \sum_{t=1}^T H_{it}^\psi$. Define the second-order approximation exactly by

$$q_{iT} = -G_i^{-1} M_{iT} \bar{\xi}_i - \frac{1}{2} G_i^{-1} \bar{H}_i^\psi \bar{\xi}_i^2. \quad (\text{B.4})$$

Its conditional mean and centered component are

$$\beta_{iT} := T \mathbb{E}_i[q_{iT}], \quad \zeta_{iT}^c := q_{iT} - \mathbb{E}_i[q_{iT}]. \quad (\text{B.5})$$

Thus

$$q_{iT} = \frac{1}{T} \beta_{iT} + \zeta_{iT}^c, \quad (\text{B.6})$$

is an identity, not an asymptotic definition.

Proposition B.3 (Second-order expansion of the oracle local branch). *Under Assumption B.1, for every fixed $r \geq 2$,*

$$\tilde{\alpha}_i - \alpha_i = \bar{\xi}_i + \frac{\beta_{iT}}{T} + \zeta_{iT}^c + r_{iT}^{(3)},$$

where

$$r_{iT}^{(3)} := (\tilde{\alpha}_i - \alpha_i) - \bar{\xi}_i - q_{iT}, \quad \left\| r_{iT}^{(3)} \right\|_{L_i^r} = O(T^{-3/2}). \quad (\text{B.7})$$

Moreover,

$$\|q_{iT}\|_{L_i^r} = O(T^{-1}), \quad |\beta_{iT}| = O(1), \quad \|\zeta_{iT}^c\|_{L_i^r} = O(T^{-1}). \quad (\text{B.8})$$

Proof. The argument first isolates the first-order influence term, then substitutes the proposed second-order term into the score, and finally absorbs the terms that multiply the third-order remainder. Let $\Delta_i^{\text{or}} = \tilde{\alpha}_i - \alpha_i$. On the selected-root event, Taylor's theorem applied to the original equation $\Psi_{iT}(\tilde{\alpha}_i) = 0$ gives

$$0 = \bar{s}_i + G_i \Delta_i^{\text{or}} + M_{iT} \Delta_i^{\text{or}} + \frac{1}{2} \bar{H}_i^\psi (\Delta_i^{\text{or}})^2 + R_{\psi, iT}(\Delta_i^{\text{or}}), \quad |R_{\psi, iT}(d)| \leq C|d|^3. \quad (\text{B.9})$$

Because $G_i \bar{\xi}_i = -\bar{s}_i$, subtracting this identity from (B.9) yields

$$G_i(\Delta_i^{\text{or}} - \bar{\xi}_i) = -M_{iT} \Delta_i^{\text{or}} - \frac{1}{2} \bar{H}_i^\psi (\Delta_i^{\text{or}})^2 - R_{\psi, iT}(\Delta_i^{\text{or}}). \quad (\text{B.10})$$

The moment bound for $\bar{s}_i$ in the proof of Lemma B.2 and the identity $\bar{\xi}_i = -G_i^{-1} \bar{s}_i$ give $\|\bar{\xi}_i\|_{L_i^p} = O(T^{-1/2})$ for every fixed $p \geq 2$. Because the lemma's bounds hold for every fixed moment, we may apply them at $2r$ and $3r$. Boundedness of $\bar{H}_i^\psi$, the Taylor remainder bound, and Hölder's inequality then give

$$\begin{aligned} \|M_{iT} \Delta_i^{\text{or}} \mathbf{1}\{\mathcal{D}_{iT}^c\}\|_{L_i^r} &\leq \|M_{iT}\|_{L_i^{2r}} \|\Delta_i^{\text{or}}\|_{L_i^{2r}} = O(T^{-1}), \\ \left\| \bar{H}_i^\psi (\Delta_i^{\text{or}})^2 \mathbf{1}\{\mathcal{D}_{iT}^c\} \right\|_{L_i^r} &\leq C \|\Delta_i^{\text{or}}\|_{L_i^{2r}}^2 = O(T^{-1}), \\ \|R_{\psi, iT}(\Delta_i^{\text{or}}) \mathbf{1}\{\mathcal{D}_{iT}^c\}\|_{L_i^r} &\leq C \|\Delta_i^{\text{or}}\|_{L_i^{3r}}^3 = O(T^{-3/2}). \end{aligned}$$

Since $|G_i| \geq c_G$, identity (B.10) therefore yields an $O(T^{-1})$ conditional norm on $\mathcal{D}_{iT}^c$. On $\mathcal{D}_{iT}$, the oracle branch sets $\Delta_i^{\text{or}} = 0$, while $\bar{\xi}_i$ is uniformly bounded. Lemma B.2 thus gives

$$\|(\Delta_i^{\text{or}} - \bar{\xi}_i)\mathbf{1}\{\mathcal{D}_{iT}\}\|_{L_i^r} \leq C\mathbb{P}(\mathcal{D}_{iT} \mid \mathcal{C}_i)^{1/r} = O(\exp(-cT/r)) = o(T^{-1}).$$

Combining the two events proves

$$\|\Delta_i^{\text{or}} - \bar{\xi}_i\|_{L_i^r} = O(T^{-1}). \quad (\text{B.11})$$

We next verify the rate of the proposed second-order term. Equation (B.4) and Hölder's inequality give

$$\begin{aligned} \|q_{iT}\|_{L_i^r} &\leq c_G^{-1} \|M_{iT}\bar{\xi}_i\|_{L_i^r} + \frac{c_G^{-1}}{2} \|\bar{H}_i^\psi \bar{\xi}_i^2\|_{L_i^r} \\ &\leq C \|M_{iT}\|_{L_i^{2r}} \|\bar{\xi}_i\|_{L_i^{2r}} + C \|\bar{\xi}_i\|_{L_i^{2r}}^2 \\ &= O(T^{-1}). \end{aligned}$$

On $\mathcal{E}_{iT}^{\text{loc}}$, substitute $\Delta_i^{\text{or}} = \bar{\xi}_i + q_{iT} + r_{iT}^{(3)}$ into (B.9). Before any cancellation, the score equation becomes

$$\begin{aligned} 0 &= \bar{s}_i + G_i \bar{\xi}_i + \left(G_i q_{iT} + M_{iT} \bar{\xi}_i + \frac{1}{2} \bar{H}_i^\psi \bar{\xi}_i^2 \right) \\ &\quad + G_i r_{iT}^{(3)} + M_{iT} (q_{iT} + r_{iT}^{(3)}) + \bar{H}_i^\psi \bar{\xi}_i (q_{iT} + r_{iT}^{(3)}) \\ &\quad + \frac{1}{2} \bar{H}_i^\psi (q_{iT} + r_{iT}^{(3)})^2 + R_{\psi, iT}(\Delta_i^{\text{or}}). \end{aligned}$$

The first pair vanishes because $G_i \bar{\xi}_i = -\bar{s}_i$. The parenthesized second-order group vanishes by (B.4). Expanding the square and collecting every term that is linear in $r_{iT}^{(3)}$ on the left gives the exact identity

$$\begin{aligned} \left(G_i + M_{iT} + \bar{H}_i^\psi \bar{\xi}_i + \bar{H}_i^\psi q_{iT} \right) r_{iT}^{(3)} &= -M_{iT} q_{iT} - \bar{H}_i^\psi \bar{\xi}_i q_{iT} - \frac{1}{2} \bar{H}_i^\psi q_{iT}^2 \\ &\quad - \frac{1}{2} \bar{H}_i^\psi (r_{iT}^{(3)})^2 - R_{\psi, iT}(\Delta_i^{\text{or}}). \end{aligned} \quad (\text{B.12})$$

Equation (B.11) and the bound for q_{iT} just established give the preliminary rate

$$\|r_{iT}^{(3)}\|_{L_i^r} \leq \|\Delta_i^{\text{or}} - \bar{\xi}_i\|_{L_i^r} + \|q_{iT}\|_{L_i^r} = O(T^{-1}).$$

All preceding bounds hold at every fixed moment, so we may use this rate and the rate for q_{iT} at $2r$. Each term on the right of (B.12) then has the following explicit order:

$$\begin{aligned} \|M_{iT} q_{iT}\|_{L_i^r} &\leq \|M_{iT}\|_{L_i^{2r}} \|q_{iT}\|_{L_i^{2r}} = O(T^{-3/2}), \\ \|\bar{H}_i^\psi \bar{\xi}_i q_{iT}\|_{L_i^r} &\leq C \|\bar{\xi}_i\|_{L_i^{2r}} \|q_{iT}\|_{L_i^{2r}} = O(T^{-3/2}), \\ \|\bar{H}_i^\psi q_{iT}^2\|_{L_i^r} &\leq C \|q_{iT}\|_{L_i^{2r}}^2 = O(T^{-2}), \\ \|\bar{H}_i^\psi (r_{iT}^{(3)})^2\|_{L_i^r} &\leq C \|r_{iT}^{(3)}\|_{L_i^{2r}}^2 = O(T^{-2}), \\ \|R_{\psi, iT}(\Delta_i^{\text{or}})\|_{L_i^r} &\leq C \|\Delta_i^{\text{or}}\|_{L_i^{3r}}^3 = O(T^{-3/2}). \end{aligned}$$

It remains to verify that the random coefficient on the left of (B.12) stays away from zero. For a fixed constant $c_0 > 0$, let $\mathcal{G}_{iT}^{(3)} = \mathcal{E}_{iT}^{\text{loc}} \cap \{|M_{iT}| + |\bar{\xi}_i| \leq c_0\}$. On this event, Equation (B.4) implies $|q_{iT}| \leq Cc_0^2$. Choose c_0 small enough that $C(c_0 + c_0^2) \leq c_G/2$. The boundedness of $\bar{H}_i^\psi$ and $|G_i| \geq c_G$ then yield the pathwise lower bound

$$\begin{aligned} |G_i + M_{iT} + \bar{H}_i^\psi \bar{\xi}_i + \bar{H}_i^\psi q_{iT}| &\geq |G_i| - C\{|M_{iT}| + |\bar{\xi}_i| + |q_{iT}|\} \\ &\geq \frac{c_G}{2}. \end{aligned}$$

Dividing (B.12) by this coefficient and using the five bounds above gives

$$\left\| r_{iT}^{(3)} \mathbf{1}\{\mathcal{G}_{iT}^{(3)}\} \right\|_{L_i^r} = O(T^{-3/2}).$$

The tail bound in Lemma B.2, together with $\bar{\xi}_i = -G_i^{-1} \bar{s}_i$, gives

$$\begin{aligned} \mathbb{P}\left((\mathcal{G}_{iT}^{(3)})^c \mid \mathcal{C}_i\right) &\leq \mathbb{P}\left((\mathcal{E}_{iT}^{\text{loc}})^c \mid \mathcal{C}_i\right) + \mathbb{P}\left(|M_{iT}| > \frac{c_0}{2} \mid \mathcal{C}_i\right) \\ &\quad + \mathbb{P}\left(|\bar{\xi}_i| > \frac{c_0}{2} \mid \mathcal{C}_i\right) \leq C \exp(-cT). \end{aligned}$$

Assumption B.1 bounds Δ_i^{or} , $\bar{\xi}_i$, q_{iT} , and hence $r_{iT}^{(3)}$ uniformly. Therefore the contribution from the complementary event satisfies

$$\left\| r_{iT}^{(3)} \mathbf{1}\{(\mathcal{G}_{iT}^{(3)})^c\} \right\|_{L_i^r} \leq C \mathbb{P}\left((\mathcal{G}_{iT}^{(3)})^c \mid \mathcal{C}_i\right)^{1/r} = O(\exp(-cT/r)) = o(T^{-3/2}).$$

Combining the two event contributions proves (B.7). Finally, (B.5) and the rate for q_{iT} give

$$\begin{aligned} |\beta_{iT}| &\leq T \mathbb{E}_i[|q_{iT}|] \leq T \|q_{iT}\|_{L_i^r} = O(1), \\ \|\zeta_{iT}^c\|_{L_i^r} &\leq \|q_{iT}\|_{L_i^r} + |\mathbb{E}_i[q_{iT}]| \leq 2 \|q_{iT}\|_{L_i^r} = O(T^{-1}). \end{aligned}$$

This proves (B.8). $\square$

Remark B.4 (Exact sample-mean specialization). *For the untrimmed sample mean, take $\psi(W_{it}, a)$ to be $Z_{it} - a$. Its unique global root is $\check{\alpha}_i$. Let $\mathcal{E}_{iT} = \{\check{\alpha}_i \in I_i\}$. The oracle local branch defined above satisfies*

$$\tilde{\alpha}_i = \check{\alpha}_i \mathbf{1}\{\mathcal{E}_{iT}\} + \alpha_i \mathbf{1}\{\mathcal{E}_{iT}^c\}. \quad (\text{B.13})$$

Moreover, $G_i = -1$, $M_{iT} = 0$, $\bar{H}_i^\psi = 0$, and $\xi_{it} = Z_{it} - \alpha_i$. Consequently $q_{iT} = \beta_{iT} = \zeta_{iT}^c = 0$, whereas the oracle-branch remainder is

$$r_{iT}^{(3)} = -\bar{\xi}_i \mathbf{1}\{\mathcal{E}_{iT}^c\}.$$

Thus $r_{iT}^{(3)}$ is supported on the exponentially rare selector event; it is not identically zero. By contrast, the raw global root obeys $\check{\alpha}_i - \alpha_i = \bar{\xi}_i$ exactly. Clipping is a separate map and requires the clipping bridge used in the growing-sieve theorem.

B.1.3 Three Fixed-Sieve Bias Channels

Let $e_\theta \in \mathbb{R}^{1+2K}$ select the theta block. Let $f_{it,K}(\rho, a) \in \mathbb{R}^{1+2K}$ be the full stacked moment and fix $\rho_{0,K,N}$. Set $A_{it,K} = \partial_a f_{it,K}(\rho_{0,K,N}, \alpha_i)$ and $H_{it,K} = \partial_{aa} f_{it,K}(\rho_{0,K,N}, \alpha_i)$, with $\mu_{A,i,K} = \mathbb{E}_i[A_{i0,K}]$ and $\mu_{H,i,K} = \mathbb{E}_i[H_{i0,K}]$. Also define $\Gamma_{\xi,i}(\ell) = \mathbb{E}_i[\xi_{i0} \xi_{i\ell}]$ and $\Omega_{\xi,i} = \sum_{\ell \in \mathbb{Z}} \Gamma_{\xi,i}(\ell)$. The sum is finite under Assumption B.1. Define

$$B_{\xi,K,N}^{\text{WD}} = \frac{1}{N} \sum_{i=1}^N \sum_{\ell \in \mathbb{Z}} \mathbb{E}_i[A_{i0,K} \xi_{i\ell}]. \quad (\text{B.14})$$

$$B_{\beta,K,N,T}^{\text{WD}} = \frac{1}{N} \sum_{i=1}^N \mu_{A,i,K} \beta_{iT}. \quad (\text{B.15})$$

$$B_{\xi\xi,K,N}^{\text{WD}} = \frac{1}{2N} \sum_{i=1}^N \mu_{H,i,K} \Omega_{\xi,i}. \quad (\text{B.16})$$

Here WD denotes the weak-dependence contribution.

Proposition B.5 (Three-channel fixed- K population bias). *Suppose Assumption B.1 holds and K is fixed. Define the exact generated-coordinate coefficient*

$$\mathcal{B}_{K,N,T}^{\text{or}} = T \mathbb{E}_{N,T}[f_{it,K}(\rho_{0,K,N}, \tilde{\alpha}_i) - f_{it,K}(\rho_{0,K,N}, \alpha_i)].$$

Then

$$\mathcal{B}_{K,N,T}^{\text{or}} = B_{\xi,K,N}^{\text{WD}} + B_{\beta,K,N,T}^{\text{WD}} + B_{\xi\xi,K,N}^{\text{WD}} + o(1). \quad (\text{B.17})$$

As $T \rightarrow \infty$, allowing $N = N_T$, the constants in the remainder are uniform in i and N . For the sample mean, (B.15) is zero. Combined with the raw-to-local bridge in Lemma B.6, (B.17) yields the raw two-channel formula in the main text.

Proof. Write the unit averages as

$$\begin{aligned} \bar{f}_{i,K}(a) &= \frac{1}{T} \sum_t f_{it,K}(\rho_{0,K,N}, a), \\ \bar{A}_{i,K} &= \frac{1}{T} \sum_t \partial_a f_{it,K}(\rho_{0,K,N}, \alpha_i), \quad \bar{H}_{i,K} = \frac{1}{T} \sum_t \partial_{aa} f_{it,K}(\rho_{0,K,N}, \alpha_i). \end{aligned}$$

Taylor's theorem and Proposition B.3 give

$$\bar{f}_{i,K}(\tilde{\alpha}_i) - \bar{f}_{i,K}(\alpha_i) = \bar{A}_{i,K} \bar{\xi}_i + \bar{A}_{i,K} \left(\frac{1}{T} \beta_{iT} + \zeta_{iT}^c \right) + \bar{A}_{i,K} r_{iT}^{(3)} + \frac{1}{2} \bar{H}_{i,K} (\Delta_i^{\text{or}})^2 + R_{f,iT}, \quad (\text{B.18})$$

where $\Delta_i^{\text{or}} = \tilde{\alpha}_i - \alpha_i$. Because K is fixed, the uniform third-derivative bound in Assumption B.1 gives $\|R_{f,iT}\| \leq C |\Delta_i^{\text{or}}|^3$. Proposition B.3 therefore implies

$$\|R_{f,iT}\|_{L_i^1} = \mathbb{E}_i[\|R_{f,iT}\|] \leq C \mathbb{E}_i[|\Delta_i^{\text{or}}|^3] = C \|\Delta_i^{\text{or}}\|_{L_i^3}^3 = O(T^{-3/2}).$$

Linear term. Expanding the two sample averages and counting pairs with lag $\ell = s - t$, stationarity gives

$$\begin{aligned} T \mathbb{E}_i[\bar{A}_{i,K} \bar{\xi}_i] &= \frac{1}{T} \sum_{t=1}^T \sum_{s=1}^T \mathbb{E}_i[A_{it,K} \xi_{is}] \\ &= \sum_{|\ell| < T} \left(1 - \frac{|\ell|}{T} \right) \mathbb{E}_i[A_{i0,K} \xi_{i\ell}]. \end{aligned} \quad (\text{B.19})$$

If $|\ell| > \ell_0$, conditional ℓ_0 -dependence and $\mathbb{E}_i[\xi_{i\ell}] = 0$ imply

$$\mathbb{E}_i[A_{i0,K} \xi_{i\ell}] = \mathbb{E}_i[A_{i0,K}] \mathbb{E}_i[\xi_{i\ell}] = 0.$$

Consequently, for $T > \ell_0$,

$$\begin{aligned} \sum_{|\ell| < T} \left(1 - \frac{|\ell|}{T} \right) \mathbb{E}_i[A_{i0,K} \xi_{i\ell}] &= \sum_{\ell \in \mathbb{Z}} \mathbb{E}_i[A_{i0,K} \xi_{i\ell}] - \frac{1}{T} \sum_{|\ell| \leq \ell_0} |\ell| \mathbb{E}_i[A_{i0,K} \xi_{i\ell}] \\ &= \sum_{\ell \in \mathbb{Z}} \mathbb{E}_i[A_{i0,K} \xi_{i\ell}] + O(T^{-1}). \end{aligned}$$

First-stage-bias term. Next use the exact split (B.6). Since β_{iT} is $\mathcal{C}_i$ -measurable, $\mathbb{E}_i[\bar{A}_{i,K}] = \mu_{A,i,K}$, and $\mathbb{E}_i[\zeta_{iT}^c] = 0$,

$$\begin{aligned} T \mathbb{E}_i[\bar{A}_{i,K} q_{iT}] &= \mathbb{E}_i[\bar{A}_{i,K}] \beta_{iT} + T \mathbb{E}_i[\bar{A}_{i,K} \zeta_{iT}^c] \\ &= \mu_{A,i,K} \beta_{iT} + T \mathbb{E}_i[(\bar{A}_{i,K} - \mu_{A,i,K}) \zeta_{iT}^c]. \end{aligned}$$

The same residue-class blocking argument used in the proof of Lemma B.2, applied coordinatewise to the fixed-dimensional vector $A_{it,K}$, gives $\|\bar{A}_{i,K} - \mu_{A,i,K}\|_{L_i^2} = O(T^{-1/2})$. Hölder's inequality and Proposition B.3 then give

$$T \|\mathbb{E}_i[(\bar{A}_{i,K} - \mu_{A,i,K}) \zeta_{iT}^c]\| \leq T \|\bar{A}_{i,K} - \mu_{A,i,K}\|_{L_i^2} \|\zeta_{iT}^c\|_{L_i^2} = O(T^{-1/2}).$$

Curvature term. Use $(\Delta_i^{\text{or}})^2 - \bar{\xi}_i^2 = (\Delta_i^{\text{or}} - \bar{\xi}_i)(\Delta_i^{\text{or}} + \bar{\xi}_i)$. Proposition B.3 gives $\|\Delta_i^{\text{or}} - \bar{\xi}_i\|_{L_i^2} = O(T^{-1})$. Lemma B.2 and the triangle inequality then give $\|\Delta_i^{\text{or}} + \bar{\xi}_i\|_{L_i^2} \leq 2 \|\Delta_i^{\text{or}}\|_{L_i^2} + \|\Delta_i^{\text{or}} - \bar{\xi}_i\|_{L_i^2} = O(T^{-1/2})$.

Since $\bar{H}_{i,K}$ is uniformly bounded,

$$\begin{aligned} T \mathbb{E}_i [\|\bar{H}_{i,K} \{(\Delta_i^{\text{or}})^2 - \bar{\xi}_i^2\}\|] &\leq CT \|\Delta_i^{\text{or}} - \bar{\xi}_i\|_{L_i^2} \|\Delta_i^{\text{or}} + \bar{\xi}_i\|_{L_i^2} \\ &= O(T^{-1/2}). \end{aligned}$$

It remains to evaluate $T \mathbb{E}_i [\bar{H}_{i,K} \bar{\xi}_i^2]$. Separate $\bar{H}_{i,K} = \mu_{H,i,K} + (\bar{H}_{i,K} - \mu_{H,i,K})$. The mean part obeys

$$T \mathbb{E}_i [\bar{\xi}_i^2] = \sum_{|\ell| < T} \left(1 - \frac{|\ell|}{T}\right) \Gamma_{\xi,i}(\ell) = \Omega_{\xi,i} + O(T^{-1}). \quad (\text{B.20})$$

The last equality follows by the same finite-window count used for (B.19).

We now make the centered-Hessian count explicit. Write $H_{it,K}^c = H_{it,K} - \mu_{H,i,K}$. Expanding all three averages gives

$$T \mathbb{E}_i [(\bar{H}_{i,K} - \mu_{H,i,K}) \bar{\xi}_i^2] = \frac{1}{T^2} \sum_{r=1}^T \sum_{s=1}^T \sum_{u=1}^T \mathbb{E}_i [H_{ir,K}^c \xi_{is} \xi_{iu}].$$

For a triple (r, s, u) , connect two indices when their distance is at most ℓ_0 . If this graph is disconnected, one factor is a singleton component and is conditionally independent of the remaining two factors. That singleton has conditional mean zero because $\mathbb{E}_i [H_{ir,K}^c] = 0$ and $\mathbb{E}_i [\xi_{it}] = 0$. The corresponding triple expectation therefore vanishes. If the graph is connected, each of s and u lies within $2\ell_0$ periods of r . Thus, for each r , there are at most $(4\ell_0 + 1)^2$ connected pairs (s, u) and hence only $O(T)$ connected triples. Uniform boundedness of $H_{ir,K}^c$ and ξ_{it} now gives

$$\|T \mathbb{E}_i [(\bar{H}_{i,K} - \mu_{H,i,K}) \bar{\xi}_i^2]\| \leq \frac{CT(4\ell_0 + 1)^2}{T^2} = O(T^{-1}).$$

Remainders. Uniform boundedness of $\bar{A}_{i,K}$, the rate for $r_{iT}^{(3)}$, and the Taylor remainder bound above imply

$$\begin{aligned} T \left\| \mathbb{E}_i [\bar{A}_{i,K} r_{iT}^{(3)}] \right\| &\leq CT \left\| r_{iT}^{(3)} \right\|_{L_i^1} = O(T^{-1/2}), \\ T \left\| \mathbb{E}_i [R_{f,iT}] \right\| &\leq T \|R_{f,iT}\|_{L_i^1} = O(T^{-1/2}). \end{aligned}$$

Combining (B.19)–(B.20) with these bounds yields, for every i ,

$$T \mathbb{E}_i [\bar{f}_{i,K}(\tilde{\alpha}_i) - \bar{f}_{i,K}(\alpha_i)] = \sum_{\ell \in \mathbb{Z}} \mathbb{E}_i [A_{i0,K} \xi_{i\ell}] + \mu_{A,i,K} \beta_{iT} + \frac{1}{2} \mu_{H,i,K} \Omega_{\xi,i} + O(T^{-1/2}).$$

Every displayed rate is uniform in i and N . Therefore, the Euclidean norm of the cross-sectional average of the final remainder is bounded by $\frac{1}{N} \sum_{i=1}^N CT^{-1/2} = CT^{-1/2} = o(1)$. Averaging the preceding display over i and using (B.14)–(B.16) gives

$$\mathcal{B}_{K,N,T}^{\text{or}} = B_{\xi,K,N}^{\text{WD}} + B_{\beta,K,N,T}^{\text{WD}} + B_{\xi\xi,K,N}^{\text{WD}} + o(1),$$

which is (B.17). □

Lemma B.6 (Raw-to-local sample-mean bridge). *Suppose Assumption 3.1 holds. For $\mathcal{E}_{iT} = \{\tilde{\alpha}_i \in I_i\}$, let $\tilde{\alpha}_i$ be the oracle local branch in (B.13), and define*

$$D_{iT} = \frac{1}{T} \sum_{t=1}^T \{f_{it,K}(\rho_{0,K,N}, \tilde{\alpha}_i) - f_{it,K}(\rho_{0,K,N}, \tilde{\alpha}_i)\}.$$

Then

$$T \left\| \frac{1}{N} \sum_{i=1}^N \mathbb{E}_i [D_{iT}] \right\| = o(1), \quad \frac{1}{N} \sum_{i=1}^N \{D_{iT} - \mathbb{E}_i [D_{iT}]\} = o_p((NT)^{-1/2}).$$

Proof. On $\mathcal{E}_{iT}$, the global root $\check{\alpha}_i$ lies in I_i and therefore equals the oracle local branch. On $\mathcal{E}_{iT}^c$, the oracle branch falls back to α_i . Hence

$$D_{iT} = \left\{ \frac{1}{T} \sum_{t=1}^T f_{it,K}(\rho_{0,K,N}, \check{\alpha}_i) - \frac{1}{T} \sum_{t=1}^T f_{it,K}(\rho_{0,K,N}, \alpha_i) \right\} \mathbf{1}\{\mathcal{E}_{iT}^c\}.$$

Because $\mathcal{A}$ is an interval and $Z_{it} \in \mathcal{A}$, $\check{\alpha}_i \in \mathcal{A}$. The fixed- K score envelope and (3.7) therefore imply, for every fixed $r \geq 1$,

$$\|D_{iT}\|_{L_i^r} \leq C\mathbb{P}(\mathcal{E}_{iT}^c \mid \mathcal{C}_i)^{1/r} \leq C \exp(-cT/r).$$

For the population bridge, Jensen's inequality gives

$$T \left\| \frac{1}{N} \sum_{i=1}^N \mathbb{E}_i[D_{iT}] \right\| \leq \frac{T}{N} \sum_{i=1}^N \mathbb{E}_i[\|D_{iT}\|] \leq CT \exp(-cT) = o(1).$$

For the centered bridge, the summands are conditionally centered and independent across units. Thus

$$\begin{aligned} \mathbb{E} \left[\left\| \frac{1}{N} \sum_{i=1}^N \{D_{iT} - \mathbb{E}_i[D_{iT}]\} \right\|^2 \middle| \mathcal{C}_N \right] &= \frac{1}{N^2} \sum_{i=1}^N \mathbb{E}_i[\|D_{iT} - \mathbb{E}_i[D_{iT}]\|^2] \\ &\leq \frac{C}{N} \exp(-cT) = o((NT)^{-1}). \end{aligned}$$

The last equality uses $T \exp(-cT) \rightarrow 0$. Conditional Chebyshev's inequality proves the centered assertion. $\square$

B.2 Sample-Mean Two-Channel Drift

Proof. Under the raw sample-mean specialization in Remark B.4, $\xi_{it} = \zeta_{it}$ and $q_{iT} = \beta_{iT} = \zeta_{iT}^c = 0$. Apply $e_\theta^\top$ to (B.17). The two long-run terms satisfy

$$e_\theta^\top B_{\xi,K,N}^{\text{WD}} = B_{\zeta,\theta,K,N}, \quad e_\theta^\top B_{\xi\xi,K,N}^{\text{WD}} = B_{\zeta\zeta,\theta,K,N}.$$

The raw and oracle-branch coefficients differ by the population selector bridge:

$$\begin{aligned} \mathcal{B}_{\theta,K,N,T}^{\text{raw}} &= e_\theta^\top \mathcal{B}_{K,N,T}^{\text{or}} + T e_\theta^\top \frac{1}{N} \sum_{i=1}^N \mathbb{E}_i[D_{iT}] \\ &= e_\theta^\top \mathcal{B}_{K,N,T}^{\text{or}} + o(1), \end{aligned}$$

where the second equality follows from Lemma B.6. Because the first-stage-bias block is zero, substituting (B.17) in the last display gives (3.8). $\square$

Remark B.7 (Connection to the main theorem). *Proposition B.5 gives the fixed- K population result along the oracle local branch. For the sample mean, the raw root is globally observable, Lemma B.6 transfers the local calculation to that root, and the nonlinear first-stage-bias channel is exactly zero. The main theorem then establishes a feasible two-channel correction with growing K .*

B.3 Buffered Within-Unit Splitting

Proof. Block independence and the linear term. For $j \in \{1, 2\}$, define

$$\begin{aligned} \bar{f}_{i,j,K}(a) &= \frac{1}{T_j} \sum_{t \in \mathcal{I}_{j,T}} f_{it,K}^\theta(\rho_{0,K,N}, a), \\ \bar{A}_{i,j,K} &= \partial_a \bar{f}_{i,j,K}(\alpha_i), \quad \bar{H}_{i,j,K} = \partial_{aa} \bar{f}_{i,j,K}(\alpha_i), \end{aligned}$$

and set $T_{-1} = T_2$, $T_{-2} = T_1$, $\mathcal{I}_{-1,T} = \mathcal{I}_{2,T}$, and $\mathcal{I}_{-2,T} = \mathcal{I}_{1,T}$. The split-coordinate error is

$$\Delta_i^{(-j)} = \check{\alpha}_i^{(-j)} - \alpha_i = \frac{1}{T_{-j}} \sum_{s \in \mathcal{I}_{-j,T}} \zeta_{is}.$$

Let $m_T = \lfloor T/2 \rfloor$. The closest training and evaluation indices satisfy

$$\min_{s \in \mathcal{I}_{2,T}, t \in \mathcal{I}_{1,T}} |s - t| = (m_T + \ell_0 + 1) - (m_T - \ell_0) = 2\ell_0 + 1 > \ell_0.$$

Define the block σ -fields

$$\begin{aligned} \mathcal{F}_{i,j}^{\text{ev}} &= \sigma\left(\{f_{it,K}^\theta(\rho_{0,K,N}, a) : t \in \mathcal{I}_{j,T}, a \in \mathcal{A}\} \right. \\ &\quad \left. \cup \{\partial_a^r f_{it,K}^\theta(\rho_{0,K,N}, a) : t \in \mathcal{I}_{j,T}, a \in I_i, r \in \{1, 2, 3\}\}\right), \\ \mathcal{F}_{i,-j}^{\text{tr}} &= \sigma(\{\zeta_{is} : s \in \mathcal{I}_{-j,T}\}). \end{aligned}$$

Assumption 3.1(i) and the displayed separation give $\mathcal{F}_{i,j}^{\text{ev}} \perp\!\!\!\perp \mathcal{F}_{i,-j}^{\text{tr}} \mid \mathcal{C}_i$.

In particular, $\bar{A}_{i,j,K}$ is $\mathcal{F}_{i,j}^{\text{ev}}$ -measurable and $\Delta_i^{(-j)}$ is $\mathcal{F}_{i,-j}^{\text{tr}}$ -measurable. Conditional centering of ζ_{is} gives $\mathbb{E}_i[\Delta_i^{(-j)}] = 0$. Hence

$$\mathbb{E}_i[\bar{A}_{i,j,K} \Delta_i^{(-j)}] = \mathbb{E}_i[\bar{A}_{i,j,K}] \mathbb{E}_i[\Delta_i^{(-j)}] = 0, \quad (\text{B.21})$$

exactly.

Split-coordinate localization and Taylor remainder. The residue-class blocking argument used in the proof of Lemma B.2, now applied to the bounded centered array ζ_{is} on the training block, gives, for $x > 0$,

$$\mathbb{P}\left(\left|\Delta_i^{(-j)}\right| > x \mid \mathcal{C}_i\right) \leq C \exp(-cT_{-j}x^2).$$

Because $T_{-j} \asymp T$, tail integration and the fixed choice $x = \delta$ imply

$$\left\|\Delta_i^{(-j)}\right\|_{L_i^4} = O(T^{-1/2}), \quad \mathbb{P}((\mathcal{E}_{i,j,T}^{\text{cf}})^c \mid \mathcal{C}_i) \leq C \exp(-cT),$$

where $\mathcal{E}_{i,j,T}^{\text{cf}} = \{|\Delta_i^{(-j)}| \leq \delta\}$. Set $D_{i,j,T} = \bar{f}_{i,j,K}(\alpha_i + \Delta_i^{(-j)}) - \bar{f}_{i,j,K}(\alpha_i)$.

The identity $\alpha_i + \Delta_i^{(-j)} = \frac{1}{T_{-j}} \sum_{s \in \mathcal{I}_{-j,T}} Z_{is}$ and convexity of $\mathcal{A}$ imply that this coordinate lies in $\mathcal{A}$ almost surely. The bounded score envelope therefore gives

$$\mathbb{E}_i[D_{i,j,T} \mathbf{1}\{(\mathcal{E}_{i,j,T}^{\text{cf}})^c\}] \leq C \mathbb{P}((\mathcal{E}_{i,j,T}^{\text{cf}})^c \mid \mathcal{C}_i) = O(\exp(-cT)).$$

Define the global Taylor remainder by

$$R_{i,j,T}^{\text{cf}} = D_{i,j,T} - \bar{A}_{i,j,K} \Delta_i^{(-j)} - \frac{1}{2} \bar{H}_{i,j,K} (\Delta_i^{(-j)})^2.$$

On $\mathcal{E}_{i,j,T}^{\text{cf}}$, Taylor's theorem and Assumption 3.1(ii) give $|R_{i,j,T}^{\text{cf}}| \leq C |\Delta_i^{(-j)}|^3$. On the complement, the score envelope and boundedness of $\bar{A}_{i,j,K}$, $\bar{H}_{i,j,K}$, and $\Delta_i^{(-j)}$ give $|R_{i,j,T}^{\text{cf}}| \leq C$. Consequently,

$$\begin{aligned} \mathbb{E}_i[|R_{i,j,T}^{\text{cf}}|] &\leq C \mathbb{E}_i[|\Delta_i^{(-j)}|^3] + C \mathbb{P}((\mathcal{E}_{i,j,T}^{\text{cf}})^c \mid \mathcal{C}_i) \\ &\leq C \left\|\Delta_i^{(-j)}\right\|_{L_i^4}^3 + C \exp(-cT) = O(T^{-3/2}). \end{aligned}$$

Curvature factorization and training-block variance. The Hessian average $\bar{H}_{i,j,K}$ is $\mathcal{F}_{i,j}^{\text{ev}}$ -measurable, whereas $(\Delta_i^{(-j)})^2$ is $\mathcal{F}_{i,-j}^{\text{tr}}$ -measurable. Conditional independence and stationarity therefore give

$$\begin{aligned} \mathbb{E}_i[\bar{H}_{i,j,K} (\Delta_i^{(-j)})^2] &= \mathbb{E}_i[\bar{H}_{i,j,K}] \mathbb{E}_i[(\Delta_i^{(-j)})^2] \\ &= \mu_{\bar{f},i,K} \mathbb{E}_i[(\Delta_i^{(-j)})^2]. \end{aligned}$$

Combining this identity, (B.21), and the remainder bound yields

$$\mathbb{E}_i[D_{i,j,T}] = \frac{1}{2} \mu_{\bar{f},i,K} \mathbb{E}_i[(\Delta_i^{(-j)})^2] + O(T^{-3/2}). \quad (\text{B.22})$$

For the contiguous training block, stationarity gives the exact count

$$\begin{aligned}
\mathbb{E}_i[(\Delta_i^{(-j)})^2] &= \frac{1}{T_{-j}^2} \sum_{s \in \mathcal{I}_{-j,T}} \sum_{r \in \mathcal{I}_{-j,T}} \mathbb{E}_i[\zeta_{is}\zeta_{ir}] \\
&= \frac{1}{T_{-j}^2} \sum_{|\ell| < T_{-j}} (T_{-j} - |\ell|) \mathbb{E}_i[\zeta_{i0}\zeta_{i\ell}], \\
T_{-j} \mathbb{E}_i[(\Delta_i^{(-j)})^2] &= \Omega_{\zeta,i} - \frac{1}{T_{-j}} \sum_{|\ell| \leq \ell_0} |\ell| \mathbb{E}_i[\zeta_{i0}\zeta_{i\ell}] = \Omega_{\zeta,i} + O(T_{-j}^{-1}).
\end{aligned} \tag{B.23}$$

The last line uses $\mathbb{E}_i[\zeta_{i0}\zeta_{i\ell}] = 0$ for $|\ell| > \ell_0$. Since $T_{-j} \asymp T$,

$$\mathbb{E}_i[(\Delta_i^{(-j)})^2] = \frac{\Omega_{\zeta,i}}{T_{-j}} + O(T^{-2}).$$

Substitution into (B.22) yields

$$\mathbb{E}_i[D_{i,j,T}] = \frac{1}{2} \mu_{\bar{f},i,K} \frac{\Omega_{\zeta,i}}{T_{-j}} + O(T^{-3/2}).$$

Aggregation. The definition of the cross-fitted average gives the exact identity

$$\bar{f}_{i,K}^{\theta,\text{cf}} - \bar{f}_{i,K}^{\theta,\text{cf},0} = \frac{T_1}{T_\circ} D_{i,1,T} + \frac{T_2}{T_\circ} D_{i,2,T}.$$

Because $T_{-1} = T_2$ and $T_{-2} = T_1$, the preceding fold-specific expansion implies

$$T \mathbb{E}_i \left[\bar{f}_{i,K}^{\theta,\text{cf}} - \bar{f}_{i,K}^{\theta,\text{cf},0} \right] = \frac{1}{2} \mu_{\bar{f},i,K} \Omega_{\zeta,i} \frac{T}{T_\circ} \left(\frac{T_1}{T_2} + \frac{T_2}{T_1} \right) + O(T^{-1/2}).$$

The fixed buffer gives $T_\circ = T - 2\ell_0$, while $T_1/T \rightarrow 1/2$ and $T_2/T \rightarrow 1/2$. Therefore,

$$\frac{T}{T_\circ} \left(\frac{T_1}{T_2} + \frac{T_2}{T_1} \right) \rightarrow 2,$$

and hence, uniformly in i and N ,

$$T \mathbb{E}_i \left[\bar{f}_{i,K}^{\theta,\text{cf}} - \bar{f}_{i,K}^{\theta,\text{cf},0} \right] = \mu_{\bar{f},i,K} \Omega_{\zeta,i} + o(1).$$

Averaging this display over i gives

$$\begin{aligned}
T \frac{1}{N} \sum_{i=1}^N \mathbb{E}_i \left[\bar{f}_{i,K}^{\theta,\text{cf}} - \bar{f}_{i,K}^{\theta,\text{cf},0} \right] &= \frac{1}{N} \sum_{i=1}^N \mu_{\bar{f},i,K} \Omega_{\zeta,i} + o(1) \\
&= 2B_{\zeta\zeta,\theta,K,N} + o(1),
\end{aligned}$$

which is (3.12). For a one-way split, Equations (B.22) and (B.23) instead give

$$T \mathbb{E}_i[D_{i,j,T}] = \frac{T}{T_{\text{tr}}} \frac{1}{2} \mu_{\bar{f},i,K} \Omega_{\zeta,i} + o(1).$$

If $T_{\text{tr}}/T \rightarrow \lambda$, averaging over i yields $\lambda^{-1} B_{\zeta\zeta,\theta,K,N} + o(1)$. Finally, Equation (B.21) shows that the derivative-influence term is exactly zero at every finite T . $\square$

B.4 Fixed-Sieve Sampling Expansion

B.4.1 Full-Stack Expansion

The expansion retains the full $(1 + 2K)$ -dimensional moment until the final theta projection. Throughout, K is fixed, $N, T \rightarrow \infty$, and $N/T \rightarrow \kappa \in (0, \infty)$. Throughout this fixed-sieve argument, $\tilde{\alpha}_i$ denotes the oracle local branch defined in Section B.1; $\check{\alpha}_i$ continues to denote the globally observable raw sample mean. The

localization objects $\mathcal{H}_K$, $\mathcal{G}_K$, (γ_K^*, δ_K^*) , and V_K^* from Section C.5 are instantiated throughout the argument with $\hat{\alpha}_i = \tilde{\alpha}_i$.

Let $\tilde{F}_{NT,K}(\rho_{0,K,N}) = \mathbb{P}_{NT} f_{it,K}(\rho_{0,K,N}, \tilde{\alpha}_i)$ and $F_{T,K,N}^{\text{or}}(\rho_{0,K,N}) = \mathbb{E}_{N,T}[f_{it,K}(\rho_{0,K,N}, \tilde{\alpha}_i)]$. At the oracle coordinate, write $f_{it,K}^0 = f_{it,K}(\rho_{0,K,N}, \alpha_i)$ and $m_{i,K} = \mathbb{E}_i[f_{i0,K}^0]$. With $\mu_{A,i,K}$ and ξ_{it} defined in Section B.1, set

$$g_{it,K} = f_{it,K}^0 - m_{i,K} + \mu_{A,i,K} \xi_{it}.$$

This vector contains the ordinary oracle fluctuation and the sampling effect of reusing the local-coordinate estimate in all periods of the same unit.

Lemma B.8 (Centered full-stack expansion). *Under Assumption B.1,*

$$\tilde{F}_{NT,K}(\rho_{0,K,N}) - F_{T,K,N}^{\text{or}}(\rho_{0,K,N}) = \mathbb{P}_{NT} g_{it,K} + o_p((NT)^{-1/2}). \quad (\text{B.24})$$

Moreover,

$$\mathbb{P}_{NT} g_{it,K} = O_p((NT)^{-1/2}). \quad (\text{B.25})$$

Proof. For each unit, define the oracle-branch displacement and the three sample averages

$$\begin{aligned} \Delta_i^{\text{or}} &= \tilde{\alpha}_i - \alpha_i, & \bar{f}_{i,K}(a) &= \frac{1}{T} \sum_{t=1}^T f_{it,K}(\rho_{0,K,N}, a), \\ \bar{A}_{i,K} &= \frac{1}{T} \sum_{t=1}^T \partial_a f_{it,K}(\rho_{0,K,N}, \alpha_i), & \bar{H}_{i,K} &= \frac{1}{T} \sum_{t=1}^T \partial_{aa} f_{it,K}(\rho_{0,K,N}, \alpha_i). \end{aligned}$$

Let $R_{f,iT}^{\text{or}}$ be the Taylor remainder defined by

$$\begin{aligned} \bar{f}_{i,K}(\tilde{\alpha}_i) - \bar{f}_{i,K}(\alpha_i) &= \bar{A}_{i,K} \bar{\xi}_i + \bar{A}_{i,K} \left(\frac{\beta_{iT}}{T} + \zeta_{iT}^c \right) + \bar{A}_{i,K} r_{iT}^{(3)} \\ &\quad + \frac{1}{2} \bar{H}_{i,K} (\Delta_i^{\text{or}})^2 + R_{f,iT}^{\text{or}}. \end{aligned}$$

This is (B.18) with the local displacement made explicit, and Taylor's theorem gives $\|R_{f,iT}^{\text{or}}\| \leq C |\Delta_i^{\text{or}}|^3$. Write $\bar{A}_{i,K}^c = \bar{A}_{i,K} - \mu_{A,i,K}$. Subtracting the conditional expectation of every Taylor term gives the exact centered expansion

$$\begin{aligned} \bar{f}_{i,K}(\tilde{\alpha}_i) - \mathbb{E}_i[\bar{f}_{i,K}(\tilde{\alpha}_i)] &= \bar{f}_{i,K}(\alpha_i) - \mathbb{E}_i[\bar{f}_{i,K}(\alpha_i)] + \mu_{A,i,K} \bar{\xi}_i \\ &\quad + \{\bar{A}_{i,K}^c \bar{\xi}_i - \mathbb{E}_i[\bar{A}_{i,K}^c \bar{\xi}_i]\} \\ &\quad + \{\bar{A}_{i,K} q_{iT} - \mathbb{E}_i[\bar{A}_{i,K} q_{iT}]\} \\ &\quad + \{\bar{A}_{i,K} r_{iT}^{(3)} - \mathbb{E}_i[\bar{A}_{i,K} r_{iT}^{(3)}]\} \\ &\quad + \frac{1}{2} \{\bar{H}_{i,K} (\Delta_i^{\text{or}})^2 - \mathbb{E}_i[\bar{H}_{i,K} (\Delta_i^{\text{or}})^2]\} \\ &\quad + \{R_{f,iT}^{\text{or}} - \mathbb{E}_i[R_{f,iT}^{\text{or}}]\}. \end{aligned}$$

The first term after the oracle-coordinate fluctuation is $\frac{1}{T} \sum_t \mu_{A,i,K} \xi_{it}$. For the remaining terms, Hölder's inequality and the fixed- K derivative bounds give

$$\begin{aligned} \|\bar{A}_{i,K}^c \bar{\xi}_i\|_{L_i^2} &\leq \|\bar{A}_{i,K}^c\|_{L_i^4} \|\bar{\xi}_i\|_{L_i^4} = O(T^{-1}), \\ \|\bar{A}_{i,K} q_{iT}\|_{L_i^2} + \|\bar{H}_{i,K} (\Delta_i^{\text{or}})^2\|_{L_i^2} &\leq C \|q_{iT}\|_{L_i^2} + C \|\Delta_i^{\text{or}}\|_{L_i^4}^2 = O(T^{-1}), \\ \|\bar{A}_{i,K} r_{iT}^{(3)}\|_{L_i^2} + \|R_{f,iT}^{\text{or}}\|_{L_i^2} &\leq C \|r_{iT}^{(3)}\|_{L_i^6} + C \|\Delta_i^{\text{or}}\|_{L_i^6}^3 = O(T^{-3/2}). \end{aligned}$$

Centering a random vector increases its L_i^2 norm by at most a factor of two. Hence the first, second, and fourth bracketed centered terms in the exact expansion have L_i^2 norm $O(T^{-1})$, whereas the third and fifth have L_i^2 norm $O(T^{-3/2})$. Here finite-range averaging gives $\|\bar{A}_{i,K}^c\|_{L_i^4} + \|\bar{\xi}_i\|_{L_i^4} = O(T^{-1/2})$, and Lemma B.2 gives $\|\Delta_i^{\text{or}}\|_{L_i^4} + \|\Delta_i^{\text{or}}\|_{L_i^6} = O(T^{-1/2})$. Proposition B.3 supplies the displayed rates for q_{iT} and $r_{iT}^{(3)}$.

Let R_{iT}^{sam} be the sum of the five bracketed centered terms in the exact expansion above. The preceding bounds imply $\|R_{iT}^{\text{sam}}\|_{L_i^2} = O(T^{-1})$. Since these remainders are conditionally centered and independent across units,

$$\mathbb{E} \left[\left\| \frac{1}{N} \sum_{i=1}^N R_{iT}^{\text{sam}} \right\|^2 \middle| \mathcal{C}_N \right] = \frac{1}{N^2} \sum_{i=1}^N \mathbb{E}_i \|R_{iT}^{\text{sam}}\|^2 = O((NT^2)^{-1}) = o((NT)^{-1}).$$

Finally, stationarity and the definition of $g_{it,K}$ give

$$\begin{aligned} & \frac{1}{N} \sum_{i=1}^N \{\bar{f}_{i,K}(\alpha_i) - \mathbb{E}_i[\bar{f}_{i,K}(\alpha_i)] + \mu_{A,i,K} \bar{\xi}_i\} \\ &= \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \{f_{it,K}^0 - m_{i,K} + \mu_{A,i,K} \xi_{it}\} = \mathbb{P}_{NT} g_{it,K}. \end{aligned}$$

This proves (B.24). It remains to record the rate of its leading term. Each $g_{it,K}$ is conditionally centered. Cross-unit independence removes all terms with distinct unit indices, while conditional ℓ_0 -dependence removes the within-unit terms with $|t-s| > \ell_0$. The fixed dimension and bounded envelope therefore give

$$\begin{aligned} \mathbb{E} \left[\|\mathbb{P}_{NT} g_{it,K}\|^2 \middle| \mathcal{C}_N \right] &= \frac{1}{N^2 T^2} \sum_{i=1}^N \sum_{t=1}^T \sum_{s=1}^T \mathbb{E}_i [g_{it,K}^\top g_{is,K}] \\ &\leq \frac{C}{N^2 T^2} \sum_{i=1}^N \sum_{t=1}^T \sum_{s: |s-t| \leq \ell_0} 1 \leq \frac{C}{NT}. \end{aligned}$$

Conditional Chebyshev's inequality proves (B.25). $\square$

Proof of Proposition 3.3. Under the raw sample-mean specialization, $\xi_{it} = \zeta_{it}$, so

$$\begin{aligned} e_\theta^\top g_{it,K} &= f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i) - \mathbb{E}_i[f_{i0,K}^\theta(\rho_{0,K,N}, \alpha_i)] + \mu_{f,i,K} \zeta_{it} \\ &= x_{\theta,it,K}. \end{aligned}$$

The definition of D_{iT} gives the exact raw-to-local centered identity

$$\begin{aligned} & \mathbb{P}_{NT} f_{it,K}^\theta(\rho_{0,K,N}, \check{\alpha}_i) - \mathbb{E}_{N,T}[f_{it,K}^\theta(\rho_{0,K,N}, \check{\alpha}_i)] \\ &= e_\theta^\top \left\{ \tilde{F}_{NT,K}(\rho_{0,K,N}) - F_{T,K,N}^{\text{or}}(\rho_{0,K,N}) \right\} + e_\theta^\top \frac{1}{N} \sum_{i=1}^N \{D_{iT} - \mathbb{E}_i[D_{iT}]\}. \end{aligned}$$

Apply $e_\theta^\top$ to (B.24). The first term on the right is $\mathbb{P}_{NT} x_{\theta,it,K} + o_p((NT)^{-1/2})$, and Lemma B.6 makes the second term $o_p((NT)^{-1/2})$. This proves (3.10). $\square$

Proposition B.9 (Full-stack score with generated-coordinate drift). *Under Assumption B.1,*

$$\tilde{F}_{NT,K}(\rho_{0,K,N}) = \mathbb{P}_{NT} g_{it,K} + \frac{B_{\xi,K,N}^{\text{WD}} + B_{\beta,K,N,T}^{\text{WD}} + B_{\xi\xi,K,N}^{\text{WD}}}{T} + o_p((NT)^{-1/2}). \quad (\text{B.26})$$

Under Assumption 3.1, define $\check{F}_{NT,K}(\rho_{0,K,N}) = \mathbb{P}_{NT} f_{it,K}(\rho_{0,K,N}, \check{\alpha}_i)$. Then $\xi_{it} = \zeta_{it}$ and

$$\check{F}_{NT,K}(\rho_{0,K,N}) = \mathbb{P}_{NT} g_{it,K} + \frac{B_{\xi,K,N}^{\text{WD}} + B_{\xi\xi,K,N}^{\text{WD}}}{T} + o_p((NT)^{-1/2}). \quad (\text{B.27})$$

Proof. We first make the population decomposition explicit. The oracle stacked population normal equations give $\mathbb{E}_{N,T}[f_{it,K}(\rho_{0,K,N}, \alpha_i)] = 0$. Therefore,

$$\begin{aligned} F_{T,K,N}^{\text{or}}(\rho_{0,K,N}) &= \mathbb{E}_{N,T}[f_{it,K}(\rho_{0,K,N}, \alpha_i)] \\ &\quad + \mathbb{E}_{N,T}[f_{it,K}(\rho_{0,K,N}, \check{\alpha}_i) - f_{it,K}(\rho_{0,K,N}, \alpha_i)] \\ &= \frac{1}{T} \mathcal{B}_{K,N,T}^{\text{or}}. \end{aligned}$$

Proposition B.5 therefore yields

$$F_{T,K,N}^{\text{or}}(\rho_{0,K,N}) = \frac{B_{\xi,K,N}^{\text{WD}} + B_{\beta,K,N,T}^{\text{WD}} + B_{\xi\xi,K,N}^{\text{WD}}}{T} + o(T^{-1}).$$

Adding the centered expansion in Lemma B.8 proves (B.26), because $o(T^{-1}) = o((NT)^{-1/2})$ when $N/T \rightarrow \kappa$.
For the raw sample mean,

$$\begin{aligned} \check{F}_{NT,K}(\rho_{0,K,N}) - \tilde{F}_{NT,K}(\rho_{0,K,N}) &= \frac{1}{N} \sum_{i=1}^N D_{iT} \\ &= \frac{1}{N} \sum_{i=1}^N \{D_{iT} - \mathbb{E}_i[D_{iT}]\} + \frac{1}{N} \sum_{i=1}^N \mathbb{E}_i[D_{iT}] \\ &= o_p((NT)^{-1/2}), \end{aligned}$$

where Lemma B.6 controls both terms in the second line. Finally, Remark B.4 gives $B_{\beta,K,N,T}^{\text{WD}} = 0$. Substitution in (B.26) proves (B.27). $\square$

B.4.2 Target-Coordinate Projection

Recall that $e_\theta \in \mathbb{R}^{1+2K}$ selects the theta block. Under the sample-mean specialization,

$$\begin{aligned} e_\theta^\top g_{it,K} &= x_{\theta,it,K}, \\ e_\theta^\top (B_{\xi,K,N}^{\text{WD}} + B_{\xi\xi,K,N}^{\text{WD}}) &= B_{\theta,K,N}^\infty. \end{aligned}$$

To project the full-stack expansion onto the theta coordinate, we impose the following additional sample-stability conditions.

Assumption B.10 (Fixed- K sample stability). *The following conditions hold:*

(i) **Good-root probability and Gram stability.** For the event $\mathcal{G}_K$ constructed in Section C.5, $\mathbb{P}(\mathcal{G}_K^c) \rightarrow 0$. On $\mathcal{G}_K$,

$$\left\| \hat{\Sigma}_{p,K} \right\|_{\text{op}} + \left\| \hat{\Sigma}_{p,K}^{-1} \right\|_{\text{op}} = O_p(1).$$

(ii) **Oracle denominator law.**

$$\mathbb{P}_{NT} u_{D,it,K}^2 - V_K = o_p(1).$$

(iii) **Bounded targets.** $0 < c_V \leq V_K \leq C_V < \infty$ and $|\theta_{0,K,N}| + \|\gamma_{0,K,N}\| + \|\delta_{0,K,N}\| = O(1)$.

These are high-level fixed- K stability conditions for the raw coordinate. The proof below derives the feasible denominator consistency needed for inversion; it does not fold that conclusion into the assumption. The exact root algebra is used only on the good-root branch. The later primitive benchmark verifies the corresponding clipped, growing-sieve localization in Corollary C.19.

Proposition B.11 (Fixed- K target miscentering). *Suppose Assumptions 3.1 and B.10 hold. Then*

$$\sqrt{NT}(\hat{\theta}_K - \theta_{0,K,N}) = \frac{1}{\sqrt{NT}} \sum_{i=1}^N \sum_{t=1}^T \psi_{\theta,it,K} + \sqrt{\frac{N}{T}} V_K^{-1} B_{\theta,K,N}^\infty + o_p(1). \quad (\text{B.28})$$

If the first term on the right converges to $\mathcal{N}(0, \sigma_{\theta,K}^2)$ and $\sqrt{N/T} V_K^{-1} B_{\theta,K,N}^\infty \rightarrow b_{\theta,K}$, then the left side converges to $\mathcal{N}(b_{\theta,K}, \sigma_{\theta,K}^2)$.

Proof. **Full-stack score and nuisance rates.** Let S_γ and S_δ select the gamma and delta blocks of a stacked vector. Under the fixed- K convention, $\hat{p}_{it,K} = p_K(X_{it}, \check{\alpha}_i)$. Equations (B.25) and (B.27) give the following rate because the fixed dimension, bounded envelope, and finite dependence range make both bias vectors $O(1)$:

$$\check{F}_{NT,K}(\rho_{0,K,N}) = O_p((NT)^{-1/2} + T^{-1}) = O_p(T^{-1}).$$

On the good-root event, their affine normal equations give

$$\begin{aligned}\widehat{\Sigma}_{p,K}(\widehat{\gamma}_K - \gamma_{0,K,N}) &= S_\gamma \check{F}_{NT,K}(\rho_{0,K,N}), \\ \widehat{\Sigma}_{p,K}(\widehat{\delta}_K - \delta_{0,K,N}) &= S_\delta \check{F}_{NT,K}(\rho_{0,K,N}).\end{aligned}$$

Assumption B.10 therefore implies $\|\widehat{\gamma}_K - \gamma_{0,K,N}\| + \|\widehat{\delta}_K - \delta_{0,K,N}\| = O_p(T^{-1})$. Set $e_\gamma = \widehat{\gamma}_K - \gamma_{0,K,N}$.

Generated-coordinate denominator and profile remainder. We next verify denominator consistency at the generated coordinate. Let $p_{it,K}^0 = p_K(X_{it}, \alpha_i)$ and define

$$\begin{aligned}d_{\alpha,it,K} &= (\widehat{p}_{it,K} - p_{it,K}^0)^\top \gamma_{0,K,N}, \\ \widetilde{u}_{D,it,K} &= D_{it} - \widehat{p}_{it,K}^\top \gamma_{0,K,N} = u_{D,it,K} - d_{\alpha,it,K}.\end{aligned}$$

On $\mathcal{G}_K$, $\widehat{u}_{D,it,K} = \widetilde{u}_{D,it,K} - \widehat{p}_{it,K}^\top e_\gamma$, and the affine normal equation gives

$$\begin{aligned}\mathbb{P}_{NT}\{\widehat{u}_{D,it,K}^2 - \widetilde{u}_{D,it,K}^2\} &= -2e_\gamma^\top \mathbb{P}_{NT} \widehat{p}_{it,K} \widehat{u}_{D,it,K} - e_\gamma^\top \widehat{\Sigma}_{p,K} e_\gamma \\ &= -e_\gamma^\top \widehat{\Sigma}_{p,K} e_\gamma.\end{aligned}$$

Expanding $\widetilde{u}_{D,it,K}^2 = (u_{D,it,K} - d_{\alpha,it,K})^2$ now yields the exact decomposition

$$\widehat{V}_K - V_K = \{\mathbb{P}_{NT} u_{D,it,K}^2 - V_K\} - 2\mathbb{P}_{NT} u_{D,it,K} d_{\alpha,it,K} + \mathbb{P}_{NT} d_{\alpha,it,K}^2 - e_\gamma^\top \widehat{\Sigma}_{p,K} e_\gamma. \quad (\text{B.29})$$

The sample-mean rate follows directly from finite-range dependence:

$$\mathbb{E} \left[\frac{1}{N} \sum_{i=1}^N (\check{\alpha}_i - \alpha_i)^2 \middle| \mathcal{C}_N \right] = \frac{1}{N} \sum_{i=1}^N \mathbb{E}_i \left[\left(\frac{1}{T} \sum_{t=1}^T \zeta_{it} \right)^2 \right] \leq \frac{C}{T}.$$

The bounded derivative of p_K and the bounded target coefficient therefore give $\mathbb{P}_{NT} d_{\alpha,it,K}^2 = O_p(T^{-1})$. Assumption B.10(ii)–(iii) gives $\mathbb{P}_{NT} u_{D,it,K}^2 = O_p(1)$, so Cauchy–Schwarz implies

$$|\mathbb{P}_{NT} u_{D,it,K} d_{\alpha,it,K}| \leq (\mathbb{P}_{NT} u_{D,it,K}^2)^{1/2} (\mathbb{P}_{NT} d_{\alpha,it,K}^2)^{1/2} = O_p(T^{-1/2}).$$

Finally, $e_\gamma^\top \widehat{\Sigma}_{p,K} e_\gamma = O_p(T^{-2})$. Every term on the right of (B.29) is therefore $o_p(1)$ on $\mathcal{G}_K$. Because $\mathbb{P}(\mathcal{G}_K^c) \rightarrow 0$ and the denominator is globally extended by construction, this proves

$$\widehat{V}_K - V_K = o_p(1). \quad (\text{B.30})$$

Set $e_\pi = (\widehat{\delta}_K - \delta_{0,K,N}) - \theta_{0,K,N} e_\gamma$. Because $\theta_{0,K,N}$ is bounded,

$$\|e_\pi\| \leq \|\widehat{\delta}_K - \delta_{0,K,N}\| + |\theta_{0,K,N}| \|e_\gamma\| = O_p(T^{-1}).$$

The fixed- K Gram has bounded operator norm. Hence the profile quadratic satisfies

$$\sqrt{NT} \left| e_\gamma^\top \widehat{\Sigma}_{p,K} e_\pi \right| \leq \sqrt{NT} \left\| \widehat{\Sigma}_{p,K} \right\|_{\text{op}} \|e_\gamma\| \|e_\pi\| = O_p(T^{-1}) = o_p(1). \quad (\text{B.31})$$

Exact target-root expansion. Apply $e_\theta^\top$ to (B.27) and insert the localized exact theta identity (5.1). On the good-root event, solving that identity gives

$$\begin{aligned}\sqrt{NT}(\widehat{\theta}_K - \theta_{0,K,N}) &= \widehat{V}_K^{-1} \frac{1}{\sqrt{NT}} \sum_{i=1}^N \sum_{t=1}^T x_{\theta,it,K} \\ &\quad + \sqrt{\frac{N}{T}} \widehat{V}_K^{-1} B_{\theta,K,N}^\infty - \sqrt{NT} \widehat{V}_K^{-1} e_\gamma^\top \widehat{\Sigma}_{p,K} e_\pi + o_p(1).\end{aligned}$$

Equations (B.30) and $V_K \geq c_V$ give $\widehat{V}_K^{-1} = V_K^{-1} + o_p(1)$. The second-moment bound (B.25), rather than a central limit theorem, gives

$$\frac{1}{\sqrt{NT}} \sum_{i=1}^N \sum_{t=1}^T x_{\theta,it,K} = O_p(1).$$

The fixed- K coefficient $B_{\theta,K,N}^\infty$ is $O(1)$ by the bounded finite-range channel definitions. Therefore, replacing $\widehat{V}_K^{-1}$ by V_K^{-1} in the first two terms costs $o_p(1)$. Equation (B.31) removes the third displayed term. Because $\psi_{\theta,it,K} = V_K^{-1}x_{\theta,it,K}$, the remaining two terms are exactly those in (B.28) on $\mathcal{G}_K$.

Global root and limiting distribution. Assumption B.10(i) gives $\mathbb{P}(\mathcal{G}_K^c) \rightarrow 0$. If R_{NT} denotes the difference between the two sides of that expansion, then, for every $\varepsilon > 0$,

$$\mathbb{P}(|R_{NT}| > \varepsilon) \leq \mathbb{P}(\mathcal{G}_K^c) + \mathbb{P}(|R_{NT}| > \varepsilon, \mathcal{G}_K) \rightarrow 0.$$

This transfers the expansion to the globally reported root. The asserted distributional statement then follows from the two additional convergences in the proposition and Slutsky's theorem. $\square$

Proof of Proposition 3.4. Assumption 3.1 specializes Assumption B.1. Equation (B.27) already incorporates the population and centered raw-to-local conclusions of Lemma B.6; under this specialization, $\xi_{it} = \zeta_{it}$ and $B_{\beta,K,N,T}^{\text{WD}} = 0$. The theta projections displayed above then give

$$e_\theta^\top g_{it,K} = x_{\theta,it,K}, \quad e_\theta^\top (B_{\xi,K,N}^{\text{WD}} + B_{\xi\xi,K,N}^{\text{WD}}) = B_{\theta,K,N}^\infty.$$

Proposition B.11 therefore becomes (3.11). The final step of its proof uses Assumption B.10(i) to transfer the good-root calculation to the globally reported root constructed in Section C.5. $\square$

C Technical Foundations for Growing-Sieve Estimation

C.1 Boundary Tensor-Spline Properties

The results below establish the deterministic spline properties used by the main panel theorem. We use standard B-spline notation and basic approximation facts from de Boor (2001); Schumaker (2007), together with the following boundary-stable tensor-product construction. Let $W^+ = [0, 1]^{d_X} \times [0, 1]$, let $d_W = d_X + 1$ be fixed, and write the last coordinate as a . Set $b_T = C_b \sqrt{\frac{\log N}{T}}$ and let the deterministic coordinates satisfy the quasi-uniform low-discrepancy condition (5.6). Let p_K be the Lebesgue- L^2 normalized, open-clamped, maximum-smoothness tensor-product B-spline basis of fixed degree $p \geq 4$, built on quasi-uniform positive knot spans of width $\mathfrak{h}_K$. Write $p_K(x, a) = \psi_{K_x}(x) \otimes \phi_{K_a}(a)$, where $K_x \asymp \mathfrak{h}_K^{-d_X}$, $K_a \asymp \mathfrak{h}_K^{-1}$, and $K \asymp \mathfrak{h}_K^{-d_W}$. When $d_X = 0$, take $K_x = 1$ and $\psi_{K_x} \equiv 1$. Throughout this subsection, $N\mathfrak{h}_K \rightarrow \infty$ and $b_T/\mathfrak{h}_K \rightarrow 0$.

We keep the continuous and panel-grid measures distinct

$$\mu_c(dw) = dw, \quad \mu_g(dw) = dx \otimes \left\{ \frac{1}{N} \sum_{i=1}^N \delta_{\alpha_i}(da) \right\}.$$

Define $G_c = \int p_K p_K^\top d\mu_c$ and $G_g = \int p_K p_K^\top d\mu_g$. Whenever the inverse exists, let

$$\Pi_K^c f = p_K^\top G_c^{-1} \int p_K f d\mu_c, \quad \Pi_{K,N}^g f = p_K^\top G_g^{-1} \int p_K f d\mu_g.$$

The arguments below compare these operators while keeping them distinct.

C.1.1 A Boundary-Stable Local Dual

Lemma C.1 (Boundary local dual). *There is a tensor-product quasi-interpolant Q_K that reproduces $S_K = \{p_K^\top b : b \in \mathbb{R}^K\}$ and satisfies $\|Q_K f\|_\infty \leq C \|f\|_\infty$, with a constant independent of K . The bound holds uniformly for interior and boundary basis functions.*

Proof. The construction proceeds in four steps. We assign each one-dimensional B-spline a positive knot cell, obtain a Gram inverse that is uniform over all interior and endpoint configurations, normalize the resulting local dual, and tensorize it. The normalization is what makes the final L^∞ bound independent of the knot width.

One-dimensional construction. It suffices to construct a one-dimensional dual and then tensorize it. Write the open-clamped knot sequence as

$$0 = \tau_{-p} = \cdots = \tau_0 < \tau_1 < \cdots < \tau_M = \tau_{M+1} = \cdots = \tau_{M+p} = 1.$$

Let B_j , $j = -p, \dots, M-1$, be the raw degree- p B-splines, and let $I_r = (\tau_r, \tau_{r+1})$, $r = 0, \dots, M-1$, be the positive cells. The active indices on I_r are $A_r = \{r-p, \dots, r\}$. For every basis index, select the cell $I_{r(j)}$ with $r(j) = \max\{j, 0\}$. If $j \geq 0$, then $j \in A_j$ and $I_j \subset \text{supp}(B_j)$. If $j < 0$, repeated left-end knots give $j \in A_0$ and $I_0 \subset \text{supp}(B_j)$. Thus every basis function has a selected positive cell, including at the boundary.

Uniform local Gram bound. For a cell I_r , set $\ell_r = (p-r)_+$, $u_r = \{p - (M-1-r)\}_+$, and $c_r = (\ell_r, u_r)$. The pair c_r records the repeated-knot configuration. This finite list also covers a coarse mesh on which one cell is close to both boundaries. Let $h_r = \tau_{r+1} - \tau_r$. After the affine change $a = \tau_r + h_r x$, the raw local Gram $H_r^B = \left[\int_{I_r} B_j(a) B_k(a) da \right]_{j,k \in A_r}$ has the form $H_r^B = h_r G_{c_r}(\rho)$. The positive neighboring span ratios in ρ lie in a common compact interval by quasi-uniformity. Zero endpoint spans are fixed by c_r and are not ratio coordinates. The restrictions $\{B_j|_{I_r} : j \in A_r\}$ are linearly independent. Continuity over the finite collection of compact configuration spaces therefore gives $\|(H_r^B)^{-1}\|_{\text{op}} \leq C h_r^{-1} \leq C \mathfrak{h}_K^{-1}$.

Normalized dual. Let $s_j = \|B_j\|_{L^2(0,1)}$ and $\phi_j = B_j/s_j$. The same compactness argument gives $c \mathfrak{h}_K^{1/2} \leq s_j \leq C \mathfrak{h}_K^{1/2}$ uniformly, including the boundary indices.

Define the raw local dual

$$d_j^B(a) = \mathbf{1}\{a \in I_{r(j)}\} \sum_{\ell \in A_{r(j)}} [(H_{r(j)}^B)^{-1}]_{j\ell} B_\ell(a),$$

$$\lambda_j^B(f) = \int_{I_{r(j)}} d_j^B(a) f(a) da.$$

Then $\lambda_j^B(B_k) = \mathbf{1}\{j = k\}$. Moreover, $\|d_j^B\|_\infty \leq C \mathfrak{h}_K^{-1}$ and its support is one cell. Therefore, $|\lambda_j^B(f)| \leq C \|f\|_{\infty, I_{r(j)}}$. The normalized functional satisfies

$$\lambda_j^\phi(f) = s_j \lambda_j^B(f), \quad \lambda_j^\phi(\phi_k) = \delta_{jk},$$

$$|\lambda_j^\phi(f)| \leq C \mathfrak{h}_K^{1/2} \|f\|_{\infty, I_{r(j)}}.$$

Because $\|\phi_j\|_\infty \leq C \mathfrak{h}_K^{-1/2}$ and at most $p+1$ splines are active at a point, $Q_K^{(1)} f = \sum_j \lambda_j^\phi(f) \phi_j$ satisfies $\|Q_K^{(1)} f\|_\infty \leq C \|f\|_\infty$. The factors $\mathfrak{h}_K^{1/2}$ and $\mathfrak{h}_K^{-1/2}$ cancel.

Tensorization. For fixed d_W , tensorize the basis and duals:

$$\Phi_j(w) = \prod_{r=1}^{d_W} \phi_{j_r}^{(r)}(w_r), \quad \Lambda_j = \lambda_{j_1}^{\phi,1} \otimes \cdots \otimes \lambda_{j_{d_W}}^{\phi,d_W}.$$

Tensor biorthogonality gives $\Lambda_j(\Phi_k) = \delta_{jk}$. Also, $|\Lambda_j(f)| \leq C \mathfrak{h}_K^{d_W/2} \|f\|_\infty$ and $\|\Phi_j\|_\infty \leq C \mathfrak{h}_K^{-d_W/2}$. At most $(p+1)^{d_W}$ tensor basis functions are active at a point. Thus $Q_K f = \sum_j \Lambda_j(f) \Phi_j$ has the asserted stability. Biorthogonality proves reproduction of S_K . $\square$

C.1.2 Continuous Riesz Bounds and Endpoint-Grid Quadrature

Lemma C.2 (Continuous Riesz and derivative inverse bounds). *There are constants $0 < c < C < \infty$, independent of K, N, T , such that*

$$c \|b\|_2^2 \leq \int_{W^+} \{b^\top p_K(w)\}^2 dw \leq C \|b\|_2^2. \quad (\text{C.1})$$

For $s_b = p_K^\top b$ and $j = 1, 2, 3, 4$,

$$\|\partial_a^j s_b\|_{L^2(W^+)}^2 \leq C \mathfrak{h}_K^{-2j} \|b\|_2^2. \quad (\text{C.2})$$

For $j = 4$, the derivative is understood cellwise, equivalently as the almost-everywhere weak derivative. The fourth derivative need not agree across knots. No lower derivative bound is asserted.

Proof. The local dual and bounded overlap give the Riesz inequalities for every $d_W \geq 1$. Constants may depend on the fixed degree p , the fixed dimension d_W , and the quasi-uniformity constants, but not on K, N, T .

Tensor-product Riesz bounds. Index the normalized tensor basis by $\mathbf{j}$ and write

$$s_b(w) = \sum_{\mathbf{j}} b_{\mathbf{j}} \Phi_{\mathbf{j}}(w), \quad \Phi_{\mathbf{j}}(w) = \prod_{\ell=1}^{d_W} \phi_{j_\ell}^{(\ell)}(w_\ell).$$

At most $(p+1)^{d_W}$ tensor basis functions are nonzero at any point. Moreover, Fubini's theorem and the one-dimensional normalization give $\|\Phi_{\mathbf{j}}\|_{L^2(W_+)} = 1$. Hence

$$\begin{aligned} |s_b(w)|^2 &\leq (p+1)^{d_W} \sum_{\mathbf{j}: \Phi_{\mathbf{j}}(w) \neq 0} |b_{\mathbf{j}}|^2 \Phi_{\mathbf{j}}(w)^2, \\ \|s_b\|_{L^2(W_+)}^2 &\leq (p+1)^{d_W} \sum_{\mathbf{j}} |b_{\mathbf{j}}|^2 = (p+1)^{d_W} \|b\|_2^2. \end{aligned}$$

For the tensor lower bound, let

$$I_{\mathbf{r}} = \prod_{\ell=1}^{d_W} I_{r_\ell}^{(\ell)}, \quad \mathbf{r}(\mathbf{j}) = (r(j_1), \dots, r(j_{d_W})).$$

Each one-dimensional dual is an integral functional supported on its selected cell. Therefore, the product kernel representing $\Lambda_{\mathbf{j}}$ is supported on $I_{\mathbf{r}(\mathbf{j})}$, and its L^1 norm is bounded by the product of the one-dimensional L^1 bounds. Thus,

$$|\Lambda_{\mathbf{j}}(f)| \leq C \mathfrak{h}_K^{d_W/2} \|f\|_{\infty, I_{\mathbf{r}(\mathbf{j})}}.$$

Tensor biorthogonality gives $b_{\mathbf{j}} = \Lambda_{\mathbf{j}}(s_b)$. On every tensor cell, s_b is a tensor polynomial of degree at most p in each coordinate. Affine rescaling to $[0, 1]^{d_W}$, fixed-dimensional polynomial norm equivalence, and $\text{Leb}(I_{\mathbf{r}}) \asymp \mathfrak{h}_K^{d_W}$ give

$$|b_{\mathbf{j}}|^2 \leq C \mathfrak{h}_K^{d_W} \|s_b\|_{\infty, I_{\mathbf{r}(\mathbf{j})}}^2 \leq C \|s_b\|_{L^2(I_{\mathbf{r}(\mathbf{j})})}^2.$$

In each coordinate, I_0 has $p+1$ preimages under $j \mapsto r(j)$, whereas every other positive cell has one preimage. Hence

$$\#\{\mathbf{j} : \mathbf{r}(\mathbf{j}) = \mathbf{r}\} = (p+1)^{\#\{\ell: r_\ell=0\}} \leq (p+1)^{d_W}.$$

Summing the coefficient bounds by selected tensor cell now yields

$$\begin{aligned} \|b\|_2^2 &\leq C \sum_{\mathbf{j}} \|s_b\|_{L^2(I_{\mathbf{r}(\mathbf{j})})}^2 \\ &= C \sum_{\mathbf{r}} \#\{\mathbf{j} : \mathbf{r}(\mathbf{j}) = \mathbf{r}\} \|s_b\|_{L^2(I_{\mathbf{r}})}^2 \\ &\leq C(p+1)^{d_W} \|s_b\|_{L^2(W_+)}^2. \end{aligned}$$

Together with the tensor upper bound, this proves (C.1).

Derivative inverse bound. Fix a tensor cell $I_{\mathbf{r}}$ and affinely rescale it to $[0, 1]^{d_W}$. Let $\tilde{s}_{\mathbf{r}}$ denote the resulting tensor polynomial and let h_{a, r_a} be the width of the cell in the a -coordinate. For $j = 1, 2, 3, 4$, the chain rule and the polynomial inverse inequality on the fixed-dimensional tensor-polynomial space give

$$\begin{aligned} \|\partial_a^j s_b\|_{L^2(I_{\mathbf{r}})}^2 &= \text{Leb}(I_{\mathbf{r}}) h_{a, r_a}^{-2j} \|\partial_{u_a}^j \tilde{s}_{\mathbf{r}}\|_{L^2([0, 1]^{d_W})}^2 \\ &\leq C \text{Leb}(I_{\mathbf{r}}) h_{a, r_a}^{-2j} \|\tilde{s}_{\mathbf{r}}\|_{L^2([0, 1]^{d_W})}^2 \\ &= C h_{a, r_a}^{-2j} \|s_b\|_{L^2(I_{\mathbf{r}})}^2 \leq C \mathfrak{h}_K^{-2j} \|s_b\|_{L^2(I_{\mathbf{r}})}^2. \end{aligned}$$

Maximum smoothness and $p \geq 4$ ensure that the derivatives through order three agree across knots. The fourth derivative exists on every cell and defines an almost-everywhere weak derivative, although it may jump at a knot. Since the tensor cells partition W^+ up to a Lebesgue-null set, summing the preceding inequality and applying the upper Riesz bound give

$$\begin{aligned} \|\partial_a^j s_b\|_{L^2(W^+)}^2 &= \sum_r \|\partial_a^j s_b\|_{L^2(I_r)}^2 \\ &\leq C \mathfrak{h}_K^{-2j} \sum_r \|s_b\|_{L^2(I_r)}^2 = C \mathfrak{h}_K^{-2j} \|s_b\|_{L^2(W^+)}^2 \\ &\leq C \mathfrak{h}_K^{-2j} \|b\|_2^2. \end{aligned}$$

This proves (C.2).

The partition of unity gives $1 \in S_K$, so a nonzero coefficient vector represents $s_b \equiv 1$ while $\partial_a^j s_b = 0$ for every $j \geq 1$. Hence no positive lower derivative bound holds on the full coefficient space. $\square$

For a fixed integer m_q , let $\mathcal{Q}_{m_q, K} = \{q : q \geq 0, q \in C[0, 1], q|_{I_r} \in \mathcal{P}_{m_q}(I_r)\}$.

Lemma C.3 (Quasi-uniform coordinate quadrature). *For every $q \in \mathcal{Q}_{m_q, K}$,*

$$\left| \frac{1}{N} \sum_{i=1}^N q(\alpha_i) - \int_0^1 q(a) da \right| \leq C \left\{ \frac{1}{N \mathfrak{h}_K} + \frac{b_T}{\mathfrak{h}_K} \right\} \int_0^1 q(a) da. \quad (\text{C.3})$$

Consequently, with $\varepsilon_{N, T, K} = C\{(N \mathfrak{h}_K)^{-1} + b_T/\mathfrak{h}_K\}$, the continuous and grid Grams satisfy

$$-\varepsilon_{N, T, K} G_c \preceq G_g - G_c \preceq \varepsilon_{N, T, K} G_c. \quad (\text{C.4})$$

The result allows arbitrary grid-knot alignment and permits empty knot cells.

Proof. The argument first converts cellwise polynomial norm equivalence into a total-variation bound. It then separates the discrepancy on the interior design interval from the two omitted boundary strips. Finally, it applies the scalar quadrature bound to quadratic spline forms to obtain the Loewner comparison. Because $b_T/\mathfrak{h}_K \rightarrow 0$ and $\mathfrak{h}_K \leq 1$, we have $b_T \rightarrow 0$. Hence N, T may be taken sufficiently large that $4b_T \leq 1/2$.

Cellwise variation and boundary mass. Fix $q \in \mathcal{Q}_{m_q, K}$. For a positive knot cell $I_r = (\tau_r, \tau_{r+1})$, let $h_r = \tau_{r+1} - \tau_r$ and set $\tilde{q}_r(u) = q(\tau_r + h_r u)$ for $u \in [0, 1]$. Finite-dimensional norm equivalence on $\mathcal{P}_{m_q}([0, 1])$ and boundedness of differentiation on this space, with the derivative inequality trivial when $m_q = 0$, give

$$\|\tilde{q}_r\|_{\infty, [0, 1]} \leq C \|\tilde{q}_r\|_{L^1(0, 1)}, \quad \|\tilde{q}_r'\|_{L^1(0, 1)} \leq C \|\tilde{q}_r\|_{L^1(0, 1)}.$$

The nonnegativity of q is essential here: it gives $\|\tilde{q}_r\|_{L^1(0, 1)} = \int_0^1 \tilde{q}_r(u) du$ without cancellation. Changing variables between I_r and $[0, 1]$ therefore gives

$$\begin{aligned} \|q\|_{\infty, I_r} &= \|\tilde{q}_r\|_{\infty, [0, 1]} \leq C \int_0^1 \tilde{q}_r(u) du = C h_r^{-1} \int_{I_r} q(a) da, \\ \int_{I_r} |q'(a)| da &= \int_0^1 |\tilde{q}_r'(u)| du \leq C \int_0^1 \tilde{q}_r(u) du = C h_r^{-1} \int_{I_r} q(a) da. \end{aligned} \quad (\text{C.5})$$

Because q is continuous across the knots, its total variation has no knot-jump terms. Thus (C.5) and $h_r \asymp \mathfrak{h}_K$ imply

$$\text{TV}_{[0, 1]}(q) = \sum_r \int_{I_r} |q'(a)| da \leq C \sum_r h_r^{-1} \int_{I_r} q(a) da \leq C \mathfrak{h}_K^{-1} \int_0^1 q(a) da. \quad (\text{C.6})$$

Set $I_T = [2b_T, 1 - 2b_T]$ and $B_T = [0, 2b_T] \cup [1 - 2b_T, 1]$. For every knot cell,

$$|B_T \cap I_r| \leq \min\{h_r, 4b_T\}, \quad \frac{|B_T \cap I_r|}{h_r} \leq \min\left\{1, \frac{4b_T}{h_r}\right\} \leq C \frac{b_T}{\mathfrak{h}_K}.$$

The factor 4 is the total length of the two boundary strips and is absorbed into C . Using the exact cell intersections and the first inequality in (C.5) gives

$$\begin{aligned}
\int_{B_T} q(a) \, da &= \sum_r \int_{B_T \cap I_r} q(a) \, da \\
&\leq \sum_r |B_T \cap I_r| \|q\|_{\infty, I_r} \\
&\leq C \sum_r \frac{|B_T \cap I_r|}{h_r} \int_{I_r} q(a) \, da \\
&\leq C \frac{b_T}{\mathfrak{h}_K} \int_0^1 q(a) \, da.
\end{aligned} \tag{C.7}$$

Because the calculation uses $B_T \cap I_r$, it also covers a boundary strip that cuts through a knot cell.

Interior discrepancy. Let

$$\nu = \frac{1}{N} \sum_{i=1}^N \delta_{\alpha_i} - \text{Unif}(I_T).$$

This signed measure has total mass zero, is supported in $I_T \subset (0, 1)$, and has cumulative function $F_\nu = F_N - F_T^I$. In particular, $F_\nu(0) = F_\nu(1) = 0$. Since q is continuous and of bounded variation, Stieltjes integration by parts gives

$$\int_{[0,1]} q(a) \, \nu(da) = [q(a)F_\nu(a)]_{a=0}^{a=1} - \int_{[0,1]} F_\nu(a) \, dq(a) = - \int_{[0,1]} F_\nu(a) \, dq(a).$$

Thus the boundary term vanishes explicitly. The total-variation inequality, the discrepancy condition (5.6), and Equation (C.6) imply

$$\begin{aligned}
\left| \frac{1}{N} \sum_{i=1}^N q(\alpha_i) - \frac{1}{1-4b_T} \int_{I_T} q(a) \, da \right| &\leq \|F_\nu\|_\infty \text{TV}_{[0,1]}(q) \\
&\leq \frac{C}{N\mathfrak{h}_K} \int_0^1 q(a) \, da.
\end{aligned}$$

To pass from the interior-uniform integral to the full integral, first write the exact identity

$$\frac{1}{1-4b_T} \int_{I_T} q(a) \, da - \int_0^1 q(a) \, da = \frac{4b_T}{1-4b_T} \int_{I_T} q(a) \, da - \int_{B_T} q(a) \, da.$$

Because $1-4b_T \geq 1/2$, Equation (C.7) and $\mathfrak{h}_K \leq 1$ give

$$\begin{aligned}
\left| \frac{1}{1-4b_T} \int_{I_T} q(a) \, da - \int_0^1 q(a) \, da \right| &\leq \frac{4b_T}{1-4b_T} \int_{I_T} q(a) \, da + \int_{B_T} q(a) \, da \\
&\leq Cb_T \int_0^1 q(a) \, da + C \frac{b_T}{\mathfrak{h}_K} \int_0^1 q(a) \, da \\
&\leq C \frac{b_T}{\mathfrak{h}_K} \int_0^1 q(a) \, da.
\end{aligned}$$

Applying the triangle inequality to the empirical, interior-uniform, and full-integral quantities now gives

$$\begin{aligned}
\left| \frac{1}{N} \sum_{i=1}^N q(\alpha_i) - \int_0^1 q(a) \, da \right| &\leq \left| \frac{1}{N} \sum_{i=1}^N q(\alpha_i) - \frac{1}{1-4b_T} \int_{I_T} q(a) \, da \right| \\
&\quad + \left| \frac{1}{1-4b_T} \int_{I_T} q(a) \, da - \int_0^1 q(a) \, da \right| \\
&\leq C \left\{ \frac{1}{N\mathfrak{h}_K} + \frac{b_T}{\mathfrak{h}_K} \right\} \int_0^1 q(a) \, da.
\end{aligned}$$

This proves (C.3). Empty knot cells and grid cells not aligned with knot cells are already covered by the bounded-variation identity, so no occupancy assumption at the knot-cell level is needed.

Gram comparison. Fix $c \in \mathbb{R}^K$ and set

$$q_c(a) = \int_{[0,1]^{d_X}} \{c^\top p_K(x, a)\}^2 dx.$$

This function is nonnegative. On each a -cell I_r , there are coefficient functions $\beta_{\ell,r}$ such that

$$\begin{aligned} c^\top p_K(x, a) &= \sum_{\ell=0}^p \beta_{\ell,r}(x) a^\ell, \\ q_c(a) &= \sum_{\ell=0}^p \sum_{m=0}^p a^{\ell+m} \int_{[0,1]^{d_X}} \beta_{\ell,r}(x) \beta_{m,r}(x) dx. \end{aligned}$$

Thus $q_c|_{I_r} \in \mathcal{P}_{2p}(I_r)$. Continuity of the tensor B-splines across the a -knots implies continuity of q_c , and hence $q_c \in \mathcal{Q}_{2p,K}$.

The definitions of μ_g , μ_c , G_g , and G_c give

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^N q_c(\alpha_i) &= \frac{1}{N} \sum_{i=1}^N \int_{[0,1]^{d_X}} c^\top p_K(x, \alpha_i) p_K(x, \alpha_i)^\top c dx = c^\top G_g c, \\ \int_0^1 q_c(a) da &= \int_{W^+} c^\top p_K(w) p_K(w)^\top c dw = c^\top G_c c. \end{aligned}$$

Applying (C.3) with $m_q = 2p$ therefore gives, for every $c \in \mathbb{R}^K$,

$$-\varepsilon_{N,T,K} c^\top G_c c \leq c^\top (G_g - G_c) c \leq \varepsilon_{N,T,K} c^\top G_c c.$$

Since the matrices are symmetric and c is arbitrary, the definition of Loewner order gives (C.4). The relative quadratic-form bound has a constant independent of K . $\square$

C.1.3 Projection Stability and Derivative Approximation

Lemma C.4 (Projection stability and derivative approximation). *For all sufficiently large N, T , both Π_K^c and $\Pi_{K,N}^g$ are well defined and reproduce S_K . For $\mu \in \{\mu_c, \mu_g\}$ and $r = 0, 1, 2, 3$,*

$$\|\partial_a^r \Pi_\mu f\|_\infty \leq C \mathfrak{h}_K^{-r} \|f\|_\infty, \quad (\text{C.8})$$

where $\Pi_{\mu_c} = \Pi_K^c$ and $\Pi_{\mu_g} = \Pi_{K,N}^g$. If $f \in C^{\tau_f}(W^+)$, $3 < \tau_f \leq p+1$, then, separately for $\Pi \in \{\Pi_K^c, \Pi_{K,N}^g\}$,

$$\|\partial_a^r (f - \Pi f)\|_\infty \leq C \mathfrak{h}_K^{\tau_f - r}, \quad r = 0, 1, 2, 3. \quad (\text{C.9})$$

Here C^{τ_f} denotes the usual isotropic Hölder class: if $\tau_f = m + \beta$ with $0 < \beta < 1$, derivatives of order m are β -Hölder, while the integer case has the usual C^m meaning. Membership on W^+ is defined through extension to a fixed open rectangle containing W^+ . The constant in (C.9) may depend on the C^{τ_f} -norm of this fixed extension, but not on K, N, T . Writing $\partial_{a,\text{pw}}^4$ for the fourth derivative on knot-cell interiors, the stability conclusion also gives

$$\|\partial_{a,\text{pw}}^4 \Pi_\mu f\|_{\infty,\text{pw}} \leq C \mathfrak{h}_K^{-4} \|f\|_\infty. \quad (\text{C.10})$$

Consequently, the continuous third derivative is globally Lipschitz in the a -coordinate:

$$\sup_x \sup_{a \neq a'} \frac{|\partial_a^3 \Pi_\mu f(x, a') - \partial_a^3 \Pi_\mu f(x, a)|}{|a' - a|} \leq C \mathfrak{h}_K^{-4} \|f\|_\infty. \quad (\text{C.11})$$

The suprema in (C.10) are over cell interiors, and the points in (C.11) are restricted to W^+ . No fourth-derivative approximation claim is made.

Proof. Local basis bounds. Let $R_j = \text{supp}(p_{K,j}) = R_{x,j} \times R_{a,j}$. Each one-dimensional support covers at most $p + 1$ positive knot spans. Quasi-uniformity therefore gives

$$\text{Leb}(R_{x,j}) \leq C\mathfrak{h}_K^{d_X}, \quad \text{Leb}(R_{a,j}) \leq C\mathfrak{h}_K, \quad \text{Leb}(R_j) \leq C\mathfrak{h}_K^{d_W}.$$

When $d_X = 0$, the first bound is interpreted as $\text{Leb}(R_{x,j}) = 1$. The lower spacing condition in (5.6) implies

$$\#\{i : \alpha_i \in R_{a,j}\} \leq 1 + \frac{N}{c_\alpha} \text{Leb}(R_{a,j}) \leq C(N\mathfrak{h}_K + 1).$$

Consequently, using $N\mathfrak{h}_K \rightarrow \infty$ in the last step,

$$\begin{aligned} \mu_g(R_j) &= \text{Leb}(R_{x,j}) \frac{1}{N} \#\{i : \alpha_i \in R_{a,j}\} \\ &\leq C\mathfrak{h}_K^{d_X} \left(\mathfrak{h}_K + \frac{1}{N} \right) \leq C\mathfrak{h}_K^{d_W}. \end{aligned}$$

The same affine-scaling argument for the normalized tensor basis gives, uniformly over boundary and interior indices,

$$\begin{aligned} \|\partial_a^r p_{K,j}\|_\infty &\leq C\mathfrak{h}_K^{-d_W/2-r}, \quad r = 0, 1, 2, 3, \\ \|\partial_{a,\text{pw}}^4 p_{K,j}\|_{\infty,\text{pw}} &\leq C\mathfrak{h}_K^{-d_W/2-4}. \end{aligned}$$

Thus, for $\mu \in \{\mu_c, \mu_g\}$,

$$\left| \int p_{K,j} f \, d\mu \right| \leq \|f\|_\infty \|p_{K,j}\|_\infty \mu(R_j) \leq C\mathfrak{h}_K^{d_W/2} \|f\|_\infty.$$

Inverse-Gram row sums. Let c_0 and C_0 be the constants in Lemma C.2, so

$$c_0 I_K \preceq G_c \preceq C_0 I_K.$$

Equation (C.4) gives

$$(1 - \varepsilon_{N,T,K}) G_c \preceq G_g \preceq (1 + \varepsilon_{N,T,K}) G_c.$$

Because $\varepsilon_{N,T,K} = o(1)$, for all sufficiently large N, T , $\varepsilon_{N,T,K} \leq 1/2$ and hence

$$\frac{c_0}{2} I_K \preceq G_g \preceq \frac{3C_0}{2} I_K.$$

Thus G_g is then positive definite and $\Pi_{K,N}^g$ is well defined. After decreasing the lower constant and increasing the upper constant if necessary, both Gram matrices satisfy

$$cI_K \preceq G_\mu \preceq CI_K, \quad \mu \in \{\mu_c, \mu_g\}.$$

Index the tensor basis by its rectangular multi-index set $\mathcal{J}_K \subset \mathbb{Z}^{d_W}$ and set $L_p = p + 1$. If two tensor B-spline supports intersect, their indices differ by at most L_p in every coordinate. Therefore, the nonzero pattern of each G_μ is contained in the tensor-lattice graph that joins distinct $\mathbf{j}, \mathbf{k} \in \mathcal{J}_K$ whenever $\|\mathbf{j} - \mathbf{k}\|_\infty \leq L_p$. Let $d_L(\mathbf{j}, \mathbf{k})$ denote distance in this lattice graph. Every edge changes each coordinate by at most L_p , while moving every coordinate toward its target by at most L_p indices per step gives a path inside the rectangular index set. Hence

$$d_L(\mathbf{j}, \mathbf{k}) = \left\lceil \frac{\|\mathbf{j} - \mathbf{k}\|_\infty}{L_p} \right\rceil.$$

Set

$$a_G = \frac{2}{c + C}, \quad R_\mu = I_K - a_G G_\mu, \quad \rho = \frac{C - c}{C + c} < 1.$$

For every eigenvalue λ of G_μ ,

$$1 - a_G \lambda \in \left[1 - \frac{2C}{c + C}, 1 - \frac{2c}{c + C} \right] = [-\rho, \rho].$$

Since G_μ and R_μ are symmetric, $\|R_\mu\|_{\text{op}} \leq \rho < 1$. Moreover, $I_K - R_\mu = a_G G_\mu$, so the operator-norm convergent Neumann series gives

$$G_\mu^{-1} = a_G (I_K - R_\mu)^{-1} = a_G \sum_{n=0}^{\infty} R_\mu^n.$$

The off-diagonal nonzero pattern of R_μ is contained in the same lattice graph. Expanding a matrix power gives

$$(R_\mu^n)_{j\mathbf{k}} = \sum_{j_1, \dots, j_{n-1}} (R_\mu)_{jj_1} \cdots (R_\mu)_{j_{n-1}\mathbf{k}}.$$

Every nonzero summand describes a length- n graph walk; a diagonal factor only leaves the current index unchanged. Therefore, $(R_\mu^n)_{j\mathbf{k}} = 0$ whenever $n < d_L(j, \mathbf{k})$, and

$$\begin{aligned} |(G_\mu^{-1})_{j\mathbf{k}}| &\leq a_G \sum_{n=d_L(j, \mathbf{k})}^{\infty} |(R_\mu^n)_{j\mathbf{k}}| \\ &\leq a_G \sum_{n=d_L(j, \mathbf{k})}^{\infty} \|R_\mu^n\|_{\text{op}} \leq \frac{a_G}{1-\rho} \rho^{d_L(j, \mathbf{k})}. \end{aligned}$$

For a fixed j , let $\mathcal{S}_s(j) = \{\mathbf{k} \in \mathcal{J}_K : d_L(j, \mathbf{k}) = s\}$. For $s \geq 1$, the distance identity places this shell inside the lattice annulus

$$\{\mathbf{k} \in \mathbb{Z}^{d_W} : L_p(s-1) < \|j - \mathbf{k}\|_\infty \leq L_p s\}.$$

Boundary truncation can only reduce cardinality. The difference-of-powers formula therefore gives

$$\#\mathcal{S}_s(j) \leq (2L_p s + 1)^{d_W} - \{2L_p(s-1) + 1\}^{d_W} \leq C(1+s)^{d_W-1}.$$

Also, $\#\mathcal{S}_0(j) = 1$. Summing the entrywise inverse bound over the shells yields

$$\begin{aligned} \sum_{\mathbf{k} \in \mathcal{J}_K} |(G_\mu^{-1})_{j\mathbf{k}}| &\leq \frac{a_G}{1-\rho} \sum_{s=0}^{\infty} \#\mathcal{S}_s(j) \rho^s \\ &\leq C \sum_{s=0}^{\infty} (1+s)^{d_W-1} \rho^s \leq C. \end{aligned}$$

The constants are uniform in j and K . Hence

$$\sup_j \sum_{\mathbf{k}} |(G_\mu^{-1})_{j\mathbf{k}}| \leq C. \quad (\text{C.12})$$

Stability and reproduction. For a bounded f , define

$$m_\mu(f) = \int p_K f \, d\mu, \quad b_\mu(f) = G_\mu^{-1} m_\mu(f).$$

The local moment bound above gives $\|m_\mu(f)\|_\infty \leq C \mathfrak{h}_K^{d_W/2} \|f\|_\infty$. Together with (C.12), it implies, for every j ,

$$|\{b_\mu(f)\}_j| \leq \sum_{\mathbf{k}} |(G_\mu^{-1})_{j\mathbf{k}}| |\{m_\mu(f)\}_{\mathbf{k}}| \leq C \mathfrak{h}_K^{d_W/2} \|f\|_\infty.$$

At every w , at most $(p+2)^{d_W}$ closed tensor supports contain w . Using the basis derivative bound gives, for $r = 0, 1, 2, 3$,

$$\begin{aligned} |\partial_a^r \Pi_\mu f(w)| &\leq \sum_{j: w \in \text{supp}(p_{K,j})} |\{b_\mu(f)\}_j| |\partial_a^r p_{K,j}(w)| \\ &\leq C \mathfrak{h}_K^{d_W/2} \|f\|_\infty \mathfrak{h}_K^{-d_W/2-r} = C \mathfrak{h}_K^{-r} \|f\|_\infty. \end{aligned}$$

Taking the supremum over w proves (C.8). The identical calculation on each knot-cell interior, with $r = 4$, proves (C.10). To obtain the global consequence, fix x and $a < a'$, partition $[a, a']$ at every intervening knot, and apply the fundamental theorem of calculus on each open knot cell. Maximum smoothness and $p \geq 4$ make $\partial_a^3 \Pi_\mu f$ continuous across the knots, so the cellwise increments telescope. Therefore,

$$\begin{aligned} |\partial_a^3 \Pi_\mu f(x, a') - \partial_a^3 \Pi_\mu f(x, a)| &\leq \int_a^{a'} |\partial_{a, \text{pw}}^4 \Pi_\mu f(x, u)| \, du \\ &\leq C \mathfrak{h}_K^{-4} \|f\|_\infty |a' - a|, \end{aligned}$$

which proves (C.11).

If $s = p_K^\top b \in S_K$, then

$$\begin{aligned} \int p_K s \, d\mu &= \int p_K p_K^\top \, d\mu \, b = G_\mu b, \\ \Pi_\mu s &= p_K^\top G_\mu^{-1} G_\mu b = p_K^\top b = s. \end{aligned}$$

Thus each well-defined projection reproduces S_K .

Derivative approximation. Extend f to the fixed surrounding rectangle and set $k_f = \lceil \tau_f \rceil - 1$. Since $3 < \tau_f \leq p + 1$, we have $3 \leq k_f \leq p$. For a point $w \in W^+$, let

$$P_w(z) = \sum_{|\gamma| \leq k_f} \frac{\partial^\gamma f(w)}{\gamma!} (z - w)^\gamma$$

be the degree- k_f Taylor polynomial of the extension. Every coordinate degree of P_w is at most $k_f \leq p$, so $P_w|_{W^+} \in S_K$. Lemma C.1 therefore gives $Q_K P_w = P_w$.

For each coordinate ℓ , choose a positive cell $I_{q_\ell(w)}^{(\ell)}$ whose closure contains w_ℓ . Define the closed-support active set and the selected dual cell by

$$\mathcal{A}_K(w) = \{j : w \in \text{supp}(\Phi_j)\}, \quad J_j = \prod_{\ell=1}^{d_W} I_{r(j_\ell)}^{(\ell)}.$$

The one-dimensional support relation and the selection rule $r(j_\ell) = \max\{j_\ell, 0\}$ imply, for $j \in \mathcal{A}_K(w)$,

$$|r(j_\ell) - q_\ell(w)| \leq p + 1, \quad \ell = 1, \dots, d_W.$$

Quasi-uniformity then gives

$$\sup_{z \in J_j} \|z - w\|_\infty \leq C \mathfrak{h}_K, \quad \#\mathcal{A}_K(w) \leq (p + 2)^{d_W}.$$

Thus all selected dual cells relevant to derivatives at w lie in one fixed $O(\mathfrak{h}_K)$ patch around w . The Hölder–Taylor remainder on this patch satisfies

$$\sup_{j \in \mathcal{A}_K(w)} \|f - P_w\|_{\infty, J_j} \leq C \mathfrak{h}_K^{\tau_f}.$$

The product-kernel construction in Lemma C.1 is supported on J_j . Its local dual bound and the preceding remainder therefore give, for every $j \in \mathcal{A}_K(w)$,

$$\begin{aligned} |\Lambda_j(f - P_w)| &\leq C \mathfrak{h}_K^{d_W/2} \|f - P_w\|_{\infty, J_j} \leq C \mathfrak{h}_K^{d_W/2 + \tau_f}, \\ \|\partial_a^r \Phi_j\|_\infty &\leq C \mathfrak{h}_K^{-d_W/2 - r}, \quad r = 0, 1, 2, 3. \end{aligned}$$

Because $r \leq 3 \leq k_f$, the Taylor polynomial matches the relevant derivative at its center: $\partial_a^r(f - P_w)(w) = 0$. Differentiating the quasi-interpolant and using $Q_K P_w = P_w$ now yields

$$\begin{aligned} \partial_a^r(f - Q_K f)(w) &= \partial_a^r(f - P_w)(w) - \sum_{j \in \mathcal{A}_K(w)} \Lambda_j(f - P_w) \partial_a^r \Phi_j(w) \\ &= - \sum_{j \in \mathcal{A}_K(w)} \Lambda_j(f - P_w) \partial_a^r \Phi_j(w). \end{aligned}$$

Combining the two preceding bounds and the fixed cardinality of $\mathcal{A}_K(w)$ gives

$$|\partial_a^r(f - Q_K f)(w)| \leq C \mathfrak{h}_K^{d_W/2 + \tau_f} \mathfrak{h}_K^{-d_W/2 - r} \leq C \mathfrak{h}_K^{\tau_f - r}.$$

Taking the supremum over w proves

$$\|\partial_a^r(f - Q_K f)\|_\infty \leq C \mathfrak{h}_K^{\tau_f - r}, \quad r = 0, 1, 2, 3. \quad (\text{C.13})$$

Set $g = f - Q_K f$. Because $Q_K f \in S_K$ and each projection reproduces S_K ,

$$\Pi_\mu f = \Pi_\mu(Q_K f + g) = Q_K f + \Pi_\mu g, \quad f - \Pi_\mu f = g - \Pi_\mu g.$$

Equations (C.8) and (C.13) then give

$$\begin{aligned} \|\partial_a^r(f - \Pi_\mu f)\|_\infty &\leq \|\partial_a^r g\|_\infty + \|\partial_a^r \Pi_\mu g\|_\infty \\ &\leq C \mathfrak{h}_K^{\tau_f - r} + C \mathfrak{h}_K^{-r} \|g\|_\infty \\ &\leq C \mathfrak{h}_K^{\tau_f - r} + C \mathfrak{h}_K^{-r} \mathfrak{h}_K^{\tau_f} \leq C \mathfrak{h}_K^{\tau_f - r}. \end{aligned}$$

This proves (C.9) for each projection separately. $\square$

For later reference, the benchmark nuisance projections are $m_{K,N} = \Pi_{K,N}^g m_0$, $h_{K,N}^0 = \Pi_{K,N}^g h_0$, and $g_{K,N} = \theta_0 m_{K,N} + h_{K,N}^0$. These are panel-grid projections. The continuous projections $\Pi_K^c m_0$ and $\Pi_K^c h_0$ remain separate comparison objects.

C.2 Generated-Coordinate Clipping and Path Localization

Let ζ_{it} be the bounded, centered finite-range process in Assumption 5.2, and set $\Delta_i = \frac{1}{T} \sum_{t=1}^T \zeta_{it}$, $\check{\alpha}_i = \alpha_i + \Delta_i$, and $\hat{\alpha}_i = \Pi_{[b_T, 1-b_T]}(\check{\alpha}_i)$. Write $d_i = \hat{\alpha}_i - \alpha_i$ and $r_i^{\text{cl}} = d_i - \Delta_i$. Let c_{fm} denote the weighted-Hoeffding constant for this bounded process. Then $\mathbb{P}(|\Delta_i| > x) \leq 2 \exp(-c_{\text{fm}} T x^2)$. Set $a_0 := c_{\text{fm}} C_b^2 > 1$ and $\lambda_{\zeta,T} := T \mathbb{E}[\Delta_i^2] = \sum_{\ell \in \Lambda_{q_0}} \left(1 - \frac{|\ell|}{T}\right) \gamma_\zeta(\ell)$.

Lemma C.5 (Clipped-coordinate facts). *For $\mathcal{E}_T = \{\max_{1 \leq i \leq N} |\Delta_i| \leq b_T\}$, $\Pr(\mathcal{E}_T^c) \leq 2N^{1-a_0} \rightarrow 0$. On $\mathcal{E}_T$, $d_i = \Delta_i$, $r_i^{\text{cl}} = 0$, and $\hat{\alpha}_i = \check{\alpha}_i$. Globally, $\hat{\alpha}_i \in [b_T, 1-b_T]$, every path $\alpha_i + s d_i$, $s \in [0, 1]$, stays in this interval, $|d_i| \leq |\Delta_i|$, and $|r_i^{\text{cl}}| \leq |\Delta_i| \mathbf{1}\{|\Delta_i| > b_T\}$. For $q = 1, \dots, 8$ and nonnegative q_1, q_2 with $q_2 \geq 1$ and $q_1 + q_2 \leq 8$,*

$$\begin{aligned} \max \left\{ \mathbb{E}[|\Delta_i|^q \mathbf{1}\{|\Delta_i| > b_T\}], \mathbb{E}[|r_i^{\text{cl}}|^q] \right\} &\leq C_q b_T^q N^{-a_0}, \\ \mathbb{E}[|\Delta_i|^{q_1} |r_i^{\text{cl}}|^{q_2}] &\leq C_{q_1, q_2} b_T^{q_1 + q_2} N^{-a_0}. \end{aligned} \quad (\text{C.14})$$

For $q = 2, \dots, 8$, $\mathbb{E}|d_i|^q \leq C_q T^{-q/2}$. Moreover,

$$|\mathbb{E} d_i| \leq C b_T N^{-a_0}, \quad \left| \mathbb{E} d_i^2 - \frac{\lambda_{\zeta,T}}{T} \right| \leq C b_T^2 N^{-a_0}. \quad (\text{C.15})$$

Proof. The coordinate-design condition places $\alpha_i \in [2b_T, 1-2b_T]$. Hence $|\Delta_i| \leq b_T$ implies $\check{\alpha}_i \in [b_T, 1-b_T]$. Therefore, clipping does not bind. The union bound gives $\Pr(\mathcal{E}_T^c) \leq 2N \exp(-c_{\text{fm}} T b_T^2) = 2N^{1-a_0}$. Projection onto an interval is nonexpansive. Thus $|d_i| \leq |\Delta_i|$. Clipping binds only when $|\Delta_i| > b_T$. Direct inspection at either endpoint gives the stated bound for r_i^{cl} . The segment between $\alpha_i \in [2b_T, 1-2b_T]$ and $\hat{\alpha}_i \in [b_T, 1-b_T]$ remains inside $[b_T, 1-b_T]$.

For $q \leq 8$, the layer-cake identity gives

$$\begin{aligned} \mathbb{E}[|\Delta_i|^q \mathbf{1}\{|\Delta_i| > b_T\}] &= b_T^q \Pr(|\Delta_i| > b_T) \\ &\quad + q \int_{b_T}^\infty x^{q-1} \Pr(|\Delta_i| > x) dx. \end{aligned}$$

The first term is $O(b_T^q N^{-a_0})$. With $y = \sqrt{T}x$, the integral is bounded by $C b_T^{q-2} T^{-1} N^{-a_0} \leq C b_T^q N^{-a_0}$ because $T b_T^2 = C_b^2 \log N$. The bounds for r_i^{cl} and the mixed moments follow from $|r_i^{\text{cl}}| \leq |\Delta_i| \mathbf{1}\{|\Delta_i| > b_T\}$. The bounded sample-mean inequality gives $\mathbb{E}|\Delta_i|^q \leq C_q T^{-q/2}$. Nonexpansiveness transfers it to d_i . Finally, use $d_i = \Delta_i + r_i^{\text{cl}}$, $\mathbb{E} \Delta_i = 0$, and $\mathbb{E} \Delta_i^2 = \lambda_{\zeta,T}/T$ together with the sharp tail and mixed-moment bounds to prove (C.15). $\square$

C.3 Finite-Range Moment Tools

Lemma C.6 (Finite-memory moment bounds). *Let $\{Q_{it} : t \in \mathbb{Z}\}$ be centered, uniformly bounded in L^8 , and r -dependent for a fixed r . Uniformly in i ,*

$$\left\| \frac{1}{T} \sum_{t=1}^T Q_{it} \right\|_4 \leq CT^{-1/2}. \quad (\text{C.16})$$

$$\text{Var}_i \left(\frac{1}{T} \sum_{t \in I_T(\ell)} \{Q_{it}Q_{i,t+\ell} - \mathbb{E}_i(Q_{it}Q_{i,t+\ell})\} \right) \leq \frac{C}{T}. \quad (\text{C.17})$$

for every fixed ℓ .

Proof. Split the time indices into $r + 1$ residue classes. Within each class the summands are independent. Expanding the fourth moment of each class sum leaves only index configurations in which every index is paired. Their number is $O(T^2)$. Hence the fourth moment of the normalized full sum is $O(T^{-2})$, which proves (C.16). For (C.17), the lag-product sequence is $(r + |\ell|)$ -dependent and has a bounded second moment. Expanding its variance leaves $O(T)$ nonzero covariance pairs. Division by T^2 gives the stated bound. $\square$

Lemma C.7 (Joint finite-memory transfer). *Every fixed-lag product of the oracle arrays in Assumption 5.2, evaluated at the oracle coordinate, is r_0 -dependent for a fixed r_0 . If a scalar component has envelope a_K and a bounded fourth moment after division by a_K , then*

$$\left\| \frac{1}{T} \sum_t \{Q_{it,K} - \mathbb{E}_i Q_{it,K}\} \right\|_4 \leq Ca_K T^{-1/2}. \quad (\text{C.18})$$

The same order holds for a fixed collection of lag products. Consequently, the oracle moment and Gram bounds used in Sections C.4 and D.2 remain valid under the joint law once every nonzero expectation is centered explicitly.

Below, the lemma is applied componentwise to the oracle moment, Gram, and fixed-lag product arrays after every nonzero expectation is centered explicitly.

Proof. Each oracle observation is measurable with respect to the current value of the finite-range process. A product involving finitely many fixed lags is therefore measurable with respect to a fixed-width time block. Such blocks are independent once they are separated by more than a fixed number of periods. Color the time indices by their residue class modulo $r_0 + 1$. Each color class is an independent sum. The fourth-moment pairing argument used in Lemma C.6 gives $\mathbb{E}_i |\sum_t (Q_{it,K} - \mathbb{E}_i Q_{it,K})|^4 \leq Ca_K^4 T^2$. Division by T^4 proves (C.18). A finite number of lags changes only r_0 and the constant. This argument is applied to oracle arrays only. Generated-coordinate and estimated-root arrays are instead decomposed into oracle arrays, coordinate shifts, and root errors before the lemma is applied. $\square$

C.4 Nuisance Estimation and Gram-Matrix Results

The results below add the stochastic inputs to the clipping and spline bounds established above. We maintain Assumption 5.2. The array $\{X_{it}\}$ is independent and uniformly distributed on $[0, 1]^{d_x}$ over (i, t) and is independent of the deterministic quasi-uniform coordinate design and every disturbance process. The bounded process (ζ_{it}, v_{it}) is jointly stationary and q_0 -dependent, while the bounded q_0 -dependent outcome-error process is independent of the joint (ζ, v) process and of X_{it} . In particular, ζ_{it} and v_{it} may be correlated. The data satisfy $Z_{it} = \alpha_i + \zeta_{it}$, $D_{it} = m_0(X_{it}, \alpha_i) + v_{it}$, and $Y_{it} = \theta_0 D_{it} + h_0(X_{it}, \alpha_i) + \varepsilon_{it}$, where $0 < \sigma_v^2 = \mathbb{E}[v_{it}^2] < \infty$. Write $\gamma_{v\zeta}(\ell) = \mathbb{E}[v_{i0}\zeta_{i\ell}]$ and $\Omega_{v\zeta} = \sum_{\ell \in \Lambda_{q_0}} \gamma_{v\zeta}(\ell)$, and write $\gamma_\zeta(\ell) = \mathbb{E}[\zeta_{i0}\zeta_{i\ell}]$ and $\Omega_\zeta = \sum_{\ell \in \Lambda_{q_0}} \gamma_\zeta(\ell)$. The functions obey $m_0 \in C^{\tau_m}(W^+)$ and $h_0 \in C^{\tau_h}(W^+)$, where $3 < \tau_m, \tau_h \leq p + 1$.

Use the generated coordinate, clipping event $\mathcal{E}_T$, and clipped moments from Lemma C.5. In addition to the deterministic growth conditions above, maintain

$$\frac{N}{T} \longrightarrow \kappa_0 \in (0, \infty), \quad \frac{K \log K}{NT} \longrightarrow 0. \quad (\text{C.19})$$

$$m = \max\{d_W, 3\}, \quad \bar{r}_{\eta,K} = \frac{\mathfrak{h}_K^{-m/2}}{T}.$$

C.4.1 Generated-Coordinate Perturbation and Nuisance Expansion

Let $m_K = m_{K,N} = \Pi_{K,N}^g m_0$ and $g_K = \Pi_{K,N}^g(\theta_0 m_0 + h_0) = \theta_0 m_K + h_{K,N}^0$. Define $u_{D,it,K} = D_{it} - m_K(X_{it}, \alpha_i)$ and $u_{Y,it,K} = Y_{it} - g_K(X_{it}, \alpha_i)$. For $b \in \{\gamma, \delta\}$, use the shorthand $(R_{\gamma,it,K}, \ell_{\gamma,K}) = (u_{D,it,K}, m_K)$ and $(R_{\delta,it,K}, \ell_{\delta,K}) = (u_{Y,it,K}, g_K)$, and let $f_{it,K}^b = p_K(X_{it}, \alpha_i) R_{b,it,K}$. Its first three coordinate derivatives are

$$\begin{aligned} A_{it,K}^b &= p_{K,a} R_{b,it,K} - p_K \ell_{b,K,a}, \\ H_{it,K}^b &= p_{K,aa} R_{b,it,K} - 2p_{K,a} \ell_{b,K,a} - p_K \ell_{b,K,aa}, \\ T_{it,K}^b &= p_{K,aaa} R_{b,it,K} - 3p_{K,aa} \ell_{b,K,a} - 3p_{K,a} \ell_{b,K,aa} - p_K \ell_{b,K,aaa}, \end{aligned} \tag{C.20}$$

where every term on the right is evaluated at (X_{it}, α_i) . Replacing α_i by a generic a defines the corresponding path arrays. Stack the gamma and delta blocks as $f_{\eta,it,K}$, $A_{\eta,it,K}$, $H_{\eta,it,K}$, and $T_{\eta,it,K}$. Write their means as $\mu_{f,\eta,i,K}$, $\mu_{A,\eta,i,K}$, $\mu_{H,\eta,i,K}$, and $\mu_{T,\eta,i,K}$. The panel-grid projection normal equations give $\frac{1}{N} \sum_i \mu_{f,\eta,i,K} = 0$.

Lemma C.8 (Fixed-coordinate nuisance derivative arrays). *For $r = 0, 1, 2, 3, 4$, uniformly over fixed a and bounded scalar functions $q(\cdot, a)$, write $p_{K,a^r} = \partial_a^r p_K$, with $p_{K,a^0} = p_K$. Then*

$$\begin{aligned} \mathbb{E}_X \|p_{K,a^r}(X, a)\|_2^2 &\leq CK \mathfrak{h}_K^{-2r}, \\ \|\mathbb{E}_X [p_{K,a^r}(X, a) q(X, a)]\|_2 &\leq C \mathfrak{h}_K^{-r-1/2} \|q(\cdot, a)\|_\infty. \end{aligned} \tag{C.21}$$

For $r = 4$, these bounds concern cell interiors and hence the cellwise, almost-everywhere derivative. The nuisance arrays defined above satisfy

$$\begin{aligned} \frac{1}{N} \sum_i \mathbb{E}_i \|A_{\eta,it,K} - \mu_{A,\eta,i,K}\|_2^2 &\leq CK \mathfrak{h}_K^{-2}, \\ \frac{1}{N} \sum_i \mathbb{E}_i \|H_{\eta,it,K} - \mu_{H,\eta,i,K}\|_2^2 &\leq CK \mathfrak{h}_K^{-4}, \\ \frac{1}{N} \sum_i \mathbb{E}_i \|T_{\eta,it,K} - \mu_{T,\eta,i,K}\|_2^2 &\leq CK \mathfrak{h}_K^{-6}, \end{aligned} \tag{C.22}$$

and

$$\sup_i \|\mu_{A,\eta,i,K}\|_2 \leq C \mathfrak{h}_K^{-1/2}, \quad \sup_i \|\mu_{H,\eta,i,K}\|_2 \leq C \mathfrak{h}_K^{-3/2}, \quad \sup_i \|\mu_{T,\eta,i,K}\|_2 \leq C \mathfrak{h}_K^{-5/2}. \tag{C.23}$$

Proof. The tensor representation $p_K(x, a) = \psi_{K_x}(x) \otimes \phi_{K_a}(a)$ gives

$$\begin{aligned} \mathbb{E}_X \|p_{K,a^r}(X, a)\|_2^2 &= \mathbb{E}_X \|\psi_{K_x}(X)\|_2^2 \left\| \phi_{K_a}^{(r)}(a) \right\|_2^2 \\ &\leq CK_x \mathfrak{h}_K^{-1-2r} \leq CK \mathfrak{h}_K^{-2r}. \end{aligned}$$

Here local support and normalized affine scaling give the middle inequality; for $r = 4$ it is read on cell interiors. For the second bound in (C.21), duality and the x -coordinate Riesz upper bound imply, for every unit vector $c \in \mathbb{R}^{K_x}$,

$$|\mathbb{E}_X [(c^\top \psi_{K_x}(X)) q(X, a)]| \leq \|c^\top \psi_{K_x}\|_{L^2} \|q(\cdot, a)\|_{L^2} \leq C \|q(\cdot, a)\|_\infty.$$

Thus

$$\|\mathbb{E}_X [\psi_{K_x}(X) q(X, a)]\|_2 \leq C \|q(\cdot, a)\|_\infty.$$

Combining this inequality with $\left\| \phi_{K_a}^{(r)}(a) \right\|_2 \leq C \mathfrak{h}_K^{-r-1/2}$ proves the asserted weighted-mean bound.

For $b \in \{\gamma, \delta\}$, put

$$q_\gamma = m_0, \quad q_\delta = \theta_0 m_0 + h_0, \quad e_{b,K} = q_b - \ell_{b,K}.$$

Then $R_{\gamma,it,K} = v_{it} + e_{\gamma,K}(X_{it}, \alpha_i)$ and $R_{\delta,it,K} = \theta_0 v_{it} + \varepsilon_{it} + e_{\delta,K}(X_{it}, \alpha_i)$. Let $\tau_* = \min\{\tau_m, \tau_h\} > 3$. Lemma C.4 gives

$$\|e_{b,K}\|_\infty \leq C\mathfrak{h}_K^{\tau_*}, \quad \|\partial_a^j \ell_{b,K}\|_\infty \leq C, \quad j = 1, 2, 3.$$

The innovations are bounded and centered, so the derivative table yields

$$\begin{aligned} \mu_{A,b,i,K} &= \mathbb{E}_X[p_{K,a}e_{b,K} - p_K\ell_{b,K,a}], \\ \mu_{H,b,i,K} &= \mathbb{E}_X[p_{K,aa}e_{b,K} - 2p_{K,a}\ell_{b,K,a} - p_K\ell_{b,K,aa}], \\ \mu_{T,b,i,K} &= \mathbb{E}_X[p_{K,aaa}e_{b,K} - 3p_{K,aa}\ell_{b,K,a} - 3p_{K,a}\ell_{b,K,aa} - p_K\ell_{b,K,aaa}], \end{aligned}$$

where all functions on the right are evaluated at (X, α_i) . Applying the weighted-mean bound term by term gives

$$\begin{aligned} \|\mu_{A,b,i,K}\|_2 &\leq C\{\mathfrak{h}_K^{\tau_*-3/2} + \mathfrak{h}_K^{-1/2}\} \leq C\mathfrak{h}_K^{-1/2}, \\ \|\mu_{H,b,i,K}\|_2 &\leq C\{\mathfrak{h}_K^{\tau_*-5/2} + \mathfrak{h}_K^{-3/2} + \mathfrak{h}_K^{-1/2}\} \leq C\mathfrak{h}_K^{-3/2}, \\ \|\mu_{T,b,i,K}\|_2 &\leq C\{\mathfrak{h}_K^{\tau_*-7/2} + \mathfrak{h}_K^{-5/2} + \mathfrak{h}_K^{-3/2} + \mathfrak{h}_K^{-1/2}\} \leq C\mathfrak{h}_K^{-5/2}. \end{aligned}$$

Stacking the two blocks proves (C.23).

Finally, boundedness of $R_{b,it,K}$ and of the first three derivatives of $\ell_{b,K}$, together with the first bound in (C.21), gives

$$\mathbb{E}_i \|A_{b,it,K}\|_2^2 \leq CK\mathfrak{h}_K^{-2}, \quad \mathbb{E}_i \|H_{b,it,K}\|_2^2 \leq CK\mathfrak{h}_K^{-4}, \quad \mathbb{E}_i \|T_{b,it,K}\|_2^2 \leq CK\mathfrak{h}_K^{-6}.$$

Centering can only decrease each vector second moment. Stacking the blocks and averaging over i proves (C.22). $\square$

Define the unit averages

$$\begin{aligned} \bar{A}_{\eta,i,K} &= \frac{1}{T} \sum_t A_{\eta,it,K}, \quad \bar{H}_{\eta,i,K} = \frac{1}{T} \sum_t H_{\eta,it,K}, \quad \bar{A}_{\eta,i,K}^c = \bar{A}_{\eta,i,K} - \mu_{A,\eta,i,K}, \\ \mathcal{T}_{\eta,iT,K}(d) &= \frac{1}{T} \sum_t \int_0^1 (1-s)^2 T_{\eta,it,K}(\alpha_i + sd) ds. \end{aligned}$$

The leading nuisance influence is

$$z_{\eta,it,K} = f_{\eta,it,K} - \mu_{f,\eta,i,K} + \mu_{A,\eta,i,K} \zeta_{it}, \quad \bar{Z}_{\eta,K} = \frac{1}{NT} \sum_{i,t} z_{\eta,it,K}.$$

Since $\Delta_i = \frac{1}{T} \sum_t \zeta_{it}$,

$$\bar{Z}_{\eta,K} = \frac{1}{NT} \sum_{i,t} f_{\eta,it,K} + \frac{1}{N} \sum_i \mu_{A,\eta,i,K} \Delta_i. \quad (\text{C.24})$$

Thus the $\mu_A \Delta_i$ term is part of the leading influence and is not placed in a Taylor remainder.

Because $f_{\eta,it,K}$ and ζ_{is} are generally dependent, the exact clipped nuisance-bias coefficient must be centered under their joint law:

$$B_{\eta,K}^{J,\text{cl}} = \frac{T}{N} \sum_i \mathbb{E}_i[d_i \bar{A}_{\eta,i,K}] + \frac{T}{2N} \sum_i \mathbb{E}_i[d_i^2 \bar{H}_{\eta,i,K}] + \frac{T}{2N} \sum_i \mathbb{E}_i[d_i^3 \mathcal{T}_{\eta,iT,K}(d_i)].$$

Define three remainders, each centered at its exact joint-law expectation:

$$\begin{aligned} R_{\eta,1,K}^J &= \frac{1}{N} \sum_i [d_i \bar{A}_{\eta,i,K} - \mu_{A,\eta,i,K} \Delta_i - \mathbb{E}_i\{d_i \bar{A}_{\eta,i,K} - \mu_{A,\eta,i,K} \Delta_i\}], \\ R_{\eta,2,K}^J &= \frac{1}{2N} \sum_i [d_i^2 \bar{H}_{\eta,i,K} - \mathbb{E}_i(d_i^2 \bar{H}_{\eta,i,K})], \\ R_{\eta,3,K}^J &= \frac{1}{2N} \sum_i [d_i^3 \mathcal{T}_{\eta,iT,K}(d_i) - \mathbb{E}_i\{d_i^3 \mathcal{T}_{\eta,iT,K}(d_i)\}], \end{aligned}$$

and let $R_{\alpha,\eta,K}^J$ be their sum. These exact joint centers are essential: in general $\mathbb{E}_i[d_i \bar{A}_{\eta,i,K}] \neq \mathbb{E}_i[d_i] \mu_{A,\eta,i,K}$.

For the target coefficients $\eta_{0,K,N} = (\gamma_{0,K,N}^\top, \delta_{0,K,N}^\top)^\top$, define the generated-coordinate nuisance moment

$$\hat{F}_{\eta,NT,K}(\eta_{0,K,N}) = \left(\frac{\frac{1}{NT} \sum_{i,t} \hat{p}_{it,K}(D_{it} - \hat{p}_{it,K}^\top \gamma_{0,K,N})}{\frac{1}{NT} \sum_{i,t} \hat{p}_{it,K}(Y_{it} - \hat{p}_{it,K}^\top \delta_{0,K,N})} \right), \quad \hat{p}_{it,K} = p_K(X_{it}, \hat{\alpha}_i).$$

Lemma C.9 (Exact clipped nuisance expansion). *If $\mathfrak{h}_K^{-m}/T \rightarrow 0$, then*

$$\hat{F}_{\eta,NT,K}(\eta_{0,K,N}) = \bar{Z}_{\eta,K} + \frac{1}{T} B_{\eta,K}^{J,\text{cl}} + R_{\alpha,\eta,K}^J. \quad (\text{C.25})$$

Moreover,

$$\mathbb{E} \|\bar{Z}_{\eta,K}\|_2^2 \leq \frac{CK}{NT}. \quad (\text{C.26})$$

$$R_{\alpha,\eta,K}^J = o_p(\sqrt{K}/T). \quad (\text{C.27})$$

$$\|B_{\eta,K}^{J,\text{cl}}\|_2 = O(\mathfrak{h}_K^{-3/2}). \quad (\text{C.28})$$

Proof. Exact expansion and centering. For each nuisance block, Taylor's theorem along $a = \alpha_i + sd_i$ gives

$$\begin{aligned} f_{\eta,it,K}(\hat{\alpha}_i) &= f_{\eta,it,K} + d_i A_{\eta,it,K} + \frac{d_i^2}{2} H_{\eta,it,K} \\ &\quad + \frac{d_i^3}{2} \int_0^1 (1-s)^2 T_{\eta,it,K}(\alpha_i + sd_i) ds. \end{aligned}$$

After averaging over t , subtract the leading coordinate term $\mu_{A,\eta,i,K} \Delta_i$ and use $d_i = \Delta_i + r_i^{\text{cl}}$:

$$d_i \bar{A}_{\eta,i,K} - \mu_{A,\eta,i,K} \Delta_i = d_i \bar{A}_{\eta,i,K}^c + \mu_{A,\eta,i,K} r_i^{\text{cl}}.$$

For each of these three quantities, add and subtract its exact conditional expectation. Because $\mathbb{E}_i \Delta_i = 0$, $\mathbb{E}_i[d_i \bar{A}_{\eta,i,K} - \mu_{A,\eta,i,K} \Delta_i] = \mathbb{E}_i[d_i \bar{A}_{\eta,i,K}^c]$. Thus the uncentered expectations sum to $\frac{1}{T} B_{\eta,K}^{J,\text{cl}}$, while the centered parts are exactly $R_{\eta,1,K}^J$, $R_{\eta,2,K}^J$, and $R_{\eta,3,K}^J$. Equation (C.24) supplies the remaining leading term, so the resulting identity is (C.25).

Leading moments. The six fixed-coordinate bounds needed below are (C.22) and (C.23) from Lemma C.8. The $r = 0$ case of (C.21), bounded residuals, and centering give

$$\begin{aligned} \sup_i \mathbb{E}_i \|f_{\eta,it,K} - \mu_{f,\eta,i,K}\|_2^2 &\leq CK, \\ \sup_i \mathbb{E}_i \|\mu_{A,\eta,i,K} \zeta_{it}\|_2^2 &\leq C \mathfrak{h}_K^{-1} \leq CK. \end{aligned}$$

Consequently, $\sup_i \mathbb{E}_i \|z_{\eta,it,K}\|_2^2 \leq CK$. To verify (C.26), expand the second moment of $\bar{Z}_{\eta,K}$. Cross-unit covariance terms vanish by unit independence, and within a unit finite-range dependence leaves only $O(T)$ nonzero pairs. Thus the vector second moment satisfies

$$\mathbb{E} \|\bar{Z}_{\eta,K}\|_2^2 = \frac{1}{N^2 T^2} \sum_{i=1}^N \sum_{t=1}^T \sum_{s=1}^T \mathbb{E}_i [z_{\eta,it,K}^\top z_{\eta,is,K}] \leq \frac{CK}{NT}.$$

This is also the conclusion obtained from Lemma C.7.

Joint derivative and curvature centers. The relevant joint centers need not vanish. Put $a_{\eta,i,K}(X) = (p_{K,a}(X, \alpha_i)^\top, \theta_0 p_{K,a}(X, \alpha_i)^\top)^\top$. In the derivative table, the coefficients of v_{it} in the gamma and delta blocks are $p_{K,a}$ and $\theta_0 p_{K,a}$, respectively. Terms that depend only on X factor from the centered ζ , and the independent outcome-error terms also have zero expectation. Hence independence of X from the joint innovation process gives

$$B_{\zeta,\eta,K}^J := \sum_{\ell \in \Lambda_{q_0}} \frac{1}{N} \sum_i \mathbb{E}_i [A_{\eta,i0,K} \zeta_{i\ell}] = \Omega_{v\zeta} \frac{1}{N} \sum_i \mathbb{E}_X a_{\eta,i,K}(X).$$

Tensor local support gives $\|\mathbb{E}_X p_{K,a}(X, a)\|_2 \leq C\mathfrak{h}_K^{-3/2}$. Hence,

$$\|B_{\zeta,\eta,K}^J\|_2 \leq C\mathfrak{h}_K^{-3/2}.$$

The coefficient of v_{it} in the gamma and delta Hessian blocks is, respectively, $p_{K,aa}$ and $\theta_0 p_{K,aa}$. Condition (5.7), after a time shift, removes these two $v\zeta\zeta$ cross terms. The independent, centered outcome error removes the corresponding $\varepsilon\zeta\zeta$ terms. Every remaining Hessian term is a function of X times a disturbance factor independent of X , so

$$\mathbb{E}_i[H_{\eta,it,K}\zeta_{is}\zeta_{ir}] = \mu_{H,\eta,i,K} \mathbb{E}_i[\zeta_{is}\zeta_{ir}].$$

To count the surviving pairs, write

$$\begin{aligned} T \mathbb{E}_i[\Delta_i \bar{A}_{\eta,i,K}] &= \frac{1}{T} \sum_{s=1}^T \sum_{t=1}^T \mathbb{E}_i[A_{\eta,it,K}\zeta_{is}], \\ \frac{T}{2} \mathbb{E}_i[\Delta_i^2 \bar{H}_{\eta,i,K}] &= \frac{1}{2T^2} \sum_{r=1}^T \sum_{s=1}^T \sum_{t=1}^T \mathbb{E}_i[H_{\eta,it,K}\zeta_{is}\zeta_{ir}]. \end{aligned}$$

For a lag ℓ , exactly $T - |\ell|$ pairs remain inside the sample. Using the factorization above therefore yields

$$\begin{aligned} \frac{T}{N} \sum_i \mathbb{E}_i[\Delta_i \bar{A}_{\eta,i,K}] &= B_{\zeta,\eta,K}^J - \frac{1}{TN} \left\{ \sum_{\ell \in \Lambda_{q_0}} |\ell| \gamma_{v\zeta}(\ell) \right\} \sum_i \mathbb{E}_X a_{\eta,i,K}(X), \\ \frac{T}{2N} \sum_i \mathbb{E}_i[\Delta_i^2 \bar{H}_{\eta,i,K}] &= \frac{\Omega_\zeta}{2N} \sum_i \mu_{H,\eta,i,K} - \frac{1}{2TN} \left\{ \sum_{\ell \in \Lambda_{q_0}} |\ell| \gamma_\zeta(\ell) \right\} \sum_i \mu_{H,\eta,i,K}. \end{aligned} \tag{C.29}$$

Both finite- T endpoint terms are $O(\mathfrak{h}_K^{-3/2}/T)$, while the two leading centers are $O(\mathfrak{h}_K^{-3/2})$. Thus the joint derivative expectation need not vanish, but it has the order required by the nuisance rate: after the score's T^{-1} scaling, $\frac{1}{T}O(\mathfrak{h}_K^{-3/2}) = O(\bar{r}_{\eta,K})$ because $m = \max\{d_W, 3\} \geq 3$.

Replacing Δ_i by d_i in the first two lines costs at most

$$\begin{aligned} \left\| \frac{T}{N} \sum_i \mathbb{E}_i[(d_i - \Delta_i) \bar{A}_{\eta,i,K}] \right\|_2 &\leq CT\mathfrak{h}_K^{-3/2} b_T N^{-a_0}, \\ \left\| \frac{T}{2N} \sum_i \mathbb{E}_i[(d_i^2 - \Delta_i^2) \bar{H}_{\eta,i,K}] \right\|_2 &\leq CT\mathfrak{h}_K^{-5/2} b_T^2 N^{-a_0}. \end{aligned}$$

After division by $\mathfrak{h}_K^{-3/2}$, these are bounded by $Tb_T N^{-a_0} = o(1)$ and $(b_T/\mathfrak{h}_K)Tb_T N^{-a_0} = o(1)$.

Third-order path and centered fluctuations. For the third-order path, separate its fixed-coordinate mean, centered fixed-coordinate average, and path displacement. On each knot-cell interior, differentiation of the third row in (C.20) gives, block by block,

$$\begin{aligned} \partial_{a,\text{pw}} T_{it,K}^b &= p_{K,aaaa} R_{b,it,K} - 4p_{K,aaa} \ell_{b,K,a} - 6p_{K,aa} \ell_{b,K,aa} \\ &\quad - 4p_{K,a} \ell_{b,K,aaa} - p_K \ell_{b,K,aaaa}. \end{aligned}$$

Here D_{it} and Y_{it} are held fixed along the path, so $\partial_a R_{b,it,K} = -\ell_{b,K,a}$. The first three derivatives of $\ell_{b,K}$ are uniformly bounded by derivative approximation, while (C.10) gives $\|\ell_{b,K,aaaa}\|_{\infty,\text{pw}} \leq C\mathfrak{h}_K^{-4}$. Let

$$\mathcal{F}_{\text{dist},T} = \sigma\{(\zeta_{is}, v_{is}, \varepsilon_{is}) : 1 \leq i \leq N, 1 \leq s \leq T\}.$$

For every fixed realization of this bounded disturbance array, (C.21) applied to the five terms yields

$$\text{ess sup}_a \left\| \mathbb{E}[\partial_{a,\text{pw}} T_{it,K}^b(a) \mid \mathcal{F}_{\text{dist},T}] \right\|_2 \leq C\mathfrak{h}_K^{-9/2}.$$

The path array $T_{it,K}^b(a)$ is continuous because all spline derivatives through order three agree across knots. Partitioning any path segment at its intervening knots and integrating the preceding cellwise derivative therefore gives the global bound

$$\|\mathbb{E}[T_{it,K}^b(a') - T_{it,K}^b(a) \mid \mathcal{F}_{\text{dist},T}]\|_2 \leq C|a' - a|\mathfrak{h}_K^{-9/2}. \quad (\text{C.30})$$

Thus Lemma C.8, (C.30), and Lemma C.5 give

$$T \left\| \frac{1}{N} \sum_i \mathbb{E}_i[d_i^3 \mathcal{T}_{\eta,iT,K}(d_i)] \right\|_2 \leq C \left\{ \mathfrak{h}_K^{-5/2} T^{-1/2} + \frac{\sqrt{K} \mathfrak{h}_K^{-3}}{T} + \frac{\mathfrak{h}_K^{-9/2}}{T} \right\} + o(\mathfrak{h}_K^{-3/2}) = o(\mathfrak{h}_K^{-3/2}). \quad (\text{C.31})$$

The three displayed terms follow by first bounding the fixed mean by $C\mathfrak{h}_K^{-5/2} \mathbb{E}|d_i|^3$. The centered average has second moment $CK\mathfrak{h}_K^{-6}/T$. The path displacement is bounded by $C|d_i|\mathfrak{h}_K^{-9/2}$ by (C.30). Multiplying by $T\mathbb{E}|d_i|^3$, $T(\mathbb{E}|d_i|^6)^{1/2}$, and $T\mathbb{E}|d_i|^4$, respectively, gives the three terms in (C.31). Relative to $\mathfrak{h}_K^{-3/2}$, their exponents are $\mathfrak{h}_K^{-1}T^{-1/2}$, $\mathfrak{h}_K^{-(dw+3)/2}/T$, and $\mathfrak{h}_K^{-3}/T$. All vanish because $(dw+3)/2 \leq m$ and $\mathfrak{h}_K^{-m}/T \rightarrow 0$. Replacing Δ_i^3 by d_i^3 in the centered fixed-coordinate term adds at most $C\sqrt{K}\mathfrak{h}_K^{-3}T^{1/2}b_T^3N^{-a_0/2} = o(\mathfrak{h}_K^{-3/2})$ by the clipped-tail moment and $b_T/\mathfrak{h}_K \rightarrow 0$. Equations (C.29)–(C.31) prove (C.28).

We finally control the centered fluctuations. Define on the full sample space

$$\begin{aligned} S_{\eta,1,K}^J &= \frac{1}{N} \sum_i [\Delta_i \bar{A}_{\eta,i,K}^c - \mathbb{E}_i(\Delta_i \bar{A}_{\eta,i,K}^c)], \\ S_{\eta,2,K}^J &= \frac{1}{2N} \sum_i [\Delta_i^2 \bar{H}_{\eta,i,K} - \mathbb{E}_i(\Delta_i^2 \bar{H}_{\eta,i,K})]. \end{aligned}$$

On $\mathcal{E}_T$, the random, uncentered terms agree because $d_i = \Delta_i$. Using $r_i^{\text{cl}} = d_i - \Delta_i$, the remaining exact center differences are

$$\begin{aligned} R_{\eta,1,K}^J - S_{\eta,1,K}^J &= -\frac{1}{N} \sum_i \mathbb{E}_i[r_i^{\text{cl}} \bar{A}_{\eta,i,K}], \\ R_{\eta,2,K}^J - S_{\eta,2,K}^J &= -\frac{1}{2N} \sum_i \mathbb{E}_i[(d_i^2 - \Delta_i^2) \bar{H}_{\eta,i,K}]. \end{aligned}$$

Conditional on the disturbances, the weighted-mean bound in (C.21) gives orders $C\mathfrak{h}_K^{-3/2}$ and $C\mathfrak{h}_K^{-5/2}$ for the two averaged arrays. The clipped-tail moments from Lemma C.5 therefore imply, on $\mathcal{E}_T$,

$$\begin{aligned} \|R_{\eta,1,K}^J - S_{\eta,1,K}^J\|_2 &\leq C\mathfrak{h}_K^{-3/2} b_T N^{-a_0}, \\ \|R_{\eta,2,K}^J - S_{\eta,2,K}^J\|_2 &\leq C\mathfrak{h}_K^{-5/2} b_T^2 N^{-a_0}. \end{aligned}$$

The finite-range product bound in Lemma C.7 and cross-unit independence give

$$\begin{aligned} \mathbb{E} \|S_{\eta,1,K}^J\|_2^2 &\leq \frac{CK\mathfrak{h}_K^{-2}}{NT^2}, \\ \mathbb{E} \|S_{\eta,2,K}^J\|_2^2 &\leq \frac{CK\mathfrak{h}_K^{-4}}{NT^3} + \frac{C\mathfrak{h}_K^{-3}}{NT^2}, \\ \mathbb{E} \|R_{\eta,3,K}^J\|_2^2 &\leq \frac{CK\mathfrak{h}_K^{-6}}{NT^3}. \end{aligned} \quad (\text{C.32})$$

In the second line, the two terms correspond, respectively, to $\Delta_i^2(\bar{H}_{\eta,i,K} - \mu_{H,\eta,i,K})$ and the centered product $\mu_{H,\eta,i,K}\{\Delta_i^2 - \mathbb{E}_i \Delta_i^2\}$. The third line uses the global local-support envelope $\|\mathcal{T}_{\eta,iT,K}(d_i)\|_2 \leq C\sqrt{K}\mathfrak{h}_K^{-3}$ and $\mathbb{E}|d_i|^6 \leq CT^{-3}$. To verify the comparison with $(\sqrt{K}/T)^2 = K/T^2$ term by term, divide the four variance orders in (C.32) by K/T^2 :

$$\begin{aligned} \frac{K\mathfrak{h}_K^{-2}/(NT^2)}{K/T^2} &= \frac{\mathfrak{h}_K^{-2}}{N}, & \frac{K\mathfrak{h}_K^{-4}/(NT^3)}{K/T^2} &= \frac{\mathfrak{h}_K^{-4}}{NT}, \\ \frac{\mathfrak{h}_K^{-3}/(NT^2)}{K/T^2} &= \frac{\mathfrak{h}_K^{-3}}{NK}, & \frac{K\mathfrak{h}_K^{-6}/(NT^3)}{K/T^2} &= \frac{\mathfrak{h}_K^{-6}}{NT}. \end{aligned}$$

Each ratio vanishes because $N/T \rightarrow \kappa_0$, $K \asymp \mathfrak{h}_K^{-dw}$, $m = \max\{d_W, 3\}$, and $\mathfrak{h}_K^{-m}/T \rightarrow 0$. The two clipping centers are smaller by $a_0 > 1$ and $b_T/\mathfrak{h}_K \rightarrow 0$, while $\Pr(\mathcal{E}_T^c) \rightarrow 0$. This proves (C.27). $\square$

For the clipped-coordinate specialization, set $s_{D,K} = \frac{1}{NT} \sum_{i,t} \hat{p}_{it,K} D_{it}$ and $s_{Y,K} = \frac{1}{NT} \sum_{i,t} \hat{p}_{it,K} Y_{it}$. Use the threshold c_Σ chosen in (5.9), and define $\mathcal{H}_K = \{\lambda_{\min}(\hat{\Sigma}_{p,K}) \geq c_\Sigma\}$. On $\mathcal{H}_K$, define the affine nuisance roots $\gamma_K^* = \hat{\Sigma}_{p,K}^{-1} s_{D,K}$ and $\delta_K^* = \hat{\Sigma}_{p,K}^{-1} s_{Y,K}$. Their residuals are $u_{D,it}^* = D_{it} - \hat{p}_{it,K}^\top \gamma_K^*$ and $u_{Y,it}^* = Y_{it} - \hat{p}_{it,K}^\top \delta_K^*$, and $V_K^* = \frac{1}{NT} \sum_{i,t} (u_{D,it}^*)^2$.

C.4.2 Gram-Matrix Concentration

Define the panel-grid population Gram, oracle sample Gram, and feasible sample Gram by

$$\begin{aligned}\Sigma_{p,K}^{\text{grid}} &= \frac{1}{N} \sum_{i=1}^N \mathbb{E}_X [p_K(X, \alpha_i) p_K(X, \alpha_i)^\top], \\ \tilde{\Sigma}_{p,K} &= \frac{1}{NT} \sum_{i,t} p_K(X_{it}, \alpha_i) p_K(X_{it}, \alpha_i)^\top, \\ \hat{\Sigma}_{p,K} &= \frac{1}{NT} \sum_{i,t} p_K(X_{it}, \hat{\alpha}_i) p_K(X_{it}, \hat{\alpha}_i)^\top.\end{aligned}$$

Lemma C.10 (Population and oracle-sample Gram). *There are constants $0 < c < C < \infty$ such that, for all sufficiently large N, T, K ,*

$$cI_K \preceq \Sigma_{p,K}^{\text{grid}} \preceq CI_K. \quad (\text{C.33})$$

Moreover,

$$\|\tilde{\Sigma}_{p,K} - \Sigma_{p,K}^{\text{grid}}\|_{\text{op}} = O_p\left(\sqrt{\frac{K \log K}{NT}} + \frac{K \log K}{NT}\right) = o_p(1). \quad (\text{C.34})$$

For $j = 1, 2, 3$, the continuous derivative Gram $G_{a^j,K} = \int p_{K,a^j} p_{K,a^j}^\top$ satisfies

$$\lambda_{\max}(G_{a^j,K}) \leq C \mathfrak{h}_K^{-2j}.$$

No positive lower bound is asserted for a derivative Gram on the full coefficient space.

Proof. Deterministic spline geometry controls the population eigenvalues, and matrix concentration transfers those bounds to the oracle sample Gram. The population Gram is G_g from Section C.1. The continuous Riesz inequality (C.1), the Loewner comparison (C.4), and $\varepsilon_{N,T,K} = o(1)$ prove (C.33). The derivative claim follows by taking the supremum over $\|b\|_2 = 1$ in (C.2).

Conditional on the deterministic grid, the oracle summands are independent over (i, t) . Local support and L^2 normalization imply $\|p_K(X_{it}, \alpha_i)\|_2^2 \leq CK$. Set

$$p_{it}^0 = p_K(X_{it}, \alpha_i), \quad A_{it}^0 = p_{it}^0 p_{it}^{0\top}, \quad M_i^0 = \mathbb{E}_X A_{it}^0, \quad Y_{it}^0 = \frac{A_{it}^0 - M_i^0}{NT}.$$

Then $\sum_{i,t} Y_{it}^0 = \tilde{\Sigma}_{p,K} - \Sigma_{p,K}^{\text{grid}}$, and the Y_{it}^0 are independent, self-adjoint, and mean zero. Since $\|A_{it}^0\|_{\text{op}} \leq CK$ and $\|M_i^0\|_{\text{op}} \leq CK$,

$$\|Y_{it}^0\|_{\text{op}} \leq \frac{CK}{NT}.$$

The crucial variance calculation is made only after centering and summing. Because $(A_{it}^0)^2 = \|p_{it}^0\|_2^2 A_{it}^0 \preceq CK A_{it}^0$,

$$\begin{aligned}\sum_{i,t} \mathbb{E}_X [(Y_{it}^0)^2] &= \frac{1}{(NT)^2} \sum_{i,t} \{\mathbb{E}_X [(A_{it}^0)^2] - (M_i^0)^2\} \\ &\preceq \frac{CK}{(NT)^2} \sum_{i,t} M_i^0 = \frac{CK}{NT} \Sigma_{p,K}^{\text{grid}} \preceq \frac{CK}{NT} I_K.\end{aligned}$$

Thus the matrix-variance proxy is $CK/(NT)$. Notice that this argument does not assert the generally false unit-level bound $M_i^0 \preceq CI_K$. Matrix Bernstein's inequality therefore yields

$$\Pr\left(\left\|\tilde{\Sigma}_{p,K} - \Sigma_{p,K}^{\text{grid}}\right\|_{\text{op}} > C\left\{\sqrt{\frac{K \log K}{NT}} + \frac{K \log K}{NT}\right\}\right) \rightarrow 0.$$

This is (C.34); its right-hand side is $o(1)$ by (C.19). $\square$

Lemma C.11 (Generated-coordinate Gram perturbation). *The clipped feasible Gram obeys*

$$\left\|\hat{\Sigma}_{p,K} - \tilde{\Sigma}_{p,K}\right\|_{\text{op}} = O_p(b_T/\mathfrak{h}_K) + o_p(1). \quad (\text{C.35})$$

Consequently, $\left\|\hat{\Sigma}_{p,K} - \Sigma_{p,K}^{\text{grid}}\right\|_{\text{op}} = o_p(1)$.

Proof. The proof separates the deterministic coordinate displacement from sample fluctuation in X . Write $\Delta = (\Delta_1, \dots, \Delta_N)^\top$. On $\mathcal{E}_T$, Lemma C.5 gives $\hat{\alpha}_i = \alpha_i + \Delta_i$ and $|\Delta_i| \leq b_T$. For $s_b(x, a) = b^\top p_K(x, a)$, the fundamental theorem of calculus gives

$$s_b(x, \hat{\alpha}_i) - s_b(x, \alpha_i) = d_i \int_0^1 \partial_a s_b(x, \alpha_i + s d_i) ds.$$

The quadratic-form difference is therefore

$$s_b(x, \hat{\alpha}_i)^2 - s_b(x, \alpha_i)^2 = \{s_b(x, \hat{\alpha}_i) - s_b(x, \alpha_i)\} \times \{s_b(x, \hat{\alpha}_i) + s_b(x, \alpha_i)\}.$$

Cauchy-Schwarz along the Taylor path gives, uniformly over $\|b\|_2 = 1$,

$$\begin{aligned} \left|b^\top \{\mathbb{E}(\hat{\Sigma}_{p,K} \mid \Delta) - \Sigma_{p,K}^{\text{grid}}\}b\right| &\leq b_T \left\{\frac{1}{N} \sum_i \int_0^1 \mathbb{E}_X[\{\partial_a s_b(X, \alpha_i + s d_i)\}^2] ds\right\}^{1/2} \\ &\quad \times \left\{\frac{1}{N} \sum_i \mathbb{E}_X[\{s_b(X, \hat{\alpha}_i) + s_b(X, \alpha_i)\}^2]\right\}^{1/2}. \end{aligned}$$

The shifted-grid occupancy argument below, the derivative inverse inequality, and the order-zero Riesz bound make the two squared energies $O(\mathfrak{h}_K^{-2} \|b\|_2^2)$ and $O(\|b\|_2^2)$, respectively. Hence

$$\sup_{\|b\|_2=1} \left|b^\top \{\mathbb{E}(\hat{\Sigma}_{p,K} \mid \Delta) - \Sigma_{p,K}^{\text{grid}}\}b\right| \leq C \frac{b_T}{\mathfrak{h}_K},$$

on $\mathcal{E}_T$. Indeed, if a shifted point lies in an interval J of length $O(\mathfrak{h}_K)$, then its original coordinate lies in the b_T -enlargement of J . The lower spacing bound in (5.6) therefore limits the number of such units to $O\{N(\mathfrak{h}_K + b_T) + 1\} = O(N\mathfrak{h}_K)$. Thus the same finite-overlap integrals used in Section C.1 apply along the entire Taylor path.

Because the generated coordinates depend only on the Z process and this process is independent of X , conditional on the generated coordinates the X_{it} remain independent. Set

$$\hat{A}_{it} = \hat{p}_{it,K} \hat{p}_{it,K}^\top, \quad \hat{M}_i = \mathbb{E}_X(\hat{A}_{it} \mid \Delta), \quad \hat{Y}_{it} = \frac{\hat{A}_{it} - \hat{M}_i}{NT}.$$

Conditional on Δ , the $\hat{Y}_{it}$ are independent, self-adjoint, and mean zero, and local support gives $\left\|\hat{Y}_{it}\right\|_{\text{op}} \leq CK/(NT)$. On $\mathcal{E}_T$, the mean shift bound above and (C.33) imply

$$\left\|\mathbb{E}(\hat{\Sigma}_{p,K} \mid \Delta)\right\|_{\text{op}} \leq C + C \frac{b_T}{\mathfrak{h}_K} \leq C.$$

Moreover, using $\widehat{A}_{it}^2 \preceq CK\widehat{A}_{it}$ and centering before summing,

$$\begin{aligned}\sum_{i,t} \mathbb{E}(\widehat{Y}_{it}^2 \mid \Delta) &= \frac{1}{(NT)^2} \sum_{i,t} \{\mathbb{E}_X(\widehat{A}_{it}^2 \mid \Delta) - \widehat{M}_i^2\} \\ &\preceq \frac{CK}{(NT)^2} \sum_{i,t} \widehat{M}_i = \frac{CK}{NT} \mathbb{E}(\widehat{\Sigma}_{p,K} \mid \Delta).\end{aligned}$$

Thus, uniformly on $\mathcal{E}_T$, the conditional variance proxy is $CK/(NT)$. Conditional matrix Bernstein therefore gives

$$\left\| \widehat{\Sigma}_{p,K} - \mathbb{E}(\widehat{\Sigma}_{p,K} \mid \Delta) \right\|_{\text{op}} = O_p \left(\sqrt{\frac{K \log K}{NT}} + \frac{K \log K}{NT} \right) = o_p(1)$$

with constants uniform over the realized coordinates in $\mathcal{E}_T$. The three pieces enter through the exact decomposition

$$\begin{aligned}\widehat{\Sigma}_{p,K} - \widetilde{\Sigma}_{p,K} &= \{\widehat{\Sigma}_{p,K} - \mathbb{E}(\widehat{\Sigma}_{p,K} \mid \Delta)\} - \{\widetilde{\Sigma}_{p,K} - \Sigma_{p,K}^{\text{grid}}\} \\ &\quad + \{\mathbb{E}(\widehat{\Sigma}_{p,K} \mid \Delta) - \Sigma_{p,K}^{\text{grid}}\}.\end{aligned}$$

Integrating the conditional probability bound and adding $\Pr(\mathcal{E}_T^c) = o(1)$ proves (C.35). The final claim follows from Lemma C.10 and $b_T/\mathfrak{h}_K \rightarrow 0$. $\square$

Lemma C.12 (Feasible ordinary-Gram event). *With the constants in (C.33),*

$$\Pr \left(\frac{c}{2} I_K \preceq \widehat{\Sigma}_{p,K} \preceq 2CI_K \right) \rightarrow 1.$$

Proof. By Lemmas C.10 and C.11, the feasible Gram differs from $\Sigma_{p,K}^{\text{grid}}$ by $o_p(1)$ in operator norm. Weyl's inequality and (C.33) prove the result. $\square$

C.4.3 Affine Nuisance-Root Rate

Lemma C.13 (Affine nuisance-root rate). *Under the preceding assumptions and $\mathfrak{h}_K^{-m}/T \rightarrow 0$, $\Pr(\mathcal{H}_K^c) \rightarrow 0$ and, on $\mathcal{H}_K$,*

$$\|\gamma_K^* - \gamma_{0,K,N}\|_2 + \|\delta_K^* - \delta_{0,K,N}\|_2 = O_p(\bar{r}_{\eta,K}). \quad (\text{C.36})$$

$$\sqrt{NT} \bar{r}_{\eta,K}^2 = o(1). \quad (\text{C.37})$$

Proof. The proof first converts the two affine normal equations into a single coefficient identity and then bounds its three score components. The event $\mathcal{H}_K$ has probability approaching one by Lemma C.12. On this event, subtracting the target coefficients from the two affine normal equations gives

$$\begin{pmatrix} \gamma_K^* - \gamma_{0,K,N} \\ \delta_K^* - \delta_{0,K,N} \end{pmatrix} = \begin{pmatrix} \widehat{\Sigma}_{p,K}^{-1} & 0 \\ 0 & \widehat{\Sigma}_{p,K}^{-1} \end{pmatrix} \widehat{F}_{\eta,NT,K}(\eta_{0,K,N}).$$

The inverse block matrix has bounded operator norm on $\mathcal{H}_K$. By Lemma C.9, the three terms on the right satisfy

$$\begin{aligned}\|\bar{Z}_{\eta,K}\|_2 &= O_p \left(\sqrt{\frac{K}{NT}} \right) = O_p \left(\frac{\sqrt{K}}{T} \right), \\ \frac{1}{T} \|B_{\eta,K}^{J,\text{cl}}\|_2 &= O \left(\frac{\mathfrak{h}_K^{-3/2}}{T} \right), \\ \|R_{\alpha,\eta,K}^J\|_2 &= o_p \left(\frac{\sqrt{K}}{T} \right).\end{aligned}$$

Because $K \asymp \mathfrak{h}_K^{-d_W}$ and $m = \max\{d_W, 3\}$, both $\sqrt{K}/T$ and $\mathfrak{h}_K^{-3/2}/T$ are bounded by a constant multiple of $\bar{r}_{\eta,K}$. This proves (C.36). Finally, $\sqrt{NT} \bar{r}_{\eta,K}^2 \asymp \mathfrak{h}_K^{-m}/T$, which proves (C.37). $\square$

C.4.4 Second-Derivative and Hessian Bounds

Set the following quantities, with $b_1, b_2 \in \mathbb{R}^K$:

$$\begin{aligned}\widehat{G}_{aa,K} &= \frac{1}{NT} \sum_{i,t} p_{K,aa}(X_{it}, \widehat{\alpha}_i) p_{K,aa}(X_{it}, \widehat{\alpha}_i)^\top, \\ Q_i(b_1, b_2) &= \frac{1}{T} \sum_{t=1}^T \{p_{K,a}(X_{it}, \widehat{\alpha}_i)^\top b_1\} \{p_{K,a}(X_{it}, \widehat{\alpha}_i)^\top b_2\}.\end{aligned}$$

Lemma C.14 (Second-derivative Gram and simultaneous products). *If, in addition,*

$$\frac{K_x \log(NK_x)}{T} \longrightarrow 0, \quad (\text{C.38})$$

then there is a finite C_G such that

$$\Pr \left\{ \lambda_{\max}(\widehat{G}_{aa,K}) \leq C_G \mathfrak{h}_K^{-4} \right\} \longrightarrow 1. \quad (\text{C.39})$$

The same event construction gives the coefficient-uniform bound

$$\sup_{\|b_1\|_2 = \|b_2\|_2 = 1} \left\{ \frac{1}{N} \sum_{i=1}^N Q_i(b_1, b_2)^2 \right\}^{1/2} = O_p(\mathfrak{h}_K^{-5/2}). \quad (\text{C.40})$$

Because the supremum is taken before coefficients are substituted, Equation (C.40) applies to coefficient errors estimated from the same sample.

Proof. We first establish a unitwise X -Gram event, then use tensor-product geometry for the coefficient-uniform product bound and matrix Bernstein for the second-derivative Gram. Reshape $b \in \mathbb{R}^K$ as a $K_x \times K_a$ matrix B and write $d_K(a) = \partial_a \phi_{K_a}(a)$. The tensor basis gives

$$p_{K,a}(x, a)^\top b = \psi_{K_x}(x)^\top B d_K(a). \quad (\text{C.41})$$

For each unit, define $\widehat{G}_{x,i,K} = \frac{1}{T} \sum_t \psi_{K_x}(X_{it}) \psi_{K_x}(X_{it})^\top$. Matrix Bernstein (Tropp, 2012), followed by a union bound over i , gives

$$\max_{1 \leq i \leq N} \lambda_{\max}(\widehat{G}_{x,i,K}) = O_p(1), \quad (\text{C.42})$$

under (C.38).

On $\mathcal{E}_T$, $|\widehat{\alpha}_i - \alpha_i| \leq b_T = o(\mathfrak{h}_K)$. Thus any interval of length $O(\mathfrak{h}_K)$ contains at most $CN\mathfrak{h}_K$ shifted grid points. For $F_B(a) = \|B d_K(a)\|_2^2$, one-dimensional local support and inverse inequalities yield, uniformly in B ,

$$\begin{aligned}\sup_a F_B(a) &\leq C \mathfrak{h}_K^{-3} \|B\|_F^2, \\ \int_0^1 F_B(a) da &\leq C \mathfrak{h}_K^{-2} \|B\|_F^2, \\ \int_0^1 F_B(a)^2 da &\leq C \mathfrak{h}_K^{-5} \|B\|_F^4.\end{aligned}$$

The shifted-grid occupancy bound transfers the last inequality to

$$\frac{1}{N} \sum_i F_B(\widehat{\alpha}_i)^2 \leq C \mathfrak{h}_K^{-5} \|B\|_F^4.$$

Combining this display, (C.41), and (C.42) with

$$Q_i(b_1, b_2) = d_K(\widehat{\alpha}_i)^\top B_1^\top \widehat{G}_{x,i,K} B_2 d_K(\widehat{\alpha}_i),$$

gives, on the uniform X -Gram event,

$$\begin{aligned} |Q_i(b_1, b_2)| &\leq C F_{B_1}(\hat{\alpha}_i)^{1/2} F_{B_2}(\hat{\alpha}_i)^{1/2}, \\ \frac{1}{N} \sum_i Q_i(b_1, b_2)^2 &\leq C \left\{ \frac{1}{N} \sum_i F_{B_1}(\hat{\alpha}_i)^2 \right\}^{1/2} \left\{ \frac{1}{N} \sum_i F_{B_2}(\hat{\alpha}_i)^2 \right\}^{1/2} \\ &\leq C \mathfrak{h}_K^{-5} \|B_1\|_F^2 \|B_2\|_F^2. \end{aligned}$$

Taking square roots and using $\|B_j\|_F = \|b_j\|_2$ proves (C.40).

For the second-derivative Gram, write $\Delta = (\Delta_1, \dots, \Delta_N)^\top$. The inverse inequality in Section C.1 and shifted-grid occupancy give

$$\lambda_{\max}\{\mathbb{E}(\hat{G}_{aa,K} \mid \Delta)\} \leq C \mathfrak{h}_K^{-4} \quad \text{on } \mathcal{E}_T,$$

Local support and derivative normalization imply $\|p_{K,aa}(w)\|_2^2 \leq CK \mathfrak{h}_K^{-4}$. Define

$$A_{it}^{aa} = p_{K,aa}(X_{it}, \hat{\alpha}_i) p_{K,aa}(X_{it}, \hat{\alpha}_i)^\top, \quad M_i^{aa} = \mathbb{E}_X(A_{it}^{aa} \mid \Delta).$$

Then $(A_{it}^{aa})^2 \preceq CK \mathfrak{h}_K^{-4} A_{it}^{aa}$. Centering first and summing as in the ordinary-Gram proof gives the normalized envelope and variance proxy

$$R_{NTK} \leq C \frac{K \mathfrak{h}_K^{-4}}{NT}, \quad \sigma_{NTK}^2 \leq C \frac{K \mathfrak{h}_K^{-8}}{NT}.$$

Indeed, the latter follows from

$$\frac{1}{(NT)^2} \sum_{i,t} \{\mathbb{E}_X[(A_{it}^{aa})^2 \mid \Delta] - (M_i^{aa})^2\} \preceq \frac{CK \mathfrak{h}_K^{-4}}{NT} \mathbb{E}(\hat{G}_{aa,K} \mid \Delta).$$

Matrix Bernstein therefore gives

$$\left\| \hat{G}_{aa,K} - \mathbb{E}(\hat{G}_{aa,K} \mid \Delta) \right\|_{\text{op}} = O_p \left[\mathfrak{h}_K^{-4} \left\{ \sqrt{\frac{K \log K}{NT}} + \frac{K \log K}{NT} \right\} \right] = o_p(\mathfrak{h}_K^{-4}).$$

Equation (C.19), this bound, and $\Pr(\mathcal{E}_T^c) = o(1)$ prove (C.39). $\square$

The uniformity in (C.40) has the following direct consequence. If $b_{1,K}$ and $b_{2,K}$ are arbitrary random coefficient vectors computed from the same sample and $\|b_{j,K}\|_2 = O_p(\bar{r}_{\eta,K})$, then

$$\left\{ \frac{1}{N} \sum_{i=1}^N Q_i(b_{1,K}, b_{2,K})^2 \right\}^{1/2} = O_p(\mathfrak{h}_K^{-5/2} \bar{r}_{\eta,K}^2).$$

Indeed, on the event in (C.40), homogeneity gives the right side after multiplying the unit-sphere supremum by $\|b_{1,K}\|_2 \|b_{2,K}\|_2$. No conditioning on the estimated coefficients is used.

C.4.5 Denominator Control and Localization Inputs

Define $\tilde{u}_{D,it,K} = D_{it} - \hat{p}_{it,K}^\top \gamma_{0,K,N}$ and $e_\gamma^* = \gamma_K^* - \gamma_{0,K,N}$, and let

$$d_{\alpha,it,K} = d_i \int_0^1 \partial_a m_K(X_{it}, \alpha_i + s d_i) ds.$$

Then $\tilde{u}_{D,it,K} = u_{D,it,K} - d_{\alpha,it,K}$.

Lemma C.15 (Affine-root denominator). *On $\mathcal{H}_K$,*

$$\frac{1}{NT} \sum_{i,t} \{(u_{D,it}^*)^2 - (\tilde{u}_{D,it,K})^2\} = -e_\gamma^{*\top} \hat{\Sigma}_{p,K} e_\gamma^*. \quad (\text{C.43})$$

Consequently,

$$V_K^* - V_K = \left\{ \frac{1}{NT} \sum_{i,t} u_{D,it,K}^2 - V_K \right\} - \frac{2}{NT} \sum_{i,t} u_{D,it,K} d_{\alpha,it,K} + \frac{1}{NT} \sum_{i,t} d_{\alpha,it,K}^2 - e_\gamma^{*\top} \widehat{\Sigma}_{p,K} e_\gamma^*, \quad (\text{C.44})$$

If, in addition, $\mathfrak{h}_K^{-m}/T \rightarrow 0$, then

$$V_K^* - V_K = O_p\left((NT)^{-1/2} + \mathfrak{h}_K^{\tau_m}(NT)^{-1/2} + T^{-1} + \bar{r}_{\eta,K}^2\right). \quad (\text{C.45})$$

Proof. The proof begins with two exact square expansions and then controls the four resulting terms. This order keeps the normal-equation cancellation separate from the stochastic bounds.

Exact decomposition. The affine gamma equation gives $\frac{1}{NT} \sum_{i,t} \widehat{p}_{it,K} u_{D,it}^* = 0$. Since $u_{D,it}^* = \widetilde{u}_{D,it,K} - \widehat{p}_{it,K}^\top e_\gamma^*$, the difference of squares is

$$\begin{aligned} \frac{1}{NT} \sum_{i,t} \{(u_{D,it}^*)^2 - (\widetilde{u}_{D,it,K})^2\} &= -\frac{2}{NT} \sum_{i,t} u_{D,it}^* \widehat{p}_{it,K}^\top e_\gamma^* - e_\gamma^{*\top} \widehat{\Sigma}_{p,K} e_\gamma^* \\ &= -e_\gamma^{*\top} \widehat{\Sigma}_{p,K} e_\gamma^*. \end{aligned}$$

The cross term in the first equality is zero by the affine normal equation, which proves (C.43). The identity $\widetilde{u}_{D,it,K} = u_{D,it,K} - d_{\alpha,it,K}$ and a second exact square expansion then give (C.44).

Oracle fluctuation. It remains to bound the four terms without using a theta root. Write $u_{D,it,K} = v_{it} + e_{m,it,K}$, where $e_{m,it,K} = m_0(X_{it}, \alpha_i) - m_K(X_{it}, \alpha_i)$. The summands are bounded and finite-range dependent over t , and units are independent. Lemma C.7 therefore gives

$$\frac{1}{NT} \sum_{i,t} u_{D,it,K}^2 - V_K = O_p((NT)^{-1/2}).$$

Approximation-error cross term. Lemma C.4 gives

$$\begin{aligned} \|e_{m,\cdot,K}\|_\infty &\leq C \mathfrak{h}_K^{\tau_m}, \\ \max_{r \in \{1,2\}} \sup_{\substack{x \in [0,1]^{d_X} \\ a \in [b_T, 1-b_T]}} |\partial_a^r m_K(x, a)| &\leq C. \end{aligned}$$

On $\mathcal{E}_T$, $d_i = \Delta_i$, and Taylor's theorem makes the path term explicit

$$d_{\alpha,it,K} = \Delta_i m_{K,a}(X_{it}, \alpha_i) + \Delta_i^2 \int_0^1 (1-s) m_{K,aa}(X_{it}, \alpha_i + s\Delta_i) ds.$$

Thus the linear part carries the possibly nonzero v - ζ covariance, whereas the quadratic path is $O_p(T^{-1})$ after averaging because the second derivative is bounded and $\frac{1}{N} \sum_i \Delta_i^2 = O_p(T^{-1})$. The $e_m d_\alpha$ cross term contains only X and the first-stage mean. Put $q_{it,K} = e_{m,it,K} m_{K,a}(X_{it}, \alpha_i)$, $\bar{q}_{i,K} = \frac{1}{T} \sum_t q_{it,K}$, and $\mu_{q,i,K} = \mathbb{E}_X q_{it,K}$. The linear path splits into $\frac{1}{N} \sum_i \Delta_i \mu_{q,i,K}$ and $\frac{1}{N} \sum_i \Delta_i (\bar{q}_{i,K} - \mu_{q,i,K})$. Since $|\mu_{q,i,K}| \leq C \mathfrak{h}_K^{\tau_m}$ and the Δ_i are centered and independent across units,

$$\mathbb{E} \left[\left\{ \frac{1}{N} \sum_i \Delta_i \mu_{q,i,K} \right\}^2 \right] \leq \frac{C \mathfrak{h}_K^{2\tau_m}}{N^2} \sum_i \mathbb{E}_i \Delta_i^2 \leq \frac{C \mathfrak{h}_K^{2\tau_m}}{NT}.$$

For the second sum, let $\mathcal{F}_{\zeta,T} = \sigma\{\zeta_{it} : 1 \leq i \leq N, 1 \leq t \leq T\}$ and condition on $\mathcal{F}_{\zeta,T}$. Independence of X over (i, t) and the projection-error bound give

$$\begin{aligned} \mathbb{E}_X \left[\left\{ \frac{1}{N} \sum_i \Delta_i (\bar{q}_{i,K} - \mu_{q,i,K}) \right\}^2 \mid \mathcal{F}_{\zeta,T} \right] &\leq \frac{C \mathfrak{h}_K^{2\tau_m}}{N^2 T} \sum_i \Delta_i^2 \\ &= O_p\left(\frac{\mathfrak{h}_K^{2\tau_m}}{NT^2}\right). \end{aligned}$$

The quadratic Taylor path is bounded explicitly by

$$\left| \frac{1}{NT} \sum_{i,t} e_{m,it,K} \Delta_i^2 \int_0^1 (1-s) m_{K,aa}(X_{it}, \alpha_i + s\Delta_i) ds \right| \leq C \mathfrak{h}_K^{\tau_m} \frac{1}{N} \sum_i \Delta_i^2$$

$$= O_p(\mathfrak{h}_K^{\tau_m}/T) = o_p((NT)^{-1/2}).$$

Finally, let $R_{m,K}^{\text{cl}}$ denote the difference between the actual cross term and the preceding on- $\mathcal{E}_T$ expansion. The clipping-tail moment in Lemma C.5 gives

$$R_{m,K}^{\text{cl}} = O_p(\mathfrak{h}_K^{\tau_m} b_T N^{-a_0/2}) = o_p((NT)^{-1/2}).$$

Therefore

$$\frac{1}{NT} \sum_{i,t} e_{m,it,K} d_{\alpha,it,K} = O_p\{\mathfrak{h}_K^{\tau_m} (NT)^{-1/2}\} + o_p((NT)^{-1/2}).$$

The clipped mean is negligible by Lemma C.5.

Treatment-coordinate cross term. Because v and ζ may be correlated, the vd_α cross term can have a nonzero mean. On $\mathcal{E}_T$, a path expansion gives

$$\frac{1}{NT} \sum_{i,t} v_{it} d_{\alpha,it,K} = \frac{1}{N} \sum_i \Delta_i \{\bar{m}_{a,i,K} \bar{v}_i + \tilde{V}_{a,iT,K}\} + O_p(T^{-1}), \quad (\text{C.46})$$

where $\bar{m}_{a,i,K} = \mathbb{E}_X m_{K,a}(X, \alpha_i)$ and $\bar{v}_i = \frac{1}{T} \sum_t v_{it}$, and $\tilde{V}_{a,iT,K} = \frac{1}{T} \sum_t v_{it} \{m_{K,a}(X_{it}, \alpha_i) - \bar{m}_{a,i,K}\}$. Let $\mathcal{I}_{iT} = \sigma\{(\zeta_{it}, v_{it}) : 1 \leq t \leq T\}$ denote the joint innovation path. Conditional on $\mathcal{I}_{iT}$, each summand inside $\tilde{V}_{a,iT,K}$ is centered over X , whereas Δ_i is $\mathcal{I}_{iT}$ -measurable. Independence of X over time and boundedness of v_{it} and $m_{K,a}$ give

$$\mathbb{E}_X[\tilde{V}_{a,iT,K} \mid \mathcal{I}_{iT}] = 0, \quad \mathbb{E}_X[\tilde{V}_{a,iT,K}^2 \mid \mathcal{I}_{iT}] \leq \frac{C}{T}.$$

Iterated expectation therefore gives the conditional centering and the required product moment:

$$\mathbb{E}_i[\Delta_i \tilde{V}_{a,iT,K}] = 0,$$

$$\mathbb{E}_i[(\Delta_i \tilde{V}_{a,iT,K})^2] = \mathbb{E}_i\left[\Delta_i^2 \mathbb{E}_X[\tilde{V}_{a,iT,K}^2 \mid \mathcal{I}_{iT}]\right] \leq \frac{C}{T} \mathbb{E}_i \Delta_i^2 \leq \frac{C}{T^2}.$$

Exact finite-range pair counting gives the nonzero mean

$$\mathbb{E}_i[\Delta_i \bar{v}_i] = \frac{1}{T} \left\{ \Omega_{v\zeta} - \frac{1}{T} \sum_{\ell \in \Lambda_{q_0}} |\ell| \gamma_{v\zeta}(\ell) \right\}, \quad \text{Var}_i(\Delta_i \bar{v}_i) \leq C/T^2.$$

Since $\sup_i |\bar{m}_{a,i,K}| \leq C$, cross-unit averaging in (C.46) yields

$$\frac{1}{NT} \sum_{i,t} v_{it} d_{\alpha,it,K} = O(T^{-1}) + O_p((T\sqrt{N})^{-1}).$$

Conclusion. The clipping complement is smaller by the tail bounds in Lemma C.5. The bounded path derivative and $\frac{1}{N} \sum_i d_i^2 = O_p(T^{-1})$ imply $\frac{1}{NT} \sum_{i,t} d_{\alpha,it,K}^2 = O_p(T^{-1})$. In addition, Lemmas C.13 and C.12 give $e_\gamma^{*\top} \widehat{\Sigma}_{p,K} e_\gamma^* = O_p(\bar{r}_{\eta,K}^2)$. Substitution in (C.44) proves (C.45). $\square$

Lemma C.16 (Oracle denominator). *The finite-sieve denominator satisfies*

$$V_K = \mathbb{E}_{N,T}[u_{D,it,K}^2] = \sigma_v^2 + O(\mathfrak{h}_K^{2\tau_m}).$$

Hence there is a constant $c_V > 0$ such that $V_K \geq c_V$ for all sufficiently large N, T, K .

Proof. The partially linear equation and the grid-projection definition give $u_{D,it,K} = v_{it} + e_{m,it,K}$. Since v_{it} is centered and independent of X_{it} and the deterministic grid, $V_K = \sigma_v^2 + \mathbb{E}_{N,T}[e_{m,it,K}^2]$. Lemma C.4 bounds the last term by $C \mathfrak{h}_K^{2\tau_m}$. Because $\sigma_v^2 > 0$, the lower bound follows for a fixed $c_V < \sigma_v^2$ and all sufficiently large indices. $\square$

C.5 Estimator Localization and Exact Identities

C.5.1 Global Estimator and the Regular-Root Event

The clipped-coordinate affine objects analyzed above coincide on $\mathcal{H}_K$ with the generic construction below. The generic version supplies the off-event extension needed for the globally reported estimator and also covers the fixed- K raw coordinate in Section B.4. It is conditional on a chosen globally defined coordinate array $\{\hat{\alpha}_i\}_{i=1}^N$: the fixed- K application uses $\hat{\alpha}_i = \check{\alpha}_i$, whereas the growing-sieve application uses the clipped coordinate of Section C.2.

At the generated coordinates, let $\hat{p}_{it,K} = p_K(X_{it}, \hat{\alpha}_i)$ and $\hat{\Sigma}_{p,K} = \mathbb{P}_{NT} \hat{p}_{it,K} \hat{p}_{it,K}^\top$, and set $s_{D,K} = \mathbb{P}_{NT} \hat{p}_{it,K} D_{it}$ and $s_{Y,K} = \mathbb{P}_{NT} \hat{p}_{it,K} Y_{it}$. Choose a fixed threshold $c_\Sigma > 0$ satisfying the regular-Gram requirement for the application; under the primitive benchmark, use the choice in Equation (5.9). Define $\mathcal{H}_K = \{\lambda_{\min}(\hat{\Sigma}_{p,K}) \geq c_\Sigma\}$. Only on $\mathcal{H}_K$, define the affine nuisance roots $\gamma_K^* = \hat{\Sigma}_{p,K}^{-1} s_{D,K}$ and $\delta_K^* = \hat{\Sigma}_{p,K}^{-1} s_{Y,K}$. Their residuals and denominator are

$$u_{D,it}^* = D_{it} - \hat{p}_{it,K}^\top \gamma_K^*, \quad u_{Y,it}^* = Y_{it} - \hat{p}_{it,K}^\top \delta_K^*, \quad V_K^* = \mathbb{P}_{NT} (u_{D,it}^*)^2.$$

Extend $(\gamma_K^*, \delta_K^*, V_K^*)$ by fixed deterministic values on $\mathcal{H}_K^c$. This extension makes the tuple and subsequent events measurable. The two affine normal equations hold on $\mathcal{H}_K$ by construction:

$$\mathbb{P}_{NT} \hat{p}_{it,K} u_{D,it}^* = 0, \quad \mathbb{P}_{NT} \hat{p}_{it,K} u_{Y,it}^* = 0. \quad (\text{C.47})$$

The localization argument uses only the two high-probability inputs collected next.

Assumption C.17 (High-level localization conditions). *The regular-root event satisfies*

$$\mathbb{P}(\mathcal{H}_K^c) \longrightarrow 0, \quad (\text{C.48})$$

and, for every $\varepsilon > 0$,

$$\mathbb{P}(|V_K^* - V_K| > \varepsilon, \mathcal{H}_K) \longrightarrow 0. \quad (\text{C.49})$$

The oracle denominator satisfies $V_K \geq c_V > 0$ for all sufficiently large indices.

Corollary C.19 verifies this assumption for the clipped-coordinate primitive tensor-spline benchmark.

Define the good-root event $\mathcal{G}_K = \mathcal{H}_K \cap \{V_K^* \geq c_V/2\}$.

Proposition C.18 (Good-root probability). *Under Assumption C.17,*

$$\mathbb{P}(\mathcal{G}_K^c) \longrightarrow 0.$$

Proof. If $V_K^* < c_V/2$ on $\mathcal{H}_K$, then $|V_K^* - V_K| \geq c_V/2$. Therefore $\mathbb{P}(\mathcal{G}_K^c) \leq \mathbb{P}(\mathcal{H}_K^c) + \mathbb{P}(V_K^* < c_V/2, \mathcal{H}_K)$. The second term is bounded by $\mathbb{P}(|V_K^* - V_K| \geq c_V/2, \mathcal{H}_K)$, and both terms converge to zero by (C.48)–(C.49). $\square$

Use the threshold c_V chosen in (5.9) and recall $\mathcal{G}_K = \mathcal{H}_K \cap \{V_K^* \geq c_V/2\}$.

Corollary C.19 (Primitive localization inputs). *Under Assumption 5.2 and the maintained deterministic growth conditions stated above, suppose the thresholds satisfy (5.9) and $\mathfrak{h}_K^{-m}/T \rightarrow 0$. Then the clipped-coordinate specialization satisfies Assumption C.17. Consequently,*

$$\Pr(\mathcal{G}_K^c) \longrightarrow 0. \quad (\text{C.50})$$

Proof. The admissible Gram threshold and Lemma C.12 give $\Pr(\mathcal{H}_K^c) \rightarrow 0$. Moreover,

$$\bar{r}_{\eta,K}^2 = \frac{\mathfrak{h}_K^{-m}}{T^2} = \frac{1}{T} \frac{\mathfrak{h}_K^{-m}}{T} \longrightarrow 0.$$

Because $N/T \rightarrow \kappa_0 \in (0, \infty)$ and $0 < \mathfrak{h}_K \leq 1$ eventually, $(NT)^{-1/2} \rightarrow 0$, $\mathfrak{h}_K^m (NT)^{-1/2} \rightarrow 0$, and $T^{-1} \rightarrow 0$. Thus every term on the right side of (C.45) vanishes. Lemma C.15 therefore gives, for every $\varepsilon > 0$,

$$\Pr(|V_K^* - V_K| > \varepsilon, \mathcal{H}_K) \longrightarrow 0.$$

Finally, Lemma C.16 and $c_V < \sigma_v^2$ give $V_K \geq c_V$ eventually. These are exactly the three claims in Assumption C.17, so Proposition C.18 proves (C.50). $\square$

Fix a deterministic admissible vector $\rho_K^\dagger = (\theta_K^\dagger, \gamma_K^{\dagger\top}, \delta_K^{\dagger\top})^\top$. On $\mathcal{G}_K$, define $\theta_K^* = \mathbb{P}_{NT} u_{D,it}^* u_{Y,it}^* / V_K^*$, and set $\theta_K^* = \theta_K^\dagger$ on $\mathcal{G}_K^c$. Then

$$\mathbb{P}_{NT} u_{D,it}^* (u_{Y,it}^* - \theta_K^* u_{D,it}^*) = 0, \quad (\text{C.51})$$

on $\mathcal{G}_K$. Since $\mathcal{G}_K \subseteq \mathcal{H}_K$, combining (C.47) and (C.51) gives, on $\mathcal{G}_K$,

$$\mathbb{P}_{NT} f_{it,K}((\theta_K^*, \gamma_K^{*\top}, \delta_K^{*\top})^\top, \hat{\alpha}_i) = 0.$$

This tuple is the unique solution in $\mathbb{R}^{1+2K}$ to the full stacked system at the fixed generated coordinates on $\mathcal{G}_K$. Indeed, the lower two blocks of any solution are $s_{D,K} - \hat{\Sigma}_{p,K} \gamma = 0$ and $s_{Y,K} - \hat{\Sigma}_{p,K} \delta = 0$. The Gram lower bound therefore forces $\gamma = \gamma_K^*$ and $\delta = \delta_K^*$. After these coefficients are fixed, the first block equals $\mathbb{P}_{NT} u_{D,it}^* u_{Y,it}^* - \theta V_K^*$, whose coefficient on θ is $-V_K^* \leq -c_V/2 < 0$; hence it uniquely forces $\theta = \theta_K^*$.

Define the globally reported root by

$$\hat{\rho}_K = \begin{cases} (\theta_K^*, \gamma_K^{*\top}, \delta_K^{*\top})^\top, & \text{on } \mathcal{G}_K, \\ \rho_K^\dagger, & \text{on } \mathcal{G}_K^c. \end{cases}$$

This piecewise definition is measurable on the whole sample space.

C.5.2 Profile Identity and Transfer from the Acceptance Event

The construction above satisfies the three normal equations used in Lemma 5.1. Thus (5.1) and the corrected decomposition (5.5) apply on $\mathcal{G}_K$. On its complement, the reported corrected estimator is $\theta_K^\dagger$. Multiplying the right side of (5.5) by $\mathbf{1}\{\mathcal{G}_K\}$ and adding $\mathbf{1}\{\mathcal{G}_K^c\}(\theta_K^\dagger - \theta_{0,K,N})$ therefore gives the whole-sample-space identity.

Remark C.20 (Which population coefficient enters the identity). *Equation (5.5) uses one specified clipped-coordinate coefficient. If an intermediate proof works with a raw finite- T coefficient or an infinite-lag approximation, its difference from $\mathcal{B}_{\theta,K}^{\text{cl}}$ must enter once, through the separate R_{bridge} term in Section D.2. Accordingly, any transition between these coefficients must enter explicitly through the bridge remainder.*

Remark C.21 (Removing localization indicators). *For any remainder R_K controlled on $\mathcal{G}_K$ and any $\varepsilon > 0$, $\mathbb{P}(|R_K| > \varepsilon) \leq \mathbb{P}(\mathcal{G}_K^c) + \mathbb{P}(|R_K| > \varepsilon, \mathcal{G}_K)$. Proposition C.18 therefore transfers every probability-order bound proved on $\mathcal{G}_K$ to the globally reported estimator.*

D Score Expansion, Bias Correction, and Limit Theory

D.1 Sieve-Target Bias and the Theta-Score Expansion

The score expansion below uses the bounded joint finite-range specialization in Assumption 5.2. Units are independent across i . The within-unit processes are stationary, q_0 -dependent, and independent of the covariate array $\{X_{it}\}$, which is independent and identically distributed over (i, t) . The bounded outcome-error process is independent of the joint (ζ, v) process and of X_{it} . We maintain $N/T \rightarrow \kappa_0 \in (0, \infty)$.

Let

$$\Delta_i = \frac{1}{T} \sum_{t=1}^T \zeta_{it}, \quad \check{\alpha}_i = \alpha_i + \Delta_i, \quad \hat{\alpha}_i = \Pi_{[b_T, 1-b_T]}(\check{\alpha}_i),$$

where $b_T = C_b \sqrt{\log N/T}$. Write $d_i = \hat{\alpha}_i - \alpha_i$ and $r_i^{\text{cl}} = d_i - \Delta_i$. The clipping concentration bound gives, for some $a_0 > 1$ and every fixed $q \leq 6$,

$$\begin{aligned} \mathbb{P}(\mathcal{E}_T^c) &\leq 2N^{1-a_0}, & \mathcal{E}_T &= \left\{ \max_i |\Delta_i| \leq b_T \right\}, \\ \sup_i \mathbb{E}_i |d_i|^q &\leq C_q T^{-q/2}, & \sup_i \mathbb{E}_i |r_i^{\text{cl}}|^q &\leq C_q b_T^q N^{-a_0}. \end{aligned} \quad (\text{D.1})$$

On $\mathcal{E}_T$, $d_i = \Delta_i$ for every unit. The scalar score and its first three coordinate derivatives have bounded envelopes under the stated spline smoothness and approximation conditions.

We use one elementary consequence of finite-range dependence repeatedly. If $\bar{q}_i$ and $\bar{R}_i$ are sample means of bounded, centered components of a jointly r_0 -dependent array for some fixed r_0 , then, for $j = 1, 2, 3$,

$$\mathbb{E}_i \left| \bar{q}_i^j \bar{R}_i - \mathbb{E}_i(\bar{q}_i^j \bar{R}_i) \right|^2 \leq CT^{-(j+1)}. \quad (\text{D.2})$$

To see this, write $\|U\|_{p,i} = (\mathbb{E}_i |U|^p)^{1/p}$ and partition the time indices into their $r_0 + 1$ residue classes. Each class is an independent sum, so bounded-moment inequalities give

$$\|\bar{q}_i\|_{2(j+1),i} + \|\bar{R}_i\|_{2(j+1),i} = O(T^{-1/2}).$$

Hölder's inequality then yields

$$\left\| \bar{q}_i^j \bar{R}_i \right\|_{2,i} \leq \|\bar{q}_i\|_{2(j+1),i}^j \|\bar{R}_i\|_{2(j+1),i} = O(T^{-(j+1)/2}).$$

Centering changes the L^2 norm by at most a factor of two, which proves (D.2). A fixed collection of lagged products obeys the same bound after increasing r_0 by its fixed lag span.

Proposition 2.1 applies because the benchmark makes v centered and independent of X , and ε centered and independent of (v, X) . Lemma C.4 supplies the two grid-projection rates. Hence (2.8) makes the target difference negligible whenever $\sqrt{NT} \mathfrak{h}_K^{\tau_m + \tau_h} \rightarrow 0$.

At the finite-sieve target, the residuals used below are

$$u_{D,it,K} = v_{it} + e_{m,it,K}, \quad q_{it,K} = \varepsilon_{it} + e_{h,it,K} - \Delta_{\theta,K,N} u_{D,it,K}.$$

D.1.1 Seven-Term Expansion for the Estimated-Coordinate Score

Suppress the fixed arguments $(\rho_{0,K,N}, \alpha_i)$ and write

$$\begin{aligned} f_{it,K}^\theta &= f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i), & A_{it,K}^\theta &= \partial_a f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i), \\ H_{it,K}^\theta &= \partial_{aa} f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i), & T_{it,K}^\theta(\rho, a) &= \partial_{aaa} f_{it,K}^\theta(\rho, a). \end{aligned}$$

Set $\lambda_{i,K} = \mathbb{E}_i[A_{i0,K}^\theta] = \mu_{f,i,K}$ and $\mu_{H,i,K} = \mathbb{E}_i[H_{i0,K}^\theta] = \mu_{\ddot{f},i,K}$, and define

$$\bar{A}_{i,K}^c = \frac{1}{T} \sum_t (A_{it,K}^\theta - \lambda_{i,K}), \quad \bar{H}_{i,K}^c = \frac{1}{T} \sum_t (H_{it,K}^\theta - \mu_{H,i,K}). \quad (\text{D.3})$$

For a scalar d , define the exact third-derivative path average

$$\mathcal{T}_{iT,K}(d) = \frac{1}{T} \sum_{t=1}^T \int_0^1 (1-s)^2 T_{it,K}^\theta(\rho_{0,K,N}, \alpha_i + sd) ds.$$

Set $M_{i,K}(d) = \mathbb{E}_i[\mathcal{T}_{iT,K}(d)]$. The exact clipped finite- T coefficient is

$$\mathcal{B}_{\theta,K}^{\text{cl}} := \frac{T}{N} \sum_i \mathbb{E}_i[d_i \{\lambda_{i,K} + \bar{A}_{i,K}^c\}] + \frac{T}{2N} \sum_i \mathbb{E}_i[d_i^2 \{\mu_{H,i,K} + \bar{H}_{i,K}^c\}] + \frac{T}{2N} \sum_i \mathbb{E}_i[d_i^3 \mathcal{T}_{iT,K}(d_i)]. \quad (\text{D.4})$$

Because the score derivatives and coordinate error may be dependent, Equation (D.4) retains the exact joint-product expectations.

Let $m_{\theta,i,K} = \mathbb{E}_i[f_{i0,K}^\theta]$ and define

$$x_{\theta,it,K} = f_{it,K}^\theta - m_{\theta,i,K} + \lambda_{i,K} \zeta_{it}, \quad \bar{Z}_{\theta,K} = \mathbb{P}_{NT} x_{\theta,it,K}.$$

The remainder consists of three first-order or clipping terms, two second-order terms, and two third-order

terms. The seven centered terms are

$$\begin{aligned}
R_{\theta,1\Delta,c,K}^J &= \frac{1}{N} \sum_i \{ \Delta_i \bar{A}_{i,K}^c - \mathbb{E}_i(\Delta_i \bar{A}_{i,K}^c) \}, \\
R_{\theta,1cl,c,K}^J &= \frac{1}{N} \sum_i \{ r_i^{cl} \bar{A}_{i,K}^c - \mathbb{E}_i(r_i^{cl} \bar{A}_{i,K}^c) \}, \\
R_{\theta,1cl,m,K}^J &= \frac{1}{N} \sum_i \lambda_{i,K} \{ r_i^{cl} - \mathbb{E}_i r_i^{cl} \}, \\
R_{\theta,2c,K}^J &= \frac{1}{2N} \sum_i \{ d_i^2 \bar{H}_{i,K}^c - \mathbb{E}_i(d_i^2 \bar{H}_{i,K}^c) \}, \\
R_{\theta,2m,K}^J &= \frac{1}{2N} \sum_i \mu_{H,i,K} \{ d_i^2 - \mathbb{E}_i d_i^2 \}, \\
R_{\theta,3c,K}^J &= \frac{1}{2N} \sum_i \left[d_i^3 \{ \mathcal{T}_{i,K}(d_i) - M_{i,K}(d_i) \} - \mathbb{E}_i \left[d_i^3 \{ \mathcal{T}_{i,K}(d_i) - M_{i,K}(d_i) \} \right] \right], \\
R_{\theta,3m,K}^J &= \frac{1}{2N} \sum_i \{ d_i^3 M_{i,K}(d_i) - \mathbb{E}_i[d_i^3 M_{i,K}(d_i)] \}.
\end{aligned}$$

Every displayed term is centered at its exact joint-law expectation. For compactness below, let

$$r_{\theta,F,K}^J := r_{\theta,F,K}^J = R_{\theta,1\Delta,c,K}^J + R_{\theta,1cl,c,K}^J + R_{\theta,1cl,m,K}^J + R_{\theta,2c,K}^J + R_{\theta,2m,K}^J + R_{\theta,3c,K}^J + R_{\theta,3m,K}^J.$$

Proposition D.1 (Exact target-score expansion and remainder rate). *Define*

$$\hat{F}_{\theta,NT,K}(\rho_{0,K,N}) = \mathbb{P}_{NT} f_{it,K}^\theta(\rho_{0,K,N}, \hat{\alpha}_i).$$

Then

$$\begin{aligned}
\hat{F}_{\theta,NT,K}(\rho_{0,K,N}) &= \bar{Z}_{\theta,K} + \frac{1}{T} \mathcal{B}_{\theta,K}^{cl} + R_{\theta,1\Delta,c,K}^J + R_{\theta,1cl,c,K}^J + R_{\theta,1cl,m,K}^J \\
&\quad + R_{\theta,2c,K}^J + R_{\theta,2m,K}^J + R_{\theta,3c,K}^J + R_{\theta,3m,K}^J.
\end{aligned} \tag{D.5}$$

Moreover,

$$\sqrt{NT} |r_{\theta,F,K}| = o_p(1). \tag{D.6}$$

Proof. Exact Taylor expansion. Taylor's theorem, with the third-order remainder written explicitly, gives the following unit-by-unit expansion:

$$\frac{1}{T} \sum_t f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i + d_i) = \frac{1}{T} \sum_t f_{it,K}^\theta + d_i \{ \lambda_{i,K} + \bar{A}_{i,K}^c \} + \frac{d_i^2}{2} \{ \mu_{H,i,K} + \bar{H}_{i,K}^c \} + \frac{d_i^3}{2} \mathcal{T}_{i,K}(d_i). \tag{D.7}$$

First-order split. The population normal equation implies $\frac{1}{N} \sum_i m_{\theta,i,K} = 0$, while $\mathbb{E}_i \Delta_i = 0$. Hence $\bar{Z}_{\theta,K} = \mathbb{P}_{NT} f_{it,K}^\theta + \frac{1}{N} \sum_i \lambda_{i,K} \Delta_i$. Since $d_i = \Delta_i + r_i^{cl}$,

$$d_i \{ \lambda_{i,K} + \bar{A}_{i,K}^c \} - \lambda_{i,K} \Delta_i = \Delta_i \bar{A}_{i,K}^c + r_i^{cl} \bar{A}_{i,K}^c + \lambda_{i,K} r_i^{cl}. \tag{D.8}$$

Adding and subtracting the exact expectation of each term in (D.8), and doing the analogous mean/centered splits for the second- and third-order terms, turns (D.7) into (D.5).

Second-moment bounds. Each remainder is centered unit by unit, so all cross-unit covariance terms vanish by unit independence. Within a unit, (D.2), (D.1), and the bounded scalar derivative envelopes give

$$\begin{aligned}
\mathbb{E} |R_{\theta,1\Delta,c,K}^J|^2 &\leq \frac{C}{NT^2}, \\
\mathbb{E} |R_{\theta,1cl,c,K}^J|^2 + \mathbb{E} |R_{\theta,1cl,m,K}^J|^2 &\leq \frac{Cb_T^2 N^{-a_0}}{N}, \\
\mathbb{E} |R_{\theta,2c,K}^J|^2 + \mathbb{E} |R_{\theta,2m,K}^J|^2 &\leq \frac{C}{NT^2}, \\
\mathbb{E} |R_{\theta,3c,K}^J|^2 + \mathbb{E} |R_{\theta,3m,K}^J|^2 &\leq \frac{C}{NT^3}.
\end{aligned} \tag{D.9}$$

Conclusion. For example, the first line is the $j = 1$ case of (D.2). The last line follows directly from the bounded third-derivative path and $\mathbb{E}_i |d_i|^6 \leq CT^{-3}$. After multiplication by NT , the four right-hand sides in (D.9) are bounded by C/T , $CTb_T^2 N^{-a_0}$, C/T , and C/T^2 , respectively. Because $Tb_T^2 N^{-a_0} = C_b^2(\log N)N^{-a_0} \rightarrow 0$, Chebyshev's inequality and the triangle inequality prove (D.6). $\square$

D.2 Bias-Correction Consistency

The results below establish consistency of the correction numerator under the same bounded joint finite-range specialization as Section D.1. The retained lag set is exactly

$$\Lambda_{q_0} = \{-q_0, \dots, q_0\}, \quad I_T(\ell) = \{t : 1 \leq t \leq T, 1 \leq t + \ell \leq T\}.$$

Throughout this subsection, q_0 is fixed, so the correction sums over the fixed lag set Λ_{q_0} .

The proof separates four sources of correction error: oracle channel fluctuations, regular plug-in error, fitted second-derivative error, and the finite- T bridge between the analytical and exact clipped targets.

D.2.1 Population Targets and Plug-In Decomposition

Retain the oracle derivatives from Section D.1 and write

$$\tilde{A}_{it,K}^\theta = A_{it,K}^\theta - \lambda_{i,K}, \quad \Gamma_{i,K}^A(\ell) = \mathbb{E}_i[\tilde{A}_{i0,K}^\theta \zeta_{i\ell}].$$

The active derivative and curvature targets are

$$B_{\zeta,\theta,K}^J = \frac{1}{N} \sum_i \sum_{\ell \in \Lambda_{q_0}} \Gamma_{i,K}^A(\ell) = \Omega_{v\zeta} \bar{a}_{v,K}. \quad (\text{D.10})$$

$$B_{\zeta\zeta,\theta,K}^J = \frac{\Omega_\zeta}{2N} \sum_i \mu_{H,i,K}.$$

where

$$\begin{aligned} \bar{a}_{v,K} &= \frac{1}{N} \sum_i \mathbb{E}_X [2\Delta_{\theta,K,N} m_{K,N,a}(X, \alpha_i) - h_{K,N,a}^0(X, \alpha_i)], \\ \Omega_{v\zeta} &= \sum_{\ell \in \Lambda_{q_0}} \gamma_{v\zeta}(\ell), \quad \Omega_\zeta = \sum_{\ell \in \Lambda_{q_0}} \gamma_\zeta(\ell). \end{aligned}$$

For later rate statements, set $\tau_* = \min\{\tau_m, \tau_h\}$. The equality in (D.10) follows from the exact derivative decomposition

$$A_{it,K}^\theta = b_{it,K}^A + \{2\Delta_{\theta,K,N} m_{K,N,a} - h_{K,N,a}^0\} v_{it} - m_{K,N,a} \varepsilon_{it}, \quad (\text{D.11})$$

because X is independent of the joint (v, ζ) process and the outcome error is independent and centered.

Let $\hat{A}_{it,K}^\theta$ and $\hat{H}_{it,K}^\theta$ be the two plug-in derivatives in the feasible correction. To define exact derivative partitions, put $\hat{\pi}_K = \hat{\delta}_K - \hat{\theta}_K \hat{\gamma}_K$ and $\pi_{0,K,N} = \delta_{0,K,N} - \theta_{0,K,N} \gamma_{0,K,N}$. At $a = \hat{\alpha}_i$, for $r = 0, 1, 2$, define

$$\begin{aligned} M_{it,K}^{(r)} &= \partial_a^r p_K(X_{it}, a)^\top \gamma_{0,K,N}, & \Delta M_{it,K}^{(r)} &= \partial_a^r p_K(X_{it}, a)^\top (\hat{\gamma}_K - \gamma_{0,K,N}), \\ P_{it,K}^{(r)} &= \partial_a^r p_K(X_{it}, a)^\top \pi_{0,K,N}, & \Delta P_{it,K}^{(r)} &= \partial_a^r p_K(X_{it}, a)^\top (\hat{\pi}_K - \pi_{0,K,N}). \end{aligned}$$

Set

$$\begin{aligned} q_{it,K}^0 &= Y_{it} - \theta_{0,K,N} D_{it} - P_{it,K}^{(0)}, & u_{D,it,K}^0 &= D_{it} - M_{it,K}^{(0)}, \\ \Delta q_{it,K} &= -(\hat{\theta}_K - \theta_{0,K,N}) D_{it} - \Delta P_{it,K}^{(0)}. \end{aligned}$$

Then the fitted residuals obey the exact identities

$$\hat{u}_{D,it,K} = u_{D,it,K}^0 - \Delta M_{it,K}^{(0)}, \quad \hat{q}_{it,K} = q_{it,K}^0 + \Delta q_{it,K}. \quad (\text{D.12})$$

For any coordinate a , define the oracle derivative paths

$$A_{it,K}^{\theta,0}(a) = \partial_a f_{it,K}^\theta(\rho_{0,K,N}, a), \quad H_{it,K}^{\theta,0}(a) = \partial_{aa} f_{it,K}^\theta(\rho_{0,K,N}, a).$$

The feasible derivative formulas therefore satisfy

$$\begin{aligned} \widehat{A}_{it,K}^\theta &= -(M_{it,K}^{(1)} + \Delta M_{it,K}^{(1)})(q_{it,K}^0 + \Delta q_{it,K}) - (u_{D,it,K}^0 - \Delta M_{it,K}^{(0)})(P_{it,K}^{(1)} + \Delta P_{it,K}^{(1)}), \\ \widehat{H}_{it,K}^\theta &= -(M_{it,K}^{(2)} + \Delta M_{it,K}^{(2)})(q_{it,K}^0 + \Delta q_{it,K}) - (u_{D,it,K}^0 - \Delta M_{it,K}^{(0)})(P_{it,K}^{(2)} + \Delta P_{it,K}^{(2)}) \\ &\quad + 2(M_{it,K}^{(1)} + \Delta M_{it,K}^{(1)})(P_{it,K}^{(1)} + \Delta P_{it,K}^{(1)}). \end{aligned} \quad (\text{D.13})$$

Define

$$\begin{aligned} E_{it,K}^A &= A_{it,K}^{\theta,0}(\widehat{\alpha}_i) - A_{it,K}^\theta - M_{it,K}^{(1)} \Delta q_{it,K} - \Delta M_{it,K}^{(1)} q_{it,K}^0 \\ &\quad - \Delta M_{it,K}^{(1)} \Delta q_{it,K} - u_{D,it,K}^0 \Delta P_{it,K}^{(1)} + \Delta M_{it,K}^{(0)} P_{it,K}^{(1)} \\ &\quad + \Delta M_{it,K}^{(0)} \Delta P_{it,K}^{(1)}. \end{aligned} \quad (\text{D.14})$$

$$\begin{aligned} E_{it,K}^H &= H_{it,K}^{\theta,0}(\widehat{\alpha}_i) - H_{it,K}^\theta - M_{it,K}^{(2)} \Delta q_{it,K} + \Delta M_{it,K}^{(0)} P_{it,K}^{(2)} \\ &\quad + 2M_{it,K}^{(1)} \Delta P_{it,K}^{(1)} + 2P_{it,K}^{(1)} \Delta M_{it,K}^{(1)} + 2\Delta M_{it,K}^{(1)} \Delta P_{it,K}^{(1)}. \end{aligned}$$

$$C_{it,K}^H = -\Delta M_{it,K}^{(2)} \{q_{it,K}^0 + \Delta q_{it,K}\} - (u_{D,it,K}^0 - \Delta M_{it,K}^{(0)}) \Delta P_{it,K}^{(2)}. \quad (\text{D.15})$$

Subtract the oracle formulas at $\widehat{\alpha}_i$ from (D.13). In the Hessian difference, place the two terms containing $\Delta M^{(2)}$ or $\Delta P^{(2)}$ in C^H and place every remaining term in E^H . Term-by-term subtraction then gives the exact partitions

$$\widehat{A}_{it,K}^\theta - A_{it,K}^\theta = E_{it,K}^A, \quad \widehat{H}_{it,K}^\theta - H_{it,K}^\theta = E_{it,K}^H + C_{it,K}^H. \quad (\text{D.16})$$

Thus C^H contains exactly the fitted second-coordinate-derivative terms. The array E^H contains the coordinate displacement and all remaining parameter terms. Define

$$S_{H,i,K} = \frac{1}{T} \sum_t H_{it,K}^\theta - \mu_{H,i,K}, \quad d_{H,i,K}^E = \frac{1}{T} \sum_t E_{it,K}^H, \quad d_{H,i,K}^C = \frac{1}{T} \sum_t C_{it,K}^H.$$

Write $\widehat{\zeta}_{it} = Z_{it} - \widehat{\alpha}_i = \zeta_{it} - d_i$ and set $\widehat{B}_{\theta,K} = \widehat{B}_{\zeta,\theta,K} + \widehat{B}_{\zeta\zeta,\theta,K}$, where

$$\widehat{\Omega}_{\zeta,i} = \sum_{\ell \in \Lambda_{q_0}} \frac{1}{T} \sum_{t \in I_T(\ell)} \widehat{\zeta}_{it} \widehat{\zeta}_{i,t+\ell}, \quad \omega_{i,K} = \widehat{\Omega}_{\zeta,i} - \Omega_{\zeta}.$$

$$\widehat{B}_{\zeta,\theta,K} = \frac{1}{N} \sum_i \sum_{\ell \in \Lambda_{q_0}} \frac{1}{T} \sum_{t \in I_T(\ell)} \widehat{A}_{it,K}^\theta \widehat{\zeta}_{i,t+\ell},$$

$$\widehat{B}_{\zeta\zeta,\theta,K} = \frac{1}{2N} \sum_i \left(\frac{1}{T} \sum_t \widehat{H}_{it,K}^\theta \right) \widehat{\Omega}_{\zeta,i}.$$

Lemma C.5 and Corollary C.19 imply that the localization event below has probability approaching one.

$$\mathcal{L}_K = \mathcal{E}_T \cap \mathcal{G}_K.$$

$$\mathbb{P}(\mathcal{L}_K^c) \leq \mathbb{P}(\mathcal{E}_T^c) + \mathbb{P}(\mathcal{G}_K^c) \longrightarrow 0. \quad (\text{D.17})$$

On $\mathcal{L}_K$, clipping is inactive and all fitted normal equations are valid. Define the following six disjoint remainder components:

$$R_{A,c}^J = \frac{1}{N} \sum_i \sum_{\ell \in \Lambda_{q_0}} \frac{1}{T} \sum_{t \in I_T(\ell)} \widetilde{A}_{it,K}^\theta (\zeta_{i,t+\ell} - \Delta_i) - B_{\zeta,\theta,K}^J. \quad (\text{D.18})$$

$$R_\lambda^J = \frac{1}{N} \sum_i \lambda_{i,K} \sum_{\ell \in \Lambda_{q_0}} \frac{1}{T} \sum_{t \in I_T(\ell)} (\zeta_{i,t+\ell} - \Delta_i).$$

$$R_{A,p}^J = \frac{1}{N} \sum_i \sum_{\ell \in \Lambda_{q_0}} \frac{1}{T} \sum_{t \in I_T(\ell)} E_{it,K}^A(\zeta_{i,t+\ell} - \Delta_i). \quad (\text{D.19})$$

$$R_{H,\text{or}}^J = \frac{1}{2N} \sum_i \{ \mu_{H,i,K} \omega_{i,K} + \Omega_\zeta S_{H,i,K} + S_{H,i,K} \omega_{i,K} \}.$$

$$R_{H,E}^J = \frac{1}{2N} \sum_i \{ \Omega_\zeta d_{H,i,K}^E + d_{H,i,K}^E \omega_{i,K} \}.$$

$$R_{H,C}^J = \frac{1}{2N} \sum_i \{ \Omega_\zeta d_{H,i,K}^C + d_{H,i,K}^C \omega_{i,K} \}.$$

In particular, $R_{A,c}^J$ is centered around the active target $B_{\zeta,\theta,K}^J$. It is not centered around zero.

D.2.2 Boundary Terms and the Finite-Sample Bridge

The constant-derivative component is not eliminated by within-unit demeaning. Let $J_T(\ell) = \{t+\ell : t \in I_T(\ell)\}$. Since $\Delta_i = \frac{1}{T} \sum_s \zeta_{is}$, the following representation is exact on $\mathcal{L}_K$:

$$R_\lambda^J = \frac{1}{N} \sum_i \lambda_{i,K} \sum_{s=1}^T c_{s,T} \zeta_{is},$$

$$c_{s,T} = \frac{1}{T} \sum_{\ell \in \Lambda_{q_0}} \left\{ \mathbf{1}\{s \in J_T(\ell)\} - \frac{T - |\ell|}{T} \right\}.$$

For an index farther than q_0 periods from either boundary, $c_{s,T} = \{\sum_{\ell \in \Lambda_{q_0}} |\ell|\}/T^2$. Direct counting gives

$$\sum_{s=1}^T c_{s,T}^2 = O(T^{-2}). \quad (\text{D.20})$$

Thus the fixed-lag boundary mismatch is small but generally nonzero.

For later use, the centering of $R_{A,c}^J$ can also be made explicit. Set

$$U_{A,K} = \frac{1}{N} \sum_i \sum_{\ell \in \Lambda_{q_0}} \frac{1}{T} \sum_{t \in I_T(\ell)} \left\{ \tilde{A}_{it,K}^\theta \zeta_{i,t+\ell} - \Gamma_{i,K}^A(\ell) \right\},$$

$$G_{A,i,K}^c = \sum_{\ell \in \Lambda_{q_0}} \frac{1}{T} \sum_{t \in I_T(\ell)} \tilde{A}_{it,K}^\theta,$$

$$Q_{A,K} = \frac{1}{N} \sum_i \sum_{\ell \in \Lambda_{q_0}} \frac{|\ell|}{T} \Gamma_{i,K}^A(\ell), \quad D_{A,K} = -\frac{1}{N} \sum_i \Delta_i G_{A,i,K}^c.$$

Pair counting gives the exact identity

$$R_{A,c}^J = U_{A,K} - Q_{A,K} + D_{A,K}. \quad (\text{D.21})$$

This display isolates sample fluctuation, the finite-pair boundary, and feasible demeaning. None is counted in R_λ^J .

To connect the analytical targets to the exact coefficient in Section D.1, define

$$R_{\text{bridge}}^J = \mathcal{B}_{\theta,K}^{\text{cl}} - B_{\zeta,\theta,K}^J - B_{\zeta\zeta,\theta,K}^J. \quad (\text{D.22})$$

To state the bridge, let

$$\begin{aligned} R_{1,\text{cl}}^J &= \frac{T}{N} \sum_i \mathbb{E}_i[(d_i - \Delta_i)\{\lambda_{i,K} + \bar{A}_{i,K}^c\}], \\ R_{2,\text{cl}}^J &= \frac{T}{2N} \sum_i \mathbb{E}_i[(d_i^2 - \Delta_i^2)\{\mu_{H,i,K} + \bar{H}_{i,K}^c\}], \\ R_{3,\text{path}}^J &= \frac{T}{2N} \sum_i \mathbb{E}_i[d_i^3 \mathcal{T}_{iT,K}(d_i)]. \end{aligned}$$

Condition (5.7), after a time shift, implies

$$\mathbb{E}_i[H_{it,K}^\theta \zeta_{is} \zeta_{ir}] = \mu_{H,i,K} \mathbb{E}_i[\zeta_{is} \zeta_{ir}]. \quad (\text{D.23})$$

Using (D.11), (D.23), and finite-range pair counts give

$$\begin{aligned} T \mathbb{E}_i [\Delta_i \{\lambda_{i,K} + \bar{A}_{i,K}^c\}] &= \mathbb{E}_X[2\Delta_{\theta,K,N} m_{K,N,a} - h_{K,N,a}^0] \times \left\{ \Omega_{v\zeta} - \frac{1}{T} \sum_{\ell \in \Lambda_{q_0}} |\ell| \gamma_{v\zeta}(\ell) \right\}, \\ \frac{T}{2} \mathbb{E}_i [\Delta_i^2 \{\mu_{H,i,K} + \bar{H}_{i,K}^c\}] &= \frac{\mu_{H,i,K}}{2} \left\{ \Omega_\zeta - \frac{1}{T} \sum_{\ell \in \Lambda_{q_0}} |\ell| \gamma_\zeta(\ell) \right\}. \end{aligned}$$

Consequently,

$$R_{\text{bridge}}^J = -\frac{1}{T} \left\{ \sum_{\ell \in \Lambda_{q_0}} |\ell| \gamma_{v\zeta}(\ell) \right\} \bar{a}_{v,K} - \frac{1}{2TN} \left\{ \sum_{\ell \in \Lambda_{q_0}} |\ell| \gamma_\zeta(\ell) \right\} \sum_i \mu_{H,i,K} + R_{1,\text{cl}}^J + R_{2,\text{cl}}^J + R_{3,\text{path}}^J. \quad (\text{D.24})$$

D.2.3 Correction-Error Identity and Remainder Bounds

Define the three plug-in norms

$$\begin{aligned} e_{A,K}^2 &= \mathbb{P}_{NT}(E_{it,K}^A)^2, & e_{H,E,K}^2 &= \frac{1}{N} \sum_i (d_{H,i,K}^E)^2, \\ \kappa_{\text{closure},K}^2 &= \frac{1}{N} \sum_i (d_{H,i,K}^C)^2. \end{aligned}$$

Lemma D.2 (Preliminary root and plug-in bounds). *Under Assumption 5.2, suppose $\mathfrak{h}_K^{-m}/T \rightarrow 0$, $\mathfrak{h}_K^{-2} \bar{r}_{\eta,K} \rightarrow 0$, and $K_x \log(NK_x)/T \rightarrow 0$. On $\mathcal{L}_K$,*

$$\left| \hat{\theta}_K - \theta_{0,K,N} \right| = O_p(T^{-1}), \quad \|\hat{\eta}_K - \eta_{0,K,N}\| = O_p(\bar{r}_{\eta,K}). \quad (\text{D.25})$$

$$e_{A,K} + e_{H,E,K} = O_p\left(T^{-1/2} + \mathfrak{h}_K^{-1} \bar{r}_{\eta,K}\right). \quad (\text{D.26})$$

$$\kappa_{\text{closure},K} = O_p(\mathfrak{h}_K^{-2} \bar{r}_{\eta,K}) = o_p(1). \quad (\text{D.27})$$

Proof. Preliminary root rate. Lemma C.13 gives the nuisance part of (D.25). For theta, substitute the score expansion from Proposition D.1 into the localized root identity. On $\mathcal{G}_K$, this gives

$$\hat{\theta}_K - \theta_{0,K,N} = \hat{V}_K^{-1} \left\{ \bar{Z}_{\theta,K} + \frac{1}{T} \mathcal{B}_{\theta,K}^{\text{cl}} + r_{\theta,F,K} - e_\gamma^\top \hat{\Sigma}_{p,K} e_\pi \right\}.$$

The leading scalar score is $O_p((NT)^{-1/2}) = O_p(T^{-1})$, the coordinate-expansion remainder is $o_p(T^{-1})$, and the profile quadratic is $O_p(\bar{r}_{\eta,K}^2) = o_p(T^{-1})$. Equations (D.24) and (C.14), together with the bounded derivative paths, give $\left| R_{1,\text{cl}}^J \right| \leq CTb_T N^{-a_0}$, $\left| R_{2,\text{cl}}^J \right| \leq CTb_T^2 N^{-a_0}$, and $\left| R_{3,\text{path}}^J \right| \leq CT^{-1/2}$. Thus $\mathcal{B}_{\theta,K}^{\text{cl}} = O(1)$. The

denominator is bounded away from zero by Lemmas C.15 and C.16. Solving the root identity therefore gives $|\hat{\theta}_K - \theta_{0,K,N}| = O_p(T^{-1}) = O_p(\bar{r}_{\eta,K})$.

Derivative Gram and coefficient errors. By the Riesz bounds and $\|\gamma_{0,K,N}\| = O(1)$, $\|\hat{\pi}_K - \pi_{0,K,N}\| = O_p(\bar{r}_{\eta,K})$. For the first derivative, define $\hat{G}_{a,K} = \mathbb{P}_{NT} p_{K,a}(X_{it}, \hat{\alpha}_i) p_{K,a}(X_{it}, \hat{\alpha}_i)^\top$. The tensor and shifted-grid argument used for Lemma C.14 gives

$$\lambda_{\max}(\hat{G}_{a,K}) = O_p(\mathfrak{h}_K^{-2}). \quad (\text{D.28})$$

Indeed, conditional on the generated coordinates, shifted-grid occupancy and the derivative inverse inequality bound the conditional mean by $C\mathfrak{h}_K^{-2}$. Local support gives the summand envelope $CK\mathfrak{h}_K^{-2}$ and matrix-variance proxy $CK\mathfrak{h}_K^{-4}/(NT)$. Matrix Bernstein therefore bounds the deviation by $O_p[\mathfrak{h}_K^{-2}\{\sqrt{K \log K/(NT)} + K \log K/(NT)\}] = o_p(\mathfrak{h}_K^{-2})$. The ordinary generated Gram, (D.28), and the second-derivative event in Lemmas C.11 and C.14 now give, for $r = 0, 1, 2$,

$$\|\Delta M^{(r)}\|_{NT,2} + \|\Delta P^{(r)}\|_{NT,2} = O_p(\mathfrak{h}_K^{-r} \bar{r}_{\eta,K}). \quad (\text{D.29})$$

Moreover, $\sqrt{K} \bar{r}_{\eta,K} = o(1)$ because $(d_W + m)/2 \leq m$ and $\mathfrak{h}_K^{-m}/T \rightarrow 0$. Local support, bounded data, and the preliminary theta rate therefore give

$$\begin{aligned} \|\Delta M^{(0)}\|_\infty + \|\Delta P^{(0)}\|_\infty &= O_p(\sqrt{K} \bar{r}_{\eta,K}) = o_p(1), \\ \|\Delta q\|_{NT,2} &= O_p(\bar{r}_{\eta,K}), \quad \|\Delta q\|_\infty = o_p(1). \end{aligned} \quad (\text{D.30})$$

Derivative plug-in bound. The coordinate displacement is controlled directly by the Hessian path:

$$A_{it,K}^{\theta,0}(\hat{\alpha}_i) - A_{it,K}^\theta = d_i \int_0^1 H_{it,K}^{\theta,0}(\alpha_i + s d_i) ds.$$

The oracle Hessian path is uniformly bounded, so

$$\|A^{\theta,0}(\hat{\alpha}) - A^\theta\|_{NT,2}^2 \leq \frac{C}{N} \sum_i d_i^2 = O_p(T^{-1}).$$

Thus (D.14), (D.29), and $\frac{1}{N} \sum_i d_i^2 = O_p(T^{-1})$ imply

$$\begin{aligned} \|A^{\theta,0}(\hat{\alpha}) - A^\theta\|_{NT,2} &= O_p(T^{-1/2}), \\ \|M^{(1)} \Delta q\|_{NT,2} + \|\Delta M^{(0)} P^{(1)}\|_{NT,2} &= O_p(\bar{r}_{\eta,K}), \\ \|\Delta M^{(1)} q^0\|_{NT,2} + \|u_D^0 \Delta P^{(1)}\|_{NT,2} &= O_p(\mathfrak{h}_K^{-1} \bar{r}_{\eta,K}), \\ \|\Delta M^{(1)} \Delta q\|_{NT,2} + \|\Delta M^{(0)} \Delta P^{(1)}\|_{NT,2} &= o_p(\mathfrak{h}_K^{-1} \bar{r}_{\eta,K}). \end{aligned}$$

Consequently,

$$e_{A,K} = O_p\left(T^{-1/2} + \mathfrak{h}_K^{-1} \bar{r}_{\eta,K}\right). \quad (\text{D.31})$$

Regular Hessian plug-in bound. For the regular Hessian part, the corresponding path identity is

$$H_{it,K}^{\theta,0}(\hat{\alpha}_i) - H_{it,K}^\theta = d_i \int_0^1 \partial_{aaa} f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i + s d_i) ds.$$

The bounded third-derivative path first gives

$$\begin{aligned} \left\{ \frac{1}{N} \sum_i \left(\frac{1}{T} \sum_t d_i \int_0^1 \partial_{aaa} f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i + s d_i) ds \right)^2 \right\}^{1/2} &\leq C \left(\frac{1}{N} \sum_i d_i^2 \right)^{1/2} \\ &= O_p(T^{-1/2}). \end{aligned}$$

For the four linear coefficient-error terms, Jensen's inequality within unit and the fitted-derivative bounds in (D.29)–(D.30) give

$$\begin{aligned} & \left\{ \frac{1}{N} \sum_i \left[\frac{1}{T} \sum_t \{ |M_{it,K}^{(2)} \Delta q_{it,K}| + |\Delta M_{it,K}^{(0)} P_{it,K}^{(2)}| \} \right]^2 \right\}^{1/2} \\ & \leq \|M^{(2)} \Delta q\|_{NT,2} + \|\Delta M^{(0)} P^{(2)}\|_{NT,2} = O_p(\bar{r}_{\eta,K}), \\ & \left\{ \frac{1}{N} \sum_i \left[\frac{1}{T} \sum_t \{ |M_{it,K}^{(1)} \Delta P_{it,K}^{(1)}| + |P_{it,K}^{(1)} \Delta M_{it,K}^{(1)}| \} \right]^2 \right\}^{1/2} \\ & \leq \|M^{(1)} \Delta P^{(1)}\|_{NT,2} + \|P^{(1)} \Delta M^{(1)}\|_{NT,2} = O_p(\mathfrak{h}_K^{-1} \bar{r}_{\eta,K}). \end{aligned}$$

The last equalities use uniform boundedness of the oracle derivative paths. The only remaining term is the within-unit average of the product of the two fitted first derivatives. The coefficient-uniform bound (C.40) gives

$$\left\{ \frac{1}{N} \sum_i \left(\frac{1}{T} \sum_t \Delta M_{it,K}^{(1)} \Delta P_{it,K}^{(1)} \right)^2 \right\}^{1/2} = O_p(\mathfrak{h}_K^{-5/2} \bar{r}_{\eta,K}^2).$$

This is $o_p(\mathfrak{h}_K^{-1} \bar{r}_{\eta,K})$ because

$$\mathfrak{h}_K^{-3/2} \bar{r}_{\eta,K} = \frac{\mathfrak{h}_K^{-(m+3)/2}}{T} \leq \frac{\mathfrak{h}_K^{-m}}{T} = o(1), \quad m \geq 3.$$

Thus

$$e_{H,E,K} = O_p\left(T^{-1/2} + \mathfrak{h}_K^{-1} \bar{r}_{\eta,K}\right),$$

which completes (D.26).

Second-derivative closure. Finally, (D.15), the uniform boundedness of the fitted level residuals, Jensen's inequality within unit, and the second-derivative Gram event (C.39) give the explicit bound

$$\kappa_{\text{closure},K} \leq C_p \{ \|\Delta M^{(2)}\|_{NT,2} + \|\Delta P^{(2)}\|_{NT,2} \} = O_p(\mathfrak{h}_K^{-2} \bar{r}_{\eta,K}).$$

The assumed second-derivative plug-in rate makes this $o_p(1)$ and proves (D.27). $\square$

Proposition D.3 (Exact correction-error decomposition). *On $\mathcal{L}_K$,*

$$\widehat{V}_K^{-1} \widehat{B}_{\theta,K} - V_K^{-1} \mathcal{B}_{\theta,K}^{\text{cl}} = (\widehat{V}_K^{-1} - V_K^{-1}) \mathcal{B}_{\theta,K}^{\text{cl}} + \widehat{V}_K^{-1} \left[R_{A,c}^J + R_{\lambda}^J + R_{A,p}^J + R_{H,\text{or}}^J + R_{H,E}^J + R_{H,C}^J - R_{\text{bridge}}^J \right]. \quad (\text{D.32})$$

The decomposition separates denominator error, derivative-statistic error, curvature-statistic error, and the finite-sample bridge. Under the rate conditions in (5.14),

$$\widehat{V}_K^{-1} \widehat{B}_{\theta,K} - V_K^{-1} \mathcal{B}_{\theta,K}^{\text{cl}} = o_p(1). \quad (\text{D.33})$$

Proof. Derivative statistic. First expand the derivative statistic using $\widehat{A}^\theta = A^\theta + E^A$ and $A^\theta = \widetilde{A}^\theta + \lambda$. Definitions (D.18)–(D.19) give exactly

$$\widehat{B}_{\zeta,\theta,K} - B_{\zeta,\theta,K}^J = R_{A,c}^J + R_{\lambda}^J + R_{A,p}^J. \quad (\text{D.34})$$

Curvature statistic. Next use

$$\begin{aligned} \frac{1}{T} \sum_t \widehat{H}_{it,K}^\theta &= \mu_{H,i,K} + S_{H,i,K} + d_{H,i,K}^E + d_{H,i,K}^C, \\ \widehat{\Omega}_{\zeta,i} &= \Omega_\zeta + \omega_{i,K}. \end{aligned}$$

Subtracting the target $\mu_{H,i,K}\Omega_\zeta$ before grouping makes every cross term visible:

$$\begin{aligned} (\mu_{H,i,K} + S_{H,i,K} + d_{H,i,K}^E + d_{H,i,K}^C)(\Omega_\zeta + \omega_{i,K}) - \mu_{H,i,K}\Omega_\zeta &= \mu_{H,i,K}\omega_{i,K} + \Omega_\zeta S_{H,i,K} + S_{H,i,K}\omega_{i,K} \\ &\quad + d_{H,i,K}^E(\Omega_\zeta + \omega_{i,K}) + d_{H,i,K}^C(\Omega_\zeta + \omega_{i,K}). \end{aligned}$$

The first row after the equality defines $R_{H,\text{or}}^J$. The two factors in the last row define $R_{H,E}^J$ and $R_{H,C}^J$, respectively. Consequently,

$$\widehat{B}_{\zeta,\theta,K} - B_{\zeta,\theta,K}^J = R_{H,\text{or}}^J + R_{H,E}^J + R_{H,C}^J. \quad (\text{D.35})$$

Combining (D.34), (D.35), and (D.22), and then adding the denominator difference, proves (D.32).

Derivative and curvature bounds. We next show that each term is negligible. Fixed-range product bounds and cross-unit independence give

$$\begin{aligned} U_{A,K} &= O_p((NT)^{-1/2}), & Q_{A,K} &= O(T^{-1}), \\ |\mathbb{E}_i[\Delta_i G_{A,i,K}^c]| &\leq C/T, & \text{Var}_i(\Delta_i G_{A,i,K}^c) &\leq C/T^2. \end{aligned}$$

Therefore (D.21) and (D.20) imply

$$\begin{aligned} R_{A,c}^J &= O_p((NT)^{-1/2} + T^{-1}), \\ R_\lambda^J &= O_p\left(\frac{\|\lambda_K\|_{2,N}}{T\sqrt{N}}\right), \quad \|\lambda_K\|_{2,N} = O(\mathfrak{h}_K^{\tau_K^*}). \end{aligned} \quad (\text{D.36})$$

Take the conditional mean in (D.11). The innovation terms vanish. Lemma C.4 bounds both approximation errors in $b_{it,K}^A$ by $C\mathfrak{h}_K^{\tau_K^*}$. This gives the last bound. The joint mean generated by $\Delta_i G_{A,i,K}^c$ supplies the T^{-1} term in $R_{A,c}^J$.

For the curvature statistic, let $\widehat{\Omega}_{\zeta,i}^\Delta$ denote the same statistic as $\widehat{\Omega}_{\zeta,i}$, but with the unclipped residual $\zeta_{it} - \Delta_i$. Expanding the two demeaned factors gives the exact identity

$$\begin{aligned} \widehat{\Omega}_{\zeta,i}^\Delta - \Omega_\zeta &= \sum_{\ell \in \Lambda_{q_0}} \frac{1}{T} \sum_{t \in I_T(\ell)} \{\zeta_{it}\zeta_{i,t+\ell} - \gamma_\zeta(\ell)\} - \frac{1}{T} \sum_{\ell \in \Lambda_{q_0}} |\ell| \gamma_\zeta(\ell) \\ &\quad - \Delta_i \sum_{\ell \in \Lambda_{q_0}} \frac{1}{T} \sum_{t \in I_T(\ell)} (\zeta_{it} + \zeta_{i,t+\ell}) + \Delta_i^2 \sum_{\ell \in \Lambda_{q_0}} \left(1 - \frac{|\ell|}{T}\right). \end{aligned}$$

The first term is centered and has L^4 norm $O(T^{-1/2})$ by the fixed-range fourth-moment count. The deterministic boundary term is $O(T^{-1})$. The last two terms have L^4 norm $O(T^{-1})$, because both Δ_i and each truncated mean in the third term have uniformly bounded L^8 norm of order $T^{-1/2}$. Therefore,

$$\left|\mathbb{E}_i(\widehat{\Omega}_{\zeta,i}^\Delta - \Omega_\zeta)\right| = O(T^{-1}), \quad \left\|\widehat{\Omega}_{\zeta,i}^\Delta - \Omega_\zeta\right\|_{4,i} = O(T^{-1/2}).$$

Because $\widehat{\zeta}_{it} = (\zeta_{it} - \Delta_i) - r_i^{\text{cl}}$, expanding each lag product and using the fixed lag set gives the pathwise bound

$$\left|\widehat{\Omega}_{\zeta,i} - \widehat{\Omega}_{\zeta,i}^\Delta\right| \leq C\{|r_i^{\text{cl}}| + |r_i^{\text{cl}}|^2\}.$$

Boundedness and (D.1) transfer the preceding bounds to $\omega_{i,K}$. Together with the same finite-range calculation for $S_{H,i,K}$, they give

$$\begin{aligned} \mathbb{E}_i \omega_{i,K} &= O(T^{-1}), & \mathbb{E}_i \omega_{i,K}^2 &= O(T^{-1}), \\ \mathbb{E}_i S_{H,i,K} &= 0, & \mathbb{E}_i S_{H,i,K}^2 &= O(T^{-1}), \end{aligned} \quad (\text{D.37})$$

and

$$\|S_{H,i,K}\|_{4,i} + \|\omega_{i,K}\|_{4,i} = O(T^{-1/2}).$$

The exact factorization behind the remaining covariance calculation is also visible without introducing quadratic-form coefficients. For every retained pair $(t, t + \ell)$,

$$\begin{aligned} (\zeta_{it} - \Delta_i)(\zeta_{i,t+\ell} - \Delta_i) &= \zeta_{it}\zeta_{i,t+\ell} - \frac{1}{T} \sum_{r=1}^T (\zeta_{it}\zeta_{ir} + \zeta_{i,t+\ell}\zeta_{ir}) \\ &\quad + \frac{1}{T^2} \sum_{r=1}^T \sum_{s=1}^T \zeta_{ir}\zeta_{is}. \end{aligned}$$

Multiplying this identity by $S_{H,i,K} = T^{-1} \sum_u (H_{iu,K}^\theta - \mu_{H,i,K})$ and applying (D.23) term by term, while using $\mathbb{E}_i S_{H,i,K} = 0$ for the constant Ω_ζ , gives

$$\mathbb{E}_i \left[S_{H,i,K} \{ \widehat{\Omega}_{\zeta,i}^\Delta - \Omega_\zeta \} \right] = 0.$$

Thus (D.1), (D.37), and Hölder's inequality imply

$$\begin{aligned} \left| \mathbb{E}_i [S_{H,i,K} (\widehat{\Omega}_{\zeta,i} - \widehat{\Omega}_{\zeta,i}^\Delta)] \right| &\leq \|S_{H,i,K}\|_{2,i} \left\| \widehat{\Omega}_{\zeta,i} - \widehat{\Omega}_{\zeta,i}^\Delta \right\|_{2,i} \\ &\leq CT^{-1/2} \{b_T N^{-a_0/2} + b_T^2 N^{-a_0/2}\} = o(T^{-1}). \end{aligned}$$

Consequently,

$$\begin{aligned} \|S_{H,i,K} \omega_{i,K}\|_{2,i} &\leq \|S_{H,i,K}\|_{4,i} \|\omega_{i,K}\|_{4,i} = O(T^{-1}), \\ \text{Var}_i(S_{H,i,K} \omega_{i,K}) &\leq \mathbb{E}_i[S_{H,i,K}^2 \omega_{i,K}^2] = O(T^{-2}). \end{aligned}$$

The units are independent, while $\mu_{H,i,K}$ and Ω_ζ are uniformly bounded. The preceding mean and variance bounds therefore imply

$$|\mathbb{E} R_{H,\text{or}}^J| \leq \frac{C}{T}, \quad \text{Var}(R_{H,\text{or}}^J) \leq \frac{C}{NT}.$$

Consequently,

$$R_{H,\text{or}}^J = O_p\{T^{-1} + (NT)^{-1/2}\}. \quad (\text{D.38})$$

Plug-in terms. The plug-in bounds are pathwise and therefore do not require independence between the fitted derivatives and the innovations. For the derivative component, Cauchy–Schwarz gives

$$\begin{aligned} |R_{A,p}^J| &\leq \sum_{\ell \in \Lambda_{q_0}} \left\{ \frac{1}{NT} \sum_i \sum_{t \in I_T(\ell)} (E_{it,K}^A)^2 \right\}^{1/2} \times \left\{ \frac{1}{NT} \sum_i \sum_{t \in I_T(\ell)} (\zeta_{i,t+\ell} - \Delta_i)^2 \right\}^{1/2} \\ &\leq C e_{A,K}, \end{aligned}$$

because the lag set is fixed and the second factor is bounded. Likewise, boundedness of $\widehat{\Omega}_{\zeta,i}$ gives

$$\begin{aligned} |R_{H,E}^J| &\leq C \left\{ \frac{1}{N} \sum_i (d_{H,i,K}^E)^2 \right\}^{1/2} = C e_{H,E,K}, \\ |R_{H,C}^J| &\leq C \left\{ \frac{1}{N} \sum_i (d_{H,i,K}^C)^2 \right\}^{1/2} = C \kappa_{\text{closure},K}. \end{aligned}$$

Thus

$$R_{A,p}^J = O_p(e_{A,K}), \quad R_{H,E}^J = O_p(e_{H,E,K}), \quad R_{H,C}^J = O_p(\kappa_{\text{closure},K}). \quad (\text{D.39})$$

Lemma D.2 proves that all three right-hand sides are $o_p(1)$. The regular and closure bounds remain separate because the latter uses the second-derivative Gram event.

Finite-sample bridge. Finally, (D.1) and the bounded derivative paths give

$$|R_{1,\text{cl}}^J| \leq CT b_T N^{-a_0}, \quad |R_{2,\text{cl}}^J| \leq CT b_T^2 N^{-a_0}, \quad |R_{3,\text{path}}^J| \leq CT^{-1/2}. \quad (\text{D.40})$$

Because $N/T \rightarrow \kappa_0$ and $a_0 > 1$, the three bounds vanish. Equation (D.24) thus gives $R_{\text{bridge}}^J = o(1)$.

Denominator replacement. The analytical targets are bounded. The bridge therefore gives $\mathcal{B}_{\theta,K}^{\text{cl}} = O(1)$. The preceding denominator calculation gives $\widehat{V}_K - V_K = o_p(1)$ with both denominators bounded away from zero. Substitution of (D.36), (D.38), (D.39), and (D.40) into the exact identity (D.32) proves (D.33) on $\mathcal{L}_K$. Equation (D.17) transfers this probability order to the globally reported estimator and correction. $\square$

D.3 Joint Finite-Range Bias Calculations

The analysis derives the bias formulas under joint bounded finite-range dependence, allowing correlation between the measurement shock and treatment residual while maintaining an independent outcome-error process. These identities feed into the variance limit below and the theorem proofs in Appendix E.

D.3.1 Long-Run Covariances

Stationarity and q_0 -dependence imply

$$\gamma_{v\zeta}(\ell) = 0, \quad \gamma_{\zeta}(\ell) = 0, \quad |\ell| > q_0,$$

and hence

$$\Omega_{v\zeta} = \sum_{\ell \in \Lambda_{q_0}} \gamma_{v\zeta}(\ell), \quad \Omega_{\zeta} = \sum_{\ell \in \Lambda_{q_0}} \gamma_{\zeta}(\ell).$$

Example 5.3 gives closed-form formulas for these objects under finite moving-average filters, including the MA(1) design used in the simulations.

D.3.2 Derivative and Curvature Terms

Write $\delta_K = \Delta_{\theta,K,N}$ and suppress (X_{it}, α_i) on grid-projection functions. At the finite-sieve target,

$$u_{D,it,K} = v_{it} + e_{m,it,K}, \quad q_{it,K} := u_{Y,it,K} - \theta_{0,K,N} u_{D,it,K} = \varepsilon_{it} + e_{h,it,K} - \delta_K u_{D,it,K}.$$

Lemma D.4 (Derivative and Hessian decomposition). *The coordinate derivative of the theta score satisfies*

$$\begin{aligned} A_{it,K}^{\theta} &:= \dot{f}_{it,K}^{\theta}(\rho_{0,K,N}, \alpha_i) = b_{it,K}^A + a_{it,K}^A v_{it} - m_{K,N,a} \varepsilon_{it}, \\ b_{it,K}^A &= -m_{K,N,a} e_{h,it,K} - e_{m,it,K} h_{K,N,a}^0 + 2\delta_K e_{m,it,K} m_{K,N,a}. \end{aligned} \quad (\text{D.41})$$

$$a_{it,K}^A = 2\delta_K m_{K,N,a} - h_{K,N,a}^0. \quad (\text{D.42})$$

The second derivative satisfies

$$\begin{aligned} H_{it,K}^{\theta} &:= \ddot{f}_{it,K}^{\theta}(\rho_{0,K,N}, \alpha_i) = b_{it,K}^H + a_{it,K}^H v_{it} - m_{K,N,aa} \varepsilon_{it}, \\ b_{it,K}^H &= -m_{K,N,aa} e_{h,it,K} - e_{m,it,K} h_{K,N,aa}^0 + 2m_{K,N,a} h_{K,N,a}^0 - 2\delta_K m_{K,N,a}^2 + 2\delta_K e_{m,it,K} m_{K,N,aa}. \end{aligned} \quad (\text{D.43})$$

$$a_{it,K}^H = 2\delta_K m_{K,N,aa} - h_{K,N,aa}^0. \quad (\text{D.44})$$

Consequently,

$$\mathbb{E}_i[A_{i0,K}^{\theta} \zeta_{i\ell}] = \mathbb{E}_X[a_{i0,K}^A] \gamma_{v\zeta}(\ell). \quad (\text{D.45})$$

$$\mathbb{E}_i[H_{it,K}^{\theta} \zeta_{is} \zeta_{ir}] = \mu_{\dot{f},i,K} \mathbb{E}_i[\zeta_{is} \zeta_{ir}]. \quad (\text{D.46})$$

Proof. The proof first differentiates the score while holding (Y_{it}, D_{it}) fixed, and then substitutes the target residuals. Since $\partial_a u_{D,it,K} = -m_{K,N,a}$ and $\partial_a q_{it,K} = \delta_K m_{K,N,a} - h_{K,N,a}^0$, the product rule gives

$$\begin{aligned} A_{it,K}^{\theta} &= -m_{K,N,a} q_{it,K} + u_{D,it,K} (\delta_K m_{K,N,a} - h_{K,N,a}^0) \\ &= -m_{K,N,a} \varepsilon_{it} - m_{K,N,a} e_{h,it,K} + 2\delta_K u_{D,it,K} m_{K,N,a} - u_{D,it,K} h_{K,N,a}^0 \\ &= b_{it,K}^A + a_{it,K}^A v_{it} - m_{K,N,a} \varepsilon_{it}, \end{aligned}$$

where the last equality uses $u_{D,it,K} = v_{it} + e_{m,it,K}$ and groups the terms without v_{it} into $b_{it,K}^A$. This proves the decomposition in (D.41) and the loading formula in (D.42).

Differentiate the first line once more before substituting:

$$\begin{aligned} H_{it,K}^\theta &= -m_{K,N,aa}q_{it,K} - 2m_{K,N,a}(\delta_K m_{K,N,a} - h_{K,N,a}^0) + u_{D,it,K}(\delta_K m_{K,N,aa} - h_{K,N,aa}^0) \\ &= -m_{K,N,aa}\varepsilon_{it} - m_{K,N,aa}e_{h,it,K} + 2\delta_K u_{D,it,K}m_{K,N,aa} \\ &\quad - u_{D,it,K}h_{K,N,aa}^0 - 2\delta_K m_{K,N,a}^2 + 2m_{K,N,a}h_{K,N,a}^0 \\ &= b_{it,K}^H + a_{it,K}^H v_{it} - m_{K,N,aa}\varepsilon_{it}. \end{aligned}$$

This proves (D.43)–(D.44) and displays the two product-rule cross terms explicitly.

For (D.45), multiply the first decomposition by $\zeta_{i\ell}$. Terms in $b_{it,K}^A$ depend only on X and have a centered ζ factor; the outcome-error term has zero expectation by independence and centering. Thus only the $v\zeta$ term remains, giving (D.45).

Condition (5.7), after shifting all three dates by $-t$, gives $\mathbb{E}_i[v_{it}\zeta_{is}\zeta_{ir}] = 0$ for every (t, s, r) . The independent outcome-error process is centered as well. Only $b_{it,K}^H$ remains after multiplication by two ζ terms. Independence of X and the disturbance process then factors its expectation, proving (D.46). $\square$

Summing (D.45) over lags and using (D.46) give

$$B_{\zeta,\theta,K}^J = \Omega_{v\zeta} \frac{1}{N} \sum_i \mathbb{E}_X[a_{i0,K}^A] = \Omega_{v\zeta} \bar{a}_{v,K}. \quad (\text{D.47})$$

$$B_{\zeta\zeta,\theta,K}^J = \frac{\Omega_\zeta}{2N} \sum_i \mu_{f,i,K}. \quad (\text{D.48})$$

D.3.3 Finite-Sample and Clipping Corrections

Let $\bar{A}_{\theta,i,K} = \frac{1}{T} \sum_t A_{it,K}^\theta$ and $\bar{H}_{\theta,i,K} = \frac{1}{T} \sum_t H_{it,K}^\theta$. Pair counting gives

$$T \mathbb{E}_i[\Delta_i \bar{A}_{\theta,i,K}] = \mathbb{E}_X[a_{i0,K}^A] \left[\Omega_{v\zeta} - \frac{1}{T} \sum_{\ell \in \Lambda_{q_0}} |\ell| \gamma_{v\zeta}(\ell) \right]. \quad (\text{D.49})$$

$$\frac{T}{2} \mathbb{E}_i[\Delta_i^2 \bar{H}_{\theta,i,K}] = \frac{\mu_{f,i,K}}{2} \left[\Omega_\zeta - \frac{1}{T} \sum_{\ell \in \Lambda_{q_0}} |\ell| \gamma_\zeta(\ell) \right].$$

Indeed, a lag- ℓ pair occurs $T - |\ell|$ times and all covariances outside Λ_{q_0} vanish.

For an exact third-order Taylor formula, define

$$\mathcal{T}_{\theta,iT,K}(d) = \frac{1}{T} \sum_t \int_0^1 (1-s)^2 \partial_a^3 f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i + sd) ds.$$

Then the clipped coefficient in (5.3) is exactly

$$\mathcal{B}_{\theta,K}^{\text{cl}} = \frac{T}{N} \sum_i \mathbb{E}_i[d_i \bar{A}_{\theta,i,K}] + \frac{T}{2N} \sum_i \mathbb{E}_i[d_i^2 \bar{H}_{\theta,i,K}] + \frac{T}{2N} \sum_i \mathbb{E}_i[d_i^3 \mathcal{T}_{\theta,iT,K}(d_i)].$$

Define

$$\begin{aligned} R_{1,\text{cl},K}^J &= \frac{T}{N} \sum_i \mathbb{E}_i[(d_i - \Delta_i) \bar{A}_{\theta,i,K}], \\ R_{2,\text{cl},K}^J &= \frac{T}{2N} \sum_i \mathbb{E}_i[(d_i^2 - \Delta_i^2) \bar{H}_{\theta,i,K}], \\ R_{3,\text{path},K}^J &= \frac{T}{2N} \sum_i \mathbb{E}_i[d_i^3 \mathcal{T}_{\theta,iT,K}(d_i)]. \end{aligned}$$

The clipped-tail bounds in Section C.2 imply

$$|R_{1,\text{cl},K}^J| \leq CTb_T N^{-a_0}, \quad |R_{2,\text{cl},K}^J| \leq CTb_T^2 N^{-a_0}, \quad |R_{3,\text{path},K}^J| \leq CT^{-1/2}, \quad (\text{D.50})$$

where $a_0 = c_{\text{fm}} C_b^2 > 1$. Combining (D.49)–(D.50) gives the exact bridge

$$\begin{aligned} \mathcal{B}_{\theta,K}^{\text{cl}} - B_{\zeta,\theta,K}^J - B_{\zeta\zeta,\theta,K}^J &= -\frac{1}{T} \left\{ \sum_{\ell \in \Lambda_{q_0}} |\ell| \gamma_{v\zeta}(\ell) \right\} \bar{a}_{v,K} - \frac{1}{2TN} \left\{ \sum_{\ell \in \Lambda_{q_0}} |\ell| \gamma_{\zeta}(\ell) \right\} \sum_i \mu_{\dot{f},i,K} \\ &\quad + R_{1,\text{cl},K}^J + R_{2,\text{cl},K}^J + R_{3,\text{path},K}^J \\ &= o(1). \end{aligned} \quad (\text{D.51})$$

The first-stage nonlinear-bias channel is zero because the raw coordinate is an exact sample mean.

D.4 Variance Estimation and the Cluster Central Limit Theorem

D.4.1 Feasible-to-Oracle Comparison

We first compare the feasible fixed-lag variance estimator with its oracle counterpart. We then show that the oracle estimator converges to the finite- T variance target. The argument uses the fixed lag set Λ_{q_0} throughout.

For an array q_{it} , write $\|q\|_{NT,2}^2 = \mathbb{P}_{NT} q_{it}^2$, $\bar{q}_i = \frac{1}{T} \sum_{t=1}^T q_{it}$, and $\tilde{q}_{it} = q_{it} - \bar{q}_i$. The centered oracle score is $\chi_{\theta,it,K} = V_K^{-1} x_{\theta,it,K}$. For the feasible comparison, set $m_{\theta,i,K} = \mathbb{E}_i[f_{i0,K}^\theta(\rho_{0,K,N}, \alpha_i)]$ and define the uncentered oracle version

$$\chi_{\theta,it,K}^\circ = V_K^{-1} \{f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i) + \mu_{\dot{f},i,K} \zeta_{it}\} = \chi_{\theta,it,K} + V_K^{-1} m_{\theta,i,K}. \quad (\text{D.52})$$

The last term is constant within unit. Therefore, $\tilde{\chi}_{\theta,it,K}^\circ = \tilde{\chi}_{\theta,it,K}$ exactly. The fixed-lag oracle HAC statistic is

$$\hat{\sigma}_{\theta,\text{or},K}^2 = \frac{1}{N} \sum_{i=1}^N \sum_{\ell \in \Lambda_{q_0}} \frac{1}{T} \sum_{t \in I_T(\ell)} \tilde{\chi}_{\theta,it,K} \tilde{\chi}_{\theta,i,t+\ell,K}.$$

Define its finite- T target by

$$\sigma_{\theta,K,N}^2 = \frac{1}{N} \sum_{i=1}^N \sum_{\ell \in \Lambda_{q_0}} \left(1 - \frac{|\ell|}{T}\right) \text{Cov}_i(\chi_{\theta,i0,K}, \chi_{\theta,i\ell,K}). \quad (\text{D.53})$$

Lemma D.5 (Feasible-to-oracle score comparison after demeaning). *Under Assumption 5.2 and the rate conditions in (5.14),*

$$e_{\chi,K}^\circ := \|\hat{\chi}_{\theta,it,K} - \chi_{\theta,it,K}^\circ\|_{NT,2} = O_p\left(T^{-1/2} + \mathfrak{h}_K^{-1} \bar{r}_{\eta,K} + \left|\hat{V}_K - V_K\right|\right) = o_p(1). \quad (\text{D.54})$$

Since the unit constant in (D.52) disappears exactly and within-unit demeaning is an L^2 contraction,

$$\left\| \tilde{\chi}_{\theta,it,K} - \tilde{\chi}_{\theta,it,K}^\circ \right\|_{NT,2} \leq e_{\chi,K}^\circ. \quad (\text{D.55})$$

Proof. Preliminary estimator rate. Equation (D.25) gives $\hat{\theta}_K - \theta_{0,K,N} = O_p(T^{-1})$ on $\mathcal{L}_K$ under the present rate conditions.

Numerator decomposition. For the feasible comparison, work first on $\mathcal{L}_K = \mathcal{E}_T \cap \mathcal{G}_K$. Put

$$f_{it,K}^{\theta,0}(a) = f_{it,K}^\theta(\rho_{0,K,N}, a), \quad \bar{E}_{i,K}^A = \frac{1}{T} \sum_t E_{it,K}^A.$$

By (D.3) and the exact plug-in partition (D.16),

$$\bar{\dot{f}}_{i,K} = \frac{1}{T} \sum_t \hat{A}_{it,K}^\theta = \lambda_{i,K} + \bar{A}_{i,K}^c + \bar{E}_{i,K}^A.$$

Define $\varphi_{\theta, it, K}^\circ = f_{it, K}^{\theta, 0}(\alpha_i) + \lambda_{i, K} \zeta_{it}$ and the four numerator components

$$\begin{aligned} R_{it, K}^{\text{score}} &= f_{it, K}^\theta(\hat{\rho}_K, \hat{\alpha}_i) - f_{it, K}^{\theta, 0}(\alpha_i), \\ R_{it, K}^{\text{der, or}} &= \bar{A}_{i, K}^c \zeta_{it}, & R_{it, K}^{\text{der, pl}} &= \bar{E}_{i, K}^A \zeta_{it}, \\ R_{it, K}^{\text{dm}} &= -(\lambda_{i, K} + \bar{A}_{i, K}^c + \bar{E}_{i, K}^A) d_i. \end{aligned}$$

Since $\hat{\zeta}_{it} = \zeta_{it} - d_i$, these definitions give the finite-sample identity

$$\hat{\varphi}_{\theta, it, K} - \varphi_{\theta, it, K}^\circ = R_{it, K}^{\text{score}} + R_{it, K}^{\text{der, or}} + R_{it, K}^{\text{der, pl}} + R_{it, K}^{\text{dm}}. \quad (\text{D.56})$$

The residual identities in (D.12) further yield the exact score decomposition

$$R_{it, K}^{\text{score}} = f_{it, K}^{\theta, 0}(\hat{\alpha}_i) - f_{it, K}^{\theta, 0}(\alpha_i) + u_{D, it, K}^0 \Delta q_{it, K} - q_{it, K}^0 \Delta M_{it, K}^{(0)} - \Delta M_{it, K}^{(0)} \Delta q_{it, K}.$$

Componentwise bounds. The bounded coordinate derivative, the clipping moments, and (D.30) imply

$$\|R^{\text{score}}\|_{NT, 2} = O_p(T^{-1/2} + \bar{r}_{\eta, K}).$$

Finite-range sample-mean bounds give

$$\left\{ \frac{1}{N} \sum_i (\bar{A}_{i, K}^c)^2 \right\}^{1/2} = O_p(T^{-1/2}).$$

Bounded ζ , Jensen's inequality, and (D.31) then give

$$\|R^{\text{der, or}}\|_{NT, 2} = O_p(T^{-1/2}), \quad \|R^{\text{der, pl}}\|_{NT, 2} \leq C e_{A, K} = O_p\left(T^{-1/2} + \mathfrak{h}_K^{-1} \bar{r}_{\eta, K}\right).$$

The demeaning component does not require independence between d_i and the estimated derivatives. On $\mathcal{L}_K$, $\max_i |d_i| \leq b_T$, so

$$\begin{aligned} \|R^{\text{dm}}\|_{NT, 2} &\leq \left\{ \frac{1}{N} \sum_i (\lambda_{i, K} d_i)^2 \right\}^{1/2} + b_T \left\{ \frac{1}{N} \sum_i (\bar{A}_{i, K}^c)^2 \right\}^{1/2} + b_T \left\{ \frac{1}{N} \sum_i (\bar{E}_{i, K}^A)^2 \right\}^{1/2} \\ &= O_p(T^{-1/2} + b_T e_{A, K}), \end{aligned} \quad (\text{D.57})$$

where the last empirical norm is at most $e_{A, K}$ by Jensen's inequality. Because $b_T = o(1)$ and $\mathfrak{h}_K < 1$ eventually, (D.56)–(D.57) imply

$$\|\hat{\varphi}_{\theta, it, K} - \varphi_{\theta, it, K}^\circ\|_{NT, 2} = O_p\left(T^{-1/2} + \mathfrak{h}_K^{-1} \bar{r}_{\eta, K}\right).$$

Denominator replacement. Finally, denominator replacement is isolated by the exact identity

$$\hat{\chi}_{\theta, it, K} - \chi_{\theta, it, K}^\circ = (\hat{V}_K^{-1} - V_K^{-1}) \varphi_{\theta, it, K}^\circ + \hat{V}_K^{-1} (\hat{\varphi}_{\theta, it, K} - \varphi_{\theta, it, K}^\circ). \quad (\text{D.58})$$

The oracle numerator has bounded empirical second moment. On $\mathcal{L}_K$, both denominators are bounded away from zero and $|\hat{V}_K^{-1} - V_K^{-1}| \leq C |\hat{V}_K - V_K|$ by Lemmas C.15 and C.16. Thus (D.58) proves the stated rate in (D.54). The assumed second-derivative plug-in rate $\mathfrak{h}_K^{-2} \bar{r}_{\eta, K} = o(1)$ implies $\mathfrak{h}_K^{-1} \bar{r}_{\eta, K} = o(1)$, and the same denominator lemmas give $|\hat{V}_K - V_K| = o_p(1)$. Equation (D.17) extends the result from $\mathcal{L}_K$ to the full sample space.

Within-unit demeaning. Finally, (D.52) gives $\tilde{\chi}_{\theta, it, K}^\circ = \tilde{\chi}_{\theta, it, K}$ exactly, and the orthogonal-projection inequality for subtracting unit means proves (D.55). $\square$

Proposition D.6 (Consistency of the fixed-lag variance estimator). *Under Assumption 5.2 and the rate conditions in (5.14),*

$$\hat{\sigma}_{\theta, K}^2 - \hat{\sigma}_{\theta, \text{or}, K}^2 = o_p(1). \quad (\text{D.59})$$

$$\hat{\sigma}_{\theta, \text{or}, K}^2 - \sigma_{\theta, K, N}^2 = o_p(1). \quad (\text{D.60})$$

Consequently, $\hat{\sigma}_{\theta, K}^2 - \sigma_{\theta, K, N}^2 = o_p(1)$.

Proof. The proof makes two lag-by-lag comparisons. The first expands the feasible lag product around the oracle lag product and controls the three resulting terms by empirical Cauchy–Schwarz. The second compares the oracle lag product with its covariance target; an exact demeaning identity isolates the two boundary means. Because the retained lag set is fixed, the lagwise bounds can then be summed without changing their order.

Feasible-to-oracle comparison. Work first on $\mathcal{L}_K$. For each $\ell \in \Lambda_{q_0}$, define the aggregate feasible and oracle lag statistics

$$\begin{aligned}\widehat{\Gamma}_K(\ell) &= \frac{1}{NT} \sum_i \sum_{t \in I_T(\ell)} \widetilde{\chi}_{\theta, it, K} \widetilde{\chi}_{\theta, i, t+\ell, K}, \\ \widehat{\Gamma}_{\text{or}, K}(\ell) &= \frac{1}{NT} \sum_i \sum_{t \in I_T(\ell)} \widetilde{\chi}_{\theta, it, K} \widetilde{\chi}_{\theta, i, t+\ell, K}.\end{aligned}\tag{D.61}$$

Let $r_{it, K} = \widetilde{\chi}_{\theta, it, K} - \widetilde{\chi}_{\theta, it, K}$. Direct multiplication gives the exact three-term identity

$$\widehat{\Gamma}_K(\ell) - \widehat{\Gamma}_{\text{or}, K}(\ell) = \frac{1}{NT} \sum_i \sum_{t \in I_T(\ell)} \{r_{it, K} \widetilde{\chi}_{\theta, i, t+\ell, K} + \widetilde{\chi}_{\theta, it, K} r_{i, t+\ell, K} + r_{it, K} r_{i, t+\ell, K}\}.$$

Cauchy–Schwarz, restriction to $I_T(\ell)$, and (D.55) imply, for every retained lag,

$$\left| \widehat{\Gamma}_K(\ell) - \widehat{\Gamma}_{\text{or}, K}(\ell) \right| \leq 2e_{\chi, K}^\circ \|\widetilde{\chi}_{\theta, it, K}\|_{NT, 2} + (e_{\chi, K}^\circ)^2.\tag{D.62}$$

The oracle norm is $O_p(1)$, while Lemma D.5 gives $e_{\chi, K}^\circ = o_p(1)$. Because Λ_{q_0} is fixed and the kernels equal one on this set, summing (D.62) proves (D.59) on $\mathcal{L}_K$.

Oracle-to-target comparison. For the oracle comparison, define the undemeaned lag statistic and its two boundary means by

$$\begin{aligned}U_K(\ell) &= \frac{1}{NT} \sum_i \sum_{t \in I_T(\ell)} \chi_{\theta, it, K} \chi_{\theta, i, t+\ell, K}, \\ a_{i, K}(\ell) &= \frac{1}{T} \sum_{t \in I_T(\ell)} \chi_{\theta, it, K}, & b_{i, K}(\ell) &= \frac{1}{T} \sum_{t \in I_T(\ell)} \chi_{\theta, i, t+\ell, K}.\end{aligned}$$

Stationarity and the centered oracle score give

$$\mathbb{E} U_K(\ell) = \frac{1}{N} \sum_i \left(1 - \frac{|\ell|}{T}\right) \text{Cov}_i(\chi_{\theta, i0, K}, \chi_{\theta, i\ell, K}).\tag{D.63}$$

The lag-product bound in Lemma C.6 and independence across units yield

$$U_K(\ell) - \mathbb{E} U_K(\ell) = O_p((NT)^{-1/2}).\tag{D.64}$$

Expanding both demeaned factors in (D.61) gives the second exact identity

$$\widehat{\Gamma}_{\text{or}, K}(\ell) - U_K(\ell) = -\frac{1}{N} \sum_i \bar{\chi}_{\theta, i, K} \{a_{i, K}(\ell) + b_{i, K}(\ell)\} + \left(1 - \frac{|\ell|}{T}\right) \frac{1}{N} \sum_i \bar{\chi}_{\theta, i, K}^2.\tag{D.65}$$

The finite-range mean bound implies, for every fixed ℓ ,

$$\frac{1}{N} \sum_i \{\bar{\chi}_{\theta, i, K}^2 + a_{i, K}(\ell)^2 + b_{i, K}(\ell)^2\} = O_p(T^{-1}).$$

Thus Cauchy–Schwarz in (D.65) gives $\widehat{\Gamma}_{\text{or}, K}(\ell) - U_K(\ell) = O_p(T^{-1})$. Combining this with (D.63) and (D.64), then summing the fixed number of retained lags, proves (D.60) on $\mathcal{L}_K$. Finally, Equation (D.17) extends both comparisons to the full sample space. $\square$

D.4.2 Variance Target and Limit

Lemma D.7 (Oracle score and variance limit). *Under Assumption 5.2 and $\mathfrak{h}_K \rightarrow 0$,*

$$\frac{1}{N} \sum_{i=1}^N \sum_{\ell \in \Lambda_{q_0}} |\text{Cov}_i(x_{\theta, i0, K}, x_{\theta, i\ell, K}) - \text{Cov}_i(v_{i0}\varepsilon_{i0}, v_{i\ell}\varepsilon_{i\ell})| = o(1), \quad (\text{D.66})$$

and $V_K \rightarrow \gamma_v(0)$. Hence

$$\begin{aligned} \sigma_{\theta, K, N}^2 &\longrightarrow \sigma_{\theta, J}^2, \\ \sigma_{\theta, J}^2 &:= \frac{\sum_{\ell \in \Lambda_{q_0}} \gamma_v(\ell) \gamma_\varepsilon(\ell)}{\gamma_v(0)^2}. \end{aligned} \quad (\text{D.67})$$

Proof. At the grid target, $u_D = v + e_m$ and $u_Y - \theta_{0, K, N} u_D = \varepsilon + e_h - \Delta_{\theta, K, N} u_D$. Consequently, the exact score difference is

$$\begin{aligned} f_{it, K}^\theta(\rho_{0, K, N}, \alpha_i) - v_{it}\varepsilon_{it} &= e_{m, it, K}\varepsilon_{it} + v_{it}e_{h, it, K} + e_{m, it, K}e_{h, it, K} \\ &\quad - \Delta_{\theta, K, N}(v_{it} + e_{m, it, K})^2. \end{aligned}$$

Bounded disturbances, the two projection rates, and Proposition 2.1 make each term on the right negligible in finite-grid L^2 :

$$\frac{1}{N} \sum_{i=1}^N \mathbb{E}_i \left[|f_{i0, K}^\theta(\rho_{0, K, N}, \alpha_i) - v_{i0}\varepsilon_{i0}|^2 \right] = o(1).$$

The outcome error is centered and independent of v , so $\mathbb{E}_i[v_{i0}\varepsilon_{i0}] = 0$ and

$$m_{\theta, i, K} = \mathbb{E}_i \left[f_{i0, K}^\theta(\rho_{0, K, N}, \alpha_i) - v_{i0}\varepsilon_{i0} \right].$$

Jensen's inequality therefore supplies the unit-centering bound

$$\frac{1}{N} \sum_{i=1}^N |m_{\theta, i, K}|^2 \leq \frac{1}{N} \sum_{i=1}^N \mathbb{E}_i \left[|f_{i0, K}^\theta(\rho_{0, K, N}, \alpha_i) - v_{i0}\varepsilon_{i0}|^2 \right] = o(1).$$

The derivative decomposition in Section D.3 also gives $N^{-1} \sum_i \left| \mu_{f, i, K} \right|^2 = o(1)$. Since ζ is bounded, the full centered-score difference is

$$x_{\theta, it, K} - v_{it}\varepsilon_{it} = \{f_{it, K}^\theta(\rho_{0, K, N}, \alpha_i) - v_{it}\varepsilon_{it}\} - m_{\theta, i, K} + \mu_{f, i, K}\zeta_{it}.$$

Consequently,

$$\begin{aligned} &\frac{1}{N} \sum_{i=1}^N \mathbb{E}_i \left[|x_{\theta, i0, K} - v_{i0}\varepsilon_{i0}|^2 \right] \\ &\leq \frac{3}{N} \sum_{i=1}^N \mathbb{E}_i \left[|f_{i0, K}^\theta(\rho_{0, K, N}, \alpha_i) - v_{i0}\varepsilon_{i0}|^2 \right] + \frac{3}{N} \sum_{i=1}^N |m_{\theta, i, K}|^2 + \frac{C}{N} \sum_{i=1}^N |\mu_{f, i, K}|^2 = o(1). \end{aligned}$$

Because both scores are centered, their covariance difference has the exact two-term decomposition

$$\begin{aligned} &\text{Cov}_i(x_{\theta, i0, K}, x_{\theta, i\ell, K}) - \text{Cov}_i(v_{i0}\varepsilon_{i0}, v_{i\ell}\varepsilon_{i\ell}) \\ &= \text{Cov}_i(x_{\theta, i0, K} - v_{i0}\varepsilon_{i0}, x_{\theta, i\ell, K}) + \text{Cov}_i(v_{i0}\varepsilon_{i0}, x_{\theta, i\ell, K} - v_{i\ell}\varepsilon_{i\ell}). \end{aligned}$$

Their second moments are uniformly bounded. Averaging the last display over i and applying Cauchy-Schwarz therefore gives

$$\begin{aligned} &\frac{1}{N} \sum_{i=1}^N |\text{Cov}_i(x_{\theta, i0, K}, x_{\theta, i\ell, K}) - \text{Cov}_i(v_{i0}\varepsilon_{i0}, v_{i\ell}\varepsilon_{i\ell})| \\ &\leq C \left\{ \frac{1}{N} \sum_{i=1}^N \mathbb{E}_i |x_{\theta, i0, K} - v_{i0}\varepsilon_{i0}|^2 \right\}^{1/2} = o(1). \end{aligned}$$

Summing this bound over the fixed lag set proves (D.66). Similarly, the denominator obeys

$$\begin{aligned} V_K &= \gamma_v(0) + \frac{2}{N} \sum_{i=1}^N \mathbb{E}_i[v_{i0}e_{m,i0,K}] + \frac{1}{N} \sum_{i=1}^N \mathbb{E}_i[e_{m,i0,K}^2] \\ &= \gamma_v(0) + o(1). \end{aligned}$$

Assumption 5.2(i)–(ii) makes the cross term zero, and the projection-error bound makes the last term $o(1)$. The ε process is independent of the joint (v, ζ) process, so $\text{Cov}(v_{i0}\varepsilon_{i0}, v_{i\ell}\varepsilon_{i\ell}) = \gamma_v(\ell)\gamma_\varepsilon(\ell)$. Substitution in (D.53), followed by $T \rightarrow \infty$, gives

$$\begin{aligned} \sigma_{\theta,K,N}^2 &= \frac{1}{V_K^2} \frac{1}{N} \sum_{i=1}^N \sum_{\ell \in \Lambda_{q_0}} \left(1 - \frac{|\ell|}{T}\right) \text{Cov}_i(x_{\theta,i0,K}, x_{\theta,i\ell,K}) \\ &\longrightarrow \frac{1}{\gamma_v(0)^2} \sum_{\ell \in \Lambda_{q_0}} \gamma_v(\ell)\gamma_\varepsilon(\ell), \end{aligned}$$

which proves (D.67). For nondegeneracy, define the spectral-density trigonometric polynomials

$$f_c(\omega) = \frac{1}{2\pi} \sum_{\ell \in \Lambda_{q_0}} \gamma_c(\ell) e^{-i\ell\omega}, \quad c \in \{v, \varepsilon\}.$$

They are continuous and nonnegative. Since $\gamma_v(0) > 0$, neither is identically zero. A nonzero trigonometric polynomial has a zero set of Lebesgue measure zero. Since scalar covariance functions satisfy $\gamma_c(-\ell) = \gamma_c(\ell)$, Parseval's identity gives

$$\sum_{\ell \in \Lambda_{q_0}} \gamma_v(\ell)\gamma_\varepsilon(\ell) = 2\pi \int_{-\pi}^{\pi} f_v(\omega)f_\varepsilon(\omega) d\omega > 0.$$

Hence $\sigma_{\theta,J}^2 > 0$. □

D.4.3 Cluster Central Limit Theorem

Proposition D.8 (Scalar cluster central limit theorem). *Under Assumption 5.2 and $\mathfrak{h}_K \rightarrow 0$,*

$$N^{-1/2} \sum_{i=1}^N \Psi_{\theta,iT,K} \rightsquigarrow \mathcal{N}(0, \sigma_{\theta,J}^2). \quad (\text{D.68})$$

Proof. Assumption 5.2(i)–(ii) makes the clusters independent across i . The definition of $x_{\theta,it,K}$ gives $\mathbb{E}_i \Psi_{\theta,iT,K} = 0$. Bounded disturbances, projection stability, the bounded score-derivative envelopes, and $V_K \geq \sigma_v^2 > 0$ imply

$$\sup_{i,N,T,K} \mathbb{E}_i |\chi_{\theta,it,K}|^8 \leq C.$$

Moreover, $\chi_{\theta,it,K}$ is stationary and q_0 -dependent within each unit. Lemma C.6, applied to this centered array, therefore gives

$$\sup_{i,N,T,K} \mathbb{E}_i |\Psi_{\theta,iT,K}|^4 \leq C.$$

For all sufficiently large T , stationarity and q_0 -dependence identify the cluster variance exactly:

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^N \text{Var}_i(\Psi_{\theta,iT,K}) &= \frac{1}{N} \sum_{i=1}^N \frac{1}{T} \sum_{t=1}^T \sum_{s=1}^T \text{Cov}_i(\chi_{\theta,it,K}, \chi_{\theta,is,K}) \\ &= \frac{1}{N} \sum_{i=1}^N \sum_{\ell \in \Lambda_{q_0}} \left(1 - \frac{|\ell|}{T}\right) \text{Cov}_i(\chi_{\theta,i0,K}, \chi_{\theta,i\ell,K}) \\ &= \sigma_{\theta,K,N}^2 \longrightarrow \sigma_{\theta,J}^2 > 0, \end{aligned}$$

where the limit follows from Lemma D.7. Hence the Lyapunov ratio satisfies

$$\frac{\sum_i \mathbb{E}_i |N^{-1/2} \Psi_{\theta, iT, K}|^4}{\{\sum_i \text{Var}_i(N^{-1/2} \Psi_{\theta, iT, K})\}^2} \leq \frac{C/N}{(\sigma_{\theta, K, N}^2)^2} \rightarrow 0.$$

The triangular-array Lyapunov theorem proves (D.68). $\square$

D.5 Proof of the Active-Channel Result

Proof. Equation (D.47), obtained from Lemma D.4, gives $B_{\zeta, \theta, K}^J = \Omega_{v\zeta} \bar{a}_{v, K}$. It remains to establish the limit of $\bar{a}_{v, K}$.

Let $q_K(a) = \int_{[0,1]^{d_X}} h_{K, N, a}^0(x, a) dx$. Lemma C.4 gives a uniform bound on q'_K , so q_K has uniformly bounded total variation. Applying integration by parts to the discrepancy bound in (5.6), and accounting for the two omitted boundary strips, yields an $O(N^{-1} + b_T)$ approximation error. Uniform derivative approximation and quasi-uniform coordinate quadrature give

$$\left| \frac{1}{N} \sum_i \mathbb{E}_X h_{K, N, a}^0(X, \alpha_i) - \int_{[0,1]^{d_X}} \int_0^1 h_{0, a}(x, a) da dx \right| \leq \|h_{K, N, a}^0 - h_{0, a}\|_\infty + C(b_T + N^{-1}) = o(1).$$

Also, $|\frac{1}{N} \sum_i \mathbb{E}_X m_{K, N, a}(X, \alpha_i)| = O(1)$. Proposition 2.1 gives $\delta_K \rightarrow 0$. Therefore

$$\begin{aligned} \bar{a}_{v, K} &= 2\delta_K \frac{1}{N} \sum_i \mathbb{E}_X m_{K, N, a}(X, \alpha_i) - \frac{1}{N} \sum_i \mathbb{E}_X h_{K, N, a}^0(X, \alpha_i) \\ &\rightarrow - \int_{[0,1]^{d_X}} \{h_0(x, 1) - h_0(x, 0)\} dx = -\mathfrak{b}_h. \end{aligned} \tag{D.69}$$

The last equality is the fundamental theorem of calculus in a . Combining (D.47) and (D.69) completes the proof. $\square$

D.6 Proof of the Primitive Bias Limit

Proof. Proposition 5.4 proves the derivative-influence limit in the first line of the proposition. For the curvature channel, the exact Hessian decomposition in (D.43)–(D.44) gives

$$\begin{aligned} \frac{1}{N} \sum_i \mu_{j, i, K} &= \frac{1}{N} \sum_i \mathbb{E}_X [-m_{K, N, aa} e_{h, it, K} - e_{m, it, K} h_{K, N, aa}^0 + 2m_{K, N, a} h_{K, N, a}^0 \\ &\quad - 2\Delta_{\theta, K, N} m_{K, N, a}^2 + 2\Delta_{\theta, K, N} e_{m, it, K} m_{K, N, aa}]. \end{aligned}$$

The projection bounds in Section C.1 imply uniform convergence of the first coordinate derivatives and uniform boundedness of the displayed second derivatives. In particular, the first, second, and fifth terms obey the explicit product bounds

$$\begin{aligned} \left| \frac{1}{N} \sum_i \mathbb{E}_X [m_{K, N, aa} e_{h, it, K}] \right| &\leq \|m_{K, N, aa}\|_\infty \|e_{h, \cdot, K}\|_\infty = O(\mathfrak{h}_K^{\tau_h}), \\ \left| \frac{1}{N} \sum_i \mathbb{E}_X [e_{m, it, K} h_{K, N, aa}^0] \right| &\leq \|e_{m, \cdot, K}\|_\infty \|h_{K, N, aa}^0\|_\infty = O(\mathfrak{h}_K^{\tau_m}), \\ \left| \frac{2\Delta_{\theta, K, N}}{N} \sum_i \mathbb{E}_X [e_{m, it, K} m_{K, N, aa}] \right| &\leq 2|\Delta_{\theta, K, N}| \|e_{m, \cdot, K}\|_\infty \|m_{K, N, aa}\|_\infty = o(1). \end{aligned}$$

Proposition 2.1 gives $\Delta_{\theta, K, N} \rightarrow 0$, so the fourth term is bounded by $2|\Delta_{\theta, K, N}| \|m_{K, N, a}\|_\infty^2 = o(1)$.

It remains to transfer the signed product $m_{K, N, a} h_{K, N, a}^0$ from the coordinate grid to the continuous design. Define the two nonnegative functions

$$q_{\pm, K}(a) = \int_{[0,1]^{d_X}} \{m_{K, N, a}(x, a) \pm h_{K, N, a}^0(x, a)\}^2 dx.$$

The first coordinate derivatives of the two projections are continuous and cellwise polynomials of degree at most $p - 1$. Hence $q_{s,K} \in \mathcal{Q}_{2(p-1),K}$ for each $s \in \{+, -\}$, so Lemma C.3 applies to both signs. The polarization identity $4m_{K,N,a}h_{K,N,a}^0 = (m_{K,N,a} + h_{K,N,a}^0)^2 - (m_{K,N,a} - h_{K,N,a}^0)^2$ therefore gives

$$\begin{aligned} & \left| \frac{1}{N} \sum_i \mathbb{E}_X [m_{K,N,a}(X, \alpha_i) h_{K,N,a}^0(X, \alpha_i)] - \int_{W^+} m_{K,N,a}(w) h_{K,N,a}^0(w) dw \right| \\ & \leq \frac{1}{4} \sum_{s \in \{+, -\}} \left| \frac{1}{N} \sum_i q_{s,K}(\alpha_i) - \int_0^1 q_{s,K}(a) da \right| \\ & \leq C \left\{ \frac{1}{N\mathfrak{h}_K} + \frac{b_T}{\mathfrak{h}_K} \right\} \int_0^1 \{q_{+,K}(a) + q_{-,K}(a)\} da = o(1). \end{aligned}$$

The last line uses (5.8) and (C.9), which imply $\int_0^1 (q_{+,K} + q_{-,K}) da = O(1)$. The same derivative approximation also gives

$$\begin{aligned} & \left| \int_{W^+} m_{K,N,a}(w) h_{K,N,a}^0(w) dw - \int_{W^+} m_{0,a}(w) h_{0,a}(w) dw \right| \\ & \leq \|m_{K,N,a} - m_{0,a}\|_\infty \|h_{K,N,a}^0\|_\infty + \|m_{0,a}\|_\infty \|h_{K,N,a}^0 - h_{0,a}\|_\infty = o(1). \end{aligned}$$

Combining these two limits with the four bounds above yields

$$\frac{1}{N} \sum_i \mu_{\tilde{f},i,K} \longrightarrow 2 \int_{W^+} m_{0,a}(w) h_{0,a}(w) dw = 2\mathfrak{c}_{mh}.$$

Combining this limit with (D.48) proves (5.11). The two channel limits and the exact decomposition of $B_{\theta,K,N}^\infty$ then give (5.12). Finally, Equation (D.51) shows that the clipped finite- T coefficient differs from the two analytical channels by $o(1)$, proving (5.13). $\square$

E Proofs of the Main Growing-Sieve Results

E.1 Supporting Rates and Limit Bridges

Proposition E.1 (Rates for estimation, correction, and variance). *Suppose Assumption 5.2 holds and the rate conditions in (5.14) are satisfied. Then*

$$\|\hat{\eta}_K - \eta_{0,K,N}\| = O_p(\bar{r}_{\eta,K}).$$

$$\sqrt{NT} \|\hat{\eta}_K - \eta_{0,K,N}\|^2 = o_p(1).$$

$$\sqrt{NT} \left| e_\gamma^\top \hat{\Sigma}_{p,K} e_\pi \right| = o_p(1). \quad (\text{E.1})$$

$$\sqrt{NT} |r_{\theta,F,K}| = o_p(1).$$

$$\hat{V}_K^{-1} \hat{B}_{\theta,K} - V_K^{-1} \mathcal{B}_{\theta,K}^{\text{cl}} = o_p(1). \quad (\text{E.2})$$

$$\hat{\sigma}_{\theta,K}^2 - \sigma_{\theta,K,N}^2 = o_p(1). \quad (\text{E.3})$$

Moreover, $\mathbb{P}(\mathcal{G}_K^c) \rightarrow 0$.

Proposition E.2 (Denominator, variance target, and cluster limit). *Suppose Assumption 5.2 holds, the rate conditions in (5.14) are satisfied, and $\mathfrak{h}_K \rightarrow 0$. Then*

$$\left| \hat{V}_K - V_K \right| = o_p(1), \quad V_K \longrightarrow \gamma_v(0) > 0. \quad (\text{E.4})$$

$$\sigma_{\theta,K,N}^2 \longrightarrow \sigma_{\theta,J}^2 > 0, \quad N^{-1/2} \sum_{i=1}^N \Psi_{\theta,iT,K} \rightsquigarrow \mathcal{N}(0, \sigma_{\theta,J}^2). \quad (\text{E.5})$$

Together with (E.3), this implies

$$\hat{\sigma}_{\theta,K}^2 \xrightarrow{p} \sigma_{\theta,J}^2, \quad \Pr(\hat{\sigma}_{\theta,K}^2 > 0) \rightarrow 1, \quad \hat{\sigma}_{\theta,K} \xrightarrow{p} \sigma_{\theta,J} > 0. \quad (\text{E.6})$$

E.1.1 Proof of Proposition E.1

Proof. Section C.4 gives $\|\hat{\eta}_K - \eta_{0,K,N}\| = O_p(\bar{r}_{\eta,K})$ and $\mathbb{P}(\mathcal{G}_K^c) \rightarrow 0$. Since $N/T \rightarrow \kappa_0$, $\sqrt{NT} \bar{r}_{\eta,K}^2 \asymp \mathfrak{h}_K^{-m}/T = o(1)$. The Gram eigenvalue bound then gives the profile quadratic in (E.1). Section D.1 proves the exact score expansion and $\sqrt{NT}|r_{\theta,F,K}| = o_p(1)$. Section D.2 proves (E.2) from its joint-centered correction-error decomposition. Proposition D.6 proves (E.3). Each conclusion is first obtained on $\mathcal{G}_K$ and then made global using $\mathbb{P}(\mathcal{G}_K^c) \rightarrow 0$. $\square$

E.1.2 Proof of Proposition E.2

Proof. Lemma C.15, the identity $\hat{V}_K = V_K^*$ on $\mathcal{G}_K$, and $\mathbb{P}(\mathcal{G}_K^c) \rightarrow 0$ from Proposition E.1 give $|\hat{V}_K - V_K| = o_p(1)$. Under $\mathfrak{h}_K \rightarrow 0$, Lemma C.16 gives $V_K \rightarrow \gamma_v(0) > 0$, and Lemma D.7 gives $\sigma_{\theta,K,N}^2 \rightarrow \sigma_{\theta,J}^2 > 0$. Finally, Proposition D.8 gives the triangular-array cluster central limit theorem. Combining the variance-target limit with (E.3) proves squared-variance consistency. Because the limit is strictly positive, the positive branch in the definition of $\hat{\sigma}_{\theta,K}$ is selected with probability tending to one, and continuity of the square root proves (E.6). $\square$

E.2 Uncorrected Estimator

Proof. Rate implications. Write $m = \max\{d_W, 3\}$ and $\mathfrak{h}_K \asymp T^{-\nu/d_W}$. The lower inequality in (5.16) gives $T\mathfrak{h}_K^{\tau_m + \tau_h} \rightarrow 0$, so Proposition 2.1 gives $\sqrt{NT}|\theta_{0,K,N} - \theta_0| = o(1)$. The upper inequalities give $\mathfrak{h}_K^{-2}\bar{r}_{\eta,K} \rightarrow 0$ and $\mathfrak{h}_K^{-m}/T \rightarrow 0$: when $d_W \geq 3$, the latter is exactly $\nu < 1$, while for $d_W < 3$ it is implied by $\nu < 2d_W/(m+4)$. Finally, $K_x \asymp T^{\nu d_X/d_W}$, so $K_x \log(NK_x)/T = o(1)$ because $d_X/d_W < 1$ and $\nu < 1$. The lower bound in (5.16) also gives $\nu > 0$, and hence $\mathfrak{h}_K \rightarrow 0$. Thus (5.14) holds, and Propositions E.1 and E.2 apply.

Exact profile expansion. Combining the exact profile identity (5.1) with the score expansion (5.4) gives, on $\mathcal{G}_K$,

$$\hat{\theta}_K - \theta_{0,K,N} = \hat{V}_K^{-1} \bar{Z}_{\theta,K} + \frac{1}{T} \hat{V}_K^{-1} \mathcal{B}_{\theta,K}^{\text{cl}} + \hat{V}_K^{-1} r_{\theta,F,K} - \hat{V}_K^{-1} e_\gamma^\top \hat{\Sigma}_{p,K} e_\pi.$$

The last two terms are $o_p((NT)^{-1/2})$. Equation (E.4) and continuity of inversion give $\hat{V}_K^{-1} - V_K^{-1} = o_p(1)$. For the sampling term, $\sqrt{NT} \bar{Z}_{\theta,K} = O_p(1)$, so

$$\sqrt{NT}(\hat{V}_K^{-1} - V_K^{-1}) \bar{Z}_{\theta,K} = o_p(1).$$

For the bias term, (D.51) gives $\mathcal{B}_{\theta,K}^{\text{cl}} = O(1)$, and therefore

$$\frac{\sqrt{NT}}{T}(\hat{V}_K^{-1} - V_K^{-1}) \mathcal{B}_{\theta,K}^{\text{cl}} = o_p(1),$$

because $\sqrt{NT}/T = O(1)$. Thus the denominator may be replaced in both the sampling term and the bias term. The target difference also contributes $\sqrt{NT}(\theta_{0,K,N} - \theta_0) = o(1)$. Therefore

$$\sqrt{NT}(\hat{\theta}_K - \theta_0) = N^{-1/2} \sum_i \Psi_{\theta,iT,K} + \sqrt{N/T} V_K^{-1} \mathcal{B}_{\theta,K}^{\text{cl}} + o_p(1).$$

Limit. Now use $N/T \rightarrow \kappa_0$, $V_K \rightarrow \gamma_v(0)$, and Proposition 5.5. This proves (5.17). The noncentral Gaussian limit follows from Proposition E.2 and Slutsky's theorem. Localization transfers the result to the globally reported estimator. $\square$

E.3 Bias-Corrected Estimator and Studentized Inference

Proof. Section E.2 verifies the rate conditions and establishes the raw representation centered at θ_0 , with the finite-sieve drift $\sqrt{N/T}V_K^{-1}\mathcal{B}_{\theta,K}^{\text{cl}}$. Subtracting the feasible correction gives, on $\mathcal{G}_K$,

$$\begin{aligned}\sqrt{NT}(\hat{\theta}_K - \theta_0) &= \sqrt{NT}(\hat{\theta}_K - \theta_0) - \sqrt{N/T}\hat{V}_K^{-1}\hat{B}_{\theta,K} \\ &= N^{-1/2} \sum_i \Psi_{\theta,iT,K} + \sqrt{N/T}\{V_K^{-1}\mathcal{B}_{\theta,K}^{\text{cl}} - \hat{V}_K^{-1}\hat{B}_{\theta,K}\} + o_p(1) \\ &= N^{-1/2} \sum_i \Psi_{\theta,iT,K} + o_p(1).\end{aligned}$$

The last equality uses (E.2) and $N/T \rightarrow \kappa_0$. The Gaussian limit and variance consistency are (E.5) and (E.6). Division by the consistent positive standard deviation gives the studentized limit. Remark C.21 transfers these conclusions to the reported estimator. $\square$

E.4 Serial-Independence Specialization

Proof. The inherited conditions ensure that Theorem 5.7 applies. Serial independence makes all nonzero-lag population covariances vanish, while mutual independence gives $\Omega_{v\zeta} = 0$. Hence the active derivative channel vanishes. Equation (5.15) reduces to $\text{Var}(\varepsilon_{it})/\text{Var}(v_{it})$, which proves the stated specialization. $\square$

F Proof of the Smooth Scalar Moment Theorem

Throughout this appendix, Assumption 6.1 holds. We first expand the observable coordinate and the target score. We then control the random-coordinate Gram matrix and derive the nuisance rate from its normal equations. The remaining steps compare the feasible lag statistics to their oracle versions and apply a central limit theorem across units. For brevity, write $h = \mathfrak{h}_K$ and omit the superscript M from estimated second-stage objects within this appendix. Every such object is the procedure in Section 6, with acceptance event $\mathcal{O}_T$ and threshold $a_T = c_{\text{tol}}T^{-1/4}$.

The growth conditions imply

$$\begin{aligned}\sqrt{T}h &\rightarrow \infty, & b_T/h &\rightarrow 0, & Nh &\rightarrow \infty, \\ K \log(2K)/(NT) &\rightarrow 0, & K_x \log(2NK_x)/T &\rightarrow 0.\end{aligned}$$

For example, $h^{-1} = o(T^{1/(dw+4)})$ bounds each power of h^{-1} in these expressions by a strictly smaller power of T than its denominator; the remaining logarithms are absorbed. The deterministic chain in Lemmas C.1, C.2, C.3, and C.4 therefore applies to the present grid and spline. In particular,

$$\begin{aligned}cI_K &\preceq \Sigma_{p,K} \preceq CI_K, & \Sigma_{p,K} &= \frac{1}{N} \sum_i \mathbb{E}_X[p_K(X, \alpha_i)p_K(X, \alpha_i)^\top], \\ \sup_{x,a} \|\partial_a^r p_K(x, a)\| &\leq C\sqrt{K}h^{-r}, & \|\partial_a^r(l_0 - \Pi_{K,N}^g l_0)\|_\infty &\leq Ch^{\tau_l - r}, \quad 0 \leq r \leq 3,\end{aligned}\tag{F.1}$$

where l_0 is m_0 or h_0 and the suprema are on W^+ . Consequently the first three coordinate derivatives of the projected functions, and of the scalar score at the population coefficients, have constant envelopes. The basis-vector envelopes have the displayed $\sqrt{K}h^{-r}$ factors; these are different bounds.

Independence of X from (v, ε) and $\mathbb{E}[v] = \mathbb{E}[\varepsilon] = \mathbb{E}[v\varepsilon] = 0$ verify the moment conditions of Proposition 2.1. Consequently, $V_K = V + \mathbb{E}_{N,T}[e_{m,it,K}^2] \geq V > 0$ and $|\Delta_{\theta,K,N}| \leq Ch^{\tau_m + \tau_h}$.

F.1 Observable Root and Score Expansion

Lemma F.1 (Observable-root expansion). *Let $\Delta_i = \check{\alpha}_i - \alpha_i$ and $d_i = \hat{\alpha}_i - \alpha_i$. There are uniform constants $C, c > 0$ such that*

$$\begin{aligned}\mathbb{P}_i(|\Delta_i| > x) &\leq C \exp(-cTx^2) + C \exp(-cT), \\ \|d_i\|_{L_i^r} + \|\bar{\xi}_i\|_{L_i^r} &\leq C_r T^{-1/2}, \quad 2 \leq r < \infty.\end{aligned}$$

The expansion (3.3) holds with the stated uniform moments. Also $\max_i |d_i| = O_p(b_T)$ and $N^{-1} \sum_i |d_i|^r = O_p(T^{-r/2})$ for every fixed $r \geq 2$.

Proof. For a centered bounded L -dependent scalar array, split time into $L+1$ residue classes. The observations in each class are independent. Hoeffding's inequality in each class and a union bound give $\mathbb{P}_i(|T^{-1} \sum_t Q_{it}| > x) \leq C \exp(-cTx^2)$. Apply this to the moment at α_i and to its first two derivatives after centering. Define

$$M_{iT} = \frac{1}{T} \sum_t \{\partial_a \psi(W_{it}, \alpha_i) - G_i\}, \quad \bar{H}_i^\psi = \frac{1}{T} \sum_t \partial_{aa} \psi(W_{it}, \alpha_i).$$

Tail integration yields $O(T^{-1/2})$ L_i^r bounds for $\bar{\xi}_i, M_{iT}, \bar{H}_i^\psi - H_i^\psi$ at every fixed finite moment r .

Use a grid of spacing T^{-2} on $\mathcal{A}$. It has $O(T^2)$ points. The preceding bound and the uniform bound on $\partial_{aa} \psi$ imply

$$\mathbb{P}_i(\mathcal{J}_i^c) \leq C \exp(-cT), \quad \mathcal{J}_i = \left\{ \sup_{a \in \mathcal{A}} |\Psi'_{iT}(a) - \mathbb{E}_i[\partial_a \psi(W_{i0}, a)]| \leq c_G/2 \right\}. \quad (\text{F.2})$$

Indeed, deviation at most $c_G/4$ at every grid point, together with a CT^{-2} interpolation error, implies the event for large T ; the grid failure probability is at most $CT^2 \exp(-cT)$. The population moment has opposite signs at the two endpoints of $\mathcal{A}$ and absolute value at least $c_G \delta$ there. Endpoint concentration preserves both signs except with probability $C \exp(-cT)$. On this endpoint event intersected with $\mathcal{J}_i$, the sample moment is strictly monotone, has exactly one zero, and

$$|\Delta_i| \leq 2c_G^{-1} |\Psi_{iT}(\alpha_i)|.$$

The fallback is bounded. These facts prove the root tail and all fixed moments. Projection onto the clipping interval is a contraction about α_i , so $|d_i| \leq |\Delta_i|$.

On the root event, Taylor's theorem gives the exact equation

$$0 = -G_i \bar{\xi}_i + G_i \Delta_i + M_{iT} \Delta_i + \frac{1}{2} \bar{H}_i^\psi \Delta_i^2 + R_{\psi,i}, \quad |R_{\psi,i}| \leq C |\Delta_i|^3,$$

$$\Delta_i - \bar{\xi}_i = -G_i^{-1} \left(M_{iT} \Delta_i + \frac{1}{2} \bar{H}_i^\psi \Delta_i^2 + R_{\psi,i} \right).$$

Hölder's inequality bounds the three terms on the last line in L_i^r by $CT^{-1}, CT^{-1}, CT^{-3/2}$. The exponentially rare complement has a bounded envelope. Thus $\|\Delta_i - \bar{\xi}_i\|_{L_i^r} = O(T^{-1})$. For this appendix use the population Hessian in the quadratic approximation:

$$q_{iT}^M = -G_i^{-1} M_{iT} \bar{\xi}_i - \frac{1}{2} G_i^{-1} H_i^\psi \bar{\xi}_i^2. \quad (\text{F.3})$$

It differs from the sample-Hessian convention in (B.4) by $O_{L_i^r}(T^{-3/2})$. Direct subtraction gives

$$\Delta_i - \bar{\xi}_i - q_{iT}^M = -G_i^{-1} \left\{ M_{iT} (\Delta_i - \bar{\xi}_i) + \frac{1}{2} \bar{H}_i^\psi (\Delta_i^2 - \bar{\xi}_i^2) + \frac{1}{2} (\bar{H}_i^\psi - H_i^\psi) \bar{\xi}_i^2 + R_{\psi,i} \right\}.$$

Because $\Delta_i^2 - \bar{\xi}_i^2 = (\Delta_i - \bar{\xi}_i)(\Delta_i + \bar{\xi}_i)$, its L_i^r norm is $O(T^{-3/2})$. Each term in braces therefore has that order. The bounded exceptional branch again contributes an exponentially small moment.

Clipping changes the raw root by at most $|\Delta_i| \mathbf{1}\{|\Delta_i| > b_T\}$. Tail integration gives

$$\mathbb{E}_i |\hat{\alpha}_i - \check{\alpha}_i|^r \leq C_r b_T^r N^{-cC_b^2} + C_r \exp(-cT).$$

For $2 \leq r \leq 8$, $cC_b^2 > 16$ and $N \asymp T$ make its L_i^r norm $o(T^{-3/2})$. Hence $d_i = \bar{\xi}_i + q_{iT}^M + O_{L_i^r}(T^{-3/2})$.

For stationary arrays U, V , the exact finite-window identity is

$$T \mathbb{E}_i [\bar{U}_i \bar{V}_i] = \sum_{|\ell| < T} (1 - |\ell|/T) \mathbb{E}_i [U_{i0} V_{i\ell}]. \quad (\text{F.4})$$

For the two products in (F.3), centering and finite dependence eliminate all $|\ell| > L$. Substitution shows $|T \mathbb{E}_i [q_{iT}^M] - \beta_i| \leq C/T$. Set $q_{iT}^c = q_{iT}^M - \mathbb{E}_i [q_{iT}^M]$ and absorb $\mathbb{E}_i [q_{iT}^M] - \beta_i/T = O(T^{-2})$ into r_{iT}^M . This proves (3.3).

Markov's inequality gives the average moment assertion; the tail bound and a union bound give the maximum assertion. For bounded deterministic weights λ_i , independence across units further gives

$$\mathbb{E} \left| \frac{\sqrt{NT}}{N} \sum_i \lambda_i q_{iT}^c \right|^2 = \frac{T}{N} \sum_i \lambda_i^2 \text{Var}_i(q_{iT}^c) \leq C/T.$$

This justifies discarding the centered quadratic term only after averaging. $\square$

Lemma F.2 (Scalar-score expansion). *Under the maintained conditions,*

$$\mathbb{P}_{NT} f_{it,K}^\theta(\rho_{0,K,N}, \hat{\alpha}_i) = \mathbb{P}_{NT} x_{it,K}^M + \frac{B_{\theta,K,N}^M}{T} + O_p\left(\frac{1}{T\sqrt{N}} + T^{-3/2}\right). \quad (\text{F.5})$$

The exact clipped population coefficient differs from $B_{\theta,K,N}^M$ by $O(T^{-1/2})$.

Proof. In this proof only, write $f_{it}(a) = f_{it,K}^\theta(\rho_{0,K,N}, a)$, $A_{it} = \partial_a f_{it}(\alpha_i)$, $H_{it} = \partial_{aa} f_{it}(\alpha_i)$, and $\lambda_i = \mathbb{E}_i[A_{it}]$, $\mu_{H,i} = \mathbb{E}_i[H_{it}]$. These arrays have bounded envelopes by (F.1). Let $r_i^0 = d_i - \bar{\xi}_i - q_{iT}^M$. Taylor's formula and addition and subtraction of the two population derivatives give the exact decomposition

$$\begin{aligned} \bar{f}_i(\hat{\alpha}_i) &= \bar{f}_i(\alpha_i) + \lambda_i \bar{\xi}_i + (\bar{A}_i - \lambda_i) \bar{\xi}_i + \lambda_i q_{iT}^M + \frac{1}{2} \mu_{H,i} \bar{\xi}_i^2 + R_i, \\ R_i &= (\bar{A}_i - \lambda_i) q_{iT}^M + \bar{A}_i r_i^0 + \frac{1}{2} (\bar{H}_i - \mu_{H,i}) \bar{\xi}_i^2 \\ &\quad + \frac{1}{2} \bar{H}_i (d_i^2 - \bar{\xi}_i^2) + \frac{d_i^3}{2T} \sum_t \int_0^1 (1-s)^2 \partial_{aaa} f_{it}(\alpha_i + sd_i) ds. \end{aligned}$$

The path remains in $[0, 1]$. Finite-dependence moment bounds and Lemma F.1 yield, term by term,

$$\begin{aligned} \mathbb{E}_i |R_i| &\leq \|\bar{A}_i - \lambda_i\|_{L_i^2} \|q_{iT}^M\|_{L_i^2} + C \|r_i^0\|_{L_i^1} + C \|\bar{H}_i - \mu_{H,i}\|_{L_i^2} \|\bar{\xi}_i\|_{L_i^4}^2 \\ &\quad + C \|d_i - \bar{\xi}_i\|_{L_i^2} \|d_i + \bar{\xi}_i\|_{L_i^2} + C \mathbb{E}_i |d_i|^3 \leq CT^{-3/2}. \end{aligned}$$

No third-moment cancellation is used here. Each of the three displayed quadratic terms preceding R_i has L_i^2 norm $O(T^{-1})$; for the first this follows from the fourth moments of its two centered averages. Their centered cross-unit average is therefore $O_p(1/(T\sqrt{N}))$. Equation (F.4) gives their expectations as the three terms in (3.4), divided by T , with total error $O(T^{-2})$. Finally, the grid projection implies $N^{-1} \sum_i \mathbb{E}_i[f_{it}(\alpha_i)] = 0$. Averaging the exact decomposition proves (F.5). Taking expectations first and multiplying by T gives the population error $O(T^{-1/2})$ separately from the centered fluctuation. $\square$

F.2 Uniform Gram Control for Domain-Preserving Shifts

Lemma F.3 (Random-coordinate Gram bound). *For $0 \leq \rho \leq h$, let*

$$\begin{aligned} \mathcal{D}_\rho &= \{d \in \mathbb{R}^N : \max_i |d_i| \leq \rho, \alpha_i + d_i \in [0, 1]\}, \\ G(d) &= \mathbb{P}_{NT} p_K(X_{it}, \alpha_i + d_i) p_K(X_{it}, \alpha_i + d_i)^\top. \end{aligned}$$

There is an event $\mathcal{E}_x$, depending only on the covariates, with $\mathbb{P}(\mathcal{E}_x) \rightarrow 1$, on which, simultaneously for every $0 \leq \rho \leq h$, $d \in \mathcal{D}_\rho$, $s \in [0, 1]$, and $c \in \mathbb{R}^K$,

$$\begin{aligned} \mathbb{P}_{NT} [c^\top \partial_a^r p_K(X_{it}, \alpha_i + sd_i)]^2 &\leq Ch^{-2r} \|c\|^2, \quad r = 0, 1, \\ \sup_{d \in \mathcal{D}_\rho} \|G(d) - G(0)\|_{\text{op}} &\leq C\{\rho/h + (\rho/h)^2\}. \end{aligned}$$

Also

$$\|G(0) - \Sigma_{p,K}\|_{\text{op}} = O_p\left(\sqrt{\frac{K \log(2K)}{NT}} + \frac{K \log(2K)}{NT}\right),$$

$$\mathbb{P}(c_* I_K \preceq \hat{\Sigma}_{p,K} \preceq C_* I_K) \longrightarrow 1$$

for fixed positive c_*, C_* .

Proof. Write the tensor factors as $\psi_{K_x}(x)$ and $\phi_{K_a}(a)$, as in Assumption 5.2(iv). Consider a pair of overlapping basis functions j, k , and put $U_{it} = p_{K,j}(X_{it}, \alpha_i) p_{K,k}(X_{it}, \alpha_i)$. Local support and normalization give $|U_{it}| \leq CK$ and

$$\mathbb{E}_X[\psi_{j_x}(X)^2 \psi_{k_x}(X)^2] \leq CK_x, \quad \#\{i : \alpha_i \in \text{supp}(\phi_{j_a} \phi_{k_a})\} \leq CNh.$$

The count follows from the minimum spacing and support length Ch . Consequently

$$\frac{1}{NT} \sum_{i,t} \mathbb{E}[U_{it}^2] \leq CK_x K_a^2 h \leq CK, \quad \sum_{i,t} \text{Var}(U_{it}) \leq CNTK.$$

This is an average second-moment bound; the second moment for a single unit can instead have order $K_x K_a^2$. Independence of the covariates and Bernstein's inequality imply, for $u \geq 1$,

$$\mathbb{P} \left(\left| \frac{1}{NT} \sum_{i,t} (U_{it} - \mathbb{E} U_{it}) \right| > C \left\{ \sqrt{\frac{Ku}{NT}} + \frac{Ku}{NT} \right\} \right) \leq 2e^{-u}.$$

There are at most CK nonzero entries and at most C entries per row. A union bound followed by $\|A\|_{\text{op}} \leq \max_j \sum_k |A_{jk}|$ proves the asserted oracle rate without an additional K factor. The same argument within each unit gives

$$G_{x,i} = \frac{1}{T} \sum_t \psi_{K_x}(X_{it}) \psi_{K_x}(X_{it})^\top, \\ \max_i \|G_{x,i} - \mathbb{E}_X[\psi_{K_x}(X) \psi_{K_x}(X)^\top]\|_{\text{op}} = O_p \left(\sqrt{\frac{K_x \log(2NK_x)}{T}} + \frac{K_x \log(2NK_x)}{T} \right) = o_p(1).$$

The continuous Riesz bound permits a fixed M_x such that $\mathcal{E}_x = \{\max_i \|G_{x,i}\|_{\text{op}} \leq M_x\}$ has probability approaching one.

On this event fix any permitted shift and put $a_i = \alpha_i + sd_i$. Each a_i stays in $[0, 1]$. If J is a full positive knot span, then

$$n_J := \#\{i : a_i \in J\} \leq 1 + \frac{N}{c_\alpha} (|J| + 2\rho) \leq CNh.$$

Assign a point at a knot to either adjacent span. This count allows shifted points to coincide. At the boundary, use the full positive span of length comparable to h , rather than its intersection with the clipping interval. Reshape c into $B \in \mathbb{R}^{K_x \times K_a}$ with $\|B\|_F = \|c\|$ and $c^\top p_K(x, a) = \psi_{K_x}(x)^\top B \phi_{K_a}(a)$. For $g(a) = B \phi_{K_a}^{(r)}(a)$, scalar polynomial norm equivalence on a full span, applied to every component, gives

$$\sup_{a \in J} \|g(a)\|^2 \leq \sum_{j=1}^{K_x} \sup_{a \in J} |g_j(a)|^2 \leq Ch^{-1} \sum_{j=1}^{K_x} \int_J |g_j(a)|^2 da, \\ \frac{1}{N} \sum_i \|B \phi_{K_a}^{(r)}(a_i)\|^2 \leq \sum_J \frac{n_J}{N} \sup_{a \in J} \|g(a)\|^2 \leq C \int_0^1 \|B \phi_{K_a}^{(r)}(a)\|^2 da \leq Ch^{-2r} \|c\|^2.$$

The last inequality applies the one-dimensional Riesz and inverse bounds to each row of B . Their constants depend on the fixed degree and knot ratios, not on K_x . The tensor quadratic form now yields

$$\mathbb{P}_{NT}[c^\top \partial_a^r p_K(X_{it}, a_i)]^2 = \frac{1}{N} \sum_i [B \phi_{K_a}^{(r)}(a_i)]^\top G_{x,i} [B \phi_{K_a}^{(r)}(a_i)] \\ \leq \frac{M_x}{N} \sum_i \|B \phi_{K_a}^{(r)}(a_i)\|^2 \leq Ch^{-2r} \|c\|^2.$$

This deterministic argument on $\mathcal{E}_x$ holds simultaneously for every coefficient and domain-preserving shift, including data-dependent shifts.

For $\|c\| = 1$, put $z_{it} = c^\top p_K(X_{it}, \alpha_i)$. The fundamental theorem of calculus and Jensen's inequality give

$$\begin{aligned}\delta z_{it} &= d_i \int_0^1 c^\top \partial_a p_K(X_{it}, \alpha_i + s d_i) ds, \\ \|\delta z\|_{NT,2}^2 &\leq \rho^2 \int_0^1 \mathbb{P}_{NT}[c^\top \partial_a p_K(X_{it}, \alpha_i + s d_i)]^2 ds \leq C(\rho/h)^2, \\ |c^\top \{G(d) - G(0)\}c| &= |\mathbb{P}_{NT}(2z\delta z + \delta z^2)| \leq 2\|z\|_{NT,2}\|\delta z\|_{NT,2} + \|\delta z\|_{NT,2}^2 \leq C\{\rho/h + (\rho/h)^2\}.\end{aligned}$$

Take suprema over c and d . By Lemma F.1, the actual shifts satisfy $\max_i |d_i| = O_p(b_T)$, and $b_T/h \rightarrow 0$. Combining the perturbation bound with the oracle rate and (F.1) proves the last assertion. When $d_X = 0$, $\psi_{K_x} \equiv 1$, $G(0) = \Sigma_{p,K}$ exactly, and $\mathcal{E}_x$ can be the whole sample space. $\square$

F.3 Nuisance Rate, Acceptance, and Exact Profiling

Lemma F.4 (Dimension-independent vector moment bounds). *Let $Q_t \in \mathbb{R}^K$ be centered, L -dependent, and satisfy $\|Q_t\| \leq A_K$. Then*

$$\mathbb{E} \left\| \frac{1}{T} \sum_t Q_t \right\|^2 \leq C_L A_K^2 / T, \quad \mathbb{E} \left\| \frac{1}{T} \sum_t Q_t \right\|^4 \leq C_L A_K^4 / T^2.$$

The constant is independent of the vector dimension and requires no independence among vector coordinates.

Proof. Expand the squared norm as $\sum_{s,t} \mathbb{E} \langle Q_s, Q_t \rangle$. Only the at most $T(2L+1)$ pairs with $|s-t| \leq L$ survive centering and independence, and each has absolute value at most A_K^2 . For the fourth moment, expand

$$\mathbb{E} \left\| \sum_t Q_t \right\|^4 = \sum_{t_1, t_2, t_3, t_4} \mathbb{E} [\langle Q_{t_1}, Q_{t_2} \rangle \langle Q_{t_3}, Q_{t_4} \rangle].$$

Join two of the four labeled dates whenever their distance is at most L , including repeated dates. An isolated vertex is independent of the other three vectors, and the integrand is linear in that centered vector, so its expectation vanishes. A graph without an isolated vertex has at most two connected components. Choose one representative date per component in at most T^2 ways. All other dates are within at most three links of a representative and have only C_L possible offsets. There are a fixed number of labeled graph types. At most $C_L T^2$ terms remain, each bounded by A_K^4 by two inner-product inequalities. Division by T^4 proves the claim without expanding vector coordinates. $\square$

Lemma F.5 (Nuisance rate and derivative consistency). *Put $r_{\eta,K}^M = \sqrt{K}h^{-2}/T$. Then $\mathbb{P}(\mathcal{O}_T) \rightarrow 1$, $\hat{V}_K - V_K = o_p(1)$, and*

$$\begin{aligned}\|\hat{\gamma}_K - \gamma_{0,K,N}\| + \|\hat{\delta}_K - \delta_{0,K,N}\| &= O_p(r_{\eta,K}^M), \\ \sup_{x,a} |\partial_a^r (\hat{l}_K - l_K)(x, a)| &= O_p(Kh^{-(r+2)}/T) = o_p(1), \quad r = 0, 1, 2,\end{aligned}$$

where $(\hat{l}_K, l_K)$ is either fitted and population projection pair. On $\mathcal{O}_T$, the exact profile identity (5.1) holds for these objects, with a profile remainder $o_p((NT)^{-1/2})$. Also $\hat{\theta}_K - \theta_{0,K,N} = O_p(T^{-1})$.

Proof. For $(R, l_K) = (D, m_{K,N})$ or $(Y, g_{K,N})$, let $F_{it,K}(a) = p_K(X_{it}, a)\{R_{it} - l_K(X_{it}, a)\}$. Write $b_{0,K,N}$ for the corresponding projection coefficient $\gamma_{0,K,N}$ or $\delta_{0,K,N}$, and $\hat{b}_K$ for its fit. These are the original nuisance normal-equation contributions. Product differentiation gives

$$\begin{aligned}F' &= p_a(R - l) - pl_a, \\ F'' &= p_{aa}(R - l) - 2p_a l_a - pl_{aa}, \\ F''' &= p_{aaa}(R - l) - 3p_{aa} l_a - 3p_a l_{aa} - pl_{aaa}.\end{aligned}$$

By (F.1), the entire vector $F^{(r)}$ has envelope $A_r = C\sqrt{K}h^{-r}$, $0 \leq r \leq 3$. Write $\mu_{F,i} = \mathbb{E}_i[F_{it,K}(\alpha_i)]$ and $\mu_{F',i} = \mathbb{E}_i[F'_{it,K}(\alpha_i)]$. Since $\sum_i \mu_{F,i} = 0$ by projection, Taylor's formula gives exactly

$$\begin{aligned} \frac{1}{N} \sum_i \bar{F}_{i,K}(\hat{\alpha}_i) &= \frac{1}{N} \sum_i (\bar{F}_{i,K} - \mu_{F,i}) + \frac{1}{N} \sum_i \mu_{F',i} \bar{\xi}_i \\ &\quad + \frac{1}{N} \sum_i (\bar{F}'_{i,K} - \mu_{F',i}) \bar{\xi}_i + \frac{1}{N} \sum_i \bar{F}'_{i,K} (d_i - \bar{\xi}_i) \\ &\quad + \frac{1}{2N} \sum_i \bar{F}''_{i,K} d_i^2 + \frac{1}{N} \sum_i R_{F,i}, \quad \|R_{F,i}\| \leq CA_3 |d_i|^3. \end{aligned}$$

Derivatives without an argument are evaluated at α_i . Independence across units and Lemma F.4 imply

$$\begin{aligned} \mathbb{E} \left\| \frac{1}{N} \sum_i (\bar{F}_{i,K} - \mu_{F,i}) \right\|^2 &\leq CA_0^2/(NT), \\ \mathbb{E} \left\| \frac{1}{N} \sum_i \mu_{F',i} \bar{\xi}_i \right\|^2 &= \frac{1}{N^2} \sum_i \|\mu_{F',i}\|^2 \mathbb{E}_i \bar{\xi}_i^2 \leq CA_1^2/(NT). \end{aligned}$$

The second line uses $\mathbb{E}_i \bar{\xi}_i = 0$, not a zero derivative mean. For the remaining four terms, the expected norms are bounded by

$$\begin{aligned} \frac{1}{N} \sum_i \|\bar{F}'_{i,K} - \mu_{F',i}\|_{L_i^2} \|\bar{\xi}_i\|_{L_i^2} &\leq CA_1/T, \\ \frac{A_1}{N} \sum_i \mathbb{E}_i |d_i - \bar{\xi}_i| &\leq CA_1/T, \\ \frac{A_2}{2N} \sum_i \mathbb{E}_i d_i^2 &\leq CA_2/T, \\ \frac{CA_3}{N} \sum_i \mathbb{E}_i |d_i|^3 &\leq CA_3/T^{3/2}. \end{aligned}$$

Markov's inequality and $N \asymp T$, $\sqrt{T}h \rightarrow \infty$ show that all six terms are $O_p(\sqrt{K}h^{-2}/T)$. This calculation permits arbitrary within-unit correlation between F and ξ . It uses the vector envelope once.

On the Gram acceptance event, the candidate coefficient error obeys the exact normal equation

$$b_K^{\text{pre}} - b_{0,K,N} = \hat{\Sigma}_{p,K}^{-1} \mathbb{P}_{NT} F_{it,K}(\hat{\alpha}_i).$$

Lemma F.3 bounds its inverse in probability and proves the candidate coefficient rate. It remains to prove acceptance of the candidate denominator before transferring this conclusion to reported objects. For the candidate treatment residual, the Gram energy bound gives

$$\begin{aligned} \|u_D^{\text{pre}} - u_{D,K}\|_{NT,2} &\leq \|\hat{p}^\top (\gamma_K^{\text{pre}} - \gamma_{0,K,N})\|_{NT,2} + C \left(\frac{1}{N} \sum_i d_i^2 \right)^{1/2} \\ &= O_p(r_{\eta,K}^M + T^{-1/2}), \\ |V_K^{\text{pre}} - \mathbb{P}_{NT} u_{D,K}^2| &\leq \|u_D^{\text{pre}} - u_{D,K}\|_{NT,2} \{2\|u_{D,K}\|_{NT,2} + \|u_D^{\text{pre}} - u_{D,K}\|_{NT,2}\} = o_p(1). \end{aligned}$$

The finite-range variance bound implies $\mathbb{P}_{NT} u_{D,K}^2 - V_K = O_p((NT)^{-1/2})$. Also $V_K = V + \mathbb{E}_{N,T}[e_{m,it,K}^2] \rightarrow V > 0$. Since the Gram test passes with probability approaching one and $a_T \rightarrow 0$, the candidate denominator passes with probability approaching one. This proves $\mathbb{P}(\mathcal{O}_T) \rightarrow 1$; only now do we transfer the probability rates to the reported coefficients and denominator.

On $\mathcal{O}_T$, the candidate construction satisfies all three profile normal equations. Lemma 5.1 therefore gives (5.1) for these coordinates and coefficients. The coefficient rate and (6.2) give

$$\sqrt{NT} |e_\gamma^\top \hat{\Sigma}_{p,K} e_\pi| = O_p(T(r_{\eta,K}^M)^2) = O_p(Kh^{-4}/T) = o_p(1).$$

Lemma F.2 now proves $\widehat{\theta}_K - \theta_{0,K,N} = O_p(T^{-1})$. Finally, for $r = 0, 1, 2$,

$$\sup_{x,a} |\partial_a^r (\widehat{l}_K - l_K)(x, a)| \leq C\sqrt{K}h^{-r} \|\widehat{b}_K - b_{0,K,N}\| = O_p(Kh^{-(r+2)}/T) = o_p(1),$$

as claimed. No global moment rate for the second-stage fallback is used. $\square$

F.4 Feasible First-Step Influence and Bias

Lemma F.6 (Truncated-inverse and lag-statistic moments). *For every fixed finite $r \geq 2$, uniformly in i ,*

$$\begin{aligned} \|\widehat{G}_i^\psi - G_i\|_{L_i^r} + \|\widehat{H}_i^\psi - H_i^\psi\|_{L_i^r} + \|(\widehat{G}_i^\psi)^\ominus - G_i^{-1}\|_{L_i^r} + \left\| \max_{t \leq T} |\widehat{\xi}_{it} - \xi_{it}| \right\|_{L_i^r} &\leq C_r T^{-1/2}, \\ \|\widehat{\Omega}_{\xi,i} - \Omega_{\xi,i}\|_{L_i^2} + \|\widehat{C}_{G\xi,i} - C_{G\xi,i}\|_{L_i^2} + \|\widehat{\beta}_i - \beta_i\|_{L_i^2} &\leq CT^{-1/2}. \end{aligned}$$

Furthermore, $\max_{i,t} |\widehat{\xi}_{it}| = O_p(1)$ and $N^{-1} \sum_i (\widehat{\beta}_i^2 + \widehat{\Omega}_{\xi,i}^2) = O_p(1)$.

Proof. The event $\mathcal{J}_i$ in (F.2) controls the sample Jacobian over the entire interval. On that event, for large T ,

$$|\widehat{G}_i^\psi| \geq c_G/2 > a_T, \quad |(\widehat{G}_i^\psi)^\ominus| \leq 2/c_G.$$

On the whole sample space only the bound $|(\widehat{G}_i^\psi)^\ominus| \leq c_{\text{tol}}^{-1} T^{1/4}$ is available. Passing the observable threshold does not itself imply a constant envelope. Bounded derivatives and Lemma F.1 give

$$\begin{aligned} |\widehat{G}_i^\psi - G_i| &\leq |M_{iT}| + C|d_i|, \\ |\widehat{H}_i^\psi - H_i^\psi| &\leq |\bar{H}_i^\psi - H_i^\psi| + C|d_i|. \end{aligned}$$

Put $I_i = (\widehat{G}_i^\psi)^\ominus$ temporarily. On $\mathcal{J}_i$,

$$|I_i - G_i^{-1}| = \frac{|\widehat{G}_i^\psi - G_i|}{|\widehat{G}_i^\psi G_i|} \leq C(|M_{iT}| + |d_i|).$$

The complementary event has probability $C \exp(-cT)$ and an envelope $C(1 + T^{1/4})$. Its L_i^r contribution is at most $C_r(1 + T^{1/4}) \exp(-cT/r)$, proving the inverse rate.

To control the maximum over dates, let $z_{it} = \psi(W_{it}, \alpha_i)$ and $\widehat{z}_{it} = \psi(W_{it}, \widehat{\alpha}_i) - T^{-1} \sum_s \psi(W_{is}, \widehat{\alpha}_i)$. The exact differences are

$$\begin{aligned} \widehat{z}_{it} - z_{it} &= \{\psi(W_{it}, \widehat{\alpha}_i) - \psi(W_{it}, \alpha_i)\} - \frac{1}{T} \sum_s \{\psi(W_{is}, \widehat{\alpha}_i) - \psi(W_{is}, \alpha_i)\} - \bar{z}_i, \\ \widehat{\xi}_{it} - \xi_{it} &= -(I_i - G_i^{-1})z_{it} - I_i(\widehat{z}_{it} - z_{it}). \end{aligned}$$

Therefore, on $\mathcal{J}_i$,

$$D_{\xi,i} := \max_t |\widehat{\xi}_{it} - \xi_{it}| \leq C(|M_{iT}| + |d_i| + |\bar{z}_i|).$$

The right side contains only unit-level averages and the coordinate error; no union bound over t is needed. Its L_i^r norm is $O(T^{-1/2})$. On $\mathcal{J}_i^c$, $D_{\xi,i} \leq C(1 + T^{1/4})$, whose moment contribution is exponentially small. This proves the influence rate even when clipping binds. The subtracted mean in (6.4) is zero when selection succeeds and clipping is inactive.

Set $s_{it} = \partial_a \psi(W_{it}, \alpha_i) - G_i$ and $\widehat{s}_{it} = \widehat{G}_{it}^\psi - \widehat{G}_i^\psi$. Their maximum difference satisfies $D_{s,i} \leq 2C|d_i| + |M_{iT}|$. For $(U, V) = (\xi, \xi)$ or (s, ξ) , the fixed-lag oracle product $U_{it}V_{i,t+\ell}$ is bounded and has dependence range at most $2L$. Thus

$$\begin{aligned} \text{Var}_i \left(\frac{1}{T} \sum_{t \in I_T(\ell)} U_{it}V_{i,t+\ell} \right) &\leq \frac{C}{T^2} \sum_{t=1}^T \#\{u : |t-u| \leq 2L\} \leq C/T, \\ \mathbb{E}_i \left[\frac{1}{T} \sum_{t \in I_T(\ell)} U_{it}V_{i,t+\ell} \right] &= (1 - |\ell|/T) \mathbb{E}_i[U_{i0}V_{i\ell}]. \end{aligned}$$

The oracle lag statistics have L_i^2 error $O(T^{-1/2})$ relative to their population sums. On $\mathcal{J}_i$, replacing one factor at a time gives

$$\begin{aligned} |\mathcal{L}_{i,T}(\widehat{\xi}, \widehat{\xi}) - \mathcal{L}_{i,T}(\xi, \xi)| &\leq CD_{\xi,i}, \\ |\mathcal{L}_{i,T}(\widehat{s}, \widehat{\xi}) - \mathcal{L}_{i,T}(s, \xi)| &\leq C(D_{s,i} + D_{\xi,i}). \end{aligned}$$

On the complementary event the whole-space envelopes, with constants allowed to depend on the fixed c_{tol} , are

$$\max_t |\widehat{\xi}_{it}| \leq CT^{1/4}, \quad |\widehat{C}_{G\xi,i}| \leq CT^{1/4}, \quad |\widehat{\Omega}_{\xi,i}| \leq CT^{1/2}, \quad |\widehat{\beta}_i| \leq CT^{3/4}.$$

Their L_i^2 contributions on $\mathcal{J}_i^c$ are these polynomials multiplied by $C \exp(-cT/2)$. On $\mathcal{J}_i$, subtract (3.2) from (6.5) and replace factors to obtain

$$\begin{aligned} |\widehat{\beta}_i - \beta_i| &\leq C(|I_i - G_i^{-1}| + |\widehat{C}_{G\xi,i} - C_{G\xi,i}| \\ &\quad + |\widehat{H}_i^\psi - H_i^\psi| + |\widehat{\Omega}_{\xi,i} - \Omega_{\xi,i}|). \end{aligned}$$

This completes all moment bounds. Markov's inequality yields the average square bound. Finally $\mathbb{P}(\cap_i \mathcal{J}_i) \geq 1 - CN \exp(-cT) \rightarrow 1$, which gives the maximum bound. Finite dependence has been applied only to the oracle arrays, never to the estimated influences using the full history. $\square$

F.5 Correction, Variance, and the Limit Distribution

Proof of Theorem 6.2. Consistency of all three channels. Use the oracle derivatives

$$A_{it} = \dot{f}_{it,K}^\theta(\rho_{0,K,N}, \alpha_i), \quad H_{it} = \ddot{f}_{it,K}^\theta(\rho_{0,K,N}, \alpha_i),$$

with unit expectations $\lambda_i, \mu_{H,i}$. Lemma F.5, bounded third derivatives of the oracle score, and $\max_i |d_i| = O_p(b_T)$ imply

$$\max_{i,t} |\widehat{A}_{it} - A_{it}| + \max_{i,t} |\widehat{H}_{it} - H_{it}| = O_p(b_T + Kh^{-4}/T + T^{-1}) = o_p(1).$$

Here one first replaces coefficients at $\widehat{\alpha}_i$, using the fitted derivatives through order two, then moves the oracle score back to α_i . For the reuse channel, the exact difference is

$$\begin{aligned} &\frac{1}{N} \sum_i \mathcal{L}_{i,T}(\widehat{A}, \widehat{\xi}) - \frac{1}{N} \sum_i \sum_{|\ell| \leq L} \mathbb{E}_i[A_{i0} \xi_{i\ell}] \\ &= \frac{1}{N} \sum_i \mathcal{L}_{i,T}(\widehat{A} - A, \widehat{\xi}) + \frac{1}{N} \sum_i \mathcal{L}_{i,T}(A, \widehat{\xi} - \xi) \\ &\quad + \frac{1}{N} \sum_i \{\mathcal{L}_{i,T}(A, \xi) - \mathbb{E}_i \mathcal{L}_{i,T}(A, \xi)\} + \frac{1}{N} \sum_i \left\{ \mathbb{E}_i \mathcal{L}_{i,T}(A, \xi) - \sum_{|\ell| \leq L} \mathbb{E}_i[A_{i0} \xi_{i\ell}] \right\}. \end{aligned}$$

The four terms have respective orders $o_p(1)$, $O_p(T^{-1/2})$, $O_p((NT)^{-1/2})$, and $O(T^{-1})$: use derivative consistency and Lemma F.6 for the first two, finite-range variance and independent units for the third, and the exact window weights for the fourth.

For the first-step channel, expand

$$\begin{aligned} \widehat{\bar{A}}_i \widehat{\beta}_i - \lambda_i \beta_i &= (\widehat{\bar{A}}_i - \bar{A}_i) \widehat{\beta}_i + (\bar{A}_i - \lambda_i) \widehat{\beta}_i + \lambda_i (\widehat{\beta}_i - \beta_i), \\ \frac{1}{N} \sum_i |(\bar{A}_i - \lambda_i) \widehat{\beta}_i| &\leq \left\{ \frac{1}{N} \sum_i (\bar{A}_i - \lambda_i)^2 \right\}^{1/2} \left\{ \frac{1}{N} \sum_i \widehat{\beta}_i^2 \right\}^{1/2} = O_p(T^{-1/2}). \end{aligned}$$

The average absolute values of the other two terms are $o_p(1)$ and $O_p(T^{-1/2})$. The same three-factor decomposition with (A, λ, β) replaced by (H, μ_H, Ω_ξ) proves curvature consistency. No independence between these fitted factors is used. Thus

$$\widehat{B}_{\theta,K,N}^M - B_{\theta,K,N}^M = o_p(1).$$

Oracle influence and positive variance. At the true coordinate,

$$\begin{aligned} u_{D,K} &= v + e_m, & q_K &= \varepsilon + e_h - \Delta_{\theta,K,N}(v + e_m), \\ \partial_a f_K^\theta &= -\partial_a m_{K,N} \{ \varepsilon + e_h - \Delta_{\theta,K,N}(v + e_m) \} \\ &\quad - (v + e_m) \{ \partial_a h_{K,N}^0 - \Delta_{\theta,K,N} \partial_a m_{K,N} \}. \end{aligned}$$

Taking expectation over (X, v, ε) eliminates the terms linear in v or ε and proves (3.5). Together with (F.1), it gives

$$\begin{aligned} \sup_i |\lambda_i| &\leq Ch^{\min\{\tau_m, \tau_h\}}, \quad \sup_i |\beta_i| \leq C, \quad |B_{\text{first},K,N}^M| \leq Ch^{\min\{\tau_m, \tau_h\}} \rightarrow 0, \\ \sup_i \|f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i) - v_{it}\varepsilon_{it}\|_{L_i^2} &\rightarrow 0, \\ \sup_i \|\chi_{it,K}^M - V^{-1}v_{it}\varepsilon_{it}\|_{L_i^2} &\rightarrow 0. \end{aligned}$$

The centered influence is defined in (6.7). The last bound also centers the oracle score, because $\mathbb{E}[v_{it}\varepsilon_{it}] = 0$. For each of the finitely many lags, Cauchy–Schwarz transfers this bound to the covariance products. Therefore

$$\sigma_{K,N,T}^2 := \frac{1}{N} \sum_i \sum_{|\ell| \leq L} (1 - |\ell|/T) \mathbb{E}_i[\chi_{i0,K}^M \chi_{i\ell,K}^M] \rightarrow \sigma_M^2 > 0.$$

Feasible variance. Define the uncentered oracle $\chi_{it,K}^\circ = V_K^{-1} \{ f_{it,K}^\theta(\rho_{0,K,N}, \alpha_i) + \lambda_i \xi_{it} \}$. It differs from $\chi_{it,K}^M$ by a unit constant, so their demeaned versions coincide exactly. The fitted-level and derivative bounds give

$$\|\hat{f} - f\|_{NT,2} = o_p(1), \quad \frac{1}{N} \sum_i (\bar{A}_i - \lambda_i)^2 = o_p(1), \quad \|\hat{\xi} - \xi\|_{NT,2} = O_p(T^{-1/2}).$$

Replacing the two factors of the coordinate term and then the denominator gives

$$\begin{aligned} \|\bar{A}\hat{\xi} - \lambda\xi\|_{NT,2} &\leq \max_{i,t} |\hat{\xi}_{it}| \left\{ \frac{1}{N} \sum_i (\bar{A}_i - \lambda_i)^2 \right\}^{1/2} + C\|\hat{\xi} - \xi\|_{NT,2} = o_p(1), \\ \|\hat{\chi} - \chi^\circ\|_{NT,2} &\leq |\hat{V}_K^{-1} - V_K^{-1}| \|f + \lambda\xi\|_{NT,2} \\ &\quad + |\hat{V}_K^{-1}| \{ \|\hat{f} - f\|_{NT,2} + \|\bar{A}\hat{\xi} - \lambda\xi\|_{NT,2} \} = o_p(1). \end{aligned}$$

Within-unit demeaning is an empirical L^2 contraction. If a tilde denotes that operation, the difference between feasible and oracle lag sums is at most

$$C\|\tilde{\chi} - \tilde{\chi}^M\|_{NT,2} \{ 2\|\tilde{\chi}^M\|_{NT,2} + \|\tilde{\chi} - \tilde{\chi}^M\|_{NT,2} \} = o_p(1).$$

For the undemeaned centered oracle χ^M , each lag sum has expectation given by $\sigma_{K,N,T}^2$ after averaging lags and units, and fluctuation $O_p((NT)^{-1/2})$. To check demeaning explicitly, suppress K, M and write the two truncated unit means at lag ℓ as $\bar{\chi}_i + r_{i\ell}^{(1)}$ and $\bar{\chi}_i + r_{i\ell}^{(2)}$, where $|r_{i\ell}^{(j)}| \leq C|\ell|/T$. If $S_{i\ell} = T^{-1} \sum_{t \in I_T(\ell)} \chi_{it} \chi_{i,t+\ell}$, direct multiplication gives

$$\begin{aligned} \tilde{S}_{i\ell} - S_{i\ell} &= -(1 + |\ell|/T) \bar{\chi}_i^2 - \bar{\chi}_i(r_{i\ell}^{(1)} + r_{i\ell}^{(2)}), \\ \mathbb{E}_i |\tilde{S}_{i\ell} - S_{i\ell}| &\leq C \mathbb{E}_i \bar{\chi}_i^2 + CT^{-1} \mathbb{E}_i |\bar{\chi}_i| \leq C/T. \end{aligned}$$

Thus demeaning changes the cross-unit lag average by $O_p(T^{-1})$. The raw feasible variance converges to $\sigma_M^2 > 0$, so its floor binds with probability tending to zero. This proves variance consistency. The demeaned and fitted arrays themselves have not been assumed to be finite dependent.

Cluster central limit theorem and final remainder. Set $\Psi_{iT,K}^M = T^{-1/2} \sum_t \chi_{it,K}^M$. These variables are independent across units and have zero mean. Lemma F.4, used for a scalar bounded oracle array, implies $\sup_i \mathbb{E}_i |\Psi_{iT,K}^M|^4 \leq C$. Their average variance is $\sigma_{K,N,T}^2 \rightarrow \sigma_M^2 > 0$. The Lyapunov ratio is bounded by

$$\frac{\sum_i \mathbb{E}_i |N^{-1/2} \Psi_{iT,K}^M|^4}{\{N^{-1} \sum_i \text{Var}_i(\Psi_{iT,K}^M)\}^2} \leq C/N \rightarrow 0.$$

Hence $N^{-1/2} \sum_i \Psi_{iT,K}^M \rightsquigarrow \mathcal{N}(0, \sigma_M^2)$. Substitute (F.5) into the exact profile identity and subtract (6.6). On $\mathcal{O}_T$ this gives

$$\begin{aligned} \sqrt{NT}(\hat{\theta}_K^{M,c} - \theta_0) &= \hat{V}_K^{-1} \frac{1}{\sqrt{NT}} \sum_{i,t} x_{it,K}^M + \sqrt{N/T} \hat{V}_K^{-1} (B_{\theta,K,N}^M - \hat{B}_{\theta,K,N}^M) \\ &\quad - \sqrt{NT} \hat{V}_K^{-1} e_\gamma^\top \hat{\Sigma}_{p,K} e_\pi + \sqrt{NT}(\theta_{0,K,N} - \theta_0) + o_p(1). \end{aligned}$$

The correction error is $o_p(1)$ by the first part of the proof. The profile term is $O_p(Kh^{-4}/T) = o_p(1)$ by Lemma F.5. The target term is $O(Th^{\tau_m + \tau_h}) = o(1)$ by the projection-error product identity. The score remainder after multiplication by $\sqrt{NT}$ is $O_p(N^{-1/2} + T^{-1/2}) = o_p(1)$. The leading centered sum is $O_p(1)$, so $\hat{V}_K^{-1}$ can be replaced by V_K^{-1} . This proves (6.8), and variance consistency gives studentization by Slutsky's theorem. The preceding arguments first proved $\mathbb{P}(\mathcal{O}_T) \rightarrow 1$, so all conclusions transfer to the globally reported procedure. Finally, omit the correction subtraction to obtain (6.9); a fixed noncentral limit additionally requires convergence of $B_{\theta,K,N}^M$, as stated. $\square$